

Fleet Management

Use **VCF Operations** as a VI administrator to manage identity and access, certificates, passwords, tags, configuration profiles, and availability of the VCF fleet.

For VCF, you use VCF Operations and in certain cases individual components for these capabilities. For vSphere Foundation, you perform the supported operations on the individual management components.

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Getting to Know VCF

To apply this documentation, you must be acquainted with the [VCF Product Overview](#), [VCF Design](#), and [VCF Release Notes](#).

Deploying or Converging to VCF and Setting Up Cloud Infrastructure

For information about deploying or converging to a VCF fleet or VCF Instance, see [Deployment, Convergence, and Upgrade](#). For information about adding infrastructure and configuring storage and networking for workloads in VCF Operations and the vSphere Client as a VI administrator, see [Building Cloud Infrastructure](#).

Managing the Lifecycle of Your VCF Platform

You can manage software binaries and update your VCF fleet and VCF Instance centrally in VCF Operations. See [Lifecycle Management](#).

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See [Administration SDKs, APIs, and CLI](#).

Configuring VCF Single Sign-On

You can set up VCF Single Sign-On (SSO) for VCF to streamline user access with centralized authentication. VCF Single Sign-On offers Enterprise Admin users the ability to utilize Single Sign-On and access VCF components such as vCenter, VCF Operations, VCF Automation, VCF Operations for logs, VCF Operations for networks, VCF Operations orchestrator, VCF Operations HCX, and NSX, with one set of credentials. VCF Single Sign-On provides a seamless login experience with a single authentication source configured via the VCF Operations console across all the VCF components except SDDC Manager and ESX.

Importance and Benefits of VCF Single Sign-On

Configuring VCF Single Sign-On equips you with:

- **Improved User Experience:** Users can log in once and gain seamless access to the VCF components they have been granted access to.
- **Enhanced Security:** Reduce password fatigue and strengthen protection against unauthorized access.
- **Centralized Management:** Manage the identity provider configuration and user and group provisioning across VCF Instances at a central location in VCF Operations.
- **Support for modern and directory based identity providers:** Support for both modern protocols such as OIDC 2.0, SAML 2.0, and directory services such as OpenLDAP and Active Directory.

Identity Providers and Protocols Supported for VCF Single Sign-On

After you deploy VCF, you can configure VCF Single Sign-On by using an external identity provider and then sign in to VCF components by using VCF Single Sign-On.

Note:

Each VCF Identity Broker can be configured with a single identity provider.

Identity Providers Supported

Modern Identity Providers:

- Okta
- Ping Identity
- Microsoft Entra ID
- Microsoft Active Directory Federation Services (ADFS)
- Any SAML 2.0 Identity Providers

Directory-Based Identity Providers:

- AD/LDAP
- OpenLDAP

Authentication Protocols Supported

- **Security Assertion Markup Language (SAML 2.0).**
A well-established XML-based protocol where the identity provider issues digitally signed XML-based assertions to the service provider for user authentication.
- **OpenID Connect protocol (OIDC).**
A modern authentication protocol built on OAuth 2.0, where the identity provider issues an ID token in JSON format for user authentication.

User/Group Provisioning

As part of configuring VCF Single Sign-On, you can create, modify, or delete users and groups in the VCF Identity Broker provisioned from the identity provider. Single Sign-On in VCF is accomplished by configuring VMware's dedicated authentication solution, VCF Identity Broker, which helps provide a centralized authentication source for the VCF components. The following protocols are supported:

- 1. **SCIM 2.0** (System for Cross-domain Identity Management)
- 2. **JIT (Just-In-Time)**
 - a. Just-In-Time Group Provisioning
 - Groups are created dynamically based on the SAML assertion when the user logs in for the first time. You can use group pre-provisioning if you know the group names from the identity provider that requires entitlements.
 - b. JIT pre-provisioning of groups
 - You create the required groups in advance to perform role assignments. Users will be assigned to these groups dynamically on their first log in based on SAML assertion. Only the groups pre-provisioned are available for role assignments.

- 3. **AD/LDAP (Active Directory over LDAP)**

Users and groups are provisioned and synced at periodic intervals. The entire directory is not synced, only users and groups that are specified. Passwords are never synced.

Note:

You can use SAML 2.0 or OIDC as an authentication source and SCIM, JIT, and AD/LDAP as user-group provisioning.

Deployment Modes of the VCF Identity Broker

When you configure VCF Single Sign-On you can either implement it in embedded mode within the management domain vCenter or deploy it in appliance mode.

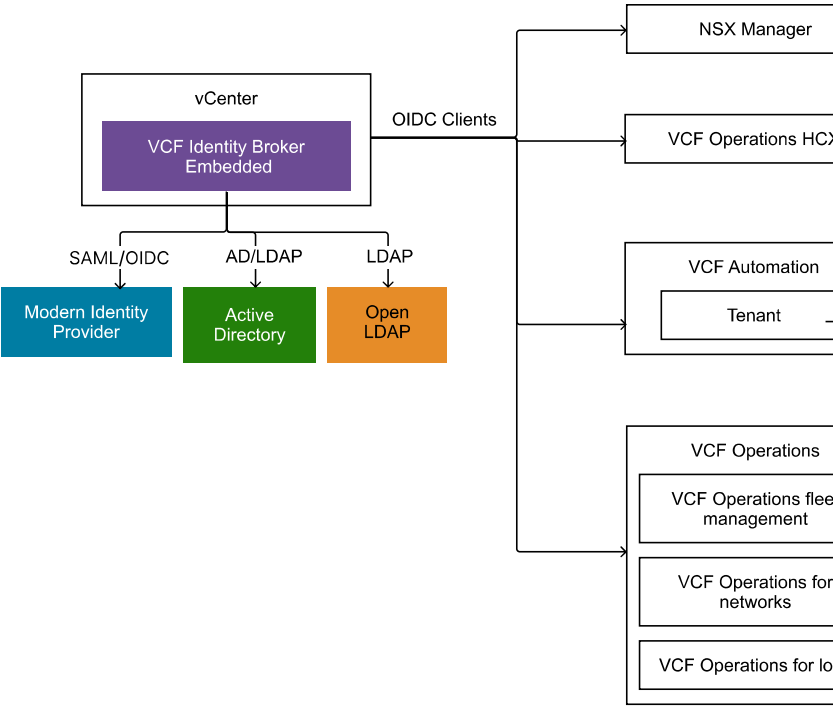
Embedded Deployment Mode

Use the embedded deployment mode if you want to use VCF Identity Broker that is embedded in the management domain vCenter and do not want to deploy a VCF Identity Broker appliance for VCF Single Sign-On configuration. You use the embedded deployment mode typically within a single VCF Instance.

In the embedded deployment mode, the VCF Identity Broker is configured in the management domain vCenter of the VCF Instance.

Figure 158: Embedded Deployment Mode

The following diagram shows the embedded deployment mode where the VCF Identity Broker is configured in the management domain vCenter.



SAML: Security Assertion Markup Language

OIDC: OpenID Connect protocol

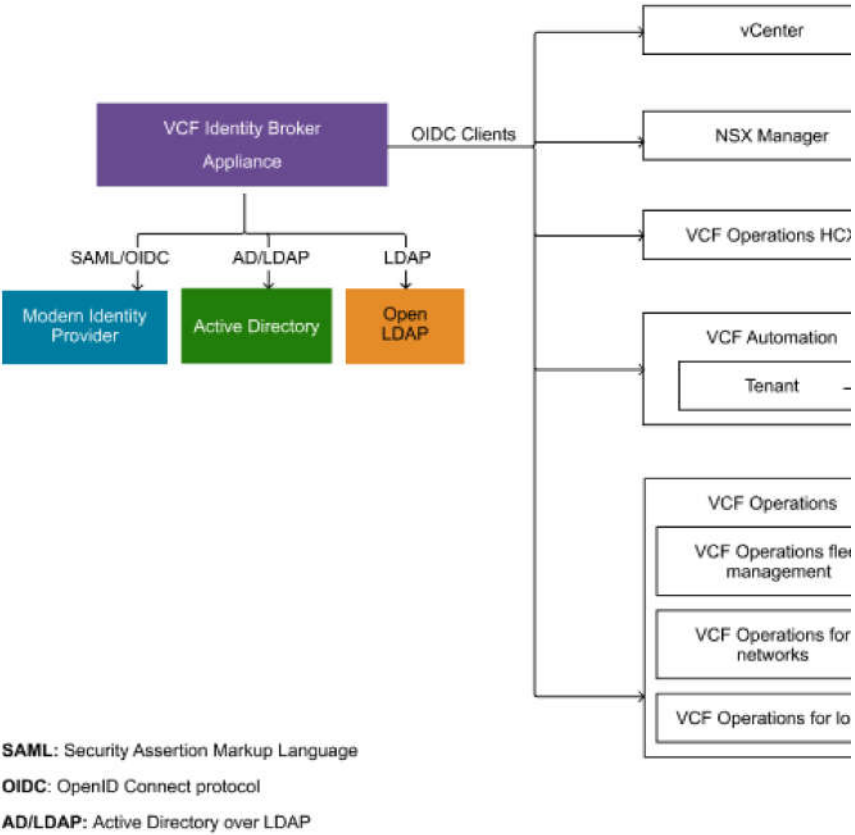
AD/LDAP: Active Directory over LDAP

Appliance Deployment Mode

VCF Identity Broker is deployed as a stand-alone appliance using VCF Operations fleet management within the management domain of the chosen VCF Instance. VCF Identity Broker appliance is a three-node cluster. Your environment must meet the minimum requirements of the VCF Identity Broker appliance deployed through VCF Operations fleet management. For more details about the minimum requirements, see [Prerequisites for Configuring Single Sign-On using the Appliance Deployment Mode](#).

Figure 159: Appliance Deployment Mode

The following diagram shows the appliance deployment mode where the VCF Identity Broker is a stand-alone instance deployed as an appliance using VCF Operations fleet management.



Distinguishing Features of the Deployment Modes

The following table provides information that distinguishes the two deployment modes: VCF Identity Broker (embedded) and VCF Identity Broker (appliance).

Distinguishing Areas	VCF Identity Broker (embedded)	VCF Identity Broker (appliance)
Deployment location	Deployed in the management domain vCenter of a VCF Instance.	Stand-alone appliance deployed using VCF Operations fleet management in a management domain.
Recommendation	Connect one VCF Identity Broker per VCF Instance.	A maximum of five VCF Instances can be connected to one VCF Identity Broker instance.
Prerequisites	For more information, see Prerequisites for Setting Up Single Sign-On .	For more information, see Prerequisites for Setting Up Single Sign-On .
Cluster size	Embedded vCenter Service	Three-node cluster
Reasons to choose a specific deployment mode	Choose the embedded deployment mode if you want to use VCF Identity Broker that is embedded in vCenter. Note: This can be a single point of failure should the management domain vCenter go down.	- Can handle single node failure. - If you need to connect more than one VCF Instance to the VCF Identity Broker, it is recommended that you use the appliance deployment mode. - If the management domain vCenter is not accessible by some VCF components, choose the appliance deployment mode.

VCF Single Sign-On Architecture

VCF allows organizations to configure VCF Single Sign-On across a fleet of VCF Instances based on your preference for manageability and availability.

You can deploy one or more VCF Identity Brokers across VCF Instances. There are one or more potential designs for VCF Single Sign-On configuration which depend on the deployment of VCF Identity Brokers across VCF Instances. For more information, see [VCF Single Sign-On Detailed Design](#).

Points to Consider and Prerequisites while Configuring VCF Single Sign-On

When you configure VCF Single Sign-On in VCF Operations, it is essential that you read and understand the following important points and ensure that you carry out the prerequisites.

Points to Consider While Setting Up VCF Single Sign-On

- **SDDC Manager UI Client authentication is not compatible with VCF Single Sign-On**
After you configure VCF Single Sign-On in the management domain vCenter, administrators are no longer able to log in to the SDDC Manager UI client with Single Sign-On users.
 - SDDC Manager UI will support logins only with local admin accounts such as @vsphere.local users.
 - The SDDC Manager APIs will support logins with VCF Single Sign-On users.
- **VCF Single Sign-On is not compatible with Enhanced Linked Mode (ELM)**
Before you configure VCF Single Sign-On, you must deactivate ELM on all participating vCenter Instances. For more information, see [Deactivate Enhanced Link Mode from vCenter Using the cmsso-util break-elm Utility](#). If you upgraded from VMware Cloud Foundation 5.x to 9.0, see [Deactivate Enhanced Link Mode \(ELM\) Using the SDDC Manager API](#).
- **Manage VCF Single Sign-On only from VCF Operations and not from the individual components**

After you configure VCF Single Sign-On from VCF Operations, you will not be able to configure any identity sources in vCenter Instances that are configured for VCF Single Sign-On.

- **PowerCLI impact**
 - PowerCLI automations will continue to work if the identity provider is configured in the management domain vCenter in embedded mode.
 - If the identity provider is configured in an alternate management domain vCenter or in appliance mode, the integration needs to be re-performed. Follow the steps in [Using Single Sign-On for PowerCLI](#).
- **Users configuring VCF Single Sign-On in VCF Operations must have access to all objects**
The 'All Objects' scope must be assigned in VCF Operations. Navigate to **Administration > Control Panel > Access Control > User Accounts/User Groups** and assign the 'All Objects' scope to the required user/group.
- **Behavior of identity and access management after setting up VCF Single Sign-On**
 - The management layer of VCF Single Sign-On will always be in the VCF Operations console.
 - You can connect a standalone vCenter directly either to an identity provider that is configured within a vCenter or integrate them as part of VCF Single Sign-On using the VCF Identity Broker in VCF Operations. However, after a vCenter is configured for Single Sign-On, you must manage its configuration in VCF Operations and not directly in the vCenter.
- **VCF Components**
 - **vCenter:** After you configure VCF Single Sign-On, any existing identity provider configurations in the participating vCenter will be overridden by the VCF Single Sign-On configuration. You may be required to perform role assignments to the required users and groups again to ensure continued access.
Other VCF components: VCF Single Sign-On is an additional authentication source to the existing authentication sources such as VMware Identity Manager and/or the component's native authentication source. You must provision users and groups again when you configure VCF Single Sign-On.
- **Single Component Log Out**
 - VCF Single Sign-On supports a single component logout mechanism. When you log out, the session ends exclusively for that particular VCF component within the designated browser.
- **Previously configured identity providers**
 - If you have previously configured an identity provider in the following combinations, the identity provider configurations are partially imported when you configure Single Sign-On. The combinations are:
 - Active Directory, OpenLDAP, or ADFS
If Active Directory, OpenLDAP, or ADFS is the configured identity provider in vCenter, when configuring VCF Single Sign-On, the configuration is partially imported to VCF Identity Broker. Only the configuration details that are not imported are required to be additionally configured.
 - Okta, Entra, or Ping
The identity provider configuration is retained if the chosen deployment mode is embedded. If the deployment mode is appliance, redo the configuration again.

Note:

The users and groups are not migrated. You must provision the users and groups again even if you have an existing identity provider configuration in place.

Prerequisites for Configuring VCF Single Sign-On

VCF Identity Broker will be configured in the management domain of the VCF instance. Ensure that the following prerequisites are met before you configure VCF Single Sign-On.

Prerequisites for Configuring VCF Single Sign-On Using the Embedded Deployment Mode

If you choose the embedded deployment mode for setting up VCF Single Sign-On, ensure that the following prerequisites are met:

- The management domain vCenter must be at version 9.0 or later and have a VMware Cloud Foundation license assigned.
- The management domain vCenter must NOT be part of Enhanced Linked Mode (ELM).

Do not activate ELM in the participating vCenter instances after configuring VCF Single Sign-On in embedded mode.

Note:

You can choose appliance as the mode of deployment if the above criteria are not met. However, the integration of vCenter with VCF Single Sign-On will not be possible at a later stage without first meeting the above criteria.

Prerequisites for Configuring VCF Single Sign-On Using the Appliance Deployment Mode

If you choose to configure VCF Single Sign-On in appliance mode, the VMware Cloud Foundation platform must meet the following minimum requirements:

- A 3-node cluster will be deployed.
- Each node will have 8 vCPUs and 16GB RAM.
- Each cluster requires a minimum of 700GB storage.

Deactivate Enhanced Link Mode from vCenter Using the cmsso-util break-elm Utility

VCF Single Sign-On is not compatible with the vCenter Enhanced Linked Mode (ELM). Before you configure VCF Single Sign-On, you must deactivate ELM on all participating vCenter Instances. As an administrator, if you want to deactivate ELM in vCenter Instances, you can do so without causing any loss to existing customer configurations such as global permissions. This procedure applies to existing customers.

The utility will only deactivate ELM and related features (for example, Single Pane of Glass (SPOG)/replication, and so on). Any shared configurations, like tags, roles, permissions and so on that are in ELM, will remain local to individual vCenter Instances after you deactivate ELM.

Note:

The new **cmsso-util break-elm** utility works only if all vCenter Instances in the ELM topology are up and running and vCenter Instances are upgraded 9.0. It is recommended that admins take offline snapshots of all vCenter instances before you deactivate the ELM. The only proper way to back up vCenter instances in ELM is to take offline snapshots of them which results in environment downtime.

If you upgraded from VMware Cloud Foundation 5.x to 9.0, see [Deactivate Enhanced Link Mode \(ELM\) Using the SDDC Manager API](#) and [Resolve Enhanced Linked Mode \(ELM\) Drift for Upgraded vCenter Instances](#).

1. Take an offline backup of the ELM topology (only recommended way to revert back to ELM).

This is a preparatory step and requires downtime.

2. SSH to any vCenter in ELM.

3. Run configuration pre-checks and fix any issues.

- a) Run `cmsso-util break-elm --mode pre-check --password <default administrator user_password>`.
- b) Fix any issues reported by the pre-check.

Output of break-elm precheck when successful:

```
Command:cmsso-util break-elm --mode pre-check --password 'ZSIY+eDe-14dukrW'
```

```
Output: Mode: pre-check This operation might take a few minutes to complete.
[Pre-check] Invoking Break ELM topology workflow...
[Pre-check] Fetch the peer nodes details...
```

```
[Pre-check] Executing break ELM on local node...
[Pre-check] Executing break ELM on all peer nodes...
Break ELM pre-checks completed successfully.
```

4. Deactivate Enhanced Link Mode.

- a) Run `cmsso-util break-elm --mode execute --password <administrator_password>`.
- b) Log out from the vSphere Client and log in again for the changes to take effect.
- c) In case of a failure, fix the cause of the failure and rerun `cmsso-util break-elm` on all nodes where the failure was reported.

You must resolve ELM drift. For more information, see [Resolve Enhanced Linked Mode \(ELM\) Drift for Upgraded vCenter Instances](#).

Resolve Enhanced Linked Mode (ELM) Drift for Upgraded vCenter Instances

If you remove vCenter instances from ELM outside of SDDC Manager, some workflows will be blocked until you reconcile the ELM drift using the SDDC Manager API.

Remove all the vCenter Instances in your VCF instance from ELM using the `cmsso-util break-elm` utility. See [Deactivate Enhanced Link Mode from vCenter Using the cmsso-util break-elm Utility](#).

Until you resolve ELM drift, you may see error messages when running certain workflows. For example:

Functionality `CREATE_DOMAIN` is not allowed because there is an ELM Drift. Ensure the ELM is completely broken, then run the `reconcile` workflow..

1. Use the SDDC Manager API to retrieve the management domain ID.

API	Description
GET /v1/domains	Find the management domain ID in the Response.

2. Validate the break ELM reconcile workflow.

API	Description
POST /v1/domains/{id}/validations	Replace {id} with the management domain ID. In the body parameter enter: <pre>{ "breakElmSpec": { "isReconcileWorkflow": true } }</pre> Find the validation ID in the Response.

3. Get the validation response.

Make sure that validation is successful.

API	Description
GET /v1/domains/{id}/validations/{validation_id}	Replace {id} with the management domain ID. Replace {validation_id} with the validation ID.

4. Resolve ELM drift across all joined domains.

API	Description
PATCH /v1/domains/{id}	Replace {id} with the management domain ID.

API	Description
	In the body parameter enter: <pre>{ "breakElmSpec": { "isReconcileWorkflow": true } }</pre> Find the task ID in the Response.

5. Monitor the task.

API	Description
GET /v1/tasks/{ <i>id</i> }	Replace { <i>id</i> } with the task ID.

Deactivate Enhanced Link Mode (ELM) Using the SDDC Manager API

If you upgraded from VMware Cloud Foundation 5.x to 9.0, you must use the SDDC Manager API to deactivate Enhanced Link Mode (ELM) for upgraded vCenter instances before you can use certain 9.0 features.

Minimum required version of the VCF components.

- SDDC Manager 9.0
- vCenter upgraded to 9.0

When you created a workload domain in VMware Cloud Foundation 5.x, you chose whether to join its vCenter to the management domain's vCenter Single Sign-On domain or to create a new vCenter Single Sign-On domain. If you joined it to the management domain's vCenter Single Sign-On domain, the workload domain vCenter was joined to the management domain vCenter in Enhanced Linked Mode (ELM).

ELM is deprecated in VMware Cloud Foundation 9.0. Before you can use new features in 9.0, such as VCF Single Sign-On and vCenter linking, you must deactivate ELM for all vCenter instances in your VCF instance.

You can use the SDDC Manager API to remove all the vCenter instances in a VCF instance from Enhanced Linked Mode (ELM), creating isolated workload domains.

Note: This procedure deactivates ELM for all vCenter instances in your VCF instance. You cannot select a subset of instances.

If you already used the `cmsso-util break-elm` utility to remove some or all of your VCF-managed vCenter instances from ELM, see [Resolve Enhanced Linked Mode \(ELM\) Drift for Upgraded vCenter Instances](#).

1. Use the SDDC Manager API to retrieve the management domain ID.

API	Description
GET /v1/domains	Find the management domain ID in the Response.

2. Validate the break ELM spec.

API	Description
POST /v1/domains/{ <i>id</i> }/validations	Replace {<i>id</i>} with the management domain ID. In the body parameter enter: <pre>{ "breakElmSpec": { "isReconcileWorkflow": false } }</pre>

API	Description
	<pre>}</pre> Find the validation ID in the Response.

3. Get the validation response.
Make sure that validation is successful.

API	Description
GET /v1/domains/{ <i>id</i> }/validations/{ <i>validation_id</i> }	Replace {<i>id</i>} with the management domain ID. Replace {<i>validation_id</i>} with the validation ID.

4. Break ELM across all joined domains.

API	Description
PATCH /v1/domains/{ <i>id</i> }	Replace {<i>id</i>} with the management domain ID. In the body parameter enter: <pre>{ "breakElmSpec": { "isReconcileWorkflow": false } }</pre> Find the task ID in the Response.

5. Monitor the task.

API	Description
GET /v1/tasks/{ <i>id</i> }	Replace { <i>id</i> } with the task ID.

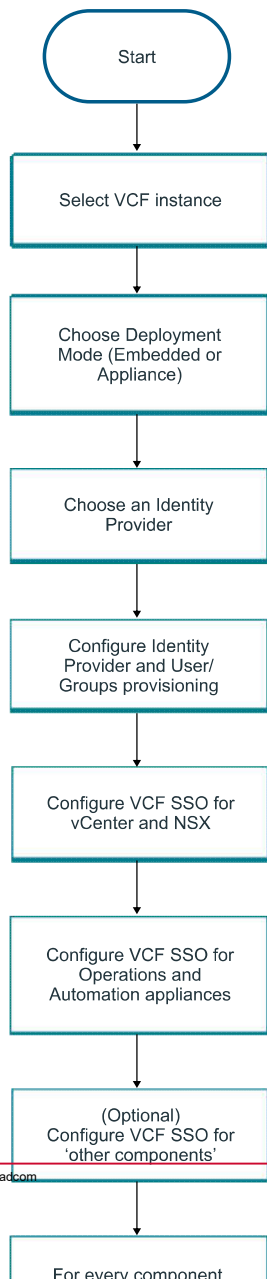
Configure a New VCF Single Sign-On for a VCF Instance

You can configure VCF Single Sign-On across components within a single VCF Instance, across a set of VCF Instances, and/or across an entire VCF fleet of VCF Instances.

Important:

Ensure that you have completed the prerequisites and read the list of important points to consider before you set up VCF Single Sign-On. For more information, see [Points to Consider and Prerequisites while Setting up Single Sign-On](#).

The following flowchart describes the high-level steps to configure VCF Single Sign-On.



Follow these high-level steps to configure VCF Single Sign-On:

1. **Select a VCF Instance for which you want to configure VCF Single Sign-On.**

For more information, see [Step 1: Select a VCF Instance](#).

2. **Choose the deployment mode you wish to use.**

Select whether you want to use the VCF Identity Broker embedded or the VCF Identity Broker appliance deployment mode. For information about selecting a deployment mode, see [Step 2: Choose the Deployment Mode](#).

3. **Select and configure the identity provider.**

Set up your preferred identity provider using modern or directory-based authentication. For more information, see [Step 3: Configure an Identity Provider](#).

4. **Configure VCF Single Sign-On for NSX and vCenter.**

For more information, see [Step 4: Configure VCF Single Sign-On for NSX and vCenter](#).

5. **Configure VCF Single Sign-On for VCF Operations and VCF Automation.**

Connect the components to the VCF Identity Broker for Single Sign-On-based access. For more information, see [Step 5: Configure VCF Single Sign-On for VCF Operations and VCF Automation](#).

6. **(Optional) Configure VCF Single Sign-On for other VCF Components.**

Connect the components for Single Sign-On-based access. For more information, see [Step 6: \(Optional\) Configure VCF Single Sign-On for other VCF Components](#).

7. **Assign required roles and permissions for users or groups.**

For more information, see [Step 7: Assigning Roles and Permissions](#).

Step 1: Select a VCF Instance

Choose the VCF Instance for which you want to configure VCF Single Sign-On.

Prerequisite

- Deploy and configure a VCF Instance with a management domain. For more information, see [Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard](#).

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. (First time only) Review and acknowledge all of the prerequisite check boxes and click **Continue**.
3. From the **Identity & Access** navigation pane, click **SSO Overview**.
4. From the **SSO Overview** page, click the **Select a VCF Instance** drop down and select the VCF Instance where you would like to configure the identity provider to enable VCF Single Sign-On.

Note:

If the identity source configuration for the specific VCF Instance is managed by another VCF Operations console, and you wish to manage the identity source configuration of the VCF Instance within the same VCF Operations console, see [Change the Identity Management of a VCF Instance Managed by Another VCF Instance](#).

Step 2: Choose the Deployment Mode

After you have selected a VCF Instance, you must choose a deployment mode for the VCF Identity Broker configuration. The options available are embedded or appliance. For more information about the VCF Single Sign-On deployment modes, see [Deployment Modes of the VCF Identity Broker](#).

Prerequisite

- Complete [Step 1: Select a VCF Instance](#).

Procedure

1. From the **Enable Single Sign-On** page, click the **Start** button against the **Choose deployment mode** option.
2. Select either **Identity broker (embedded)** or **Identity broker appliance**. For information about deploying a VCF Identity Broker (appliance), see [Deploying VCF Identity Broker](#).
3. Click **Next**.

Step 3: Configure an Identity Provider

After you select the deployment mode, you must select and configure the identity provider using SAML, OIDC, or LDAP as the authentication protocol.

For information about supported identity providers and protocols, see [Identity Providers and Protocols Supported for Single Sign-On](#).

- To configure Okta as an identity provider, see [Configure Okta as an Identity Provider](#).
- To configure Ping Identity as an identity provider, see [Configure Ping Identity as an Identity Provider](#).
- To configure Microsoft Entra ID as an identity provider, see [Configure Microsoft Entra ID as an Identity Provider](#).
- To configure Microsoft ADFS as an identity provider, see [Configure Microsoft ADFS as an Identity Provider](#).
- To configure an identity provider using Generic SAML 2.0, see [Configure an Identity Provider using Generic SAML 2.0](#).
- To configure Active Directory as an identity provider using AD/LDAP, see [Configure AD/LDAP as an Identity Provider](#).
- To configure OpenLDAP as an identity provider, see [Configure OpenLDAP as an Identity Provider](#).

Note:

If you already have an identity provider configured in the following combinations, you can use the preselected existing identity provider or choose a new identity provider in the UI workflows:

- Active Directory, OpenLDAP, or ADFS
- Okta, Microsoft Entra ID, or Ping Federate with OIDC

Configure Okta as an Identity Provider

You can configure Okta as an identity provider using SAML 2.0 or OIDC authentication protocols.

- For information about configuring Okta as an identity provider using SAML as the authentication protocol, see [Configure Okta as an Identity Provider Using SAML](#).
- For information about configuring Okta as an identity provider using OIDC as the authentication protocol, see [Configure Okta as an Identity Provider Using OIDC](#).

Configure Okta as an Identity Provider Using SAML

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure Okta as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).

Procedure

Choose an identity provider

1. From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose Identity Provider** section, select **Okta** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language) and click **Next**.
5. In the **Service Provider Registration** screen, log in to your Okta admin console and follow the steps in [Add a private Single Sign-On integration](#) to create a SAML 2.0 application in Okta.

While registering Okta, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

6. Click **Next**.
7. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (VCF Identity Broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. Note: If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • PasswordProtectedTransport: Authentication with a password sent over a secure transport (For example, HTTPS). • Password: Authentication using a password without requiring transport security. • Unspecified: If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- System for Cross-domain Identity Management (SCIM), or
- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

a) If you select **SCIM** and click **Next**, complete the configuration as follows.

- In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

b) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider. You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, see [KB 386870](#).

- In the **Directory Details** screen, enter the following details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal.

Option	Description
	• The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location and Encrypted Connections options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename <code>{directory-name}-concatenated.pem</code> when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .
Primary domain controller (Displayed when you select the Global Catalog option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, <code>hostname = DC01</code> for the domain called <code>example.com</code> .
Secondary domain controller (Displayed when you select the Global Catalog option)	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, <code>dc=example, dc=com</code> .
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

- a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
 - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
 - If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
 - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
- b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
 1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.
The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
Note:
If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with user@company.com@domain.com. For more information, see [KB 393150](#).
 2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from the list in the datagrid below.

Option	Description
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review** screen, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

11. Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset Single Sign-On'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure Okta as an Identity Provider using OIDC

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure Okta as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning

Prerequisites

- Complete [Step 2: Choose the Deployment Mode](#).
- If you configure VCF Single Sign-On using VCF Identity Broker appliance, and there are network restrictions that do not allow outbound public internet access, you must set up HTTP proxy. For more information, [KB 393078](#).

Procedure

Choose an identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.

If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

For an existing source (existing identity provider configuration)

- Choose either SAML or OIDC as the authentication protocol.

Note:

If you choose SAML, the VCF Single Sign-On configuration is considered as a new authentication source configuration. Continue from Step. 6.

- If you choose OIDC:
 - If you choose the appliance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
 - Review the details and provide the shared secret and click **Next**.
 - Identity provider name
 - Client identifier
 - OpenID address
 - Update Redirect URI in the Identity provider's OIDC application
 - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
 - Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
 - Review the SCIM push details and click **Done**. The identity provider configuration is complete.
 - If you choose the embedded deployment mode, and select OIDC:
 - The configuration details of the identity provider configuration are automatically imported from existing configuration into the VCF Identity Broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
 - Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
 - Review the SCIM details, click **Done** and then click **Finish Setup**. The identity provider configuration is complete.
- For a fresh identity provider configuration, from the **Choose Identity Provider** section, select **Okta** from the list and click **Next**.

Configure the identity provider
 - From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
 - In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
 - In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider. Example, Okta.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider for authentication.
Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process.
Redirect URI	The URL where the identity provider sends the authentication response after a user logs in.

- In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- System for Cross-domain Identity Management (SCIM), or
- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

- If you select **SCIM** and click **Next**, complete the configuration as follows.

- In the **Domain(s)** screen, enter the domains that users from your enterprise will log in with, and click **Next**.

- If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC token when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider. You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.
- If you select **AD/LDAP** and click **Next**, select **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker** and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, [KB 386870](#).

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the VCF Identity Broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, <code>name</code> .

Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called Unique Identifier in OIDC Identity Provider .
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- In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory. The directory name should contain only letters, digits, spaces, hyphens, and underscores.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	Activates LDAPs for all the connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location option)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename <code>{directory-name}-concatenated.pem</code> when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .

Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, <code>hostname = DC01</code> for the domain called <code>example.com</code> .
Secondary domain controller	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, <code>dc=example, dc=com</code> .
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read [KB 386870](#) to understand the finer details about configuring the setup.

- a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
 - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
 - If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
 - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
- b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
 1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.
The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
Note:
If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with `user@company.com@domain.com`. For more information, see [KB 393150](#).
 2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

- From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

- From the **Review screen**, review the configuration details of the identity provider.
- Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
- Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

- Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure Ping Identity as an Identity Provider

You can configure Ping Identity as an identity provider using SAML or OIDC authentication protocols.

- For information about configuring Ping Identity as an identity provider using SAML as the authentication protocol, see [Configure Ping Identity as an Identity Provider Using SAML](#). For information about configuring Ping Identity as an identity provider using OIDC as the authentication protocol, see [Configure Ping Identity as an Identity Provider Using OIDC](#).

Configure Ping Identity as an Identity Provider Using SAML

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure Ping Identity as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

- Choose an identity provider
- Configure the identity provider
- Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).

Procedure

Choose an identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.
- From the **Choose Identity Provider** section, select **Ping Identity** from the list and click **Next**.

Configure the identity provider

- From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
- In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language) and click **Next**.
- In the **Service Provider Registration** screen, log in to your Ping Identity admin console and follow the steps in [Ping Identity](#) to create a SAML 2.0 application in Ping Identity.

While registering Ping Identity, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

- Click **Next**.
- In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata	Select either URL or XML.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. Note: If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.

Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • PasswordProtectedTransport: Authentication with a password sent over a secure transport (For example, HTTPS). • Password: Authentication using a password without requiring transport security. • Unspecified: If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- System for Cross-domain Identity Management (SCIM), or
- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

a) If you select **SCIM** and click **Next**, complete the configuration as follows.

- In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

b) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain/s** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups Pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider.

You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive

- c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, [KB 386870](#).

In the **Directory Details** screen, enter the following details.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location and Encrypted Connections options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename <code>{directory-name}-concatenated.pem</code> when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid,sAMAccountName</code> .
Primary domain controller (Displayed when you select the Global Catalog option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, <code>hostname = DC01</code> for the domain called <code>example.com</code> .
Secondary domain controller (Displayed when you select the Global Catalog option)	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.

Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid</code> , <code>sAMAccountName</code> .
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, <code>dc=example,dc=com</code> .
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

- If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
 - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
 - If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
 - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
- If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

- In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.

Note:

If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with `user@company.com@domain.com`. For more information, see [KB 393150](#).

- In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.

Option	Description
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

- From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

- From the **Review screen**, review the configuration details of the identity provider.
- Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
- Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

11. Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure Ping Identity as an Identity Provider Using OIDC

After installing or upgrading to VMware Cloud Foundation, to configure VCF Single Sign-On, you can configure Ping Identity as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

- Choose an identity provider
- Configure the identity provider
- Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).
- If you configure VCF Single Sign-On using VCF Identity Broker appliance, and there are network restrictions that do not allow outbound public internet access, you must set up HTTP proxy. For more information, see [KB 393078](#).

Procedure

Choose an identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.

If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

For an existing source (existing identity provider configuration)

- Choose either SAML or OIDC as the authentication protocol.

Note:

If you choose SAML, the VCF Single Sign-On configuration is considered as a new authentication source configuration. Continue from Step. 6.

- If you choose OIDC:
 - If you choose the appliance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
 - Review the details and provide the shared secret and click **Next**.
 - Identity provider name
 - Client identifier
 - OpenID address
 - Update Redirect URI in the Identity provider's OIDC application
 - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
 - Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
 - Review the SCIM push details and click **Done**. The identity provider configuration is complete.
 - If you choose the embedded deployment mode, and select OIDC:
 - The configuration details of the identity provider configuration are automatically imported from existing configuration into the VCF Identity Broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
 - Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
 - Review the SCIM details, click **Done** and then click **Finish Setup**. The identity provider configuration is complete.

- For a fresh identity provider configuration, from the **Choose Identity Provider** section, select **Ping Identity** from the list and click **Next**.

Configure the identity provider

- From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
- In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
- In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Ping Identity.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider for authentication.
Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process.
Redirect URI	The URL where the identity provider sends the authentication response after a user logs in.

SSL Certificate	A public key certificate to verify the OIDC identity.
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- In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
 - System for Cross-domain Identity Management (SCIM), or
 - Just-In-Time Provisioning (JIT), or
 - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)
 - If you select **SCIM** and click **Next**, complete the configuration as follows.
 - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.
 - If you select **JIT** and click **Next**, complete the configuration as follows:
 - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider. You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.
- If you select **AD/LDAP** and click **Next**, select **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker** and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, see [KB 386870](#).

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the VCF Identity Broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, name .

Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called Unique Identifier in OIDC Identity Provider .
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In the In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: - The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. - The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. - The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	Activates LDAPs for all the connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location and Encrypted Connection options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename {directory-name}-concatenated.pem when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, hostname = DC01 for the domain called example.com.
Secondary domain controller	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.

Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, dc=example, dc=com.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read [KB 386870](#) to understand the finer details about configuring the setup.

- a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
 - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
 - If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
 - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
- b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
 1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.
The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
Note:
If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with user@company.com@domain.com. For more information, see [KB 393150](#).
 2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.

Option	Description
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review screen**, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

9. Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure Microsoft Entra ID as an Identity Provider

You can configure Microsoft Entra ID as an identity provider using SAML or OIDC authentication protocols.

- For information about configuring Microsoft Entra ID as an identity provider using SAML as the authentication protocol, see [Configure Microsoft Entra ID as an Identity Provider Using SAML](#).
- For information about configuring Microsoft Entra ID as an identity provider using OIDC as the authentication protocol, see [Configure Microsoft Entra ID as an Identity Provider Using OIDC](#).

Configure Microsoft Entra ID as an Identity Provider Using SAML

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure Microsoft Entra ID as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider

3. Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).

Procedure**Choose an identity provider**

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.
- From the **Choose Identity Provider** section, select **Microsoft Entra ID** from the list and click **Next**.

Configure the identity provider

- From the **Configure the Identity Provider** section, click **Configure** to set up the identity broker to integrate with the selected identity provider for user authentication.
- In the **Authentication Method** pane, select **SAML** (Security Assertion Markup Language).
- In the **Service Provider Registration** pane, log in to your Microsoft Entra ID admin console and follow the steps in [Register a SAML App in your external client](#).

While registering Microsoft Entra ID, use the metadata provided in the **Service Provider Registration** screen for **Entity ID/Metadata URL** and **Assertion Consumer Service URL**. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

- Click **Next**.
- In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider. For example, Microsoft Entra ID.
Metadata	Select either URL or XML.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (VCF Identity Broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. Note: If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.

SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. PasswordProtectedTransport: Authentication with a password sent over a secure transport (For example, HTTPS). Password: Authentication using a password without requiring transport security. Unspecified: If you don't want to enforce any particular type of authentication mechanism, you can use this option.
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- In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- System for Cross-domain Identity Management (SCIM), or
- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

- If you select **SCIM** and click **Next**, complete the configuration as follows.

- In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

- If you select **JIT** and click **Next**, complete the configuration as follows:

In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Note:

Add group claims. For more information, see the section about adding group claims to tokens in the page called [Configure group claims for applications by using Microsoft Entra ID](#).

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain/s** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

This **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.

You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

- c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, [KB 386870](#).

In the **Directory Details** screen, enter the following details.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: - The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. - The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. - The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location and Encrypted Connections options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename <code>{directory-name}-concatenated.pem</code> when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid,sAMAccountName</code> .
Primary domain controller (Displayed when you select the Global Catalog option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, <code>hostname = DC01</code> for the domain called <code>example.com</code> .
Secondary domain controller (Displayed when you select the Global Catalog option)	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.

Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid,sAMAccountName</code> .
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, <code>dc=example,dc=com</code> .
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

- a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
- If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
 - If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
 - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
- b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
- In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.
The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
Note:
If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with `user@company.com@domain.com`. For more information, see [KB 393150](#).
 - In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.

Option	Description
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

- From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

- From the **Review screen**, review the configuration details of the identity provider.
- Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
- Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

11. Click **Finish Setup**.

Note:

Special characters in the username, such as # are not allowed. The symbols that are allowed for the username are: *letters, digits (0-9), space, and (+, -, _ , .)*.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure Microsoft Entra ID as an Identity Provider Using OIDC

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure Microsoft Entra ID as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

- Choose an identity provider
- Configure the identity provider
- Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).

- If you configure VCF Single Sign-On using VCF Identity Broker appliance, and there are network restrictions that do not allow outbound public internet access, you must set up HTTP proxy. For more information, see [KB 393078](#).

Procedure

Choose an identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.

If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

For an existing source (existing identity provider configuration)

- Choose either SAML or OIDC as the authentication protocol.

Note:

If you choose SAML, the VCF Single Sign-On configuration is considered as a new authentication source configuration. Continue from Step. 6.

- If you choose OIDC:
 - If you choose the appliance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
 - Review the details and provide the shared secret and click **Next**.
 - Identity provider name
 - Client identifier
 - OpenID address
 - Update Redirect URI in the Identity provider's OIDC application
 - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
 - Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
 - Review the SCIM push details and click **Done**. The identity provider configuration is complete.
 - If you choose the embedded deployment mode, and select OIDC:
 - The configuration details of the identity provider configuration are automatically imported from existing configuration into the VCF Identity Broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
 - Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
 - Review the SCIM details, click **Done** and then click **Finish Setup**. The identity provider configuration is complete.
- For a fresh identity provider configuration, from the **Choose Identity Provider** section, select **Microsoft Entra ID** from the list and click **Next**.

Configure the identity provider

- From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
- In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
- In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Microsoft Entra ID.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider, for authentication.

Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process.
Redirect URI	The URL where the identity provider will send the authentication response after a user logs in.

6. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- System for Cross-domain Identity Management SCIM, or
- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

a) If you select **SCIM** and click **Next**, complete the configuration as follows.

- In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

b) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.

You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.

- c) If you select **AD/LDAP**, select the **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker**, and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, [KB 386870](#).

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the VCF Identity Broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, <code>name</code> .
Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called Unique Identifier in OIDC Identity Provider .

In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the option DNS server location)	Activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the option DNS server location and Encrypted Connection)	Upload the root certificate(s) required to access the domain controllers over SSL.

	All uploaded root certificates will be merged into a single file with the filename {directory-name}-concatenated.pem when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, hostname = DC01 for the domain called example.com.
Secondary domain controller	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, dc=example, dc=com.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read [KB 386870](#) to understand the finer details about configuring the setup.

- If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
 - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
 - If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
 - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
- If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

- In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.

Note:

If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with user@company.com@domain.com. For more information, see [KB 393150](#).

- In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DN that you want to sync from the list in the datagrid below.

Option	Description
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

- From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

- From the **Review screen**, review the configuration details of the identity provider.
- Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
- Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

- Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure Microsoft ADFS as an Identity Provider

You can configure Microsoft ADFS as an identity provider using SAML or OIDC authentication protocols.

- For information about configuring Microsoft ADFS as an identity provider using the SAML authentication protocol, see [Configure Microsoft ADFS as an Identity Provider Using SAML](#).
For information about configuring Microsoft ADFS as an identity provider using the OIDC authentication protocol, see [Configure Microsoft ADFS as an Identity Provider Using OIDC](#).

Configure Microsoft ADFS as an Identity Provider Using SAML

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure Microsoft ADFS as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

- Choose an identity provider
- Configure the identity provider

- Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).

Procedure

Choose an identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.
- From the **Choose Identity Provider** section, select **Microsoft Entra ADFS** from the list and click **Next**.

Configure the identity provider

- From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
- In the **Authentication Method** pane, select **SAML** (Security Assertion Markup Language).
- In the **Service Provider Registration** pane, log in to your Microsoft ADFS admin console and follow the steps in [Set up a SAML 2.0 provider with ADFS](#) to create a SAML 2.0 in Microsoft ADFS.

While registering Microsoft Entra ADFS, use the metadata provided in the **Service Provider Registration** screen for **Entity ID/Metadata URL** and **Assertion Consumer Service URL**. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

- Click **Next**.
- In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata	Select either URL or XML.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (VCF Identity Broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. Note: If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity.

	• PasswordProtectedTransport: Authentication with a password sent over a secure transport (For example, HTTPS). • Password: Authentication using a password without requiring transport security. • Unspecified: If you don't want to enforce any particular type of authentication mechanism, you can use this option.
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8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

a) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. To assign entitlements to users before their first login, add them as users to a local group and assign permissions. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider. You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

b) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, [KB 386870](#).

In the **Directory Details** screen, enter the following details.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.

Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	Activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location and Encrypted Connections options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename <code>{directory-name}-concatenated.pem</code> when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .
Primary domain controller (Displayed when you select the Global Catalog option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, <code>hostname = DC01</code> for the domain called <code>example.com</code> .
Secondary domain controller (Displayed when you select the Global Catalog option)	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, <code>dc=example, dc=com</code> .
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, click **Done**.

- If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

- In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.

Note:

If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with user@company.com@domain.com. For more information, see [KB 393150](#).

- In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

- From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

- From the **Review screen**, review the configuration details of the identity provider.
- Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
- Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

11. Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure Microsoft ADFS as an Identity Provider Using OIDC

After installing or upgrading to VCF 9.0, to configure VCF Single Sign-On, you can configure Microsoft ADFS as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

- Choose an identity provider
- Configure the identity provider
- Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).
- If you configure VCF Single Sign-On using VCF Identity Broker appliance, and there are network restrictions that do not allow outbound public internet access, you must set up HTTP proxy. For more information, see [KB 393078](#).

Procedure

Choose an identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.

If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

For an existing source (existing identity provider configuration)

- Choose either SAML or OIDC as the authentication protocol.

Note:

If you choose SAML, the VCF Single Sign-On configuration is considered as a new authentication source configuration. Continue from Step. 6.

- If you choose OIDC:
 - If you choose the appliance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
 - Review the details and provide the shared secret and click **Next**.
 - Identity provider name
 - Client identifier
 - OpenID address
 - Update Redirect URI in the Identity provider's OIDC application
 - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
 - Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
 - Review the SCIM push details and click **Done**. The identity provider configuration is complete.
 - If you choose the embedded deployment mode, and select OIDC:
 - The configuration details of the identity provider configuration are automatically imported from existing configuration into the VCF Identity Broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
 - Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
 - Review the SCIM details, click **Done** and then click **Finish Setup**. The identity provider configuration is complete.
- For a fresh identity provider configuration, from the **Choose Identity Provider** section, select **Microsoft Entra ADFS** from the list and click **Next**.

Configure the identity provider
 - From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
 - In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
 - In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Microsoft Entra ADFS.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider, for authentication.
Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider.
OpenID Address	The URL of the identity provider's OIDC configuration end point for initiating the authentication process.
Redirect URI	The URL where the identity provider will send the authentication response after a user logs in.

SSL Certificate	Public key certificate to verify OIDC identity. This field is required only if a self-signed certificate is used to sign the identity provider.
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- In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

- If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.
You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.
- If you select **AD/LDAP**, select the **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker**, and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, see [KB 386870](#).

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the VCF Identity Broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, name .

Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called Unique Identifier in OIDC Identity Provider .
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In the **In the Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: - The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. - The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. - The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	Activates LDAPS for all the connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location and Encrypted Connection options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename {directory-name}-concatenated.pem when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, hostname = DC01 for the domain called example.com.
Secondary domain controller	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.

Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, dc=example, dc=com.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

8. Configure user and group provisioning

9. In the **Configure User and Group Provisioning** screen, click **Done**.

- If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.

Note:

If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with user@company.com@domain.com. For more information, see [KB 393150](#).

2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.

Option	Description
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

- From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

- From the **Review screen**, review the configuration details of the identity provider.
- Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
- Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

- Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF SSO configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF SSO. Access the list of components under the 'Component Configuration' tab at the end of the setup. For more information, see [Step 7. Assigning Roles and Permissions](#).

Configure an Identity Provider using Generic SAML 2.0

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure an identity provider using Generic SAML 2.0 as the authentication protocol.

To set up an identity provider, you must:

- Choose an identity provider
- Configure the identity provider
- Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).

Procedure

Configure the identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.
- From the **Choose Identity Provider** section, select **Generic SAML 2.0** from the list and click **Next**.
- From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
- In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language).
- In the **Service Provider Registration** screen, log in to your Identity Provider admin console and create a SAML 2.0 application.

While configuring the application, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

- Click **Next**.
- In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (VCF Identity Broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. Note: If your identity provider is not publicly accessible or uses a self-signed certificate, VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. PasswordProtectedTransport: Authentication with a password sent over a secure transport (For example, HTTPS). Password: Authentication using a password without requiring transport security. Unspecified: If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:

- System for Cross-domain Identity Management (SCIM), or
- Just-In-Time Provisioning (JIT), or
- Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

- a) If you select **SCIM** and click **Next**, complete the configuration as follows.

- In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

- b) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- If you chose **Groups pre-provisioning** in the previous screen, then in the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider. You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.
- c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, see [KB 386870](#).
- In the **Directory Details** screen, enter the following details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include:

	• The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the DNS server location and Encrypted Connections options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename <code>{directory-name}-concatenated.pem</code> when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, <code>hostname = DC01</code> for the domain called <code>example.com</code> .
Secondary domain controller	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, <code>uid, sAMAccountName</code> .
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, <code>dc=example, dc=com</code> .
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

- a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
 - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
 - If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
 - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
- b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
- c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
 1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.
The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
Note:
If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with user@company.com@domain.com. For more information, see [KB 393150](#).
 2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the Select Base Group DN button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy. Note:

Option	Description
	For large AD groups, do not select this option. Instead, you must manually select the required groups.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review screen**, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see [Managing AD/LDAP](#).

11. Click **Finish Setup**.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both the management domain vCenter and the VCF Identity Broker appliance, based on the deployment option selected. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Reset SSO'. This practice ensures that you can revert the VCF Single Sign-On configuration to a recognized and stable state if necessary.

Important Notes on Role Assignment

Log in as a local admin to each component and assign service roles to provisioned users/groups to activate VCF Single Sign-On. Access the list of components under the 'Component Configuration' tab at the end of the setup.

12.

Configure Active Directory as an Identity Provider Using AD/LDAP

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure Active Directory as an identity provider using AD/LDAP.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).

Procedure

Choose an identity provider

- From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.

If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

For an existing source (existing identity provider configuration)

Depending on the protocol you have chosen, the following fields are prepopulated from the existing configuration in the vCenter.

- AD/LDAP
 - Directory name
 - Primary domain controller: Host and port
 - Base DN
 - Bind user name

Review the prepopulated values. If there are any values that are empty, enter the details. Proceed to Step 7 to complete configuring the identity provider.

- For a fresh identity provider configuration, from the **Choose identity provider** section, select **AD/LDAP** from the list and click **Next**.

Configure the identity provider

- From the **Configure the identity provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
- In the **Directory information** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the VCF Identity Broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: - The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. - The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. - The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. Note: When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	Activates LDAPS for all the connections to this directory.

Certificate(s) for encrypted connection (Displayed when you select the DNS server location and Encrypted Connections options)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename {directory-name}-concatenated.pem when you save the configuration. Note: Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Primary domain controller (Removed from display when you select DNS Server Location)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, hostname = DC01 for the domain called example.com.
Certificate for primary domain controller	The root certificate required to access the primary domain controller over SSL. Note: Ensure that the certificate is in PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines. This field is displayed if you type 'LDAP' in the Primary domain controller field.
Secondary domain controller (Removed from display when you select DNS Server Location)	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.
Certificate for secondary domain controller	The root certificate required to access secondary domain controller over SSL. Note: Ensure the certificate is in PEM format and include the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines. This field is displayed if you type 'LDAP' in the Secondary domain controller field.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, dc=example, dc=com.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

- In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

6. In the **Configure user and group provisioning** screen, click **Configure** to provision the users and groups from the identity provider to the VCF Identity Broker.

7. In the **Review Directory Information** screen, review the information directory information and click **Next**.

8. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

The attributes in the VCF Identity Broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.

Note:

If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with user@company.com@domain.com. For more information, see [KB 393150](#).

9. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Sync Nested Groups	Select Sync Nested Group to sync all the users from the selected group and its nested group without maintaining the group hierarchy.
Specify the base group DN	Enter the required base group DN and click the Select Base Group DN button. The list of groups that belong to the base group DN is listed in the grid below. You can search for the group name and select the group DNs that you want to provision. Note: VCF Single Sign-On allows for the specification of a single base group DN for conducting group searches. If you need to provision groups from multiple locations within Active Directory, it is recommended that you set the base group DN at a higher level in the directory hierarchy. This should encompass all the required groups, such as a parent Organizational Unit (OU) or domain root, and then selecting the necessary groups.

10. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the base user DN	Click the Select Base User DN button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from the list in the datagrid below. Note: If you want to sync the users directly from the group, do not specify a base user DN.

11. From the **Review** screen, review the configuration details of the identity provider.

12. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time. To modify the sync schedule, see [Managing AD/LDAP](#).

13. Click **Done**.

Managing AD/LDAP

You can integrate Active Directory with VCF Single Sign-On. After Active Directory is connected, you can perform several operations to manage the Active Directory connection such as view a list of active directories, add additional directories, modify sync settings, and view the sync log.

To start managing the AD/LDAP connection, navigate to the corresponding VCF Instance in which VCF Single Sign-On was configured. The **Identity Source** tab contains the relevant information pertaining to the identity provider configuration. If more than one VCF Instance is connected to the Active Directory, use the link in the **Identity Source** tab to navigate to the relevant VCF Instance. Reset VCF Single Sign-On to manage the Active Directory connection.

Prerequisite

- Configure an identity provider for AD/LDAP before you start managing it. For more information, see [Configure Active Directory as an Identity Provider Using AD/LDAP](#).

Procedure

- From the VCF Operations main menu, select **Fleet Management > Identity & Access**.
- From the **Identity & Access** pane, select the VCF Instance for which you would like to reset VCF Single Sign-On from the **VCF Instances** drop-down option.
- From the **Identity Source** tab on the right side, you see different sections such as Deployment Mode, Authentication Information, and Directory Information.

The **Deployment Mode** and **Authentication Information** sections indicate the type of deployment and connection respectively.

Directory Information

The datagrid lists the directories that have been configured. You can add, edit, or delete the directories.

Column Name	Description
Directory Name	User friendly name of the directory that was configured.
Domain	The domain name of the domain controller that was configured as part of the directory configuration.
Synced groups	The total number of groups that have been synced from the directory to the VCF Identity Broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but only 8 groups were synced, then only 8 groups are displayed.
Synced users	The total number of users that have been synced from the directory to the VCF Identity Broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but, only 8 groups were synced, then only 8 groups are displayed.
Alerts	The number of alert(s) that may have occurred during the last performed successful sync.
Sync on request	Click the link to trigger an on-demand sync from the corresponding directory in Active Directory.
Last sync status	The day and time of the last performed successful sync. If the last sync fails, the status reflects the same. Click on Learn More for more information. See the table about Sync Failure Messages for more information about sync failure errors and the reasons.

Sync Failure Messages

Error Message	Reason/Remediation
LDAP server is not reachable	This could be due to network issues, firewall blocks, an incorrect host name, or invalid BIND credentials. Verify the configuration and retry the sync.
An internal error occurred during a sync with the LDAP server.	Retry sync.
The users don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
The groups don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
Group Query Failed	The group is not found in Active Directory.

Add Additional Directories

You can add a new directory by clicking on the **Add** button. To configure a directory in AD/LDAP, see [Configure Active Directory as an Identity Provider Using AD/LDAP](#).

Edit Existing Directories

You can edit the directory by clicking on the **Edit** button. From the Directory Details page, click the **Edit** button at the top of each of the following pages to modify the existing configuration details:

- Details
- Attribute mappings
- Group provisioning
- User provisioning
- Sync Settings - For more information see, [Modify Sync Settings](#).
- Sync Log - For more information, see [View Sync Log](#).

Note:

After you modify configurations in the Attribute Mappings, Group provisioning, User provisioning, and Sync Settings screens, you are prompted with an option to trigger an on-demand sync. If you do not trigger the on-demand sync, changes to users and groups take place in the next scheduled sync.

Modify Sync Settings

Editing the directories includes modifying the sync settings. The sync schedule is set to once per week, however you can modify the sync schedule by clicking the **Edit** button in the **Sync Settings** screen. Sync schedule is configured in UTC.

View Sync Log

The sync log under each directory contains the log of each sync between VCF Identity Broker. and the configured domain controller in Active Directory. As part of this sync, users and groups are added or removed. You can view the time at which the sync was completed, the number of groups and users that were modified because of the sync, the status of the sync, and alerts. Here is a list of error messages.

Error Message	Possible Reason(s)
Invalid group attribute name	The group object has an invalid or a missing required attribute from Active Directory.
Invalid user attribute name	The user object has an invalid or missing required attribute from Active Directory.
Multiple attribute values	The attribute that was mapped to the user/group contains more than one value in Active Directory.
User or Group Query Failed	The user/group was not found in Active Directory.

Configure OpenLDAP as an Identity Provider

After installing or upgrading to VMware Cloud Foundation 9.0, to configure VCF Single Sign-On, you can configure OpenLDAP as an identity provider.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning

Prerequisite

- Complete [Step 2: Choose the Deployment Mode](#).
- Ensure that the `memberOf overlay` is activated in the OpenLDAP server. For information about activating `memberOf overlay`, see [OpenLDAP Overlays](#).

Procedure

Choose an identity provider

1. From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose Identity Provider** section, select **OpenLDAP** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the VCF Identity Broker to integrate with the selected identity provider for user authentication.
4. In the **Directory Details** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
Directory name	Enter a name for the directory.
Primary domain controller	A server machine responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, <code>hostname = DC01</code> for the domain called <code>example.com</code> .
Certificate for primary domain controller	The root certificate required to access primary domain controller over SSL. Note: Ensure the certificate is in PEM format and include the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Secondary domain controller	A backup domain controller for the VCF Identity Broker to point to if the primary domain controller is not reachable.
Certificate for secondary domain controller	The root certificate required to access secondary domain controller over SSL. Note: Ensure the certificate is in PEM format and include the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. Note: By default 'Custom Attribute' is chosen.

Custom directory search attribute for Users	The LDAP attribute used in queries to search or identify users in the directory.
Custom directory search attribute for Groups	The LDAP attribute used in queries to search or identify groups in the directory.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, <code>dc=example,dc=com</code> .
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

5. In the **LDAP Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
Groups	The search filter for obtaining group objects. For example: <code>(objectClass=group)</code>
Bind user	The search filter for obtaining the bind user object, that is, the user that can bind to the directory. For example: <code>(objectClass=person)</code>
Users	The search filter for obtaining users to sync. For example: <code>(&(objectClass=user)(objectCategory=person))</code>
Object UUID	The attribute that is used in your LDAP directory to define the UUID of a user or group. For example: <code>entryUUID</code> .

6. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

7. In the **Configure User and Group Provisioning** screen, click **Configure** to provision the users and groups from the identity provider to the VCF Identity Broker.
8. In the **Directory Review Information** screen, review the information directory information and click **Next**.
9. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

The attributes in the VCF Identity Broker directory are mapped to the LDAP Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.

Note:

If you sync the email address of your company and try to log in with that email address, you cannot do so as VCF Identity Broker expects the domain name. For example, log in with `user@company.com@domain.com`. For more information, see [KB 393150](#).

10. In the **Group Provisioning** screen, enter the relevant details or select the required option and click **Next**.

Option	Description
Specify the base group DN	Enter the details of the base group DN.
View Groups	Click this option if you want to specify the group names. You can search for the group name and select the group DN that you want to sync from the list in datagrid below. Note:

Option	Description
	You can specify only one Base DN for group searches. To provision groups from multiple locations within Active Directory, it is recommended that you set the Base DN at a higher level in the directory hierarchy that includes all the required groups such as a parent Organizational Unit (OU) or domain root and then select the necessary groups.

Only groups with a valid DN, a unique identifier, and the required attributes (such as group name and object GUID) that match the configured directory search attribute are provisioned.

11. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click this option to specify the user names that you want to sync. You can search for the user name and select the user DN that you want to sync from the datagrid. Note: If you want to sync the users directly from the group, do not specify a base user DN.
View Users	Click this option if you want to specify the user names. You can search for the user name and select the user DN that you want to sync from the list in the datagrid below.

12. From the **Review** screen, review the configuration details of the identity provider.

13. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.

14. Click **Done**.

Managing OpenLDAP

You can integrate OpenLDAP with VCF Single Sign-On. After OpenLDAP is integrated, you can perform several operations to manage the OpenLDAP connection such as view a list of directories, add additional directories, modify sync settings, and view the sync log.

To start managing the OpenLDAP connection, navigate to the corresponding VCF Instance in which VCF Single Sign-On was configured. The **Identity Source** tab contains the relevant information pertaining to the identity provider configuration. If more than one VCF Instance is connected to the OpenLDAP, use the link in the **Identity Source** tab to navigate to the relevant VCF Instance.

Procedure

- From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
- From the **Identity & Access** pane on the right side, select the VCF Instance for which you would like to reset VCF Single Sign-On from the **VCF Instances** drop-down option.
- From the **Identity Source** tab on the right side, you see different sections such as Deployment Mode, Authentication Information, and Directory Information.

The **Deployment Mode** and **Authentication Information** sections indicate the type of deployment and connection respectively.

Directory Information

The datagrid lists the directories that have been configured. You can add, edit, or delete the directories.

Column Name	Description
Directory Name	User friendly name of the directory that was configured.
Domain	The domain name of the domain controller that was configured as part of the directory configuration.
Synced groups	The total number of groups that have been synced from the directory to the VCF Identity Broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but only 8 groups were synced, then only 8 groups are displayed.
Synced users	The total number of users that have been synced from the directory to the VCF Identity Broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but only 8 groups were synced, then only 8 groups are displayed.
Alerts	The number of alert(s) that may have occurred during the last performed successful sync.
Sync on request	Click the link to trigger an on-demand sync from the corresponding directory in OpenLDAP.
Last sync status	The day and time of the last performed successful sync. If the last sync fails, the status reflects the same. Click on Learn More for more information. See the table about Sync Failure Messages for more information about sync failure errors and the reasons.

Sync Failure Messages

Error Message	Reason/Remediation
LDAP server is not reachable	This could be due to network issues, firewall blocks, an incorrect host name, or invalid BIND credentials. Verify the configuration and retry the sync.
An internal error occurred during a sync with the LDAP server.	Retry sync.
The users don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
The groups don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
Group Query Failed	The group is not found in Active Directory.

Add Additional Directories

You can add a new directory by clicking on the **Add** button. To configure a directory in OpenLDAP, see [Configure OpenLDAP as an Identity Provider](#).

Edit Existing Directories

You can edit the directory by clicking on the **Edit** button. From the **Directory Details** page, click the **Edit** button at the top of each of the following pages to modify the existing configuration details:

- Details
- Attribute mappings
- Group provisioning
- User provisioning

- Sync Settings - For more information see, [Modify Sync Settings](#).
- Sync Log - For more information, see [View Sync Log](#).

Note:

After you modify configurations in the Attribute Mappings, Group provisioning, User provisioning, and Sync Settings screens, you are prompted with an option to trigger an on-demand sync. If you do not trigger the on-demand sync, changes to users and groups take place in the next scheduled sync.

Modify Sync Settings

Editing the directories includes modifying the sync settings. The sync schedule is set to once per week, however you can modify the sync schedule by clicking the **Edit** button in the **Sync Settings** screen. Sync schedule is configured in UTC.

View Sync Log

The sync log under each directory contains the log of each sync between VCF Identity Broker. and the configured domain controller in OpenLDAP. As part of this sync, users and groups are added or removed. You can view the time at which the syn was completed, the number of groups and users that were modified because of the sync, the status of the sync, and alerts. Here is a list of error messages.

Error Message	Possible Reason(s)
Invalid group attribute name	The group object has an invalid or a missing required attribute from OpenLDAP.
Invalid user attribute name	The user object has an invalid or missing required attribute from OpenLDAP.
Multiple attribute values	The attribute that was mapped to the user/group contains more than one value in OpenLDAP.
User or Group Query Failed	The user/group was not found in OpenLDAP.

Step 4: Configure VCF Single Sign-On for NSX and vCenter

After you have configured an identity provider for VCF Single Sign-On, connect vCenter and NSX components to the VCF Identity Broker for Single Sign-On-based access.

Prerequisites

- Ensure that you have configured the identity provider for VCF Single Sign-On. For more information, see [Configure an Identity Provider](#).

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select the VCF Instance for which you are configuring VCF Single Sign-On from the **VCF Instances** drop-down option.
3. From the **Enable Single Sign-On** page, click the **Start** button against the **Configure Components** option.
4. From the **Configure clients** page, you can see the VCF Identity Broker, VCF Instance, and deployment mode used. Select the VCF components such as vCenter or NSX for which you want to connect and configure VCF Single Sign-On from the grid below.

Note:

- Only licensed VCF components that are of version 9.0 or later and are not part of the ELM ring are supported for configuring VCF Single Sign-On access.
 - After the components are configured as part of the initial setup, log in with the local admin account to each component and assign the necessary service roles to the provisioned users and groups. You can launch the components from the Access Console from the grid.
 - For information about assigning service roles, see [Assigning Roles and Permissions](#).
5. Click **Configure**.
 6. On the **Role Assignment Required** dialog, select the **I confirm that I understand the requirement to perform role assignments in order to enable SSO for the selected component(s)**. check box and click **Continue**.
 7. Click **Finish Setup**.
 8. On the **Finish Setup** dialog, click **Continue**.

Log in to VCF Operations as an administrator and assign the necessary service roles to the provisioned users and groups. For more information, see [Assigning Roles and Permissions](#).

Step 5: Configure VCF Single Sign-On for VCF Operations and VCF Automation

After you have configured an identity provider for VCF Single Sign-On, configure VCF Single Sign-On for VCF Operations and VCF Automation. Select the VCF Identity Broker with which you wish to establish a connection for VCF Single Sign-On.

- To configure VCF Single Sign-On for VCF Operations, see [Configure VCF Single Sign-On for VCF Operations](#).
- To configure VCF Single Sign-On for VCF Automation, see [Configure VCF Single Sign-On for VCF Automation](#).

Configure VCF Single Sign-On for VCF Operations

After you have configured an identity provider for VCF Single Sign-On, configure VCF Single Sign-On for VCF Operations.

Prerequisite

- Ensure that an identity provider has been configured for VCF Single Sign-On. For more information, see [Configure a New VCF Single Sign-On for a VCF Instance](#).

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane on the right side, click **VCF Management > operations appliance**.
3. From the **operations appliance** page, under **Enable Single Sign-On**, click **Continue**.
4. From the **Role Assignment Required** dialog, select the **I confirm that I understand the requirement to perform role assignments in order to enable SSO for the selected component(s)**. check box and click **Continue**.
5. From the **Configure Component** page, select the VCF Identity Broker from the **Identity Broker** drop down list, with which you want to enable VCF Single Sign-On, and click **Configure**.
6. From the **Role Assignment Required** dialog, select the **I confirm that I understand the requirement to perform role assignments in order to enable SSO for the selected component(s)**. check box and click **Continue**.

Log in to VCF Operations as an administrator and assign the necessary service roles to the provisioned users and groups. For more information, see [Assigning Roles and Permissions](#).

Configure VCF Single Sign-On for VCF Automation

After you have configured an identity provider for VCF Single Sign-On, configure VCF Single Sign-On for VCF Automation.

Prerequisite

- Ensure that an identity provider has been configured for VCF Single Sign-On. For more information, see [Configure a New VCF Single Sign-On for a VCF Instance](#).

Procedure

1. From the VCF Operations main menu, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane on the right side, click **VCF Management > automation appliance**.
3. From the **automation appliance** page, under **Enable Single Sign-On**, click **Continue**.
4. Click **Configure** to configure VCF Automation as a client to VCF Single Sign-On.
5. From the **Configure Component** page, select the VCF Identity Broker from the **Identity Broker** drop down list, with which you want to enable VCF Single Sign-On, and click **Configure**.
6. From the **Role Assignment Required** dialog, select the **I confirm that I understand the requirement to perform role assignments in order to enable SSO for the selected component(s)**. check box and click **Continue**.

Log in to VCF Operations as an administrator and assign the necessary service roles to the provisioned users and groups. For more information, see [Assigning Roles and Permissions](#).

Step 6: (Optional) Configure VCF Single Sign-On for other VCF Components

Configure VCF Single Sign-On for other VCF Components such as VCF Operations HCX, VCF Operations for networks, VCF Operations orchestrator, and VCF Operations for logs.

- To configure VCF Single Sign-On for VCF Operations HCX, [Configure VCF Single Sign-On](#).
- To configure VCF Single Sign-On for VCF Operations for networks, see [Configuring VCF SSO for VCF Operations for Networks](#).
- To configure VCF Single Sign-On for VCF Operations for logs, see [Configuring Single Sign-On for VCF Operations for logs](#).
- To configure Single Sign-On for CLI tools like PowerCLI, see [Using VCF Single Sign-On to access vCenter using PowerCLI](#).

Using VCF Single Sign-On to Access vCenter using PowerCLI

You can configure CLI and automation tools like PowerCLI to use VCF Single Sign-On. You can configure CLI and automation tools like PowerCLI to access vCenter only, using VCF Single Sign-On. If you are an existing customer upgrading to VCF 9.0, existing automation scripts using CLI tools will not be impacted if Single Sign-On is configured in the same management domain vCenter in embedded mode. However, if Single Sign-On is configured in appliance mode, or in an alternate management domain vCenter, a new OAuth client needs to be generated for PowerCLI.

Prerequisite

Ensure that VCF Single Sign-On is completely configured for at least one VCF Instance. For more information, see [Configure a New VCF Single Sign-On for a VCF Instance](#).

- From the VCF Operations navigation pane, select **Fleet Management > Identity & Access > VCF Other Components**.
- Click **Add** and enter the following information:
 - Name - User friendly name for the OIDC client.
 - Identity broker - Choose the VCF Identity Broker with which you want to configure Single Sign-On.
 - Redirect URI - The endpoint to which the VCF Identity Broker needs to redirect after authentication. The URI needs to follow the HTTP schema, and be on localhost with a free port on the client machine. For example: "http://localhost:8844/auth."
 - Post logout redirect URI - Not applicable for Single Sign-On with PowerCLI.
- Click **Generate OIDC Client**.
- Copy and store the Client ID, client secret, and Identity Broker issuer URL securely in a safe place.
- Click **Save**.
- Create an OAuth security context object for PowerCLI. For example, see the following script.


```
$oauthSecContext = New-OAuthSecurityContext -TokenEndpointUrl "https://<vcenter_fqdn>/acs/t/CUSTOMER/to-ken" -
AuthorizationEndpointUrl "https://<vcenter_fqdn>/acs/t/CUSTOMER/authorize" -RedirectUrl "http://local-
host:8844/auth" -
ClientId "powercli-native" -ClientSecret "<secret_value>"
TokenEndpointUrl
AuthorizationEndpointUrl
RedirectUrl
ClientId
ClientSecret
```

Note:

The Redirect URL should be the same as the value entered in Step 2 - Redirect URI.

- Exchange the OAuth 2.0 security context for a SAML security context.


```
$samlSecContext = New-VISamlSecurityContext -VCenterServer '<vcenter_fqdn>' -OAuthSecurityContext $oauthSecContext
```
- Connect to your vCenter using the SAML security context.


```
Connect-VIServer -Server '<vcenter_fqdn>' -SamlSecurityContext $samlSecContext
```

Note:

Single Sign-On through CLI tools is possible only through vCenter in VCF 9.0.

Step 7. Assigning Roles and Permissions

As an administrator, after you complete configuring VCF Single Sign-On in the VCF Operations console, you must log in to the individual components, such as, vCenter, NSX, VCF Automation console, and the VCF Operations console and assign the required roles and permissions for users or groups.

- Assign Service Roles in vCenter:** To know more about assigning service roles in vCenter, see [Assigning Permissions](#).
- Assign service roles in NSX:** To know more about assigning service roles in NSX, see [Manage Local User Accounts](#).
- Assign service roles in VCF Operations:** To know more about assigning service roles in VCF Operations, see [Assigning Service Roles in VCF Operations](#).
- Assign service roles in VCF Automation:** To know more about assigning service roles in VCF Automation, see [Assigning Service Roles in VCF Automation](#).

Assigning Service Roles in VCF Operations

After you complete configuring VCF Single Sign-On in the VCF Operations console, log in to VCF Operations as a local admin to assign the required roles and permissions for users or groups.

Assign service roles in the VCF Operations console if configured for VCF Single Sign-On. Service roles have to be assigned to the provisioned users/groups to be able to login with the configured identity provider to achieve Single Sign-On between the VCF components. Without these roles, users will not be able to access the respective component consoles with their enterprise credentials.

Note:

If you had chosen JIT as the user/group provisioning method without specifying any groups, then attempt login from the associated domain for the user and groups to be dynamically provisioned when you log in. This is essential before you can assign the service roles.

Procedure

- Log in to VCF Operations as a local admin.
- From the left menu, navigate to **Administration > Control Panel > Access Control**.
- From the **Access Control** page, navigate to the **User Accounts and/or User Groups** tabs and click the **horizontal ellipsis** next to the **Add** button, and select **Import from Source**.
- From the **Import from User Groups and/or Import from Users** page, select **VCF SSO** from the drop-down options against the **Import From** field.
- Use the **Search Prefix** field to search for users and/or groups and click **Finish**.
- After the user and/or group has been added, from the **Access Control** page, select the **User and/or Group**, and click the **Vertical ellipsis** and click **Edit**.
- From the section called **Assign Roles and Scope**, assign the required roles and scope to the use and/or groups and click **Save**.

Assigning Service Roles in VCF Automation

After you complete configuring VCF Single Sign-On in the VCF Operations console, log in to the VCF Automation Provider Management Portal to assign the required roles and permissions for users or groups.

Assign service roles in the VCF Automation Provider Management Portal, if configured for VCF Single Sign-On. Service roles have to be assigned to the provisioned users/groups to be able to login with the configured identity provider to

achieve Single Sign-On between VCF components. Without these roles, users will not be able to access the respective component consoles with their enterprise credentials.

Note:

If you had chosen JIT as the user/group provisioning method without specifying any groups, then attempt log in from the associated domain for the user and groups to be dynamically provisioned when you log in. This is essential before you can assign the service roles.

Procedure

1. Log in to the VCF Automation Provider Management Portal with the **system** organization and **admin** account.
2. From the left menu, click **Access Control**.
3. From the **Users** tab, click the **Import Users** option.
4. From the **Source** drop down options, select **VCF SSO**.
5. Search for and select the required users. Assign the required roles from the **Assign Role** option.

Repeat the steps to assign roles for groups.

Connect a VCF Instance to an Existing VCF Single Sign-On Configuration

After you have configured VCF Single Sign-On, you can connect to an existing Single Sign-On configuration and add the components of the VCF Instance to the pre-configured VCF Identity Broker.

Prerequisites

- Ensure that you have added a VCF Instance. For more information, see [Step 1: Select a VCF Instance](#).
- Ensure that you have configured an identity provider for VCF Single Sign-On.

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select the VCF Instance that you want to configure to an existing SSO configuration from the **VCF Instances** drop-down option. For more information about selecting a VCF Instance, see [Step 1: Select a VCF Instance](#).
3. From the **Enable Single Sign-On** page, click the **Connect to an existing SSO configuration** option and click **Continue**.
4. Select the identity broker instance from the **Identity Broker** drop down option to connect a VCF Instance to an existing VCF Single Sign-On configuration.
5. From the **Configure clients** page, select the VCF Instance components and click **Configure**.
6. From the **Role Assignment Required** dialog, select the **I confirm that I understand the requirement to perform role assignments in order to enable SSO for the selected component(s)** checkbox and click **Continue**.

Managing VCF Single Sign-On

After you configure and start using VCF Single Sign-On, you can view the existing VCF Identity Brokers, reset VCF Single Sign-On, change an identity provider, change the deployment mode, edit the configuration of an identity provider, and add or deregister existing component configurations.

- To view configured VCF Identity Brokers, see [Overview of VCF Single Sign-On Configurations](#).
- To reset VCF Single Sign-On, see [Reset VCF Single Sign-On](#).
- To change the identity provider in VCF Single Sign-On, see [Change the Identity Provider in VCF Single Sign-On](#).
- To change the deployment mode, see [Change the Deployment Mode in VCF Single Sign-On](#).
- To edit the configuration of an identity provider, see [Edit the Configuration of an Identity Provider](#).

- To carry out additional VCF Single Sign-On component configurations, see [Additional Component Configurations](#).
- To deregister existing VCF Single Sign-On component configurations, see [Deregister Existing Component Configurations](#).
- To manage the deployed VCF Identity Broker appliance, see [Deploying VCF Identity Broker](#).
- To change the identity management of a VCF Instance managed by another VCF Instance, see [Change the Identity Management of a VCF Instance Managed by Another VCF Instance](#).

Overview of VCF Single Sign-On Configurations

Use the **SSO Overview** page for a holistic overview of the VCF fleet that has been configured with VCF Single Sign-On.

From the **SSO Overview** page you can view VCF Identity Brokers that are grouped together. Click each VCF Identity Broker in the grid to view a list of VCF Instances that are connected to it. You can view information of each component that you click in the **SSO Information** pane on the right side.

In VCF Operations, you can monitor the status of VCF Identity Brokers based on parameters such as health and readiness. Health indicates if the VCF Identity Broker is up and running or not. Readiness indicates if the VCF Identity Broker is ready to accept incoming requests or not. From the VCF Operations console, navigate to **Inventory > Integrations icon > Infrastructure Management > Identity Broker health**.

Reset VCF Single Sign-On

You can reset VCF Single Sign-On that is associated with a specific VCF Instance.

For example, if you need to change the identity management of the VCF Instance, VCF Single Sign-On that was configured in the initial VCF Instance must be reset, and performed again in the desired VCF Instance.

You can choose to reset either:

- The identity provider configuration and the provisioned users/groups, or
- The entire VCF Single Sign-On configuration including the identity provider configuration, provisioned users/groups, and component configuration.

Note:

- The reset action cannot be reverted.
- If the user performing the reset of VCF Single Sign-On is part of the particular VCF Single Sign-On configuration being reset, then the session terminates immediately post reset. The user has to log in again as a local admin account user. You may want to consider logging in as a local admin user to perform the reset activity.

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select the VCF Instance for which you would like to reset VCF Single Sign-On from the **VCF Instances** drop-down option.
3. From the **Identity Source** tab, click **Reset SSO**.
4. Select either of the following two options and click **Reset**.

The reset options are:

- Deleting the identity provider configuration with the provisioned users/groups.
Note:
The components configured for VCF Single Sign-On against this VCF Identity Broker is retained. After you configure a new identity provider, you may need to assign service roles in each component to the newly provisioned users and groups.
- Deleting all Single Sign-On configurations including the identity broker configuration with the provisioned users/groups, and component configurations.
Note:

After you reset VCF Single Sign-On for the VCF Instance, you must configure VCF Single Sign-On again.

5. You are directed to the **Enable Single Sign-On** page from where you can complete the steps to configure VCF Single Sign-On. For more information about configuring VCF Single Sign-On, see, [Configure a New VCF Single Sign-On for a VCF Instance](#).

Note:

Based on the option you choose in Step 4, you either have to configure only the identity provider or complete all the steps to configure VCF Single Sign-On.

Change the Identity Provider in VCF Single Sign-On

If you have configured VCF Single Sign-On using a certain identity provider and protocol, you can change the identity provider in the future.

For example, you have configured VCF Single Sign-On using Active Directory, and wish to change to a modern identity provider like Okta, Ping Identity, or Microsoft Entra ID. To change the identity provider, you must reset VCF Single Sign-On and reconfigure the identity provider.

Note:

Since this is a partial reset, the deployment mode and client component configuration is retained. Therefore, you do not have to reconfigure the client for vCenter, VCF Operations HCX, VCF Operations, VCF Automation, VCF Operations for networks, or VCF Operations for logs to the specific VCF Identity Broker.

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select the VCF Instance for which you would like to reset VCF Single Sign-On from the **VCF Instances** drop-down option.
3. From the **Identity Source** tab on the right side, click **Reset SSO**. For more information about resetting VCF Single Sign-On, see [Reset VCF Single Sign-On](#).
4. In the **Reset SSO** dialog box that is displayed, since you are changing the identity provider, select the first option, I confirm to delete the identity provider configuration including the provisioned users and groups.

The provisioned users and groups are removed. Ensure that the users and groups are assigned the required service roles in each component after you complete configuring the identity provider. For information about assigning roles, see [Step 7. Assigning Roles and Permissions](#).
5. Click **Reset**. The **Enable Single Sign-On** screen is displayed.
6. From the **Enable Single Sign-On** screen, click the second step, **Configure Identity Provider**, to reconfigure the identity provider. For more information, see [Step 3: Configure an Identity Provider](#).

Change the Deployment Mode in VCF Single Sign-On

If you have configured VCF Single Sign-On using a certain deployment mode, you can choose another deployment mode in the future.

For example, you have chosen the 'VCF Identity Broker (embedded)' deployment mode to configure VCF Single Sign-On for a VCF Instance. If required, you can change the deployment mode to 'VCF Identity Broker (appliance)' for the specific VCF Instance. Conversely, you can change the deployment mode from 'VCF Identity Broker (appliance)' to 'VCF Identity Broker (embedded)'. To change the deployment mode, you must reset and reconfigure VCF Single Sign-On. For more information, see [Reset Single Sign-On](#).

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane on the right side, select the VCF Instance for which you would like to reset VCF Single Sign-On from the **VCF Instances** drop-down option.
3. From the **Identity Source** tab on the right side, click **Reset SSO**. For more information about resetting VCF Single Sign-On, see [Reset VCF Single Sign-on](#).
4. From the **Reset SSO** dialog box that is displayed, since you are changing the deployment mode, select the second option, I confirm to delete of all SSO-related configurations, including the identity provider configuration, the provisioned users and groups, and the component configuration.
5. Click **Reset**. The **Enable Single Sign-On** screen is displayed.
6. From the **Enable Single Sign-On** screen, reconfigure VCF Single Sign-On. For more information, see [Configure a New VCF Single Sign-On for a VCF Instance](#).

Ensure that the users and groups are assigned the required service roles in each component after you complete configuring the identity provider. For information about assigning roles, see [Step 7. Assigning Roles and Permissions](#).

Edit the Configuration of an Identity Provider

You can make changes to the identity provider configuration after you have configured VCF Single Sign-On.

The **Identity Source** tab that you access for each VCF Instance displays the deployment mode, authentication information, and directory information. If the VCF Identity Broker is deployed in the same VCF Instance, you can make changes in the associated **Identity Source** tab. If the VCF Identity Broker for the required VCF Instance is deployed

in another VCF Instance, use the hyperlink in the **Identity Source** tab to navigate to the relevant page to edit the configuration of the identity provider.

Note:

To change the identity provider or provisioning mode, you must reset VCF Single Sign-On. For more information, see [Reset VCF Single Sign-On](#).

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select the VCF Instance for which you would like to reset VCF Single Sign-On from the **VCF Instances** drop-down option.
3. If the VCF Identity Broker is Deployed in the Same VCF Instance, from the **Identity Source** tab, click **Edit**.

Note:

If the VCF Identity Broker is deployed in a different VCF Instance, from the **Identity Source** tab, click the link to the relevant VCF Instance to manage and edit the identity provider.

To configure identity providers, see [Step 3: Configure an Identity Provider](#). Navigate to the page depending on which identity provider you have configured.

Additional Component Configurations

You can connect additional vCenter and NSX components to the VCF Identity Broker for Single Sign-On-based access. You can also connect additional VCF components such as VCF Operations for logs, VCF Operations for networks, and VCF Operations HCX to the VCF Identity Broker for Single Sign-On-based access.

Procedure

- **Add additional vCenter and NSX components**
- From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
- From the **Identity & Access** pane, navigate to **VCF Instances** and select the relevant VCF Instance for which you have configured VCF Single Sign-On.
- Select the **Component Configuration** tab on the right side, and in the grid below, use the filter and search for components that are 'Not Configured'. Select them and click **Configure Component**.
- **Add additional components like VCF Operations for logs, VCF Operations for networks, and VCF Operations HCX**
- From the **Identity & Access** pane, navigate to **VCF Other Components** and follow the steps in [Step 6: \(Optional\) Configure VCF Single Sign-On for other VCF Components](#).

As an administrator, after you complete configuring VCF Single Sign-On in the VCF Operations console, you must log in to the individual components such as, vCenter , NSX, and others and assign the required roles and permissions for users or groups. For more information, see [Step 7. Assigning Roles and Permissions](#).

Deregister Existing Component Configurations

If you have configured components such as vCenter, NSX, VCF Operations, VCF Automation and so on as clients for VCF Single Sign-On, you can choose to deregister or remove these components from VCF Single Sign-On.

You can deregister the following component configurations:

- Deregister vCenter and NSX from VCF Single Sign-On. For more information, see [Deregister vCenter and NSX from VCF Single Sign-On](#).

- Deregister VCF Operations and/or VCF Automation, from VCF Single Sign-On. For more information, see [Deregister VCF Operations and/or VCF Automation from VCF Single Sign-On](#).
- Modify VCF Operations and/or VCF Automation from VCF Single Sign-On. For more information see [Modify VCF Single Sign-On for VCF Operations and/or VCF Automation](#).
- Deregister other components such as, VCF Operations HCX, VCF Operations for logs, VCF Operations for networks, and VCF Operations orchestrator from VCF Single Sign-On. For more information, see [Deregister VCF Single Sign-On for Other Components](#).

Deregister vCenter and NSX from VCF Single Sign-On

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane on the right side, select the VCF Instance for which you have configured VCF Single Sign-On from the **VCF Instances** drop-down option.
3. From the **Component Configuration** tab on the right side, select the component you want to deregister and click the **Deregister Component** button.

Deregister VCF Operations and/or VCF Automation from VCF Single Sign-On

Note:

If you have configured VCF Single Sign-On for the VCF Operations or VCF Automation, you can choose to deregister these appliances from VCF Single Sign-On. The change deactivates the previous authentication source and deletes the provisioned users and groups. This action cannot be reverted.

To deregister VCF Single Sign-On for VCF Operations, perform the following steps:

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select **operations appliance**.
3. Click **Edit**.
4. From the pop-up box that is displayed, click the **VCF SSO Enabled** toggle button to deactivate the configuration.

Note:

The existing provisioned users and groups cannot be used in VCF Operations. Reassign the VCF Operations roles to the newly provisioned users and groups from the new authentication source.

5. Click **Continue**.

To deregister VCF Single Sign-On for VCF Automation, perform the following steps:

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane on the right side, select **automation appliance**.
3. Click **Edit**.
4. From the pop-up box that is displayed, click the **VCF SSO Enabled** toggle button to deactivate the configuration.

Note:

The existing provisioned users and groups cannot be used in VCF Automation. Reassign the VCF Automation roles to the newly provisioned users and groups from the new authentication source.

5. Click **Continue**.

Modify VCF Single Sign-On for VCF Operations and/or VCF Automation

Note:

If you have configured VCF Single Sign-On for VCF Operations and/or VCF Automation, you can choose to change the identity broker for these appliances. The change deactivates the previous authentication source and the existing provisioned users and groups cannot be used in VCF Automation. You have to reassign the VCF Automation and/or VCF Operations roles to the newly provisioned users and groups from the new authentication source. This action cannot be reverted.

To modify VCF Single Sign-On for VCF Operations, perform the following steps:

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select **operations appliance**.
3. Click **Edit**.
4. From the the pop-up box that is displayed, select a different identity broker from the **Identity Broker** drop down option.
5. Click **Continue**.

Note:

The existing provisioned users and groups cannot be used in VCF Operations. Reassign the VCF Operations roles to the newly provisioned users and groups from the new authentication source.

To modify VCF Single Sign-On for VCF Automation, perform the following steps:

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane, select **automation appliance**.
3. Click **Edit**.
4. From the the pop-up box that is displayed, select a different identity broker from the **Identity Broker** drop down option.
5. Click **Continue**.

Note:

The existing provisioned users and groups cannot be used in VCF Automation. Reassign the VCF Automation roles to the newly provisioned users and groups from the new authentication source.

Deregister VCF Single Sign-On for Other Components

If you have configured VCF Single Sign-On for other components for such as VCF Operations for logs, VCF Operations for networks, VCF Operations HCX, and VCF Operations orchestrator , you can choose to deregister the component.

Procedure

1. From the VCF Operations navigation pane, select **Fleet Management > Identity & Access**.
2. From the **Identity & Access** pane on the right side, select **VCF Other components**.
The list of generic components for which you have configured VCF Single Sign-On is listed.
3. Select the component you want to deregister and click **Delete**.

Note:

Depending on the component for which this OIDC client was created, you can choose to delete the VCF Single Sign-On configuration from the respective component console for complete deregistration. For example, if this OIDC client was used to configure VCF Single Sign-On for VCF Operations HCX, after the client is deleted under **VCF Other Components**, log in to VCF Operations HCX as a local admin user and delete the VCF Single Sign-On configuration.

Change the Identity Management of a VCF Instance Managed by Another VCF Instance

If a particular VCF Instance was connected to an existing Single Sign-On configuration, the identity source configuration for that instance can be managed only from within the initial VCF Instance in which it was configured. You may want to connect the identity source configuration to a different VCF Instance or create a new Single Sign-On configuration.

Scenario #1: Connect to a Different Single Sign-On Configuration

If you wish to connect the components of a VCF Instance that is managed by a different VCF Instance to an alternate one, perform the following steps:

1. Navigate to the VCF Instance that you intend to connect to a different VCF Identity Broker.
2. Click **Component Configuration**.
3. Click **Edit** next to **Authentication Source**.
4. In the modal box that opens, from the **Identity Broker** drop down, select the desired VCF Identity Broker.
5. Click **Continue**.
6. All the registered components of this VCF Instance are now be connected to the Single Sign-On configuration of the newly selected VCF Identity Broker.

Scenario #2 - Deregister the VCF Instance From the Single Sign-On Configuration

If you wish to remove the VCF Instance from the existing Single Sign-On configuration, perform the following steps:

1. Navigate to the VCF Instance that you intend to connect to a different VCF Identity Broker.
2. Click **Component Configuration**.
3. In the grid under **Component List**, all the components belonging to this VCF component are listed.
4. Use the filter 'Status' to filter the configured components.
5. Select all the configured components and click **Deregister Component**.
6. After all the registered components have been deregistered, the page reloads to display the following options. Choose the desired option to configure Single Sign-On for this VCF Instance.
 - a. Connect to an existing Single Sign-On configuration. For more information, see [Connect a VCF Instance to an Existing VCF Single Sign-On Configuration](#).
 - b. Create a new Single Sign-On configuration. For more information, see [Configure a New VCF Single Sign-On for a VCF Instance](#).

Managing Certificates in VMware Cloud Foundation

Manage certificates for VMware Cloud Foundation components according to industry standards and the requirements of your organization. You can use the VCF Operations console to view and manage the certificates for all VMware Cloud Foundation components.

Access to all management component interfaces must be over a Secure Socket Layer (SSL) connection. During deployment, each component is assigned a certificate from a default signing Certificate Authority. You should replace the default certificates for the management domain components with trusted enterprise CA-signed certificates to provide secure access to each component.

In addition, you should replace certificates when:

- A certificate has expired or is nearing its expiration date.
- A certificate has been revoked by the issuing certificate authority.
- Optionally, when you create a new workload domain.

Use the VCF Operations console to:

- View certificates and certificate alerts.
- Enable automatic renewal of certificates.
- Configure a certificate authority (CA).
- Generate certificate signing requests (CSRs).
- Replace certificates.

You can manage certificates for the VCF Management components and for the components in a VCF Instance or VCF domain.

VCF Management Components	VCF Instance/domain Components
• Fleet Management • VCF Automation • VCF Identity Broker • VCF Operations • VCF Operations for logs • VCF Operations for networks	• ESX • NSX Manager • SDDC Manager • vCenter

For Non-Disruptive Certificate updates, you must manage the following certificates from VCF Operations. You can enable auto renewal for these certificates to avoid any disruption caused during certificate renewal.

Certificate	Component Product
ESX SSL	ESXi
vCenter machine SSL	vCenter
NSX LM and VIP	NSX
SDDC M SSL	SDDC Manager
VCF Identity Broker	VIDB
VCF Operations	VCF Operations
VCF Automation	VCF Automation
VCF Operations for logs	VCF Operations for logs
VCF Operations for networks	VCF Operations for networks
Fleet Management	VCF Operations Fleet Management

View Certificates and Certificate Alerts for VMware Cloud Foundation Components

You can view detailed information, including alerts, about the certificates for all VMware Cloud Foundation components from the VCF Operations console.

The certificate management UI includes filtering options that can help you find certificates based on type, status, component, and so on.

1. Log in to the VCF Operations console at https://<vcf_operations_fqdn> with a user assigned the Administrator role.
2. Click **Fleet Management > Certificates**.
3. Click **VCF Management** to view certificate information for the VCF Management components.
4. Click **VCF Instances** and click a VCF Instance name to view certificate information for the components in the selected VCF Instance.
Use the **Show ESX Hosts** toggle to show or hide ESX hosts.

5. Click **VCF Instances** and click a VCF domain name to view certificate information for the components in the selected VCF domain.
Use the **Show ESX Hosts** toggle to show or hide ESX hosts.
6. Click **Certificate Alerts** to view certificate-related alerts.

Configure a Certificate Authority for VMware Cloud Foundation

Before you can replace a certificate with a CA-configured certificate or activate auto-renewal for Microsoft Certificate Authority or OpenSSL certificates, you must configure a Certificate Authority (CA).

For configuring a Microsoft CA:

- Prepare the Microsoft Certificate Authority. See [Prepare Your Microsoft Certificate Authority to Allow VMware Cloud Foundation to Manage Certificates](#).
- Verify connectivity between VCF Operations and the Microsoft Certificate Authority Server. See [VMware Ports and Protocols](#).
- Verify that time is synchronized between the Microsoft Certificate Authority and the SDDC Manager appliance. Each system can be configured with a different time zone, but it is recommended that they receive their time from the same NTP source.

If you are using OpenSSL configured on the SDDC Manager appliance, there are no additional prerequisites.

VCF management components only support Microsoft Certificate Authority. VCF Instance components support both Microsoft Certificate Authority and OpenSSL.

1. Log in to the VCF Operations console at https://<vcf_operations_fqdn> with a user assigned the Administrator role.
2. Click **Fleet Management > Certificates**.
3. Click **VCF Management** or click **VCF Instances** and click a VCF Instance name.
You configure the Certificate Authority for VCF management components separately from the Certificate Authority for VCF Instance components. If you have more than one VCF Instance, you configure the Certificate Authority for each one separately.
4. Click **Configure CA**.
5. Enter the details for your Certificate Authority and click **Save**.

Table 929: Microsoft Certificate Authority

Setting	Description
CA Server URL	Specify the URL for the issuing certificate authority. This address must begin with https:// and end with certsrv . For example, https://ca.example.com/certsrv .
User Name	Enter a least privileged service account. For example, <code>svc-vcf-ca</code> .
Password	Enter the password for the least privileged service account.

Setting	Description
Template Name	Enter the issuing certificate template name. You must create this template in Microsoft Certificate Authority. For example, VMware.

Table 930: Open SSL

Setting	Description
Common Name	Specify the FQDN of the SDDC Manager appliance.
Country	Select the country where your company is registered.
Locality Name	Specify the city or the locality where your company is legally registered.
Organization Name	Specify the name under which your company is known. The listed organization must be the legal registrant of the domain name in the certificate request.
Organization Unit Name	Use this field to differentiate between the divisions within your organization with which this certificate is associated.
State	Enter the full name (do not abbreviate) of the state, province, region, or territory where your company is legally registered.

If you have more than one VCF Instance, you can repeat this procedure to configure a Certificate Authority for each one.

Prepare Your Microsoft Certificate Authority to Allow VMware Cloud Foundation to Manage Certificates

If you plan to configure a Microsoft Certificate Authority for use with VMware Cloud Foundation, there are some steps you must perform before you can do so.

In order to generate signed certificates the Microsoft Certificate Authority must be configured to allow integration with VMware Cloud Foundation.

Install Microsoft Certificate Authority Roles

Install the Certificate Authority and Certificate Authority Web Enrollment roles on the Microsoft Certificate Authority server to facilitate certificate generation from VCF Operations.

Note:

When connecting to Microsoft Active Directory Certificate Services, ensure that the Web Enrollment role is installed on the same machine where the Certificate Authority role is installed. VCF Operations can't request and sign certificates automatically if the two roles (Certificate Authority and Web Enrollment roles) are installed on different machines.

- Log in to the Microsoft Certificate Authority server by using a Remote Desktop Protocol (RDP) client.

FQDN	<i>Active Directory Host</i>
User	Active Directory administrator
Password	<i>ad_admin_password</i>

- Add roles to Microsoft Certificate Authority server.
 - Click **Start > Run**, enter *ServerManager*, and click **OK**.
 - From the **Dashboard**, click **Add roles and features** to start the **Add Roles and Features** wizard.
 - On the **Before you begin** page, click **Next**.
 - On the **Select installation type** page, click **Next**.
 - On the **Select destination server** page, click **Next**.
 - On the **Select server roles** page, under **Active Directory Certificate Services**, select **Certification Authority** and **Certification Authority Web Enrollment** and click **Next**.
 - On the **Select features** page, click **Next**.
 - On the **Confirm installation selections** page, click **Install**.

Configure the Microsoft Certificate Authority for Basic Authentication

Configure the Microsoft Certificate Authority with basic authentication to allow VCF Operation the ability to manage signed certificates.

The Microsoft Certificate Authority and IIS must be installed on the same server.

- Log in to the Active Directory server by using a Remote Desktop Protocol (RDP) client.

FQDN	<i>Active Directory Host</i>
User	Active Directory administrator
Password	<i>ad_admin_password</i>

- Add Basic Authentication to the Web Server (IIS).
 - Click **Start > Run**, enter *ServerManager*, and click **OK**.
 - From the **Dashboard**, click **Add roles and features** to start the **Add Roles and Features** wizard.
 - On the **Before you begin** page, click **Next**.
 - On the **Select installation type** page, click **Next**.
 - On the **Select destination server** page, click **Next**.
 - On the **Select server roles** page, under **Web Server (IIS) > Web Server > Security**, select **Basic Authentication** and click **Next**.
 - On the **Select features** page, click **Next**.
 - On the **Confirm installation selections** page, click **Install**.
- Configure the certificate service template and CertSrv web site, for basic authentication.
 - Click **Start > Run**, enter *Inetmgr.exe* and click **OK** to open the **Internet Information Services Application Server Manager**.
 - Navigate to *your_server > Sites > Default Web Site > CertSrv*.
 - Under **IIS**, double-click **Authentication**.
 - On the **Authentication** page, right-click **Basic Authentication** and click **Enable**.
 - In the navigation pane, select **Default Web Site**.
 - In the **Actions** pane, under **Manage Website**, click **Restart** for the changes to take effect.

Create and Add a Microsoft Certificate Authority Template

You must set up a certificate template in the Microsoft Certificate Authority. The template contains the certificate authority attributes for signing certificates for the VMware Cloud Foundation components. After you create the template, you add it to the certificate templates of the Microsoft Certificate Authority.

1. Log in to the Active Directory server by using a Remote Desktop Protocol (RDP) client.

FQDN	Active Directory Host
User	Active Directory administrator
Password	ad_admin_password

2. Click **Start > Run**, enter `certtmpl.msc`, and click **OK**.
3. In the **Certificate Template Console** window, under **Template Display Name**, right-click **Web Server** and select **Duplicate Template**.
4. In the **Properties of New Template** dialog box, click the **Compatibility** tab and configure the following values.

Setting	Value
Certification Authority	Windows Server 2008 R2
Certificate recipient	Windows 7 / Server 2008 R2

5. In the **Properties of New Template** dialog box, click the **General** tab and enter a name for example, `VMware` in the **Template display name** text box.
6. In the **Properties of New Template** dialog box, click the **Extensions** tab and configure the following.
 - a) Click **Application Policies** and click **Edit**.
 - b) Click **Server Authentication**, click **Remove**, and click **OK**.
 - c) Click **Basic Constraints** and click **Edit**.
 - d) Click the **Enable this extension** check box and click **OK**.
 - e) Click **Key Usage** and click **Edit**.
 - f) Click the **Signature is proof of origin (nonrepudiation)** check box, leave the defaults for all other options and click **OK**.
7. In the **Properties of New Template** dialog box, click the **Subject Name** tab, ensure that the **Supply in the request** option is selected, and click **OK** to save the template.
8. Add the new template to the certificate templates of the Microsoft CA.
 - a) Click **Start > Run**, enter `certsrv.msc`, and click **OK**.
 - b) In the **Certification Authority** window, expand the left pane, right-click **Certificate Templates**, and select **New > Certificate Template to Issue**.
 - c) In the **Enable Certificate Templates** dialog box, select **VMware**, and click **OK**.

Assign Certificate Management Privileges to the VMware Cloud Foundation Service Account

Before you can use the Microsoft Certificate Authority and the pre-configured template, it is recommended to configure least privilege access to the Microsoft Active Directory Certificate Services using an Active Directory user account as a restricted service account.

- Create a user account in Active Directory with Domain Users membership. For example, `svc-vcf-ca`.

1. Log in to the Microsoft Certificate Authority server by using a Remote Desktop Protocol (RDP) client.

FQDN	Active Directory Host
User	Active Directory administrator

Password	ad_admin_password

2. Configure least privilege access for a user account on the Microsoft Certificate Authority.
 - a) Click **Start > Run**, enter `certsrv.msc`, and click **OK**.
 - b) Right-click the certificate authority server and click **Properties**.
 - c) Click the **Security** tab, and click **Add**.
 - d) Enter the name of the user account and click **OK**.
 - e) In the **Permissions for** section configure the permissions and click **OK**.

Setting	Value (Allow)
Read	Deselected
Issue and Manage Certificates	Selected
Manage CA	Deselected
Request Certificates	Selected

3. Configure least privilege access for the user account on the Microsoft Certificate Authority Template.
 - a) Click **Start > Run**, enter `certtmpl.msc`, and click **OK**.
 - b) Right-click the VMware template and click **Properties**.
 - c) Click the **Security** tab, and click **Add**.
 - d) Enter the `svc-vcf-ca` service account and click **OK**.
 - e) In the **Permissions for** section configure the permissions and click **OK**.

Setting	Value (Allow)
Full Control	Deselected
Read	Selected
Write	Deselected
Enroll	Selected
Autoenroll	Deselected

Set Up Automatic Renewal of Certificates in VMware Cloud Foundation

You can enable automatic renewal of certificates for one or more VMware Cloud Foundation components.

If you are using Microsoft CA or OpenSSL certificates for auto-renewal, configure a Certificate Authority before activating auto-renewal. See [Configure a Certificate Authority for VMware Cloud Foundation](#).

VMware Cloud Foundation supports automatic renewal of Transport Layer Security (TLS) certificates for components that have VMCA, Microsoft CA, Open SSL, or self-signed certificates and support non-disruptive certificate updates.

You configure automatic renewal separately for VCF Management components and VCF Instance components.

Automatic renewal happens 60 days prior to certificate expiration.

1. Log in to the VCF Operations console at https://<vcf_operations_fqdn> with a user assigned the Administrator role.
2. Click **Fleet Management > Certificates**.
3. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or VCF domain name.
4. Use the **Activate Auto-renewal** toggle to activate auto-renewal.
5. Review the information and click **Confirm**.
Updated auto-renewal status can some time to appear in the VCF Operations console.

Manually Renew Certificates in VMware Cloud Foundation

You can manually renew Transport Layer Security (TLS) certificates for components that have VMCA, Microsoft CA, Open SSL, or self-signed certificates.

If you are using Microsoft CA or OpenSSL certificates to manually renew certificates, configure a Certificate Authority. See [Configure a Certificate Authority for VMware Cloud Foundation](#).

You cannot renew root certificates.

1. Log in to the VCF Operations console at https://<vcf_operations_fqdn> with a user assigned the Administrator role.
2. Click **Fleet Management > Certificates**.
3. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or VCF domain name.
4. Select a component with a supported certificate type and click **Renew Certificate**.
Use the **Show ESX Hosts** toggle to show or hide ESX hosts.
5. Click **Yes**.

Replace a Certificate with a CA-Signed Certificate

After you configure a certificate authority (CA) in VCF Operations, you can generate a certificate signing request (CSR) and then replace an existing certificate for a VMware Cloud Foundation component with a CA certificate.

Configure a Certificate Authority. See [Configure a Certificate Authority for VMware Cloud Foundation](#).

1. Log in to the VCF Operations console at https://<vcf_operations_fqdn> with a user assigned the Administrator role.
2. Click **Fleet Management > Certificates**.
3. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or VCF domain name.
4. Select a component click **Generate CSR** from the three-dot menu (...).
Use the **Show ESX Hosts** toggle to show or hide ESX hosts.
5. Enter the information for the CSR and click **Save**.

Option	Description
Common Name	Enter the common name.
Organization	Type the name under which your company is known. The listed organization must be the legal registrant of the domain name in the certificate request.
Organizational Unit	Use this field to differentiate between divisions within your organization with which this certificate is associated.

Option	Description
Country	Select the country name where your company is legally registered.
State/Province	Type the full name (do not abbreviate) of the state, province, region, or territory where your company is legally registered.
Locality	Type the city or locality where your company is legally registered.
Email Address	Optionally, enter a contact email address.
Host	Enter the host name.
Subject Alternative Name	Enter the subject alternative name(s).
Key Size	Select the key size (2048 bit or 4096 bit) from the drop-down menu. Note: ESX hosts can only use the 2048 bit key size.

Wait until the CSR is successfully generated before proceeding to the next step.

6. Click the three-dot menu (...) and select **Replace With Configured CA Certificate**.
7. Select your CA and click **Confirm**.
OpenSSL is only available for VCF Instance components.

Replace a Certificate with an External CA-Signed Certificate

If you do not want to configure a Certificate Authority in VCF Operations, you can generate a CSR, request a signed certificate from your external Certificate Authority, import the certificate, and then use it to replace an existing certificate for a VMware Cloud Foundation component.

1. Log in to the VCF Operations console at https://<vcf_operations_fqdn> with a user assigned the Administrator role.
2. Click **Fleet Management > Certificates**.
3. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or VCF domain name.
4. Select a component click **Generate CSR** from the three-dot menu (...).
Use the **Show ESX Hosts** toggle to show or hide ESX hosts.
5. Enter the information for the CSR and click **Save**.

Option	Description
Common Name	Enter the common name.
Organization	Type the name under which your company is known. The listed organization must be the legal registrant of the domain name in the certificate request.
Organizational Unit	Use this field to differentiate between divisions within your organization with which this certificate is associated.
Country	Select the country name where your company is legally registered.
State/Province	Type the full name (do not abbreviate) of the state, province, region, or territory where your company is legally registered.
Locality	Type the city or locality where your company is legally registered.
Email Address	Optionally, enter a contact email address.
Host	Enter the host name.

Option	Description
Subject Alternative Name	Enter the subject alternative name(s).
Key Size	Select the key size (2048 bit or 4096 bit) from the drop-down menu.

Wait until the CSR is successfully generated before proceeding to the next step.

- Click the three-dot menu (...) and select **Download CSR**.
- When the download completes, request a signed certificate from your external Certificate Authority.
- Click the three-dot menu (...) and select **Import Certificate**.
Note: Only PEM encoded certificates are supported.
- Enter a name for the certificate.
- Select a **Source** and enter the required information.

Source	Required Information
Paste Text	Copy and paste the: - Server Certificate - Certificate Authority Paste the server certificate and the certificate authority in PEM format (base64-encoded) . For example: <pre>-----BEGIN CERTIFICATE----- <certificate content> -----END CERTIFICATE-----</pre> If the Certificate Authority includes intermediate certificates, it should be in the following format: <pre>-----BEGIN CERTIFICATE----- <Intermediate certificate content> -----END CERTIFICATE----- -----BEGIN CERTIFICATE----- <Root certificate content> -----END CERTIFICATE-----</pre>
Certificate Chain	Click Browse to upload the certificate chain. Files with .crt, .cer, .pem, .p7b and .p7c extensions are supported.

- Click **Validate**.
If validation fails, resolve the issues and try again, or click **Cancel** to skip the certificate installation.
- Once the signed certificate has been validated successfully, click **Save**.
- Click **Replace With Imported Certificate**.
- Select the imported certificate from the drop-down menu and click **Replace**.

Managing Certificates in VMware vSphere Foundation

If you are using VMware vSphere Foundation, you can use the VCF Operations console to view and manage certificates for the VCF Operations component of VMware vSphere Foundation.

Managing VCF Operations Certificates

VCF Operations includes a central page where you can review authentication certificate contents.

How the Certificates Page Works

The Certificates page lets you examine certificate contents without the need to open the certificate outside of VCF Operations.

Where You Find Certificates

In the menu, click **Administration > Control Panel**, and then click the **Trusted Certificates** tile.

Certificate Tabs

The certificate tab describes columns of exceptions tabs.

Note:

The Certificate Revocation List (CRL) tab is activated only when you select the **Activate Standard Certificate Validation** under **Global Settings**. For more information, see [List of Global Settings](#).

Table 931: Certificate Tabs

Tabs	Description
Exceptions	Lists the certificate that is accepted by the VCF Operations administrator but is not certified by the Certificate Authority (CA).
CRL	A Certificate Revocation List (CRL) is a list of digital certificates that have been revoked by the issuing Certificate Authority (CA) before their scheduled expiration date and should no longer be trusted. Click the Add icon to upload the certificates.

Certificate Options

The options include a data grid for examining certificate contents.

Table 932: Certificate Options

Option	Description
Certificate Thumbprint	Unique alphanumeric string associated with the certificate
Issued By	Content associated with the issuer of the certificate, such as organization name and location
Issued To	Typically, content associated with the issuer, plus the certificate object Identifier (OID)
Expires	The date after which the certificate cannot be used for successful authentication

Importing CA Certificates in VCF Operations

Certificate Authority (CA) or root certificates are used for establishing the outgoing connections from VCF Operations. CA Certificates imported by the users will be used in the following VCF Operations domains: Authentication Sources (Active Directory (AD), Open LDAP, VMware Identity Manager), Outbound Plugins, and Adapter Endpoint.

1. In the menu, click **Administration > Control Panel**, and then click the **Trusted Certificates** tile.
2. Click **Import**.
The Import CA Certificate(s) dialog box appears. You can only import certificates that are encoded in the PEM format.
3. Click **Browse**.
4. Locate the certificate .pem file and click **Open** to load the file in the Import CA Certificate(s) dialog box.
The certificate information box appears with the certificate thumbprint, issued by, issued to and expiry date. For example, if you select a certificate that will expire in 10 days, you will receive a notification that the certificate is expiring soon.
Note: If a certificate is close to its expiry date, a corresponding notification is displayed on the **Home** page.
5. Click **Preview** and then click **Import**.
6. Optional: Click the **Vertical Ellipsis** to delete a certificate.

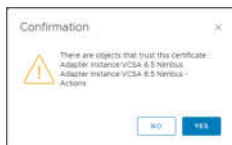
Removing an Adapter Certificate in VCF Operations

You can delete an old or expired certificate associated with an adapter.

1. In a Web browser, navigate to the VCF Operations administration interface at <https://node-FQDN-or-ip-address/ui>.
2. Log in with the administrator user name and password.
3. In the menu, click **Administration > Control Panel**, and then click the **Trusted Certificates** tile.
4. In the certificate window, select the certificate that has to be removed.
5. Click **Delete** to remove the certificate.

Note: You must own the credential used to configure the adapter to be able to delete the certificate. For more information, see [Configuring Credentials in Integrations](#).

6. If the certificate is being used by the adapter, then the following message comes up:



A certificate can be configured for one or more adapters if it is the same destination system.

7. If you delete a certificate which is already being used by another adapter, the adapter fails to connect or start. As a workaround, perform the following steps:
 - a) On the left pane, click **Administration > Integrations**.
The Accounts tab appears.
 - b) Select the particular integration and click the vertical ellipsis and then click **Edit**.
The Account Information page opens.
 - c) Click **Validate Connection**.
Note: If the adapter instance is saved using a credential that is owned by another user, you must apply your own credentials to validate the connection. For more information, see [Manage Credentials](#).
 - d) A prompt comes up asking the user to import the associated certificate. Click **OK**.
 - e) Restart the adapter from the **Accounts** tab

Custom VCF Operations Certificates

For secure VCF Operations operation, you might need to perform maintenance on authentication certificates.

Authentication certificates are for a secure machine-to-machine communication within VCF Operations itself or between VCF Operations and other systems.

By default, VCF Operations includes its own authentication certificates. The default certificates cause the browser to display a warning when you connect to the VCF Operations user interface.

Your site security policies might require that you use another certificate, or you might want to avoid the warnings caused by the default certificates. In either case, VCF Operations supports the use of your own custom certificate. You can upload your custom certificate during the initial primary node configuration or later.

Custom VCF Operations Web Certificate Requirements

A certificate used with VCF Operations must conform to certain requirements. Using a custom certificate is optional and does not affect VCF Operations features. You can also use wildcard certificates in VCF Operations.

Requirements for Custom Certificates

Custom VCF Operations certificates must meet the following requirements.

- The certificate file must include the terminal (leaf) server certificate, a private key, and all issuing certificates if the certificate is signed by a chain of other certificates.
- In the file, the leaf certificate must be first in the order of certificates. After the leaf certificate, the order does not matter.
- In the file, all certificates and the private key must be in PEM format. VCF Operations does not support certificates in PFX, PKCS12, PKCS7, or other formats.
- In the file, all certificates and the private key must be PEM-encoded. VCF Operations does not support DER-encoded certificates or private keys.
PEM-encoding is base-64 ASCII and contains legible BEGIN and END markers, while DER is a binary format. Also, file extension might not match encoding. For example, a generic .cer extension might be used with PEM or DER. To verify encoding format, examine a certificate file using a text editor.
- The file extension must be .pem.
- The private key must be generated by the RSA or DSA algorithm.
- The private key can be encrypted by a pass phrase. The generated certificate can be uploaded using the primary node configuration wizard or the administration interface.
- The REST API in this VCF Operations release supports private keys that are encrypted by a pass phrase.
- The VCF Operations certificate must have IPs and Hostnames in the Subject Alternative Name (SAN) extension.
For example, Subject Alternative Name comprises of the DNS Name: localhost and the IP Address: 127.0.0.1.

- The VCF Operations Web server on all nodes have the same certificate file, so it must be valid for all nodes. One way to make the certificate valid for multiple addresses is with multiple Subject Alternative Name (SAN) entries.
- SHA1 certificates create browser compatibility issues. Therefore, ensure that all certificates that are created and being uploaded to VCF Operations are signed using SHA2 or newer.
- The VCF Operations supports custom security certificates with key length up to 8192 bits. An error is displayed when you try to upload a security certificate generated with a stronger key length beyond 8192 bits.
Note: Fill the certificate extension fields using the UTF-8 encoding.

If VCF Operations fails to accept and apply Custom CA Certificate. For more information, see the following KB article [2046591](#).

Configure a Custom Web Certificate

You can use OpenSSL to configure an authentication certificate for use with VCF Operations. You must first generate a Certificate PEM for VCF Operations, then install the Certificate PEM in VCF Operations. The certificates applied through the VCF Operations Admin UI will be used only for securely connecting and serving the user interfaces to (external) clients. We do not update the SSL certificates used for establishing a secure connection from VCF Operations to other services like VMware Identity Manager, vCenter, and VCF Operations for logs.

Take your cluster offline before uploading the custom web certificate.

Note:

If you are using VCF Operations 9.0 or later, update your certificates from the **Fleet Management > Certificates** page.

1. Generate a Certificate PEM file for use with VCF Operations.

1. Generate a key pair by running this command:
`openssl genrsa -out key_filename.key 2048`
2. Use the key to generate a certificate signing request by running this command:
`openssl req -new -key key_filename.key -out certificate_request.csr`
3. Submit the CSR file to your Certificate Authority (CA) to obtain a signed certificate.
4. From your Certificate Authority, download the certificate and the complete issuing chain (one or more certificates). Download them in Base64 format.
5. Enter the command to create a single PEM file containing all certificates and the private key. In this step, the example certificate is *server_cert.cer* and the issuing chain is *cacerts.cer*.

Note: The order of CA's certs in the .PEM file: Cert, Private Key, Intermediate Cert and then Root Cert.

```
cat server_cert.cer key_filename.key cacerts.cer > multi_part.pem
```

In Windows replace cat with type.

The finished PEM file should look similar to the following example, where the number of CERTIFICATE sections depends on the length of the issuing chain:

```
-----BEGIN CERTIFICATE-----
(Your Primary SSL certificate: your_domain_name.crt)
-----END CERTIFICATE-----
-----BEGIN RSA PRIVATE KEY-----
(Your Private Key: your_domain_name.key)
-----END RSA PRIVATE KEY-----
-----BEGIN CERTIFICATE-----
(Your Intermediate certificate: DigiCertCA.crt)
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
(Your Root certificate: TrustedRoot.crt)
```

```
-----END CERTIFICATE-----
```

2. Install a PEM in VCF Operations.

1. In a Web browser, navigate to the VCF Operations administration interface.
`https://vrops-node-FQDN-or-ip-address/admin`
2. Log in with the admin user name and password.
3. At the upper right, click the yellow **SSL Certificate** icon.
4. In the **SSL Certificate** window, click **Install New Certificate**.
5. Click **Browse** for certificate.
6. Locate the certificate .pem file, and click Open to load the file in the **Certificate Information** text box. The certificate file must contain a valid private key and a valid certificate chain.
7. Click **Install**.

Verifying a Custom VCF Operations Web Certificate

When you upload a custom certificate file, the VCF Operations interface displays summary information for all certificates in the file.

For a valid custom certificate file, you should be able to match issuer to subject, issuer to subject, back to a self-signed certificate where the issuer and subject are the same.

In the following example, OU=MBU,O=VMware\, Inc.,CN=vc-ops-slice-32 is issued by OU=MBU,O=VMware\, Inc.,CN=vc-ops-intermediate-32, which is issued by OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84, which is issued by itself.

```
Thumbprint: 80:C4:84:B9:11:5B:9F:70:9F:54:99:9E:71:46:69:D3:67:31:2B:9C
Issuer Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-intermediate-32
Subject Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-slice-32
Subject Alternate Name:
PublicKey Algorithm: RSA
Valid From: 2015-05-07T16:25:24.000Z
Valid To: 2020-05-06T16:25:24.000Z
```

```
Thumbprint: 72:FE:95:F2:90:7C:86:24:D9:4E:12:EC:FB:10:38:7A:DA:EC:00:3A
Issuer Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84
Subject Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-intermediate-32
Subject Alternate Name: localhost,127.0.0.1
PublicKey Algorithm: RSA
Valid From: 2015-05-07T16:25:19.000Z
Valid To: 2020-05-06T16:25:19.000Z
```

```
Thumbprint: FA:AD:FD:91:AD:E4:F1:00:EC:4A:D4:73:81:DB:B2:D1:20:35:DB:F2
Issuer Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84
Subject Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84
Subject Alternate Name: localhost,127.0.0.1
PublicKey Algorithm: RSA
Valid From: 2015-05-07T16:24:45.000Z
Valid To: 2020-05-06T16:24:45.000Z
```

Sample Contents of Custom VCF Operations Web Certificates

For troubleshooting purposes, you can open a custom certificate file in a text editor and inspect its contents.

PEM Format Certificate Files

A typical PEM format certificate file resembles the following sample.

```
-----BEGIN CERTIFICATE-----
MIIFlDCCBLygAwIBAgIKFYXYUwAAAAAAGTANBgkqhkiG9w0BAQ0FADBhMRMwEQYK
CZImiZPyLGBGRYDY29tMRUwEwYKCZImiZPyLGBGRYFdm13Y3MxGDABGogJkiaJ
<snip>
vKstQJNr7z2+pTy92M6FgJz3y+daL+9ddbaMNp9fVXjHBoDLGGaLOvyD+KJ8+xb
aGJfGf9ELXM=
-----END CERTIFICATE-----
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEA415ffX694riI1RmdRLJwL6sOWa+Wf70HROltx21kZzbXbUQN
mQhTriidJ3Ro2gRbj/btSsI+OMUzotz5VRT/yeyoTC512uJEapld45RroUDHQWwJ
<snip>
DAN9hQus3832xMkAuVP/jt76dHDYyviYIbmxzMalX7LZy1MCQVg4hCH0vLsHtLh
M1rOAsz62Eht/iB61AsVCCiN3gLrX7MksYdxZcRVruGXSiH33ynA
-----END RSA PRIVATE KEY-----
-----BEGIN CERTIFICATE-----
MIIDnTCCAoWgAwIBAgIQY+j29InmdYNCs2cK1H4kPzANBgkqhkiG9w0BAQ0FADBh
MRMwEQYKCZImiZPyLGBGRYDY29tMRUwEwYKCZImiZPyLGBGRYFdm13Y3MxGDAB
<snip>
ukzUuqX7wEhc+QgJWgl41mWZBZ09gfsA9XuXBL0k17IipVhPEgwwrjQz8X68m4I99
dD5Pflf/nLRJvR9jwXl62yk=
-----END CERTIFICATE-----
```

Private Keys

Private keys can appear in different formats but are enclosed with clear BEGIN and END markers.

Valid PEM sections begin with one of the following markers.

```
-----BEGIN RSA PRIVATE KEY-----
-----BEGIN PRIVATE KEY-----
```

Encrypted private keys begin with the following marker.

```
-----BEGIN ENCRYPTED PRIVATE KEY-----
```

Bag Attributes

Microsoft certificate tools sometimes add Bag Attributes sections to certificate files. VCF Operations safely ignores content outside of BEGIN and END markers, including Bag Attributes sections.

```
Bag Attributes
Microsoft Local Key set: <No Values>
localKeyID: 01 00 00 00
Microsoft CSP Name: Microsoft RSA SChannel Cryptographic Provider
friendlyName: 1e-WebServer-8dea65d4-c331-40f4-aa0b-205c3c323f62
Key Attributes
X509v3 Key Usage: 10
-----BEGIN PRIVATE KEY-----
MIICdwIBADANBgkqhkiG9w0BAQEFAASCAmEwgGJdAgEAAoGBAKHgyfc+qcQK4yxJ
om3PuB8dYzm34Qlt81GAAnBPYe3B4Q/Oba6PV8GtWG2svIpc1/eflwGHgTU3zJxR
gkKh7i3K5tGESn8lipyKtKpbYebh+aBMqPKrNNUEK1r0M9sa3WSco3350tCc1ew
5ZkNY24BRUVYwM0HogeGhOthRn2fAgMBAAECgYABhPmGN3FSZKPDG6HJlARvT1BH
```

```
KAGVnBGHD0OmMabghFBnBKXa8LwDldgGBng1oOakEXTftkiJdB+uwkU5P4aRrO7
vGuJUtRyRCU/4fjLBduxQL/KpQfruAQaof9uWUwh5W9fEeW3g26fzVL8AFZnbXS0
7Z0AL1H3LNCld5rpOQJBANnI7vFu06bFxVF+kq6Z0JFMx7x3K4VGxgg+PFFEBEPS
UJ2LuDH5/Rc63BaxFzM/q3B3Jhehvgw61mMyxU7QSSUCQQC+VDuW3XEWJjsiU6KD
gEgPCyJ5SBePbLSukljPgidKkDNlkLgbWVytCVkTAmuoAz33kMWfqIiNcqQbUgVV
UnpzAkB7d0CPO0deSsy8kMdTmKXLKf4qSF0x55epYK/5MZhBYuA1ENrR6mmjW8ke
TDNc6IGm9sVvrFBz2n9kKYpWThrJAKEAk5R69DtW0cbkLy5MqEzOHQaup36gD11L
WMXPvUfzSYTQ5aM2rrY2/1FtSSkqUwFYh9sw8eDbqVpIV4rc6dDfcwJBALiidPT0
tz86wySJNeOiUkQm36iXVF8AckPKT9TrbC3Ho7nC8OzL7gEl1ETa4Zc86Z3wpcGF
BhhEDMHaihyVgI=
-----END PRIVATE KEY-----
Bag Attributes
localKeyID: 01 00 00 00
1.3.6.1.4.1.311.17.3.92: 00 04 00 00
1.3.6.1.4.1.311.17.3.20: 7F 95 38 07 CB 0C 99 DD 41 23 26 15 8B E8
D8 4B 0A C8 7D 93
friendlyName: cos-oc-vcops
1.3.6.1.4.1.311.17.3.71: 43 00 4F 00 53 00 2D 00 4F 00 43 00 2D 00
56 00 43 00 4D 00 35 00 37 00 31 00 2E 00 76 00 6D 00 77 00 61 00
72 00 65 00 2E 00 63 00 6F 00 6D 00 00 00
1.3.6.1.4.1.311.17.3.87: 00 00 00 00 00 00 00 00 02 00 00 00 20 00
00 00 02 00 00 00 6C 00 64 00 61 00 70 00 3A 00 00 00 7B 00 41 00
45 00 35 00 44 00 44 00 33 00 44 00 30 00 2D 00 36 00 45 00 37 00
30 00 2D 00 34 00 42 00 44 00 42 00 2D 00 39 00 43 00 34 00 31 00
2D 00 31 00 43 00 34 00 41 00 38 00 44 00 43 00 42 00 30 00 38 00
42 00 46 00 7D 00 00 00 70 00 61 00 2D 00 61 00 64 00 63 00 33 00
2E 00 76 00 6D 00 77 00 61 00 72 00 65 00 2E 00 63 00 6F 00 6D 00
5C 00 56 00 4D 00 77 00 61 00 72 00 65 00 20 00 43 00 41 00 00 00
31 00 32 00 33 00 33 00 30 00 00 00
subject=/CN=cos-oc-vcops.eng.vmware.com
issuer=/DC=com/DC=vmware/CN=VMware CA
-----BEGIN CERTIFICATE-----
MIIFWTCCBEGgAwIBAgIKSjGT5gACAAAwKjANBgkqhkiG9w0BAQUFADBhMRMwEQYK
CZImiZPyLGBGRYDY29tMRYwFAYKCZImiZPyLGBGRYGdm13YXJlMRIwEAYDVQQD
Ew1WTXdhcmUgQ0EwHhcNMjQwMjA1MTg1OTM2WmcNMjYwMjA1MTg1OTM2WjAmMSQw
```

Add a Custom Web Certificate to VCF Operations

If you did not add your own SSL/TLS certificate when configuring the VCF Operations primary node, you can still add a certificate after VCF Operations is installed.

- Create and configure the primary node.
- Verify that your certificate file meets the requirements for VCF Operations .

1. In a Web browser, navigate to the VCF Operations administration interface at <https://node-FQDN-or-ip-address/admin>.
2. Log in with the admin user name and password.
3. At the upper right, click the SSL certificate icon.
4. In the certificate window, click **Install New Certificate**.
5. Click **Browse for certificate**.
6. Locate the certificate .pem file, and click **Open** to load the file in the Certificate Information text box.
7. Click **Install**.

Managing Accounts and Passwords for VMware Cloud Foundation Components

Manage passwords for VMware Cloud Foundation according to industry standards and the requirements of your organization. You can use the VCF Operations console to update and remediate the passwords for VMware Cloud Foundation components.

Managing Passwords

Use the VCF Operations UI to:

- View password alerts and information about the components whose passwords are managed by VMware Cloud Foundation.
- Update passwords.
- Remediate passwords.

You can also use the [SDDC Manager API](#) to look up and manage credentials.

Note:

Password expiration information and connection status in VCF Operations are updated once a day. To get real-time information, use the SDDC Manager API.

For additional information on account and password management, see [Account Management Design](#).

In the context of password management actions, server refers to the component where the account resides, and client refers to the component where the account is used. VMware Cloud Foundation offers the following password management actions:

Action	Description	VCF Component UI
Update	You use the Update action to provide a new password for an account and change that password on the server side (where the account resides) and on the client side (where the account is used).	VCF Operations
Remediate	You use the Remediate action to provide the new password on the client side (where the account is used) after the password is changed on the server side (where the account resides).	VCF Operations

Action	Description	VCF Component UI
Rotate	You use the Rotate action to change the password with an auto-generated password on both server (where the account resides) and client (where the account is used) sides.	SDDC Manager

When you deploy VMware Cloud Foundation or VMware Cloud Foundation components you can choose to specify or auto-generate the initial passwords for each component. After deployment, you can use the VCF Operations console to update and remediate the passwords for the VCF Management components (which can manage multiple VCF instances) and for the components in a VCF Instance or a VCF domain.

VCF Management Components	VCF Instance / Domain Components
• Fleet Management • VCF Automation • VCF Identity Broker • VCF Operations • VCF Operations for logs • VCF Operations for networks	• ESX • NSX Manager • vCenter • SDDC Manager (backup password only)

Accounts in VMware Cloud Foundation

VMware Cloud Foundation comprises of multiple local (OS), application accounts, and service accounts. Each account has different attributes and is managed according to the specific methods outlined in the table.

VCF Component	Account	Description	Managed by
SDDC Manager	vcf	• Local appliance account • OS level access • Default expiry: 365 days	Manually by using the OS.
	root	• Local appliance account • OS level access • Default expiry: 365 days	Manually by using the OS.
	backup	• Local appliance account • OS level access • Default expiry: 365 days	VCF Operations UI or API
	admin@local	• Application account • API access • Default expiry: Never	SDDC Manager API
ESX	root	• Local appliance account • OS level access • Default expiry: Never	• VCF Operations UI or API • SDDC Manager UI or API
	svc-vcf-ESX-hostname	• Service account between SDDC Manager (client) and the ESX host (server). • Default expiry: Never	SDDC Manager UI / API
vCenter	root	• Local appliance account • OS level access • VAMI access • Default expiry: 90 days	• VCF Operations UI or API • SDDC Manager UI or API

VCF Component	Account	Description	Managed by
	administrator@vsphere.local	- vCenter Single Sign-On account - Application and API access - Additional VMware Cloud Foundation Admin account required to perform manual password rotation. - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	svc-sddc-manager-hostname-vcenter-hostname@vsphere.local	- Service account between SDDC Manager (client) and vCenter (server). - System managed - Automatically rotated every 30 days by default Administrator role - Default expiry: 90 days	SDDC Manager UI / API
	svc-nsx-manager-hostname-vcenter-hostname@vsphere.local	- Service account between NSX Manager (client) and vCenter (server). Administrator role - System managed - Automatically rotated every 30 days by default - Default expiry: 90 days	N/A
	svc-fleet-management-hostname-vcenter-hostname@vsphere.local	- Service account between VCF Operations fleet management (client) and vCenter (server) Administrator role - System managed - Automatically rotated every 30 days by default - Default expiry: None	N/A
	svc-ops-vc-ops-cluster-uid@vsphere.local	- Service account between VCF Operations fleet management (client) and vCenter (server) Administrator role - System managed. - Automatically rotated every 30 days by default. - Default expiry: None	N/A
	svc-vcfa_uid@vsphere.local	- Service account between VCF Operations (client) and vCenter (server) Administrator role - System managed - Automatically rotated every 30 days by default - Default expiry: None	N/A
	svc-vcfsp-vc-uid@vsphere.local	- Service account between VCF Services Platform (client) and vCenter (server) VCF Services Runtime role - System managed - Automatically rotated every 30 days by default. - Default expiry: None	N/A

VCF Component	Account	Description	Managed by
	svc-vcfsp-vc-admin-uid@vsphere.local	- Service account between VCF Services Platform (client) and vCenter (server) VCF Services Platform Admin role - System managed - Automatically rotated every 30 days by default - Default expiry: None	N/A
NSX Global Manager	root	- Local appliance account - OS level access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	admin	- Local appliance account - OS level access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	audit	- Local appliance account - OS level access - Read-only application level access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
NSX Local Manager	root	- Local appliance account - OS level access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	admin	- Local appliance account - OS level, API, and application access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	audit	- Local appliance account - OS level access - Read-only application level access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	svc-sddc-manager-hostname-nsx-vip-hostname-uid	- Service account between SDDC Manager (client) and NSX (server) Enterprise Admin role - System managed - Automatically rotated every 30 days by default - Default expiry: 90 days	N/A
	svc-ops-nsx-uid	- Service account between VCF Operations (client) and NSX (server). Enterprise Admin role - System managed - Default expiry: Never	N/A
	svc-ni-nsx-uid	- Service account between VCF Operations for networks (client) and NSX (server) - System managed - Default expiry: Never	N/A
	svc-vcfa_uid	Service account between VCF Automation (client) and NSX (server)	N/A

VCF Component	Account	Description	Managed by
		- System managed - Default expiry: Never	
NSX Edge	root	- Local appliance account - OS level access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	admin	- Local appliance account - OS level, API, and application access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
	audit	- Local appliance account - OS level access - Read-only application level access - Default expiry: 90 days	- VCF Operations UI or API - SDDC Manager UI or API
VCF Operations	root	- Local appliance account - OS level access - Default expiry: 90 days	VCF Operations UI or API
	admin	- Local appliance account - OS level, API, and application access - Default expiry: Never	VCF Operations UI or API
VCF Operations Collectors	root	- Local appliance account - OS level access - Default expiry: 90 days	VCF Operations UI or API
VCF Operations fleet management	root	- Local appliance account - OS level access - Default expiry: 90 days	VCF Operations UI or API
	admin@local	- API and application access - Default expiry: Never	VCF Operations UI or API
VCF Automation	root	- Local appliance account - OS level access - Default expiry: 365 days	Manually by using the OS.
	admin	- Administrative account for the VCF Automation system organization - Default expiry: Never	VCF Operations UI or API
	vmware-system-user	- Local appliance account - OS level access - Default expiry: 90 days	VCF Operations UI or API
VCF Operations for logs	root	- Local appliance account - OS level access - Default expiry: 90 days	VCF Operations UI or API
	admin	- Local appliance account - API and application access - Default expiry: Never	VCF Operations UI or API
VCF Operations for networks	root	- Local appliance account - OS level access - Default expiry: Never	Manually by using the OS.

VCF Component	Account	Description	Managed by
	admin@local	- API and application access - Default expiry: Never	VCF Operations UI or API
	support	- Local appliance account - OS level access - Default expiry: Never	VCF Operations UI or API
	consoleuser	- Local appliance account - CLI level access - Default expiry: Never	VCF Operations UI or API
VCF Operations for networks collector	support	- Local appliance account - OS level access - Default expiry: Never	VCF Operations UI or API
	consoleuser	- Local appliance account - CLI level access - Default expiry: Never	VCF Operations UI or API
VCF Identity Broker	vmware-system-user	- Local appliance account - OS level access - Default expiry: 90 days	VCF Operations UI or API
vSphere Supervisor	vmware-system-user	- Local appliance account - OS level access - Encoded in the control plane namespace secret - Default expiry: Never	Manually by using the Kubernetes Secrets.
VMware Live Recovery	root	- Local appliance account - OS level access - Default expiry: Never	Manually by using the OS.
	admin	- Local appliance account - API and application access - Default expiry: Never	Manually by using the VAMI.

Default Password Requirements for VCF Components

VCF components have default password requirements for user-managed local (OS), application, and service accounts. Use the information in the table below to change component password requirements.

VCF Component	Account	Settings		Configured in
SDDC Manager	vcf root backup	Minimum characters	12	SDDC Manager uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Alphabetic characters	2	
		Reject dictionary word	Yes	
	admin@local	Minimum characters	15	Cannot be modified

VCF Component	Account	Settings		Configured in
		Maximum characters	127	
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		A single character cannot be repeated consecutively	3	
ESX	root	Minimum characters	7	Use the vSphere Client, the ESX host Client, or the ESX CLI to configure the password settings.
		Maximum characters	40	
		A mix of at least three out of four character classes: lowercase letters, uppercase letters, numbers, and special characters	Yes	
		Reject dictionary word	Yes	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
	svc-vcf-ESX-hostname	Minimum characters	8	vCenter Single Sign-On password policy. Administrators can change the default password policy.
		Maximum characters	20	
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
vCenter	root	Minimum characters	8	vCenter uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, you can use the vCenter VAMI or you can edit the corresponding OS files directly.
		Maximum characters	20	
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
	administrator@vsphere.local	Failed log-in attempts	3	
		Lockout period (seconds)	300	
		Minimum characters	8	vCenter Single Sign-On password policy. Administrators can change the default password policy.
		Maximum characters	20	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Non-ASCII characters are allowed.		

VCF Component	Account	Settings		Configured in
NSX Global Manager NSX Local Manager NSX Edge	root admin audit	Minimum characters	12	NSX uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, you can use the NSX CLI or you can edit the corresponding OS files directly.
		Maximum characters	128	
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
VCF Operations	root admin	Minimum characters	15	VCF Operations uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	3	
VCF Operations collector	root	Minimum characters	15	VCF Operations collector uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	3	
VCF Operations fleet management	root	Lockout period (seconds)	600	VCF Operations fleet management uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Minimum characters	12	
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
	admin@local	Password history	5	Cannot be modified.
		Failed log-in attempts	3	
		Lockout period (seconds)	600	
		Minimum characters	12	
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	

VCF Component	Account	Settings		Configured in
		Lockout period (seconds)	900	
VCF Automation	root	Minimum characters	15	VCF Automation uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
	admin	Minimum characters	15	VCF Automation UI
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
	vmware-system-user	Minimum characters	15	VCF Automation uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
VCF Operations for logs	root	Minimum characters	8	VCF Operations for logs uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	3	
		Lockout period (seconds)	900	
	admin	Minimum characters	8	VCF Operations for logs UI
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	

VCF Component	Account	Settings		Configured in
VCF Operations for networks	root	Minimum characters	8	VCF Operations for networks uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
	admin@local	Minimum characters	8	VCF Operations for networks UI
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
	support consoleuser	Minimum characters	8	VCF Operations for networks uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
VCF Operations for networks collector	support consoleuser	Minimum characters	8	VCF Operations for networks collector uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	
VCF Identity Broker	vmware-system-user	Minimum characters	8	VCF Identity Broker uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Failed log-in attempts	5	
		Lockout period (seconds)	900	

VCF Component	Account	Settings		Configured in
VMware Live Recovery	root admin	Minimum characters	8	VMware Live Recovery uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.
		Maximum characters	20	
		Uppercase characters	1	
		Lowercase characters	1	
		Numerical characters	1	
		Special characters	1	
		Password history	5	
		Failed log-in attempts	3	
		Lockout period (seconds)	600	

Update Passwords for VMware Cloud Foundation Components

For security reasons, you can change passwords for the accounts that are used by VMware Cloud Foundation components. Changing these passwords periodically or when certain events occur, such as an administrator leaving your organization, reduces the likelihood of security vulnerabilities.

- Verify that there are no failed workflows. To check for failed workflows, click **Fleet Management > Tasks** in the VCF Operations console.
- Verify that no active workflows are running or are scheduled to run during the password update.
- The user's role must include the following permissions:
 - Fleet Management: Passwords:Manage.
 - Fleet Management: Passwords:View.

You can update only one password at a time.

Although individual components support different password requirements, it is recommended that you set passwords following a common set of requirements across all accounts:

- Minimum length: 15
- Maximum length: 20
- At least one lowercase letter, one uppercase letter, a number, and one of the following special characters: ! @ # \$ % ^ & * _
- Must NOT include:
 - A dictionary word
 - A palindrome
 - More than four monotonic character sequences
 - Three of the same consecutive characters

1. In the VCF Operations console, click **Fleet Management > Passwords**.
2. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or VCF domain name.
3. Select a component and click **Update Password**.
4. Enter and confirm the new password.
5. Click **Update Password**.

A message appears at the top of the page showing the progress of the operation.

Remediate Passwords for VMware Cloud Foundation Components

To resolve any errors that might occur during password changes, you use password remediation. Password remediation synchronizes the password of the account stored in the SDDC Manager with the updated password in the component. After you reset the password in a component, you must remediate the password in VCF Operations.

- Verify that no workflows are running or are scheduled to run while you remediate the password.
- Verify that the user you log in with in the VCF Operations UI has the following permissions:
 - Fleet Management: Passwords:Manage.
 - Fleet Management: Passwords:View.

Note: You can remediate the password for only one account at a time.

Although individual components support different password requirements, it is recommended that you set passwords following a common set of requirements across all accounts.

For information on updating passwords manually, see [Update Passwords for VMware Cloud Foundation Components](#).

1. In the VCF Operations console, click **Fleet Management > Passwords**.
2. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or VCF domain name.
3. Select a component and click **Remediate Password**.
4. Enter and confirm the password that was set manually on the component.
5. Click **Remediate Password**.

In case a VCF component loses connectivity to another VCF component due to synchronization issues (for example, the password for the service account changes on the server side, while the client side is not updated), you can manually remediate the service account on the client side by following the procedures below.

Created in	Used by	Procedure
vCenter	SDDC-M	To manually remediate the SDDC Manager service account in vCenter, use the vCenter Password API to reset the SDDC Manager service account password. You then use the remediate password functionality in SDDC Manager. 1. Create a session to vCenter. <pre>session=\$(curl -ks https://<vcenter_fqdn>/api/session -u 'administrator@vsphere.local:password' -X POST jq -r .)</pre> 2. Reset the service account password. <pre>curl -ks https://<vcenter_fqdn>/api/vcenter/svc-account-mgmt/password?action=reset -H "vmware-api-session-id: \$session" -H 'Content-Type: application/json' -d '{ "account_name": "sddc_m_to_vcenter_service_account" }' jq</pre> The command returns the new password. 3. Log in to SDDC Manager at <a href="https://<sddc_manager_fqdn>">https://<sddc_manager_fqdn> with a user assigned the Administrator role. 4. In the left pane, navigate to Security > Password management. 5. On the Password management page, select the vCenter service account, click the ellipsis icon, and select Remediate password. 6. In the Remediate password dialog box, enter the new password and click Remediate password. Contact Broadcom Global Support for additional assistance.
	NSX	To manually remediate the NSX service account in vCenter, you must enter the service account credentials in the NSX UI and NSX automatically repairs the integration.

		1. Log in to NSX Manager at <a href="https://<nsx_manager_fqdn>/login.jsp">https://<nsx_manager_fqdn>/login.jsp with a user assigned the Administrator role. 2. On the main navigation bar, click System. 3. In the left pane, navigate to Fabric > Compute managers. 4. Select the compute manager that has a Not registered status. 5. Click the message and click Resolve. 6. In the dialog box, enter the vCenter service account credentials and click Resolve.
	VCF Operations	To manually remediate the VCF Operations service account in vCenter, you use the VCF Operations UI.
	VCF Operations for networks	1. Log in to the VCF Operations interface at <a href="https://<vcf_operations_fqdn>">https://<vcf_operations_fqdn> with a user assigned the Administrator role. 2. In the left pane, navigate to Administration > Integrations. 3. On the Accounts tab, expand the VMware Cloud Foundation section. 4. Expand your VCF Instance and click the domain. 5. On the domain page, click the vCenter tab. 6. Next to System managed credentials, click Rotate. The existing service account gets deleted and a new service account is created. The integration between VCF components is not affected during the creation of the new service account. When VCF Operations fleet management synchronizes the inventory with VCF Operations, VCF Operations fleet management generates a new service account in vCenter through SDDC Manager.
	VCF Operations for logs	
	VCF Operations fleet management	
	VCF Operations orchestrator	To manually remediate the VCF Operations orchestrator service account in vCenter, you log in to the VCF Operations orchestrator pod as root and reset the authentication provider by using the <code>vracli vro authentication</code> CLI command: <pre>vracli vro authentication set -p vsphere \ --hostname https://FQDN_SPHERE\ --usernameadministrator@vsphere.local\ --tenantvsphere.local\ --admin-groupAdministrators\ --admin-group-domainvsphere.local</pre> Restart the VCF Operations orchestrator pod.
	VCF Automation	To manually remediate the VCF Automation service account in vCenter, use the VCF Automation UI. 1. Log in to the VCF Automation interface at <a href="https://<vcf_automation_fqdn>">https://<vcf_automation_fqdn> with a user assigned the Administrator role. 2. In the left pane, navigate to VCF Instances. 3. On the VCF Instance page, click the vertical ellipsis next to the vCenter instance, and, from the drop-down menu, select Rotate service account. VCF Automation deletes and recreates the service account.
NSX	SDDC Manager	To manually remediate the SDDC Manager service account in NSX, use the NSX API to update the SDDC Manager service account password. You then use the remediate password functionality in SDDC Manager. 1. Find the user ID for the SDDC Manager service account from the list of service accounts. <pre>curl -sk 'https://vip_nsx_manager/api/v1/node/users?filter=service' -u 'admin:admin_password' jq</pre> 2. Reset the service account password. <pre>curl -ik 'https://vip_nsx_manager/api/v1/node/users/user_id?action=reset_password' -u 'admin:admin_password' -H 'Content-Type: application/json' -d '{ "password": "new_service_account_password" }'</pre> 3. Log in to SDDC Manager at <a href="https://<sddc_manager_fqdn>">https://<sddc_manager_fqdn> with a user assigned the Administrator role. 4. In the left pane, navigate to Security > Password management.

		5. On the Password management page, select the NSX service account, click the ellipsis icon, and select Remediate password. 6. In the Remediate password dialog box, enter the password you configured and click Remediate password.
	VCF Operations	To manually remediate the VCF Operations service account in NSX, you use the VCF Operations UI.
	VCF Operations for networks	1. Log in to the VCF Operations interface at <a href="https://<vcf_operations_fqdn>">https://<vcf_operations_fqdn> with a user assigned the Administrator role. 2. In the left pane, navigate to Administration > Integrations. 3. On the Accounts tab, expand the VMware Cloud Foundation section. 4. Expand your VCF Instance and click the domain. 5. On the domain page, click the NSX tab. 6. Next to System managed credentials, click Rotate. The existing service account gets deleted and a new service account is created. The integration between VCF components is not affected during the creation of the new service account. When VCF Operations fleet management synchronizes the inventory with VCF Operations, VCF Operations fleet management generates a new service account in vCenter through SDDC Manager.
	VCF Automation	To manually remediate the VCF Automation service account in NSX, use the VCF Automation UI. VCF Automation deletes and recreates the service account. 1. Log in to the VCF Automation interface at <a href="https://<vcf_automation_fqdn>">https://<vcf_automation_fqdn> with a user assigned the Administrator role. 2. In the left pane, navigate to VCF Instances. 3. On the VCF Instance page, click the vertical ellipsis next to the NSX instance, and, from the drop-down menu, select Rotate service account.
SDDC Manager API keys	VCF Operations VCF Operations fleet management	To manually remediate the SDDC Manager API key in VCF Operations, you use the VCF Operations UI. 1. Log in to the VCF Operations interface at <a href="https://<vcf_operations_fqdn>">https://<vcf_operations_fqdn> with a user assigned the Administrator role. 2. In the left pane, navigate to Administration > Integrations. 3. On the Accounts tab, expand the VMware Cloud Foundation section. 4. Click the VCF Instance. 5. On the Account tab, in the VCF Credentials section, click Rotate. The existing API key is deleted and a API key is created. The integration between VCF components is not affected during the creation of the new service account. When VCF Operations fleet management synchronizes the inventory with VCF Operations, VCF Operations fleet management generates a new API key in SDDC Manager.
	VCF Automation	To manually remediate the SDDC Manager API key in VCF Automation, you use the VCF Automation API to delete the SDDC Manager endpoint. VCF Automation creates a new API key and adds the SDDC Manager instance during the next synchronization. Please note that this action usually takes 24 hours, during which period VCF Automation is not able to register new workload domains.

Rotate Passwords for VMware Cloud Foundation Components

Use SDDC Manager to rotate passwords for managed components or schedule automatic rotation on a recurring basis. During rotation, SDDC Manager generates a randomized password based on the component password policy and re-establishes respective component integrations.

Plan a maintenance window to avoid service interruptions if you are rotating passwords for multiple accounts.

ESX accounts can be rotated, but scheduled rotation is not supported.

SDDC Manager UI offers preset intervals for scheduled password rotation. Use the API to schedule password rotation for custom intervals.

1. Log in to SDDC Manager at https://sddc_manager_fqdn with a user assigned the **Administrator** role.
2. In the left pane, navigate to **Security > Password management**.
3. On the **Password management** page, select the account which password you want to change and click the **Rotate password** button.
4. In the **Rotate password** dialog box, click **Rotate**.

Reset Local Account Passwords for VMware Cloud Foundation Components

You can reset a forgotten or an expired local account password for some VCF components.

VCF Component	Account	Expired Password	Forgotten Password
SDDC Manager	vcf	Manually by using the OS with root permissions.	Manually by using the OS with root permissions.
	root	The password never expires.	The password cannot be reset.
	backup	Manually by using the OS with root permissions.	Manually by using the OS with root permissions.
	admin@local	The password never expires.	See Broadcom Knowledge Base Article 403099 .
ESX	root	The password never expires.	• See Broadcom Knowledge Base Article 323617. • See Broadcom Knowledge Base Article 376979.
vCenter	root	See Broadcom Knowledge Base Article 322247 .	See Broadcom Knowledge Base Article 322247 .
	administrator@vsp here.local	The password never expires.	See Broadcom Knowledge Base Article 326186 .
NSX Local Manager NSX Edge	root	See Resetting Expired Passwords .	See Resetting User Passwords .
	admin		
	audit		
VCF Operations	root	See Broadcom Knowledge Base Article 325005 .	See Broadcom Knowledge Base Article 325005 .
	admin	The password never expires.	• For UI method, see Broadcom Knowledge Base Article 326391 or Reset the VCF Operations Administrator Password from the Administration UI. • For CLI method, see Reset the VCF Operations Administrator Password from CLI.
VCF Operations cloud proxy	root	See Broadcom Knowledge Base Article 325005 .	See Broadcom Knowledge Base Article 325005 .
VCF Operations orchestrator	root	The password never expires.	The password cannot be reset.
VCF Automation	vmware-system-user	See Password Remediation for vmware-system-user .	See Password Remediation for vmware-system-user .
vSphere Supervisor	vmware-system-user	See Password Remediation for vmware-system-user .	See Password Remediation for vmware-system-user .

Use the SDDC Manager Appliance to Look Up Account Credentials

To look up the account credentials for the built-in accounts that are managed by SDDC Manager, you can log in to the SDDC Manager appliance.

This task returns information about the components for a specific VCF Instance and does not include VCF Management components.

If you prefer, you can use the `GET /v1/credentials` API to get the same information.

1. SSH in to the SDDC Manager appliance using the `vcf` user account.
2. Enter the following command:

```
lookup_passwords
```
3. Enter an entity type from the displayed list.
For example: `VCENTER`.
4. When prompted, enter a user name and password for an SDDC Manager account assigned the **ADMIN** role.
5. Optional: (Optional) Save the command output to a secure location with encryption so that you can access it later and use it to log in to the accounts as needed.

Updating Passwords for SDDC Manager Accounts

The process for updating SDDC Manager passwords varies, depending on which account you are updating.

The following table lists the SDDC Manager user accounts and links to the procedures for how to update them.

SDDC Manager Account	Update Procedure
backup	Update Passwords for VMware Cloud Foundation Components
vcf root	Update SDDC Manager Root and Super User Passwords
admin@local	Use the <code>PATCH /v1/users/local/admin</code> API to update the local user password. See Update password for local account for more information.

Update SDDC Manager Root and Super User Passwords

For security reasons, you can change passwords for the SDDC Manager root (`root`) and super user (`vcf`) accounts. Changing these passwords periodically or when certain events occur, such as an administrator leaving your organization, reduces the likelihood of security vulnerabilities.

The SDDC Manager `root` password expires after 90 days. To update an expired `root` password, see [Update an Expired SDDC Manager Root Password](#).

1. SSH in to the SDDC Manager appliance using the `vcf` user account.
2. Enter `su` to switch to the root user.
3. Enter one of the following commands:

Option	Description
<code>passwd vcf</code>	To change the super user password.
<code>passwd root</code>	To change the root password.

4. Enter and retype the new password. For example:

```
root@sddc-manager [ /home/vcf ]# passwd vcf
New password:
```



```
Retype new password:
passwd: password updated successfully
```

The password is updated.

Related Links

[Update an Expired SDDC Manager Root Password on page 2951](#)

If the SDDC Manager root (`root`) user password expires, you must update the password using the vSphere Client.

Update an Expired SDDC Manager Root Password

If the SDDC Manager root (`root`) user password expires, you must update the password using the vSphere Client.

1. In a web browser, log in to the management domain vCenter using the vSphere Client (https://<vcenter_fqdn>/ui).
2. In the inventory, browse to and select the SDDC Manager virtual machine.
3. On the **Summary** tab, click **Launch Remote Console**.
4. Click within the console window and press **Enter** on the Login menu item.
5. Type **root** as the user name and enter the current password for the root user.
6. Type `passwd root`.
7. When prompted for a new password, enter a different password than the previous one and click **Enter**.

Related Links

[Update SDDC Manager Root and Super User Passwords on page 2950](#)

For security reasons, you can change passwords for the SDDC Manager root (`root`) and super user (`vcf`) accounts. Changing these passwords periodically or when certain events occur, such as an administrator leaving your organization, reduces the likelihood of security vulnerabilities.

Managing Passwords for VMware vSphere Foundation Components

If you are using VMware vSphere Foundation, you can use the VCF Operations console to update and remediate the passwords for the VCF Operations component of VMware vSphere Foundation.

Reset the VCF Operations Administrator Password from the Administration UI

Passwords are used by users to access the product interfaces or to console sessions on cluster nodes. You might need to reset the VCF Operations administrator password as part of securing or maintaining your deployment and if you forget the admin account password.

1. In a Web browser, navigate to the VCF Operations administration interface at `https://<master-node-name> or <master-node-ip-address>/admin`.
2. Log in with the admin user name and password for the master node.
3. In the left pane, click **Administrator Settings**.
4. In the **Change Administrator Password** section, enter the current password, and enter the new password twice to ensure its accuracy.

Note: You cannot change the administrator user name.

5. Click **Save**.
6. Optionally, to recover a forgotten password, configure the **Password Recovery Settings**.

Table 933: Password Recovery Settings

Password Recovery Settings Options	Description
Your E-mail	Email id to which you want to receive the recovery email.
SMTP Server	DNS name or IP address of the SMTP server that is used to send the password recovery email.
Port	Port used for the communication. By default, 25 is used for a non-secure port and 465 for a secure port.
SSL (SMTPS)	Activate to protect the communication using the secure socket layer.
STARTTLS Encryption	Activate to switch the insecure communication starting with the TLS handshake.
Sender E-mail	The email id from which the password recovery email is sent.
User name	User name for the STMP server account, as some servers require authentication.
Password	Password for the SMTP server account.
Test	To verify the mandatory fields and make an attempt to communicate with the given SMTP server.

7. Click **Save**. Optionally, click **Reset** to enter the details again.

Reset the VCF Operations Administrator Password from CLI

You must reset the password if the admin account password is lost.

This procedure requires root account credentials.

- In VCF Operations vApp deployments, when you log in to the console of the virtual application for the first time, you are forced to set a root password.
- The VCF Operations console root password can be different than the admin account password that you set when configuring the VCF Operations primary node.

When the VCF Operations password for the built-in admin account is lost, follow these steps to reset it on vApp clusters.

1. Log in to the master node command-line console as `root`.
2. Enter the following command, and follow the prompts.

```
$VMWARE_PYTHON_BIN $VCOPS_BASE/./vmware-vcopssuite/utilities/sliceConfiguration/bin/vcopsSetAdminPassword.py --reset
```

Generate a VCF Operations Passphrase

When users need to add a node to the VCF Operations cluster, you can generate a temporary passphrase instead of giving them the primary administrator login credentials, which might be a security issue.

Create and configure the primary node.

A temporary passphrase is good for one use only.

1. In a Web browser, navigate to the VCF Operations administration interface at `https://master-node-name-or-ip-address/admin`.
2. Log in with the admin user name and password for the master node.
3. In the list of cluster nodes, select the master node.
4. From the toolbar above the list, click the option to generate a passphrase.
5. Enter a number of hours before the passphrase expires.
6. Click **Generate**.

A random alphanumeric string appears, which you can send to a user who needs to add a node.

Have the user supply the passphrase when adding a node.

Tags and Categories Overview

You can create tag categories and tags in your VCF Operations instance and use them to organize and manage your inventory objects at scale.

What Are Tags and Categories

A tag consists of a name and description that you specify. To group related tags together and manage inventory objects at scale, you create categories and tags inside the categories. When you create a category, you can specify the object types for its tags, and whether more than one tag in the category can be applied to an object. After you create the category and tags, you can apply the tags to objects in your vSphere Client inventory in order to organize and manage the objects more easily, and to tag specific inventory objects when you configure different vSphere policies.

For example, if you want to tag inventory objects with memory over-commitment, you can create a category named Compute Memory, specify that the category applies to just hosts as an object type, and then add a tag named Memory Over-Commitment. After you create the category and tag, you can assign the tag to the specific hosts from your inventory.

VCF Operations Tags and Categories

By using VCF Operations, you can manage tags from a central place for all VCF instances managed by the VCF Operations instance. You can create new tags and categories, and import existing tags and categories from workload or management domain vCenter instances.

After you create or import categories and tags in VCF Operations, you can push them to selected VCF domain vCenter instances or vCenter instances part of custom groups. You can then use the tags and categories in existing automation scripts or to tag objects in the vSphere Client. This will ensure scalability and prevent loss of tags and their associations to objects during vMotion.

By using the vSphere Client, you apply the tags as metadata to objects in the vCenter inventory. For more information, see [How Do You Assign or Remove a vSphere Tag](#).

Create a Tag Category in VCF Operations

When you create a tag category, you must also create at least one tag in that category. After you create the category, you can create and add as many tags in it as you need.

Verify that you have **Tags > Manage** permissions in VCF Operations.

You cannot create an empty category, you must create at least one tag in the category. For each category that you create, you must specify whether a single or multiple tags are allowed per object, and which object types can be associated with this category. After you create the category, you can edit the tags added to the category. You can add more tags, delete some of the existing tags, or edit them.

Note:

You can also import existing tags and categories from vCenter instances. For more information, see [Import Tags and Categories from a vCenter Instance](#).

1. Log in to VCF Operations.
2. From the left navigation menu, click **Fleet Management**, and click **Tags**.
3. Click **Create**
The **Create Category** dialog box appears.
4. Enter a name for the category. The name can consist of up to 128 characters.
Once you create the category, you cannot change its name.
5. Optional: Enter a description for the category. The description can consist of up to 1024 characters.
6. Optional: To edit the tags per object and the object types that can be associated with the category, click **Advanced Settings**.
 - a) Select the number of tags that can be assigned per associated object.

CAUTION:

If you select **Multiple**, after you create the category, you cannot change the value to **Single**.

- a) Select to which objects types the tags from the category can be assigned to.

CAUTION:

After you create the category, you can only add more object types to be associated with the category, you cannot remove the object types that you selected when creating the category. If you select **All object types**, after you create the category, you cannot change the value.

7. Create a tag to add to the category:
 - a) Enter a name for the tag.
Once you create the tag, you cannot change its name.
 - a) Enter a description for the tag.
8. Optional: To create more tags and add them to the category, click **Add Tag**, and enter a unique name and description for each tag.
9. Click **Create**.

The category appears in the table with available categories for this VCF Operations instance.

You can edit, delete, or push the newly created category to VCF domain vCenter instances or custom groups. When you delete a category, you also delete all tags added to it. For backup purposes, you can also export all tag categories and tags available in the VCF Operations instance in JSON file format.

Create a Tag in VCF Operations

Verify that you have **Tags > Manage** permissions in VCF Operations.

In VCF Operations, you can only create tags inside an existing tag category or at the time when you create a new category.

1. Log in to VCF Operations.
2. From the left navigation menu, click **Fleet Management**, and click **Tags**.
3. Click the arrow icon next to category, in which you want to create a tag.
A table with all tags added to this category appears.
4. Click **Create**.
The **Create Tag** dialog box appears.
5. Enter a unique name for the tag. The name can consist of up to 128 characters.
6. Optional: Enter a description for the tag. The description can consist of up to 1024 characters.
7. Optional: To create more tags and add them to the category, click **Add Tag**, and enter a name and description for each tag.
8. Click **Create**.

The tag appears in the table with the available tags for the category.

You can edit or delete the tag. You can also export all tag categories and tags available in the VCF Operations instance in JSON file format.

Note:

When an export is in progress, initiated by you or another user, you cannot initiate the export again before the first export is complete.

To assign tags to specific inventory objects, use the vSphere Client. For more information, see [How Do You Assign or Remove a vSphere Tag](#).

Import Tags and Categories to VCF Operations from a vCenter Instance

You can import existing tags and tag categories from your VCF domain vCenter instances to VCF Operations. Importing tags into VCF Operations enables you to manage tags and categories for more than one vCenter instance from a single place.

- Verify that the VCF domain vCenter instance from which you want to import tags and tag categories is of version 9.0.
- Verify that the VCF domain vCenter instance from which you want to import tags has a valid VCF 9.0 license.
- Verify that the VCF domain vCenter instance from which you want to import tags and tag categories is integrated with this VCF Operations instance.
- Verify that you have **Tags > Manage** permissions in VCF Operations.

After you import tags and categories to your VCF Operations instance, you can push them to other vCenter instances in order to have to have consistent tagging across you VCF Fleet.

Important:

If you cannot see the conflict details, verify that you have the REST API read privileges assigned.

When a task is in progress, you cannot initiate the same task again. For example, if another user or you initiate an import, you cannot initiate import again before the first task is complete.

1. Log in to VCF Operations.
2. From the left navigation menu, click **Fleet Management**, and click **Tags**.
3. Click **Import**.
The **Import Categories** dialog box appears, and you can see a list of all VCF domains integrated with your VCF Operations instance.

4. Select a VCF domain from which you want to import tags and tag categories, and click **Import**.
You cannot initiate an import task if the task is already in progress, you must wait for the task to complete to initiate it again.
If another user initiated an import task and it is in progress, a warning appears.

The tags and categories are imported from the VCF domain vCenter instance. If some of the categories cannot be imported, click **View Conflict Details** or **View Import Details** in the banner to view a list with the import conflict or error reasons and possible resolutions. Once you dismiss the banner notification, you cannot view it again.

You cannot see banner notifications for tasks performed by another user.

Edit a Tag Category in VCF Operations

You can edit the description of any category in VCF Operations. To edit advanced settings of the category, special conditions apply. You cannot change the category name.

- Verify that you have **Tags > Manage** permissions in VCF Operations.

1. Log in to VCF Operations.
2. From the left navigation menu, click **Fleet Management**, and click **Tags**.
3. In the table with available categories, select the check box next to the category you want to edit.

Note:

You can edit only one category at a time.

4. Click **Edit**.
5. Enter a new description for the category.
6. Optional: To change the number of tags allowed per object and the object types to be associated with tags from the category, click **Advanced Settings**, and update the values.

Important:

- If during the creation of a category, the tags allowed per object were set to Multiple, you cannot change that value to Single.
- If during the creation of a category, the associable object types were set to All object types, you cannot change that value at a later time. Any object type that you selected upon creating the category cannot be deselected at a later time.

7. Click **Save**.

Edit a Tag in VCF Operations

You can edit the description of any tag in VCF Operations. You cannot change the tag names.

- Verify that you have **Tags > Manage** permissions in VCF Operations.

1. Log in to VCF Operations.
2. From the left navigation menu, click **Fleet Management**, and click **Tags**.
3. In the table with available categories, click the double arrow button next to the category that contains the tag you want to edit.

A table with all tags added to this category appears on the right.

4. In the table with available tags, select the check box next to the tag you want to edit.

Note:

You can edit only one tag at a time.

5. Click **Edit**.

6. Enter a new description for the tag, and click **Save**.

Push Tag Categories to a VCF Domain vCenter Instance or a Custom vCenter Group

After you create or import tag categories and tags into a VCF Operations instance, you can push these tag categories to other VCF domain vCenter instances or vCenter instances that are part of custom groups. To push categories to vCenter instances part of custom groups, the custom groups must be of type *Tag Group*.

- Verify that you have vSphere Tagging privileges for the target VCF domain vCenter instance.
- Verify that the target VCF domain vCenter instance is of version 9.0.
- Verify that you have **Tags > Manage** permissions in VCF Operations.
- If you want to push tag categories to vCenter instances that are part of custom groups, verify that there are custom groups of type *Tag Group* created in the VCF Operations instance, and vCenter instances are added to those custom groups. Even if other object types are also added to the custom groups, tags and categories are pushed only to vCenter instances. For more information about creating custom object groups in VCF Operations, see [Managing Custom Object Groups in VCF Operations](#).

By using VCF Operations, you create and manage tags and categories from a single place, at scale. You can then push the created categories and the tags added to them to as many VCF domain vCenter instance as you need, instead of creating the categories manually in each vCenter instance.

When you try to push a tag category to VCF domain vCenter instances, if the tag category you try to push, has some of the same properties as a tag category in the target vCenter instance, but differences in other properties, the operations fails and the categories are displayed as conflicts. Similarly, if a tag added to the category have some of the same properties as a tag in the same category in the target vCenter instance, but differences in other properties, the operation fails and the categories are displayed as conflicts.

Important:

If you cannot see the conflict details, verify that you have the REST API read privileges assigned for the VCF Operations instance.

To proceed with pushing the categories, you must first manually resolve the conflicts by following the guidance provided in the details for each conflict. Alternatively, you can select to overwrite the category properties in the target VCF domain vCenter instance wherever a conflict is detected. The following types of conflicts can only be resolved manually, even you select the check box:

Conflict	Resolution
A category or tag with the same name and ID exists both in VCF Operations, and in the vCenter instance selected for the push operation. The value for Tags per object is set to One Tag in VCF Operations, but it is set to Multiple Tags in the vCenter instance.	In VCF Operations, change the value for the Tags per object attribute to Multiple Tags .
A category or tag with the same name and ID exists both in VCF Operations, and in the vCenter instance selected for the push operation. The Associable object types attribute in the vCenter instance contains more object types than the Associable object types attribute in VCF Operations.	In VCF Operations, edit the values for the Associable object types attribute and verify they are the same as in the vCenter instance.

Conflict	Resolution
A category or tag with the same name but a different ID exists both in VCF Operations, and in the vCenter instance selected for the push operation.	Rename the category or tag in the vCenter instance, or delete the conflicting category or tag from VCF Operations or from the vCenter instance.

When a task is in progress, you cannot initiate the same task again. For example, if another user or you initiate a push of categories, you cannot initiate push again before the first task is complete.

1. Log in to VCF Operations.
2. From the left navigation menu, click **Fleet Management**, and click **Tags**.
3. From the table with available tag categories, select the check box next to the tag categories you want to push.

Note:

You can push up to 20 tag categories at a time.

4. Click **Push Categories**.

The **Push Categories** dialog box appears.

5. Select the VCF domain vCenter instances or custom groups to which you want to push categories:
 - To push categories to specific VCF domain vCenter instances, click the **VCF Domain vCenter Instances** tab, and select VCF Instances or a VCF domain vCenter instances from the list.
 - To push categories to custom groups, click the Custom Groups tab, and select a custom group from the list.
6. Optional: If you want to overwrite the category properties in the target VCF domain vCenter instance when a conflict is detected, select the check box at the bottom of the dialog box.
7. Click **Push**.

The tag categories are pushed to the selected VCF domain vCenter instances or vCenter instances in a custom group.

You cannot see banner notifications for tasks performed by another user.

You can use the tags and categories with existing automation scripts or assign tags to inventory objects by using the vSphere Client. For more information about assigning tags to specific inventory object, see [How Do You Assign or Remove a vSphere Tag](#).

Managing your Environment Configurations

You can use VCF Operations Configuration Drifts feature to monitor and view the configuration settings of vCenter instances and vSphere clusters that have drifted from the desired configurations, without the need to manually track each change.

Configuration Drifts shows the changes in product configurations at a point in time and allows you to compare the changes to the assigned desired values. It provides a proactive method to monitor configuration changes.

Note:

This feature applies only to vCenter instances and vSphere clusters deployed in VCF 9.0 or later.

Why use Configuration Drifts

Configuration Drifts offers several benefits, including:

1. Consistency: Ensures that vCenter instances and vSphere clusters are configured consistently. This helps reduce potential issues from configuration drift.

2. Compliance: Helps ensure that vCenter instances and vSphere clusters are configured to meet organizational policies and regulations.
3. Scalability: Allows for infrastructure configuration at scale.

Managing Configuration Templates in VCF Operations

You can define a desired state based on your current environment using configuration templates. You can create, edit, or delete configuration templates in VCF Operations.

- If the target vCenter is not deployed as a part of the same VCF instance as the VCF Operations Console, ensure that the user is added to the vCenter Administrators group and that the account used to add vCenter to VCF Operations has administrator privileges.

Note:

This applies only if you select user credentials when integrating a vCenter cloud account in VCF Operations. For more information, see [Configure a vCenter Account in VCF Operations](#).

- Verify that the user has the required **Configuration Drifts** roles to access configuration templates. For more information, see the [Access Control: Roles Tab](#).
 - **Manage Privilege:** for creating templates, assigning them to policy, and running configuration drift check.
 - **View Privilege:** for viewing template content and viewing existing configuration drifts.
- Verify that your cloud account is registered to a cloud proxy or a cloud proxy group.

A configuration template is a set of configurations representing a desired state. A configuration template enables an administrator to define and monitor the configurations. The configuration schema defines the settings for the desired state.

The **Configuration Templates** tab provides an overview of all configuration templates, including the object type assigned to each template, description, number of policies assigned to each template, template author, and template creation date.

You can also manage configuration templates using a supported source control solution. For more information refer to [Managing Configuration Templates with Source Control](#).

1. To create a configuration template, from the left menu, click **Fleet Management > Configuration Drifts** and then click the **Configuration Templates** tab.
2. Click **Create Config Template**.
3. Enter a name and description for the configuration template.
4. On the **Template Setup** tab, select a **Configuration Source** vCenter from which you want to extract the configuration settings and click **Next**.
5. On the **Configuration Schema** tab, select the configuration settings that you want to include in your template and click **Next**.
6. (Optional) - On the **Assign vCenters using Policy (Optional)** tab select one or more VCF Operations policies to assign to your configuration template and then click **Next**.

Note:

Assign a policy to the configuration template. If you have not assigned any policies during the configuration template creation process, follow the steps below:

1. From the left navigation menu, click **Infrastructure Operations > Configurations > Policy Definition**.
2. Select the policy you want to assign.
3. Select **Configuration Templates**.
4. Click on the **Edit** icon and select the template you want to activate.
5. Click the drop down and click **Activated** to assign the configuration template for the policy.
6. Click **Save**.

7. On the **Review** tab, a preview of your configuration template is displayed in the right pane. Review the configuration template and then click **Create**.

You can view the template listed on the **Configuration Templates** tab. You can also edit or delete the configuration template from the same page.

Managing Configuration Templates with Source Control

Integrating the VCF Operations Configuration Drifts Templates with a source control repository allows you to control versioning, auditing, and the management of your configuration templates.

Integrating with Source Control

Integrating with a source control repository enables you to manage configuration templates under source control. This functionality also facilitates accurate configuration updates, supports change control with reviews, and integrates continuous integration and continuous development for automated testing and validation.

GitLab and GitHub are supported source control repositories.

The source control repository system must support the following Console requirements:

- Reachable by FQDN
- Access Token can be generated by the source control and used by the Console
- Accepts a Branch input field
- Accepts a Repository input field

This integration has been tested with the following Git-based source control solutions:

- **GitLab:** Version 15.x, 16.x, and 17-17.6.
- **GitHub:** Version 3.13.13.

Generating Access Tokens in GitLab (Example)

The following is a generic workflow for generating an access token in GitLab. Your experience may vary based on your environment.

1. Log in to your GitLab account.
2. Click the **Profile** icon and click **Edit profile**.
3. Click **Access Tokens**.
4. Enter the **Token name** and select the **Expiration date**.
5. Select the following scopes. Scopes set the permission levels granted to the token.
 - a. api
 - b. read_api
 - c. read_repository
 - d. write_repository
6. Click **Create personal access token**.

Generating Access Tokens in GitHub (Example)

The following is a generic workflow for generating an access token in GitHub. Your experience may vary based on your environment.

1. Log in to your GitHub account.
2. Click the **Profile** icon and click **Settings**.
3. Click **Developer settings**.
4. Expand **Personal Access Tokens** and then select **Tokens (classic)**.

5. Click **Generate new token** and then click **Generate new token (classic)**.
6. Enter the **Name** and select the **Expiration** date.
7. Select the following scopes. Scopes define the access for personal tokens.
 - a. repo
 - b. Admin:repo_hook
 - c. project
8. Click **Generate token**.

Configuring Source Control in VCF Operations

You can configure source control settings in VCF Operations to synchronize configuration templates to and from a repository.

- Verify that your VCF Operations fleet management can establish a connection to the appropriate the source control solution.
 - Ensure bi-directional HTTPS connectivity between the *VCF Operations management appliance* and the *Git repository*.
 - Verify that you are using a supported source control solution.
 - Generate access tokens to authenticate and authorize a secure connection to the repositories.
 - **GitLab**: Generate an access token for your repository following the [documentation](#) from GitLab.

Note:

The access token must have the Maintainer or Owner role. Ensure the access token has an expiration date and the following scopes:

 - Read access to the repository. (i.e. `read_repository` in GitLab)
 - Write access to the repository. (i.e. `write_repository` in GitLab)
 - Read access to the project. (i.e. `read_api` in GitLab)
 - Read and write access to the project. (i.e. `api` in Gitlab)
 - **GitHub**: Generate an access token for your repository following the [documentation](#) from GitHub.
1. To integrate source control in VCF Operations, from the left menu, click **Administration > Control Panel** and then click **Source Control**.
 2. For **Source Control Type**, select a source control solution.
 3. Enter the **Source Control Server** address where the configuration templates will be stored.
 4. Enter the access token from your source control type in the **Access Key**.
 5. Specify the **Branch** for storing or reading content.
 6. Provide the **Repository** path for storing or retrieving contents.
 7. By default, **Code Review** is enabled by default. The code review process creates a merge request for any operation on a configuration template. The content syncs to the repository after the merge request is approved.

Note:

To disable the code review process, uncheck the **Code Review** checkbox.
 8. Click **Test Connection** to verify if the integration is successful. If the **Connection Result** is successful, click **Save** to complete the integration.

After the setup is complete, you can manage the configuration templates from your source control solution.

Template Organization and Repository Structure: Configuration templates are automatically created and organized in a defined folder structure: `Operations-ConfigTemplate/config-templates/vCenter`. Each template must be stored in a unique folder in the source control repository.

Template Name and Storage: Each configuration template must have a unique name to avoid duplication.

Template Content and File Structure: For each configuration template, three files are automatically created in the repository:

1. `<templatename>-desiredState.json`: Contains the desired state of the template.
2. `<templatename>-metaData.json`: Contains metadata information like template name, version, and selected configuration.
3. `<templatename>-manifest.json`: Used to manage and maintain the template's structure and metadata.

Syncing Configuration Templates with Source Control

After you configure the source control solution in VCF Operations, sync the configuration templates with the source control repository. This topic outlines the different workflows you can follow to manage the configuration templates and integrate them with the repository.

Review, Sync, and Merge Templates from VCF Operations

Review Process Enabled

When you create, edit, or delete a template in VCF Operations, a **merge request** is created in source control and is visible in the Config Template list view. You must review and approve the request in your source control solution. After the request is approved, the template status changes to **OUT_OF_SYNC**. Sync the template using the **Sync** option in VCF Operations. The template status changes to **IN_SYNC** after the sync is complete.

Note:

The edited or deleted content is reflected only after the merge request is approved and the content is synced.

Review Process Disabled

When you edit a template, the template is directly edited in the source control repository without a request. If you delete a template from VCF Operations, the template is marked **OUT_OF_SYNC** and is directly deleted from the source control during the next sync.

Sync Existing Templates in VCF Operations

Code Review Process Enabled

When source control integration is enabled, merge **requests** are created for syncing all existing templates from VCF Operations to the source control repository. After the request is approved, the template status changes to **OUT_OF_SYNC**. Use the **Pull from GitHub/GitLab** option to sync content from the repository. The status of the template changes to **IN_SYNC** after the sync is successful.

Code Review Process Disabled

The status of all existing templates reflects as **OUT_OF_SYNC**: No request is raised. You must use the **Sync** option to sync content from source control. The status changes to **IN_SYNC** after the sync is complete.

Sync Remote Templates to VCF Operations

If you edit or delete existing templates in the source control repository, the status of the corresponding templates reflects as **OUT_OF_SYNC** in VCF Operations.

When you edit a template in the source control repository, you must **Pull from GitHub/GitLab**, and after the sync is successful, the updated template content gets reflected in VCF Operations and the status changes to **IN_SYNC**.

When you delete a template in the source control repository, the template is removed from VCF Operations.

When you configure a source control in VCF Operations, existing templates without any content in the source control repository will be created in VCF Operations as placeholders and reflect the **OUT_OF_SYNC** status. You must sync the content to populate the templates. The status changes to **IN_SYNC** after the sync is complete.

Additional Information

Merge Request Discarded

A merge request can be discarded if the changes proposed in the merge request are no longer valid or required. If a merge request is discarded or closed by the user, a warning message is displayed informing the user that the changes made to the corresponding template in VCF Operations are not merged and the current template may not reflect the desired state in source control.

You can edit the template to make the necessary updates and address the issues that caused the merge request to be discarded. After you make the update a new request is raised. This ensures that the latest template changes are submitted for review.

Drift Detection

Drift detection is skipped if the template does not exist in the source control repository. This may occur in scenarios where the template has not been synced with the source control.

Updating Source Control Configurations

Editing or Updating Branch or Repositories

You can update the branch or repository or switch to a new branch or repository. Once done, the templates from the previous source control repository displays the **OUT_OF_SYNC** status. The existing templates in VCF Operations will sync to the new branch or repository. For more information, see Sync Existing Templates in VCF Operations section.

If you delete a source control configuration, the templates stored locally in VCF Operations remain unchanged, however the status changes to **OUT_OF_SYNC**.

Note:

If you want to update your source control type, you must first delete the existing configuration.

Using the Configuration Drifts Overview Page

The Configuration Drift dashboard allows you to view and monitor vCenter and vSphere cluster drift statuses.

You can view and compare configuration drifts and identify settings that have changed.

A configuration drift notifies you of any deviation from assigned configuration template values. Use the **Configuration Enabled Objects Summary** bar to view the total number of objects and their enabled statuses. Enabled objects are comprised of all the vCenter at version 9.0 or greater that have a configuration template attached to them and include vSphere clusters with Configuration Profiles enabled.

The **vCenter Instances Drift Status** and **Cluster Drift Status** charts represent the state of the enabled vCenter instances and vSphere clusters at a given point in time. All the potential drift statuses and their definitions are listed below:

- **Not Drifted:** This state shows vCenter instances and vSphere clusters with no drifts from the desired state.
- **Drift Detected:** This state shows when vCenter instances or vSphere clusters have deviated from the desired standard configurations.
- **Drift Check Failed:** This state occurs under the following scenario's
 - vCenter instances or vSphere clusters are unavailable.
 - The drift computation is in progress
 - The drift has not been computed against that specific resource.

- **Unavailable:** This state shows a list of vCenter instances and vSphere clusters that are either
 - Below version 9.0
 - Not registered with the cloud proxy.

Additionally, you can view the **Configuration Templates** assigned to your vCenter instances and view all **Objects**.

Scheduling Drift Detection

You can automate the drift detection process and schedule drift detection in VCF Operations.

Scheduling drift detection enables you to run automatic drift checks on your vCenter instances at an interval of your choice.

1. To schedule drift checks, from the left menu, click **Fleet Management > Configuration Drifts > Schedule Drift Detection**.
2. Enter a name and description in the **Configuration Drift Check Information** pane and click **Next**.
You can schedule drifts only for vCenter object types.
3. **Select an object** from the **Select Scope** window on the right and drag it into the left Scope window to add vCenter instances and click **Next**.

If you select a VCF folder as the scope, all VCF instances belonging to the folder get automatically added to the scope.
4. Optional: Click **Preview Scope** to confirm the vCenter instances that the drift check will run on, and then click **Next**.
5. Optional: You can filter the criteria and add multiple criterias for the vCenter object type and then click **Next**.
6. Set the **Schedule** and click **Create**.

A new job is created to run scheduled drift detection in the automation central page. For more information see [Configuring Automation Jobs](#).

Viewing Drifts and Downloading Drift Reports for vCenter Instances

You can detect configuration drifts for each configuration template you have created for a specific vCenter.

To view vCenter configuration setting values and detect configuration drifts, you must:

- Ensure that you have created at least one configuration template for your vCenter.
- Ensure that the user is configured with the **Administrator** role in VCF Operations. For more information, see [Privileges Required for Configuring a Adapter Instance](#).
- Verify that you have **Configuration Drifts** privileges to access configuration templates. For more information, see [Managing Users and Access Control in VCF Operations](#).
 - **Manage Privilege:** for creating templates, assigning them to policy, and running drift check.
 - **View Privilege:** for viewing template content and viewing existing drift.
- Verify that you have the vCenter instance configured and registered with the cloud proxy and that the vCenter instance has one active policy assigned to the configuration template.

1. To view your configuration drift details, **Fleet Management > Configuration Drifts**.
2. Expand **VCF Instances**, select any **VCF** instance, and click on the **Drifts** tab.

On the **Drifts Details** page, you can view the number of enabled vCenter instances on the **vCenter Instances** tile. You can also view the number of vSphere Configuration Profiles enabled clusters on the **Clusters** tile. For more information, see [Viewing Drifts for Clusters](#).

By default, the vCenter instances tile is selected.

3. Select one or more vCenter instances.
4. Click **Detect Drift** to detect configuration drift.

This detects drifts across all the configuration templates and displays the result in the Drifts table.

5. To view the drift for each vCenter, click on the vCenter instance.
6. Select a **Config Template** to view the drift details.

If a drift is detected, the right pane displays the drifts between the **Current vCenter Value** and the **Config Template Value**. If no drift is detected for the selected template, the message **No drift found for this Config Template since last configuration drift check** status is displayed.

- **Detect Drift:** To further detect drift for a particular template in a vCenter instance.
- **View Template:** To view the configuration template.

7. Navigate to the **Drifts Details** page, select one or more vCenter instances, and click **Download Drift Status Report** to download the historical drift data.

The Config Drift Report displays the date, resource, template, and drift status for each vCenter instance. By default the report shows data for the last seven days. You can update the duration to up to 30 days from the Global Settings page. For more information, see [List of Global Settings](#).

Viewing Drifts for Clusters

You can view the drift status of your vSphere clusters in VCF Operations. The drift status is visible only for vSphere clusters that have vSphere Configuration Profiles enabled.

- Verify that you have vCenter version 9.0 or later.
- Verify that you have registered your vCenter with a cloud proxy.
- Clusters are managed with vSphere Configuration Profiles in individual vCenter instances. Enable vSphere Configuration Profiles for clusters in vSphere. For more information, see [Using vSphere Configuration Profiles](#).

Note:

There may be a delay of up to 8 hours for updates made on the cluster in vCenter instances to be reflected on the Configuration Drifts page.

1. To view the cluster drift status, from the left menu, click **Fleet Management > Configuration Drifts**.
2. Expand **VCF Instances**, select any **VCF** instance, and click on the **Drifts** tab.
3. Click the **Clusters** tile.

You can also view the number of vSphere Configuration Profiles enabled clusters. The drift status of all enabled clusters associated with the **VCF** instance is displayed.

4. You can click on a cluster to view the cluster drift in vSphere Client (vSphere Client > Inventory > DataCenters > Cluster Name > Configure > Desired State > Configuration > Compliance)

Configuration Drifts FAQ

This topic covers some frequently asked questions about Configuration Drifts in VCF Operations.

1. What are the prerequisites for setting up Configuration Drifts?
 - a. To view the prerequisites for setting up configuration drifts, please see [Managing Configuration Templates in VCF Operations](#).

2. When does Configuration Template creation fail?
 - a. Configuration Template creation can fail if the template size exceeds 9 MB. You can copy the desired state contents to a file to check the file size and ensure it is under 9 MB.
 - b. Configuration Template creation can fail if a connection issues occurs between vCenter and the configuration adapter. You can check the connection status of vCenter and the configuration adapter to ensure that there is a proper connection between the two. To troubleshoot, see [Cloud Proxy Troubleshooting](#) and [vSphere Management Pack Issues](#).
3. What should I do if there is an error while displaying drift details?
 - a. If you encounter an error while attempting to view drift details, it may be due to connectivity issues between vCenter and the configuration adapter. To troubleshoot, see [Cloud Proxy Troubleshooting](#) and [vSphere Management Pack Issues](#).
4. Why is there a mismatch in the vSphere Configuration Profiles (VCP) drift status between the vCenter Console and the VCF Operations Console?
 - a. The VCP drift status in the VCF Operations Console is updated through a polling mechanism that runs every 8 hours. Wait for a few hours to allow the next data collection cycle to update the status in the VCF Operations Console.
5. Why is the Drift Report PDF generated with "No drift details available for the selected resource/resources" message?
 - a. This message may appear if the **Detect Drift** operation has never been triggered for the selected resources. To resolve this, run the **Detect Drift** process on the relevant resources first, and then try downloading the report again.
6. How can I remediate a vCenter if I receive a notification indicating that it has drifted from a Configuration Template?
 - a. Remediation cannot be done at the VCF Operations console at this time. You must log in to the vCenter and manually revert the drifted configurations to their original values as defined in the template to make the vCenter compliant again.
 - b. If you want the drifted values to be the Configuration Template values, edit the template under Configuration Drifts to match the current configuration of the vCenter.
7. How do I troubleshoot if a vCenter Drift status is in an **Error** state?
 - a. Check the connection between VCF Operations and vCenter to ensure that vCenter is in a **Collecting** state within VCF Operations. If it is not collecting, start or resume the data collection for vCenter. This will restore proper drift detection and clear the error. To troubleshoot, see [vSphere Management Pack Issues](#).
8. How can I debug when the configuration template sync fails to and from remote repository?
 - a. In the VCF Operations navigate to **Administration > Control Panel > Source Control** and click **Test Connection** to verify if connection with the Git Repository is successful or not. Ensure the connection is successful before performing any action related to configuration drifts using Git. If the test fails, check the repository URL, credentials, and network connectivity.
9. How often is the cluster drift status updated?
 - a. The cluster drift status is updated once every 8 hours.

Configuring Management Appliances

Update the DNS Server Configuration for VCF Core Components

SDDC Manager uses DNS servers to provide name resolution for the VCF core components. You can update the DNS server information that you provided during VMware Cloud Foundation deployment.

- Verify that both forward and reverse DNS resolution are functional for each component using the updated DNS server information.
- Verify that the new DNS server is reachable from each of the VMware Cloud Foundation components.
- Verify that SDDC Manager can reach all VMware Cloud Foundation components over SSH.

- Verify that all VMware Cloud Foundation components are in an active state.

When you update the DNS server configuration, SDDC Manager performs DNS configuration updates for the following components:

- SDDC Manager
- vCenter instances
- ESX hosts
- NSX Managers
- NSX Edge nodes

If the update fails, SDDC Manager rolls back the DNS settings for the failed component. Fix the underlying issue and retry the update starting with the failed component.

Updating the DNS server configuration can take some time to complete, depending on the size of your environment. Schedule DNS updates at a time that minimizes the impact to the system users.

To update the DNS configuration for VCF management components, see [Update the DNS Configuration for VCF Management Components](#).

1. In the VCF Operations console, click **Administration** > **SDDC Manager** > **Network Settings** > **DNS Servers**.
2. Click **Edit**.
3. Review the overview information and click **Next**.
4. Review the prerequisites and click **Next**.
5. Enter a primary DNS server IP address and click **Save**.
Optionally, enter an alternative DNS server IP address.

Update the NTP Server Configuration for VCF Core Components

SDDC Manager uses NTP servers to synchronize time between the VCF core components. You can update the NTP server information that you provided during VMware Cloud Foundation deployment.

- Verify the new NTP server is reachable from all components.
- Time skew between new NTP servers is less than 5 minutes.
- Verify that SDDC Manager can reach all VMware Cloud Foundation components over SSH.

You must have at least one NTP server. When you update the NTP server configuration, SDDC Manager performs NTP configuration updates for the following components:

- SDDC Manager
- vCenter instances
- ESX hosts
- NSX Manager nodes
- NSX Edge nodes

If the update fails, SDDC Manager rolls back the NTP settings for the failed component. Fix the underlying issue and retry the update starting with the failed component.

Updating the NTP server configuration can take some time to complete, depending on the size of your environment. Schedule NTP updates at a time that minimizes the impact to the system users.

To update the NTP configuration for VCF management components, see [Update the NTP Configuration for VCF Management Components](#).

1. In VCF Operations , click **Administration** > **SDDC Manager** > **Network Settings** > **NTP Servers**.
2. Click **Edit**.
3. Review the overview information and click **Next**.
4. Review the prerequisites and click **Next**.
5. Enter one or more NTP server IP addresses and click **Save**.

Add a VASA Provider for vVols Storage in Workload Domains

To use vVols storage for workload domains, you must add a VASA provider.

Before you add the VASA provider details in VCF Operations , ensure that you have set up the VASA provider. For information about setting up a VASA provider, see the vSphere Storage documentation.

Important:

Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

1. In VCF Operations, select **Administration** > **SDDC Manager** > **Storage Settings**.
2. Click **Add VASA Provider**.
3. Enter a name for the VASA provider.
4. Enter a URL for the VASA provider.
5. Enter VASA provider user credentials.
6. Enter VASA provider storage container details.
7. Click **Save**.

Note:

You can add additional VASA user credentials and storage containers.

Update the NTP Configuration for VCF Management Components

To update the NTP configuration for your VCF management component, you can use VCF Operations fleet management. The **Update NTP Configuration** option is supported for VCF Automation and VCF Identity Broker.

Verify that you have a current backup of the nodes in your cluster. See [Backup and Restore of VMware Cloud Foundation](#)

To update the NTP configuration of VCF core components, see [Update the NTP Server Configuration for VCF Core Components](#).

1. Under **Fleet Management > Lifecycle > Components**, select the integrated component that you want to update.
2. On the component details screen that appears, click the ellipses (...) and select **Update NTP Configuration**.
3. The **Proceed to NTP Configuration Update** pop-up window appears. If you have taken a backup of your cluster, click **Proceed**.
4. Review the current configuration of your NTP servers and click **Next**.
5. On Update NTP Configuration, you can:
 - Click **Use Host Time** to change the time sync mode of your server to use the ESX NTP server.
 - Click **Add New Server** to configure a new server with a unique name and FQDN or IP address. Enter the server details, and click **Submit**.
 - To change the priority of your servers, click the up or down arrows.

When finished, click **Next**.

6. To change NTP server priority, use the up and down arrows of the NTP servers listed and click **Next**.
7. Click **RUN PRECHECK**.

Note:

If the data validation is not successful, perform the recommended actions and run the precheck again.

8. After a successful precheck run, review the summary of your add nodes request and click **Finish**.

Update the DNS Configuration for VCF Management Components

To update the DNS configuration for your VCF management component, you can use the fleet management appliance. The **Update DNS Configuration** option is supported for VCF Automation and VCF Identity Broker.

Verify that you have a current backup of the nodes in your cluster.

To update the DNS configuration of VCF core components, see [Update the DNS Server Configuration for VCF Core Components](#).

1. Under **Fleet Management > Lifecycle > Components**, select the integrated component that you want to update.
2. On the component details screen that appears, click the ellipses (...) and select **Update DNS Configuration**.
3. The **Proceed to DNS Configuration Update** pop-up window appears. If you have taken a backup of your cluster, click **Proceed**.
4. Review the current configuration of your DNS servers and click **Next**.
5. On Update DNS Configuration, you can:
 - Click **Add New Server** to configure a new server with a unique name and IP address. Enter the server details, and click **Submit**.
 - To change the priority of your servers, click the up or down arrows.

When finished, click **Next**.

6. To change DNS server priority, use the up and down arrows for the DNS servers listed and click **Next**.
7. Click **RUN PRECHECK**.

Note:

If the data validation is not successful, perform the recommended actions and run the precheck again.

8. After a successful precheck run, review the summary of your add nodes request and click **Finish**.

Configuring the NTP Settings for VMware vSphere Foundation Components

To configure the NTP configuration for your VMware vSphere Foundation component, you can use the VCF Operations administration UI.

The Network Time Protocol (NTP) setting is used to synchronize the time of your VCF Operations cluster nodes. Configure the NTP setting in VCF Operations to ensure that all the cluster nodes follow the same time protocol. Following the same time protocol allows your cluster's services to run smoothly. The primary node serves as the NTP server by default and the time of all nodes are synchronized to the primary node. You can continue using the primary node as the NTP server and add another NTP server to serve as a failover if the primary node fails. In case you do not want to use the primary node, you can add the NTP server you wish to use.

1. Log in to the VCF Operations administration interface at <https://primary-node-name-or-ip-address/admin>.
2. Click the **NTP Settings** icon.
The Global Network Time Protocol settings wizard opens. The primary node is listed as the existing NTP server.
Note: You must configure at least two NTP servers to have a back-up if one server fails.
3. Add a secondary NTP server. Enter the IP of FQDN of the NTP server you wish to use in **NTP Server Address** field.
4. Click **Add**.
The new NTP server is displayed along with the primary node.
5. Ensure that the status of the NTP server is green which means its reachable and working.
6. To delete a NTP server, click **Remove** next to the NTP server status.

Resize Storage

To increase storage for your VCF management component, you can use the fleet management appliance. The **Storage Resize** option is supported for VCF Automation and VCF Identity Broker.

Under **Fleet Management > Lifecycle > Components**, select the integrated component that you want to update.

On the component details screen for VCF Automation or VCF Identity Broker, click **Storage Resize**. The current storage configuration appears with values displayed for every volume in the cluster:

- Volume Group Name: Name of the volume group.
- Number of Volumes: Number of volumes in the group.
- Volume Group Capacity: Size of each volume in the group.
- Total Used Capacity: Storage capacity used in the group.
- Total Storage Capacity: Number of Volumes times the Volume Group Capacity.

Note:

Storage resize only works to extend your storage capacity. To reduce storage, you must redeploy the component.

To resize storage:

1. Enter a larger value for a Volume Group Capacity and click **Next**.
2. After reviewing the change, click **Resize**.

Deploying VCF Automation as a Day-N Operation

If VCF Automation was not deployed as part of the initial VMware Cloud Foundation deployment or import of an Aria Automation 8.x instance, you can add it as a Day-N operation using VCF Operations fleet management. Deployment of only a single VCF Automation instance is supported.

Download the install binaries for VCF Automation from Binary Management. See [Downloading VCF management components](#).

The following procedure shows how to deploy a new instance of VCF Automation.

- On the **Overview** tab under **Fleet Management > Lifecycle**, click **Add** on the tile for VCF Automation.
- Select the options for your deployment.
 - For the Installation type, select **New Install**.
 - Version: 9.0 (default)
 - Deployment Type: To choose the correct type, click **View Sizing Info** and review the compute nodes and resources that a Small, Medium, or Large deployment supports. For sizing guidelines, download the [Planning and Preparation Workbook](#) and click the **Management Domain Sizing** tab. Notes with general guidelines are at the far right of the page.
- If you select NEW, click **+** to create a certificate:
 - Enter an FQDN
 - The Common Name and a Subject Alternative Name must include the FQDN
- Enter the Infrastructure properties.
 - For vCenter, select the Management Domain vCenter
 - Enter information for the primary cluster.
 - Cluster
 - Folder
 - Resource Pool
 - Network
 - Datastore
- Enter the Network properties.
 - Domain Name: Top-level domain, such as example.com.
 - Domain Search Path: Top-level domain, such as example.com.
 - DNS Servers: To select the DNS server that you want to configure VCF Automation with, click **Edit Server Selection**. If the DNS server that you want is not available, click **Add New Server** to add and select it.
 - NTP: To select the NTP server that you want to VCF Automation with, click **Edit Server Selection**. If the NTP server that you want is not available, click **Add New Server** to add and select it.
 - IPv4 Details:
 - IPv4 gateway: Default gateway of the network for VCF Automation to join.
 - IPv4 netmask: Netmask of the network for VCF Automation to join.
- Enter the Component information.

Component Properties

- FQDN -- Enter the component FQDN
- Certificate is prepopulated with the value from the previously added certificate.
- Component Password -- Select one component password for two accounts:
 - vmware-system-user
 - Provider's admin account

If you do not have a password, click **Add Password** to create a new one. Password must be a minimum of 15 characters.

Cluster VIP

- Choose internal or external load balancer:
 - For internal load balancer, the FQDN should resolve to the primary VIP address.
 - For an external load balancer or others, the FQDN should resolve to the Virtual Server IP address of the load balancer.

Components

- VM Node Prefix: A unique identifier used to name component nodes. To avoid conflicts and maintain proper node identification within the cluster, ensure that the node prefix is unique.
- Internal Cluster CIDR: IP Address range used for internal cluster communication within the cluster. Choose a range that does not conflict with existing networks in infrastructure
- Primary VIP: The Primary Virtual IP Address is used for accessing component services
 - Figure 160: Primary VIP if using internal load balancer**

If using an internal load balancer, this Primary VIP Address should resolve to the component FQDN.



– **Figure 161: Primary VIP if using external load balancer**

If using an external load balancer, this Primary VIP Address and Additional VIPs should be in or added to the server pool that the load balancer can direct traffic to.



- Additional VIP addresses: This is optional. Depending on the small, medium, or large deployment type, you may add up to two additional VIPs.
- VM Node Prefix: Unique identifier used to name component nodes. Ensure this prefix is distinct for a cluster to avoid conflicts and maintain proper node identification
- Cluster Node IP Pool: The Node IP Pool is a range of IP addresses allocated to cluster nodes. These IP's ensure that each node has a unique address for internal communication within the cluster. Additional IP addresses from the pool will be used during rolling cluster upgrades. The pool must contain a minimum of four IP addresses.
- Cluster CIDR: IP address range used for internal cluster communication within the cluster. Choose a range that does not conflict with existing networks in the infrastructure.

7. Precheck: Click **Run Precheck**.
8. Summary: To trigger the deployment after reviewing the summary, click **Submit**.

Linking vCenter instances in VCF Operations

vCenter linking in VCF Operations provides you with a single pane of glass view of all the linked vCenter instances from a single vSphere Client.

- Verify that you have vCenter version 9.0 or later.
- Ensure that there is functioning forward and reverse DNS resolution between all vCenter instances in a vCenter linking group.
- Ensure that the user used to register the vCenter account in VCF Operations has Administrator privileges.

Note:

This applies only if you select user credentials when integrating the vCenter cloud account in VCF Operations. For more information, see [Configure a vCenter Account in VCF Operations](#).

- Verify that the user has the required **vCenter Linking** role assigned to access the vCenter instances. For more information, see the [Access Control: Roles Tab](#).
 - **Manage:** To add, edit, or delete the vCenter linking group.
 - **View Privilege:** To view the vCenter linking group and its members.
- Ensure that common Identity Providers (IDP) is configured on all vCenter systems. For more information, see [Identity Providers and Protocols Supported for VCF Single Sign-On](#).
- Ensure that the IDP user has the required permission for each vCenter system.

Note:

- IDP user permissions are not shared between vCenter systems and must be configured on each vCenter individually.
- Verify that the vCenter systems are not part of an Enhanced Link Mode group.

Note:

From VMware Cloud Foundation 9.0 onward, its recommended to use vCenter linking for vSphere Client single pane of glass functionality instead of Enhanced Linked Mode (ELM). If a vCenter system is part of ELM, it cannot be added to a vCenter linking group. You must deactivate ELM before proceeding. To deactivate ELM, see [How to Deactivate Enhanced Link Mode in vCenter Without Configuration Loss](#).

Functionalities previously provided by Enhanced Linked Mode in vCenter like synchronization of roles, tags, global permissions across the linked vCenter instances are not available in VCF Operations based vCenter linking.

- To configure VCF for VCF components, see [Configuring VCF Single Sign-On](#).
- To manage tags for a fleet of vCenter instances, see [Tags and Categories Overview](#).

You can link a minimum of two and a maximum of 15 vCenter instances in a group. All the vCenter instances in a linked group must be configured with common IDP users. Once linked, you can log into any one of the linked vCenter instances using the vSphere Client, and monitor and manage the inventories of all the linked vCenter instances.

1. From the left menu, click **Infrastructure Operations > Configurations**.
2. Under **Logical Groupings**, click **vCenter Linking**.
3. Click **Create Group**.
4. Enter the **Group Name** and **Description**, and then click **Next**.
5. Select the compatible vCenter instances that you want to add to the group and then click **Next**.

Note:

You can view the list of compatible and incompatible vCenter systems.

- Compatible: vCenter systems that meet the version prerequisite and are not part of any other vCenter linking group.
- Incompatible: vCenter systems that do not meet the version prerequisite or belong to another vCenter linking group.

6. Review your selection and click **Finish**.

The vCenter linking group is created.

7. If the status of any of the vCenter instances shows an error, fix the cause, and then click **Reset Linking** to troubleshoot.

Note:

Some of the common errors you may face and fix before you click **Reset Linking** as follows:

- vCenter is not reachable from VCF Operations: You must fix the connection between VCF Operations and vCenter.
- vCenter is part of Enhanced Linked Mode (ELM): You must deactivate ELM.

8. After the group is created, you can add or remove members from a group and you can also edit the group details or delete the group.

You can log in to any vSphere Client for any one of the linked vCenter systems using the IDP user that has the required permissions for all vCenter systems and view the inventory of all the vCenter systems in the group.

Collecting Data with Cloud Proxy in VCF Operations

You can use cloud proxies in VCF Operations to collect and monitor data from your on-premises data centers across different geographical locations.

Cloud proxies provide high availability within your VCF Operations environment, you can group two or more cloud proxies to form a collector group. The cloud proxy collector group ensures that there is no single point of failure in your cloud environment. If one of the cloud proxies experiences a network interruption or becomes unavailable, the other cloud proxy from the collector group takes charge and ensures that there is no downtime. All other user-initiated manual operations on the collector, such as to stop or restart the collector manually, do not result in automated re-balancing. For more information, see [Configuring Cloud Proxies in VCF Operations](#).

To configure cloud proxy using VCF Operations Fleet Management, see [Add Nodes to VCF Operations](#).

Note:

When a node within the collector group goes down, high availability failover occurs, so that data collection is not impacted. However some data loss is expected before the failover is complete.

You can also use cloud proxies to re-balance the resources in your collector group. The re-balance option is available as part of the Edit menu in the Collector Groups page.

Note:

You can use the re-balance option before vCenter initiates data collection. After the data collection starts, the re-balance option is disabled.

FIPS mode is supported in a cloud proxy. To leverage this functionality, make sure your cluster is in FIPS mode.

You can upgrade your cloud proxy. For more information, see [Upgrading](#).

You can deploy a cloud proxy based on your sizing requirements. For more information, see KB article [324340](#).

Table 934: Cloud Proxy Types

Cloud Proxy Type	VM Scale Requirements	CPU Requirements	Memory Requirements
Classic Small Cloud Proxy	Up to 16,000 VMs	2 vCPUs	8GB
Classic Large Cloud Proxy	Between 16,000 to 80,000 VMs	4 vCPUs	32GB
Unified Small Cloud Proxy	Up to 16,000 VMs	4 vCPUs	16GB
Unified Large Cloud Proxy	Between 16,000 to 80,000 VMs	8 vCPUs	48 GB

Use a classic cloud proxy in VCF Operations, to collect and monitor data from your remote data centers. In addition to data collection, you can use your cloud proxy to collect logs into the Log Cluster and to collect and upload support bundles to the Broadcom support portal site. Use unified cloud proxy for log collection or log assist. For more information on log assist, see [Log Assist](#). For more information on log collection, see [Configuring and Analyzing Logs](#).

Configuring Cloud Proxies in VCF Operations

You need cloud proxies to collect data from physical data centers. You can deploy classic or unified cloud proxies based on your requirements. For high availability or load balancing, you must deploy two or more cloud proxies in a collector group. A cloud proxy collects data from the end-point environment and uploads it to VCF Operations. Cloud proxies can support multiple vCenter accounts. For more information on cloud proxies, see [Cloud Proxy FAQ](#).

- Verify that you have an IP address and a DNS configuration for the cloud proxy.
- Verify that you have proper permissions to deploy OVF templates in vSphere. For more information, see [Deploy and Export OVF and OVA Templates](#).
- Allow outgoing HTTPS traffic for cloud proxy over port 443.
- Allow incoming traffic to cloud proxy over ports 443, 8443, 4505, and 4506 for telegraf based application monitoring.
- Allow incoming traffic to cloud proxy over port 443 for push model adapters or Suite-API on cloud proxy.
- Verify that the user has the required Cloud Proxy roles to access configuration templates. For more information, see the [Access Control: Roles Tab](#).
- Add a vCenter cloud account and provide an account with the following read and write privileges:
 - vCenter IP address or FQDN.
 - Permissions required to install a cloud proxy on the vCenter Server.
 For more information on privileges, see [Privileges Required for Configuring a Adapter Instance](#).
- Cloud proxies must have a proper DNS resolution to the VCF Operations nodes when using short/long FQDN names.

Note:

Using a firewall to restrict traffic by IP is not recommended since IPs can change without notice. Restricting traffic must be performed via FQDNs only.

Refer to the cloud proxy sizing requirements before configuring a cloud proxy. For more information, see [Collecting Data with Cloud Proxy in VCF Operations](#).

1. Log in to VCF Operations.
2. From the left menu, click **Administration > Cloud Proxies**, and then click **Add**.
3. Click **Broadcom support portal** to navigate to the Broadcom support portal, log in using your credentials and download the VMware Cloud Foundation Operations collector (Cloud proxy) OVA file. The same VMware Cloud Foundation Operations collector (Cloud proxy) OVA file can be deployed in either a classic cloud proxy configuration or a unified cloud proxy configuration.

Note:

VCF Operations supports both log collection and log assist. To collect logs, deploy the Cloud Proxy OVA using the Unified Cloud Proxy configuration. If you prefer not to use your Cloud Proxy for log collection or log assist, deploy the OVA in the Classic Cloud Proxy configuration instead. For more information on types of cloud proxy and sizing requirements, see [Collecting Data with Cloud Proxy in VCF Operations](#).

4. Navigate to your vCenter, select the name of your vSphere cluster, and select **Deploy OVF Template** from the **Actions** menu.
5. Insert the OVA link and then click **Next**.
 - Paste the cloud proxy OVA link in the **URL** field.
 - Click the **Local File** option, browse, and select the downloaded OVA file.
6. Follow the prompts to install the OVA on your vCenter.

For the most current information about sizing and scaling, see [Knowledge Base article 324340](#).
7. When prompted to enter the Unique Registration Key in the **Customize template** screen, return to the Install Cloud Proxy page in VCF Operations.
8. Optional: Activate **Data Persistence** to store data in the cloud proxy in case of connectivity issues. For more information, see [Activating Data Persistence in Cloud Proxy](#).
9. Click the **Copy Key** icon.

The unique registration key expires 24 hours after generation. To avoid using an expired key, click **Regenerate Key** before proceeding. The unique registration key is used by the cloud proxy to authenticate to VCF Operations.

Note: Upon clicking Regenerate Key, the unique registration key is refreshed and a new key is generated if you reload the cloud proxy page. A new unique registration key is generated if you activate data persistence, or log forwarding, or both.

10. Return to vSphere and paste the key in the **Unique Registration Key** text box to install the VCF Operations Cloud Proxy.
11. Optional: Select **Prefer IPv6** to use IPv6 for internal communications.
12. Optional: Set up a proxy server in the **Customize template** screen.
 - a) Enter details in the **Network Proxy IP Address** and **Network Proxy Password** properties.
Note: If you use network proxy for log forwarding, port 9543 must be open.
 - b) To activate SSL, select the **Use SSL connection to proxy** check box.
 - c) If you are using SSL, you can verify the certificate of the proxy server. Public certificate authorities are used to verify the proxy server certificate. To activate this, select the **Verify proxy's SSL cert** check box in the **Verify SSL cert** property.
 - d) You can specify the IP/FQDN URL that is used to access the system when a load balancer is used.
 - e) If you have a custom certificate authority, paste the root certificate authority in the **Custom CA** property to verify the certificate of the proxy server. Include the following lines when you copy the root certificate authority:

```
"-----BEGIN CERTIFICATE-----"
"-----END CERTIFICATE-----"
```
13. Click **Finish**.
 The deployment takes a few minutes to finish.
14. Locate the cloud proxy you just installed, select the VCF Operations Cloud Proxy, and click **Power on**.
Note: You must power on the VMware Aria Operations Cloud Appliance within 24 hours of registering it. After 24 hours, the Unique Registration Key expires, and you must delete the VMware Aria Operations Cloud Appliance and deploy another cloud proxy.
15. Return to the Cloud Proxy page in VCF Operations to view the status of the cloud proxy you just installed. For more information, see [Monitoring the Health of Cloud Proxies](#).
16. To view the accounts that are using this connection, click the Cloud Proxy.
17. Optional: To delete a cloud proxy, click the vertical ellipsis and then click **Delete**. For more information, see [Deleting Cloud Proxies](#).

Activating Data Persistence in Cloud Proxy

Use data persistence, to avoid data gaps in case of temporary connectivity issues. You can activate data persistence to allow the cloud proxy to store data that is sent to VCF Operations, if the connection fails between the cloud proxy and VCF Operations. The cloud proxy stores all the persisted data that includes metrics and properties, along with time stamps.

Cloud proxy can store data for a maximum duration of one hour. If there is lack of space or if the connection fail lasts for more than an hour, the cloud proxy rotates the stored data by deleting the oldest stored data and replacing it with the most recently collected data. In case of lack of space, you can add additional storage, for more information, see KB article [2016022](#).

After the connection is restored, the cloud proxy sends the stored data to VCF Operations before the real-time data. The stored data is displayed before the real-time data as data is displayed in the correct time series. If the connection loss was more than one hour, there can also be a gap in the displayed data.

You can activate data persistence when you deploy a new cloud proxy. For more information, see [Configuring Cloud Proxies in VCF Operations](#). You can also activate data persistence in an existing cloud proxy. For more information, see [Monitoring the Health of Cloud Proxies](#).

Monitoring the Health of Cloud Proxies

You can view the status and health of your cloud proxy after you add it in VCF Operations. You can then monitor the health and view alerts and metrics of your cloud proxy using the VCF Operations cloud proxy object.

1. Log in to VCF Operations.
2. From the left menu, click **Administration > Cloud Proxies**.
 The list of cloud proxies is displayed.

Option	Description
Name	The name of the cloud proxy.
IP	The IP address of the cloud proxy.
Status	Status of the cloud proxy. For example, the Going Online status is displayed for a few minutes when you add a new cloud proxy. After the cloud proxy is connected to VCF Operations, the status changes to Online. If the VCF Operations is not connected, the Offline status is displayed.
Version	The version used to install the cloud proxy.
Accounts	The number of accounts that are created and associated with the cloud proxy.
Type	Displays the cloud proxy type.
Collector Group	Displays if the cloud proxy is part of the collector group. If the cloud proxy is part of the collector group, the name is displayed. Click the name to view the collector group details. Note: No data is displayed if the cloud proxy is a standalone cloud proxy and not a part of the collector group.
Network Proxy Address	The network proxy address of the cloud proxy.
Network Proxy Port	The network proxy port number of the cloud proxy.
Target	Displays the target location on which the cloud proxy is deployed.
Data Persistence	Displays the status of data persistence for the cloud proxy. Data persistence activates the cloud proxy to store data if the connection fails between the cloud proxy and VCF Operations. Activated. Cloud proxy will store data. Deactivated. Cloud proxy will not store data. To activate/deactivate data persistence for multiple cloud proxies, select the cloud proxies, click the horizontal ellipsis, and then select Activate Data Persistence or Deactivate Data Persistence . To activate/deactivate data persistence for a single cloud proxy, select the cloud proxy, click the vertical ellipsis, and then select Activate Data Persistence or Deactivate Data Persistence . Note: When the connection is restored, the cloud proxy sends the stored data to VCF Operations. The stored data is displayed before the real-time data is displayed.
Log Assist	Displays if log assist is activated or deactivated for the cloud proxy.
Time Estimation	Displays the estimated time duration for which the cloud proxy persists data. Cloud proxy can store data for a maximum duration of one hour. If there is a lack of space or if the connection fail lasts for more than an hour, the cloud proxy rotates the stored data by deleting the oldest stored data and replacing it with the most recently collected data. Note: The time estimate value is displayed only if data persistence is activated.
Filter	You can search the list of cloud proxies according to the following criteria: Name

Option	Description
	- IP - Status - Version - Collector Group - Network Proxy Address - Network Proxy Port - Target - Data Persistence - Cloud Proxy Id

- Optional: Click the vertical ellipsis.
 - Click **Activate/Deactivate Data Persistence** to activate or deactivate data persistence for the cloud proxy.
 - Click **Remove** to delete the cloud proxy.
- Click a **Cloud Proxy** name.
The **Cloud Proxy Details** open.

Table 935: Cloud Proxy Details

Option	Description
Rename	Click the Rename Cloud Proxy icon to update the cloud proxy name.
Proxy ID	ID of the cloud proxy.
IP Address	IP address of the cloud proxy.
OVA Version	The OVA file version used to install the cloud proxy.
Status	Status of the cloud proxy. For example, the Getting Online status is displayed for a few minutes when you add a cloud proxy. After the cloud proxy is connected to VCF Operations, the status changes to Online. If the VCF Operations is not connected, the Offline status is displayed.
Creation Date	Date of creation of the cloud proxy.
Last Heartbeat	Last time stamp when VCF Operations ran a Health Check for this cloud proxy. When you click a cloud proxy to view its details, VCF Operations sends a heartbeat to check if the cloud proxy is still reachable.
CPU	CPU usage.
Memory	Memory usage.
Number of Cloud Accounts	Displays the number of cloud accounts using the cloud proxy. Each cloud proxy might have one or more cloud accounts. You can also view the health and status of these cloud accounts from this details pane.

- If your cloud proxy is not collecting data, you can view the health of the cloud proxy. From the left menu, click **Inventory**, select the **VCF Operations Cloud Proxy Object** from the list, and then click **Show Detail**.
For more details, see [Inventory: List of Objects](#).

- After you locate the VCF Operations cloud proxy object, you can view the object details using the Summary tab. For more information, see [Summary Tab](#).
- Use the [Alerts](#) tab to monitor the health of the cloud proxy. If there are any issues, troubleshoot them using the [Metrics](#) tab.
If your cloud proxy is not working properly, an alert is displayed.

One or more VCF Operations services on a cloud proxy are down.

To clear this alert, perform the following steps:
 - Check the network connectivity and configuration for the cloud proxy.
 - Take the cloud proxy offline and then bring it back online.
 For more information, see [Cloud Proxy FAQ](#) and [Cloud Proxy Troubleshooting](#).
Note: It is recommended that you create a notification rule for this alert so that, quick remediation steps can be taken, if necessary.
- Optional: You can use the cloud proxy command line interface for other cloud proxy related actions. For more details, see [Using the Cloud Proxy Command-Line Interface](#).

Click the **Collector Groups** tab to view the cloud proxies that are part of the collector groups. For more information on collector groups, see [Configuring Collector Groups](#).

Monitoring the Health of Cloud Proxies from the Administration UI

After you configure your cloud proxy, you can view the status, health, and upgrade history of your cloud proxy in the VCF Operations administration interface.

- Log in to the VCF Operations administration interface at <https://primary-node-fqdn/admin>.
- Click **Cloud Proxies**.

Table 936: Cloud Proxies Page Options

Option	Description
IP Address	IP address of the cloud proxy.
Name	Name of the cloud proxy.
Network Proxy Configuration	Determines whether the network proxy setting is configured or not.
Health Status	Determines the health of the cloud proxy.
Upgrade Status	Determines whether the upgrade is complete, in progress, or failed.
Last Upgrade Time	Determines when was the last upgrade done.
Version	Version number of the cloud proxy.

Click the **Expand** icon to view the upgrade history.

Table 937: Upgrade History Options

Option	Description
ID	The conventional name used to identify the PAK file. It is usually the name of the PAK file and its version numbers joined together without extensions. For example, VMware Aria-Operations-Cloud-Proxy-<number>.

Option	Description
Type	The type of upgrade used for the cloud proxy. The cloud proxy can be upgraded either automatically or manually using the command-line interface. For more information, see Using the Cloud Proxy Command-Line Interface .
Start Time	Time stamp when the upgrade started.
End Time	Time stamp when the upgrade ended.
Upgrade Status	Determines whether the upgrade is complete, in progress, or failed. Note: For a detailed view of the upgrade status, click the row to open the Info pop-up. If the cloud proxy upgrade fails due to low disk space, the <code>FAIL_NO_SPACE_FOR_PAK_DOWNLOAD</code> message appears.
Version	Version number of the cloud proxy PAK file.

Deleting Cloud Proxies

You can delete a cloud proxy if there are issues with your cloud proxy. For example, you can delete the cloud proxy from the cloud proxy page in case of incorrect deployment of your cloud proxy, or if you need to deploy your cloud proxy in another cluster, or if you no longer require a cloud proxy. After you delete a cloud proxy, data collection stops.

1. Log in to VCF Operations.
2. From the left menu, click **Administration > Cloud Proxy**.
3. Click the vertical ellipsis in front of the cloud proxy you want to delete and then click **Delete**.
The confirmation wizard appears.
4. Select the **I understand the risk** check box and then click **OK**.
5. The cloud proxy is deleted.

Upgrading Cloud Proxy

Cloud Proxies are upgraded to a compatible cluster version automatically after the cluster upgrade. Expect a downtime of one or two cycles, as cloud proxies do not collect any data during this period. Data collection resumes after the upgrade is complete. The upgrade process runs concurrently for up to ten cloud proxies. If you have more than ten cloud proxies, the upgrade process will continue sequentially and the next cloud proxy upgrade will start as soon as any one of the ten running upgrades completes or fails the upgrade process. In case the automatic upgrade fails, upgrade your cloud proxy manually using the CLI.

Note: When your cloud proxies are upgraded, the cloud proxy internal certificates also get renewed. After each upgrade, the cloud proxies will have a new CA signed certificate with a five year validity period.

For more information on what data gets collected, see [VMware vSphere Solution in VCF Operations](#).

You can manually upgrade your cloud proxy [Using the Cloud Proxy Command-Line Interface](#).

Using the Cloud Proxy Command-Line Interface

Use the cloud proxy command-line interface to upgrade your cloud proxy, generate support bundles, and view the cloud proxy health status.

You can use SSH to access the cloud proxy instance and use its command-line Interface to run the following actions:

- Manually upgrade your cloud proxy in case the automatic download of the latest binary fails. When the automatic download fails, you see a notification on the VCF Operations user interface. To manually upgrade your cloud proxy instance to the latest version, see the following KB article [80590](#).

- Generate the support bundle.
- Gather the status of the cloud proxy's health and connectivity details.

Command Line	Description
<code>cprc-cli -h, --help</code>	Displays the help message and use of the command-line interface.
<code>cprc-cli -s, --status</code>	Prints the cloud proxy life-cycle status, configuration details, upgrade-related information, and more. It is useful to catch necessary information related to support and troubleshooting, or check the connection to VCF Operations, or check the product version number, and so on.
<code>cprc-cli -u PRODUCT_PAK, --upgrade PRODUCT_PAK</code>	The cloud proxy instance is enabled for an automated upgrade by default. But if the automated upgrade fails due to any exceptional issue, use this command line to upgrade your cloud proxy instance to the desired version.
<code>cprc-cli -hsb</code> <code>cprc-cli --generate-heavy-support-bundle</code> <code>cprc-cli -sb [{false,true}]</code> <code>cprc-cli --generate-support-bundle [{false,true}]</code>	Generates the cloud proxy support bundle which is a package of logs, configurations, and status files. The support bundles are necessary for product support and troubleshooting. Generated support bundles can be found at the <code>/storage/core/vmware-vrops-cprc/support</code> location. You can generate light or heavy support bundles. Heavy support bundles include <code>journal.ctl</code> logs. To generate heavy support bundles, the following commands must be set to true. <code>cprc-cli -sb [{true}]</code> <code>cprc-cli --generate-support-bundle [{true}]</code> With the true option, the heavy support bundle is generated with <code>journal.ctl</code> logs. With the false option, the light support bundle is generated without <code>journal.ctl</code> logs. If no option is specified here, the light support bundle will be generated by default. To exclude docker logs from the support bundles during generation, include the following options in the support bundle generation commands: <code>-ed</code> <code>--exclude-docker-support-bundle</code>
<code>cprc-cli -ssb</code> <code>cprc-cli --status-support-bundle</code>	Displays the support bundle statuses.
<code>cprc-cli -rsb SUPPORT_BUNDLE, --remove-support-bundle SUPPORT_BUNDLE</code>	Removes any specified support bundle. Although generated support bundle packages can be removed using the system-embedded commands, it is recommended to use this command for that action.
<code>cprc-cli -fm, --enable-fips-mode</code>	Enables FIPS mode for cloud proxy.
<code>cprc-cli -v -d all install network services connection</code>	Runs the troubleshoot process to check for all or any one of the following issues: • Installation: Checks if VM deployment and cloud proxy installation is successful. • Network: Checks IP configurations, DNS configurations, and Host (hostname) configurations. • Service: Checks whether the <code>vmware-casa.service</code>, <code>collector.service</code>, <code>haproxy.service</code>, <code>httpd-no-rth.service</code>, and <code>httpd-south.service</code> are active and loaded based on the <code>systemctl show</code> command.

Command Line	Description
	Connection: Checks the connection to VCF Operations cluster nodes using <code>curl</code>. For the proxy server cases, checks that connection via the proxy server using its specified configuration. Use the <code>-v</code> command to generate more verbose outputs. This command generates a log file that gets stored in <code>/var/log/suport/cprc.analysis.log</code> location.
<code>-cs, --certificates-status</code>	Displays the status of the imported certificates and if the certificates have been configured correctly or not.
<code>cprc-cli -rc, --replace-certificate</code>	Allows you to replace the self signed certificates for cloud proxy. Add the full path to the new certificate to the command. The function expects a certificate full chain containing the certificate and private key. For more information, see KB article 89583 .
<code>cprc-cli -cc, --connectivity-check</code>	Allows you to run the connectivity check. You can check the HTTPS connectivity to a specified url. In case the url is not specified, it checks the HTTPS connectivity to the default prerequisites For on-prem configurations, the default prerequisites are the cluster nodes.
<code>cprc-cli -cc -r, --response</code>	Displays the response from the server during the connectivity check.
<code>cprc-cli -cc -rt, --routing</code>	Displays the routing information during the connectivity check.
<code>cprc-cli -env, --ovfenv</code>	Displays the content of <code>ovfEnv.xml</code> file.
<code>cprc-cli -rec, --recovery</code>	Runs the CPRC recovery script.
<code>cprc-cli -rh, --run-on-hosts</code>	Runs a specific script on hosts according to the configurations provided.
<code>cprc-cli -gt, --generate-template</code>	Generates the <code>template.json</code> files for <code>-rh</code> options configurations.

Using APIs with VCF Operations Cloud Proxy

You can use a browser or an HTTP client program to send requests and receive responses.

REST Client Programs

Any client application that can send HTTPS requests is an appropriate tool for developing REST applications with the VCF Operations API. REST client plug-ins are available for most browsers and many IDEs. The following open-source programs are commonly used:

- cURL. <http://curl.haxx.se>
- Postman application. <http://www.getpostman.com>

In addition, VMware provides language-specific client bindings for the VCF Operations API.

About the Schema Reference

The VCF Operations REST API documentation includes reference material for all elements, types, queries, and operations in the VCF Operations API. It also includes the schema definition files.

Swagger based API documentation is available with the product, with the capability of making REST API calls right from the landing page.

To access the API documentation, use the URL of your VCF Operations cloud proxy. For example, if the URL of your cloud proxy is <https://cloudproxy.vmware.com>, the API reference is available from: <https://cloudproxy.vmware.com/suite-api/doc/swagger-ui.html>

Note: You must use your cloud proxy IP or FDQN in the url to access the API documentation.

Language-specific client bindings are available from:

<https://cloudproxy.vmware.com/suite-api/>

Authorize VCF Operations API

To perform API calls, you must authorize your VCF Operations cloud proxy. You can perform basic authorization and enter the **Username**, **Password**, and the **Auth Source**, or perform token based authorization and enter the **Ops Token (apikey)** value, and then click **Authorize**.

About the VCF Operations API Examples

All examples include HTTP requests and responses. These examples show the workflow and content associated with operations such as creating and querying for information about objects in your monitored environment.

Example request bodies are in JSON. Request headers required by the VCF Operations API are included in example requests that are not fragments of a larger example.

Most example responses show only those elements and attributes that are relevant to the operation being discussed. Ellipses (...) indicate omitted content within response bodies.

Cloud Proxy FAQ

This topic covers some frequently asked questions about VCF Operations cloud proxy.

Configuration

- What are the prerequisites for setting up a cloud proxy account?
Prerequisites are given in the topic, [Configuring Cloud Proxies in VCF Operations](#).
- What does one-way connection mean?
Outbound connections are initiated from cloud proxy to VCF Operations, over `https/443`. Cloud proxy can also facilitate vCenter actions.
- Which ports should be opened?
The most up-to-date technical information about ports can be found on [Ports and Protocol](#).
- Which ports should be allowed for incoming traffic to cloud proxy?
Allow port 443 https protocol for push model adapters like application monitoring or Suite-API on cloud proxy. Allow ports 4505, 4506, and 8443 via TCP protocol for application monitoring. Allow the VRRP protocol for intercommunication between cloud proxies in a application monitoring high availability activated collector group.
- How do I edit environment settings for cloud proxy?
You can edit vApp options. For more information, see [Edit OVF Details for a Virtual Machine](#).
- How are certificates managed?
Certificates are managed by cloud proxies. But for any additional proxy servers with SSL communication, you need to provide certificate(s).
- What credential is used to login to cloud proxy?
You can login as the "root" user. You are expected to set a new password on the first login to cloud proxy VM. SSH access is disabled by default, so the first login must be done via the vCenter console. You can run the following command to start SSH service:

```
systemctl start sshd
```

```
systemctl enable sshd
```

To reset password, see the VMware KB Article, [2001476](#).

8. Will I be notified if the connection between cloud proxy and VCF Operations breaks down?

You can configure alerts/notifications on the *VCF Operations cloud proxy* object. For more information, see [Monitoring the Health of Cloud Proxies](#).

VCF Operations automatically generates notifications for the following scenarios:

- Cloud proxy is not reachable.
 - Cloud proxy is nearing sizing limits.
9. How do I change account for cloud proxy?
You can edit vApp options. For more information, see [Edit OVF Details for a Virtual Machine](#).
 10. How can I check the status of cloud proxy?
For more information, see [Monitoring the Health of Cloud Proxies](#).
 11. Should I use Remote Collector or cloud proxy for monitoring?

VMware recommends that you use cloud proxy to take advantage of the latest enhancements. Also, application monitoring, HA of collector groups, and data persistence are only supported through cloud proxy.

Sizing

1. How should I size the cloud proxy?
For information on sizing, see the VMware KB article [85832](#).

Upgrade

1. How do I upgrade cloud proxy?
Cloud proxy is upgraded automatically. In case the upgrade fails, see the VMware KB article [80590](#).

High Availability

1. Is high availability supported?
Cloud proxy supports high availability. You can add multiple cloud proxies to a collector group. If the collecting cloud proxy fails or gets disconnected, collection can be picked up by another proxy in the group.
Note: Since the failover is initiated after a period of 10 minutes, few collection cycles are lost.

To troubleshoot cloud proxy issues, see [Cloud Proxy Troubleshooting](#).

Cloud Proxy Troubleshooting

Cloud proxy troubleshooting steps are provided to help you easily resolve issues that you may come across in VCF Operations.

Before you proceed with troubleshooting, see the [Cloud Proxy FAQ](#).

Installation and/or First Boot Failure

To verify the issue, check if `/var/log/firstboot` contains a file named "Succeeded".

If not, the following problems could result in VCF Operations installation and/or first boot failure:

1. Unique Registration Key used while deploying Cloud Proxy is invalid. To verify, check the cloud proxy console.
Solution: Redeploy cloud proxy.

Cloud Proxy VM is running, but the status is Offline in VCF Operations.

Name	IP	Status	Version	Accounts	Network Proxy Address	Network Proxy Port
CP_TG	10.192.198.5	Offline	8.6.0.51997631	2 accounts	-	-

To verify the connection, use the following commands: (For the complete list of commands, please see [Using the Cloud Proxy Command-Line Interface](#).)

```
# Overall status of cloud proxy: cprc-cli -s
```

```
# Ping itself:
ip addr
ping <address>
```

```
# Ping gateway:
ip route
ping <gateway>
```

```
# Verify the connection outside the cloud proxy.
ping 8.8.8.8
```

Note: If you are using a network proxy, use the `/opt/vmware/share/vami/vami_config_net option#5` command to ensure you have the correct configuration for the testings.

The following problems could result in VCF Operations displaying the status of cloud proxy as offline.

1. Incorrect network proxy information in cloud proxy configuration.

To verify the connection via a network proxy, use the following:

```
curl -vvv --proxy http(s)://proxy_user:proxy_pass@proxy_ip:proxy_port -H 'Accept: application/json' -H 'Content-Type: application/json' -X GET https://<gateway url>/casa/security/ping (gateway url example - 10238.gw.dev.vrops-ops.com)
```

To ignore SSL validation for a proxy server, use `curl --proxy-insecure`. With SSL validation the customer can provide Proxy Server certificate during cloud proxy deployment or re-configuration so that provided certificate from customer can be used to check the connection with curl with SSL certificate validation.

Solution:

1. SSH to the cloud proxy VM and set the `connectretry` to 0 in `/storage/db/vmware-vrops-cprc/configuration/cprc.configuration` to ensure that the cloud proxy retries to connect.
2. Shutdown the cloud proxy VM.
3. Update the network proxy configurations from the vCenter VM options using the vApp options.
4. Boot the cloud proxy VM.

2. Required ports are not open.

To verify:

```
openssl s_client -showcerts -connect {address}:443
```

```
curl -v telnet://{address}:443
```

```
# Or, change the address to the machine you want to check:
```



```
python -c "import socket; print(socket.socket(socket.AF_INET, socket.SOCK_STREAM).connect_ex(('127.0.0.1', 443)))"
```

If you get a !=0 response, the server is not listening to the port.

Solution:

- 1. SSH to the cloud proxy VM and set connectretry to 0 in /storage/db/vmware-vrops-cprc/configuration/cprc.configuration to ensure that the cloud proxy retries to connect.
- 2. Provide port access as mentioned in the prerequisite section of [Configuring Cloud Proxies in VCF Operations](#)
- 3. Boot the cloud proxy VM.

3. Invalid certificate.

To verify:

```
openssl s_client -showcerts -connect {address}:443
```

Solution:

- 1. SSH to the cloud proxy VM and set connectretry to 0 in /storage/db/vmware-vrops-cprc/configuration/cprc.configuration to ensure that the cloud proxy retries to connect.
- 2. Follow the steps mentioned in VMware KB Article, [83698](#).
- 4. The logs folder /storage/log is running out of partition space.
Solution: Remove log files to ensure that enough space is available. Note that this is an exceptional case. In normal conditions, log files are auto archived.

5. One or more of the following services are down: httpd-north.service , haproxy.service and collector.service .

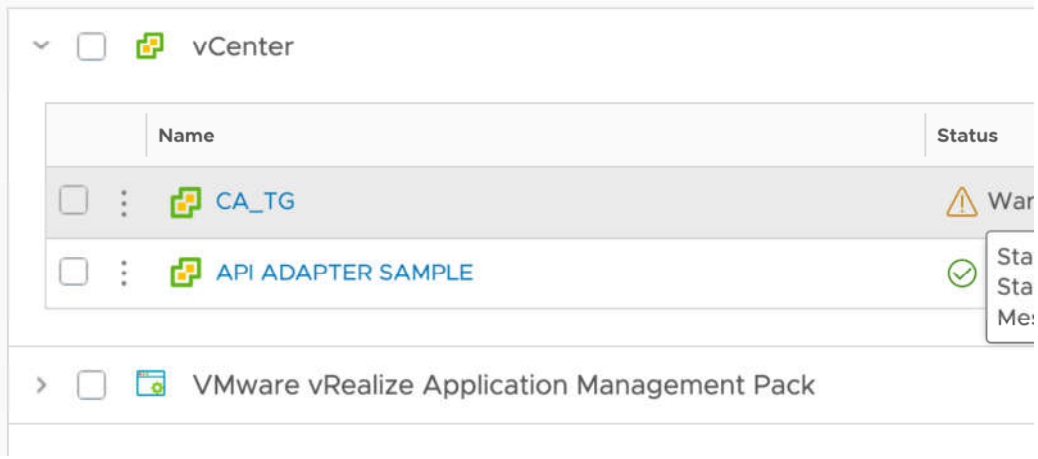
Solution:

- Check service status by running the following command: systemctl status <service name>.
- To start service, use the following command: systemctl start <service name>.

6. Unique Registration Key expired.

Solution: Redeploy cloud proxy with new Unique Registration Key.

Cloud proxy is online, and state of Cloud Account is Collecting, but the status is Object Down.



The following problem could result in VCF Operations displaying the state of Cloud Account as Collecting , while the status is Object Down .

- 1. Incorrect account credentials.
Solution: Check and update the credentials used while setting up the cloud account.

Cloud proxy status is stuck in Going Online.



It can take up to 20 mins on first reboot, for the cloud proxy to be registered and come online. Wait for the specified time to see if cloud proxy comes online. If it still does not come online, one or more of the following services are down: httpd-north.service , haproxy.service , and collector.service .

Solution:

- 1. Check service status by running the following command: systemctl status <service name>
- 2. To start service, use the following command: systemctl start <service name>.

Cloud proxy does not upgrade automatically, after the upgrade of VCF Operations

There could be a few possible reasons why cloud proxy does not upgrade automatically after an upgrade of VCF Operations.

1. High network latency leading to PAK download failure. Latency of >500ms is not supported.
Solution: See the VMware KB article [80590](#) on how to manually upgrade cloud proxy via CLI.
2. Upgrade status is stuck at `Running` since the previous upgrade had failed.
Solution: Follow the steps given below to change the upgrade status.
 1. Stop the casa service: `systemctl stop vmware-casa.service.`
 2. Change the upgrade status from `RUNNING` to `NONE` in the following files:


```
./storage/db/vmware-vrops-cprc/status/cprc.upgrade.status
./storage/db/vmware-vrops-cprc/status/cprc.pak.status
```
 3. See the VMware KB article [80590](#) and run the manual upgrade.

Cloud proxy gets disconnected at regular intervals

There could be a few possible reasons why cloud proxy gets disconnected at regular intervals.

1. Check the network connectivity and latency.
2. Check if the cloud proxy VM can reach the DNS and use NSlookup to validate the DNS connectivity.

Configuring Collector Groups

VCF Operations uses collectors like cloud proxies to manage cloud account processes such as gathering metrics from objects. You can select a collector or a collector group when configuring a cloud account.

If there are cloud proxies in your environment, you can create a collector group, and add the cloud proxies to the group. When you assign an account to a collector group, that account can use any collector in the group. Use collector groups to achieve resiliency in cases where the collector experiences network interruption or becomes unavailable. If this occurs, and the collector is part of a group, the total workload is redistributed among all the collectors in the group, reducing the workload on each collector. For more information, see [Configuring Cloud Proxies in VCF Operations](#).

You can add, edit, or remove collector groups in VCF Operations, and re-balance your cloud accounts.

Re-balancing a Cloud Account

Re-balancing of your cloud accounts is not intended to provide equally distributed cloud accounts across each collector in the collector group. The re-balancing action considers the number of resources that each cloud account collects to determine the re-balancing placement. The re-balancing happens at the cloud account, which can result in several small cloud accounts on a single collector, and a single huge cloud account on another collector, in your VCF Operations instance.

Re-balancing your collector groups can add a significant load on the entire cluster. Moving cloud accounts from one collector to another collector requires that VCF Operations stops the cloud account and all its resources on the source collector, then starts them on the target collector.

If a collector fails to respond or loses connectivity to the cluster, VCF Operations starts automated re-balancing in the collector group. All other user-initiated manual operations on the collector, such as stopping or restarting the collector manually, do not result in automated re-balancing.

If one of the collectors fails to respond, or if it loses network connectivity, VCF Operations performs automated re-balancing. In cases of automated re-balancing, to properly re-balance the collector group, you must have spare capacity on the collectors in the collector group.

Where You Manage Collector Groups

From the left menu, click **Administration > Cloud Proxies**, and then click the **Collector Groups** tab.

Table 938: Collector Group Summary Grid

Options	Description
Collector Group toolbar	To manage collector groups, use the toolbar icons. • Add: Add a collector group • Click the Vertical Ellipses to perform any one of the following actions: – Edit: Modify the collector group by adding or removing cloud proxies. You can also edit a collector group to activate or deactivate application monitoring high availability. – Delete: Remove the selected collector group. – Re-balance collector group: Re-balance one collector group at a time. If you have permission to manage clusters, you can re-balance the workload across the collectors and the cloud proxies in the collector group. The re-balance action moves objects from one collector group to another to re-balance the number of objects on each collector in the collector group. If a disk re-balance is already in progress, the collector re-balance does not run – Retry collector group configuration: In case the application monitoring high availability activated collector group configuration fails, you can retry the configuration. – The actions are disabled when the application monitoring high availability activated collector group configuration is in progress. After the configuration is complete you can perform the actions.
Name	The name given to the collector group when the collector group is created or updated.
Collector Group ID	The collector group ID created when the collector group is created.
Description	Description given to the collector group when the collector group is created or updated.
Application Monitoring HA	Displays the application monitoring high availability status of the collector group. • Activated: The collector group can be used for application monitoring. • Deactivated: The collector group cannot be used for application monitoring. Note: For default collector groups, high availability for application monitoring is deactivated by default.
HA Status	Displays the application monitoring high availability of the collector group. • OK: The collector group high availability configuration is successful. • In Progress: The collector group high availability configuration is in progress. This status is displayed when adding, deleting, updating, or deactivating the collector group. • Empty: If high availability is not configured, no status is displayed. Note: If the high availability status of the cloud proxy fails, the HA status of the failed cloud proxy is displayed. In case there are multiple cloud proxies in a collector group with failed HA status, the first failed message is displayed in the Collectors Group page.
Virtual IP	Virtual IP address of the collector group. Note: Pick the Virtual IP address that any of the cloud proxies in a collector group can own and receive traffic from. The Virtual IP address must be in the same subnet as the physical address of the cloud proxies. After you configure the Virtual IP address, ping it from a different network to ensure that it is routable.
Filters	Search the list of collector groups according to the following criteria: • Name • Description • Collector Name • Collector IP Address • Application Monitoring HA

Options	Description
	- HA Status - Group Virtual IP - Collector Group Id

Click a collector group to view the collector group details.

Table 939: Collector Group Details Grid

Detail Grid Options	Description
Collector Group Name	Displays the name of the collector group. Click Edit to modify the collector group details. Click View Description to view the collector group description.
Application Monitoring High Availability Status	Displays the status of the application monitoring high availability.
Virtual IP	Displays the virtual IP added during the collector group creation.
Name	Name given to the cloud proxy when the cloud proxy was created. Click the name to view the cloud proxy details. For more information on cloud proxies, see Monitoring the Health of Cloud Proxies .
IP Address	IP address of the cloud proxy
Status	Status of the cloud proxy: online or offline.
HA Status	Displays the cloud proxy high availability status. - Ok: The cloud proxy application monitoring high availability configuration is successful. - Activation In Progress: Displayed when the cloud proxy is added to a application monitoring HA activated collector group or when the creation of a new application monitoring HA activated collector group is in progress. - Deactivation In Progress: Displayed when the deactivation of a application monitoring HA activated collector group is in progress. - Removal In Progress: Displayed when the removal of a cloud proxy from a application monitoring HA activated collector group is in progress. - Group Delete In Progress: Displayed when the deletion of the entire application monitoring HA activated collector group is in progress. - Empty: If the cloud proxy is not part of a collector group with high availability, no status is displayed. - If the cloud proxy high availability configuration fails, the failure reason is displayed. For example, "Keepalived is failed" or "Apache south is failed".
Type	Displays the cloud proxy type.
Number of Objects	Displays the number of objects collected by the cloud proxy.
Number of cloud accounts	Displays the number of cloud accounts using the cloud proxy.

Adding a Collector Group

Create a new collector group from the available cloud proxies in your VCF Operations environment. A cloud proxy can only be added to only one collector group at a time.

Where You Add New Collector Groups?

From the left menu, click **Administration > Cloud Proxies**, and then click the **Collector Groups** tab. Click **Add**.

Table 940: Add New Collector Group

Option	Description
Name	Name of the collector group.
Description	Description of the collector group.
Application Monitoring High Availability	Activate this option to use high availability in application monitoring using collector groups.
Virtual IP	If you activate high availability for application monitoring, enter the virtual IP address. Note: Pick the Virtual IP address that any of the cloud proxies in a collector group can own and receive traffic from. The Virtual IP address must be in the same subnet as the physical address of the cloud proxies. After you configure the Virtual IP address, ping it from a different network to ensure that it is routable.
Activate for load balancing	Select this option to activate load balancing for HA activated collector group for open source Telegraf agents.
Collectors added to this collector group	Displays the collectors that are assigned to this collector group. To assign collectors, double click a collector, or drag and drop it from the Select Collectors section to add it to the collector group. Note: You cannot assign a collector that is part of another collector group. To reassign a collector to another collector group, you must remove it from the existing collector group and then reassign it. To remove a collector from the collector group, click the Remove Collector icon.
Select Collectors	Displays a list of cloud proxies in your VCF Operations environment together with their collector name, IP address, type, collector group, and status. Collectors that are assigned to this collector group appear with a check mark before the collector name. Note: Cloud proxies with log forwarding cannot be added to collector groups.
Filters	You can search the list of collectors according to the following criteria: - Collector Name - IP address - Collector Group Name - Status

Editing Collector Groups

Edit a collector group in VCF Operations by adding cloud proxies or removing the collectors that you no longer require from an existing group.

Where You Edit a Collector Group?

From the left menu, click **Administration > Cloud Proxies**, and then click the **Collector Groups** tab. Click the vertical ellipsis and then select **Edit**. The Edit Collector Group page opens. You can update the collector group details and then click **Save**.

Table 941: Edit Collector Group Options

Option	Description
Name	Name given to the collector group when the collector group is created.
Description	Description given to the collector group when the collector group is created.
Application Monitoring High Availability	Activate this option to use high availability in application monitoring using collector groups. If high availability in application monitoring is activated, you can deactivate it.
Virtual IP	If you activate high availability for application monitoring, enter the virtual IP. Note: You cannot edit the virtual IP address for a collector group with high availability for application monitoring activated.
Activate for load balancing	Select this option to activate load balancing for HA activated collector group for open source Telegraf agents.
Collectors added to this collector group	Displays the collectors that are assigned to this collector group. To assign collectors, double click a collector, or drag and drop it from the Select Collector section to add it to the collector group. Note: You cannot assign a collector that is part of another collector group. To reassign a collector to another collector group, you must remove it from the existing collector group and then reassign it. To remove a collector from the collector group, click the Remove Collector icon.
Select Collectors	Displays a list of the available cloud proxies in your VCF Operations environment together with their name, IP address, type, collector group, and status. Collectors that are assigned to this collector group appear with a check mark before the collector name. Note: Cloud proxies with log forwarding cannot be added to collector groups.
All Filters	Filter the list of collectors according to the following criteria: - Collector Name - Collector Group Name - IP Address - Status

Managing Cloud Proxy Support Bundles from the Administration UI

VCF Operations support bundles contain log and configuration files that help troubleshoot a VCF Operations cloud proxy issue.

Use the Support Bundles (Cloud Proxy) page to create the support bundle on cloud proxy.

Log in to the VCF Operations administration interface at <https://master-node-name-or-ip-address/admin> and then click **Support > Support Bundles (Cloud Proxy)**.

Support Bundle (Cloud Proxy) Options

The options include toolbar and data grid options.

Use the toolbar options to add, download, or remove items.

Table 942: Support Bundle (Cloud Proxy) Toolbar Options

Option	Description
Add	Open a dialog box that guides you through the process of creating a support bundle on cloud proxy. Select the cloud proxy and then click OK to create a support bundle on the selected cloud proxy. The support bundle is created under the following directory / storage/core/vmware-vrops-cprc/support.
Delete	Remove the selected support bundle.
Download	Download the support bundle in ZIP format.
Reload	Refresh the list of support bundles.

Use the data grid options to view item details.

Table 943: Support Bundle (Cloud Proxy) Data Grid Options

Option	Description
Bundle	System-generated identifier for the support bundle.
Cloud Proxy Name	The name of the cloud proxy on which the support bundle is created.
Date and Time Created	Time when support bundle creation began.
Status	Progress of support bundle creation.
File Size	The size of the support bundles.

Note: Generation and download of support bundles through the Support Bundles (Cloud Proxy) page works only if the cloud proxy is connected to the cluster.

If there is a disconnect between the cloud proxy and VCF Operations, you can generate a support bundle on cloud proxy manually.

Open an SSH connection with the cloud proxy appliance and run the following command:

- For 8.3 and 8.4 version: `$> cprc-cli -sb`
- For 8.5 and later versions: `$> cprc-cli -sb IS_HEAVY`, where `IS_HEAVY` should be specified as `true` or `false`

With `cprc-cli -sb true`, the support bundle is generated with `journalctl` logs. With `cprc-cli -sb false`, the support bundle is generated without `journalctl` logs. The support bundle is created in the following path: `directory /storage/db/vmware-vrops-cprc/support`.

For adding some extra files into the support bundle, you need to modify configuration file available here: `/storage/db/vmware-vrops-cprc/configuration/cprc.support.bundle.configuration`. Add the path of needed file into `cprc.support.bundle.configuration` in the `files:` section. Then generate the support bundle with the `cprc-cli -sb` or `cprc-cli -sb false` command, based on the version you are on.

For example if you want to add the `/var/log/firstboot/vcopssuitevm.log` file into the support bundle, add this path into the `files:` section of `/storage/db/vmware-vrops-cprc/configuration/cprc.support.bundle.configuration`. Then generate the support bundle with either the `cprc-cli -sb` or the `cprc-cli -sb false` command, based on the version you are on.

Once done, you can download or delete these support bundles from the Support Bundles (Cloud Proxy) page.

Data Protection and Recovery of the VCF Fleet

You can use VMware Live Site Recovery and VMware vSphere Replication to provide support for planned migration or disaster recovery of the VCF fleet and its managed VCF Instances.

For more information about data protection and recovery of the VCF fleet, see the [Site Protection and Recovery for VMware Cloud Foundation](#) validated solution.

Shutdown and Startup of VMware Cloud Foundation

Shutting down VMware Cloud Foundation, for example, during hardware or power maintenance, and then starting it up must be done in a way that prevents data loss or appliance malfunction, and supports collection of troubleshooting data.

You follow a strict order and steps for shutdown and startup of the VMware Cloud Foundation management components.

Shutting Down VMware Cloud Foundation

To avoid data loss and maintain the SDDC components operational, you follow a specific order when shutting down the management virtual machines in VCF.

- Verify that you have complete backups of all management components.
- Verify that the management virtual machines are not running on snapshots.
- If a vSphere Storage APIs for Data Protection (VADP) based backup solution is running on the management clusters, verify that the solution is properly shut down by following the vendor guidance.
- To reduce the startup time before you shut down the management virtual machines, migrate the vCenter instance for the management domain to the first VMware ESX host in the default management cluster in the management domain.

You shut down the customer workloads and the management components for the workload domains before you shut down the components for the management domain. If you have multiple VCF instances you start with the instances that does not run VCF Operations and VCF Automation. The VCF instance running VCF Operations must be the last instance to shut down.

If the NSX Manager cluster and NSX Edge cluster are shared with other workload domains, shut down the NSX Manager and NSX Edge clusters as part of the shutdown of the first workload domain.

Shut Down a Workload Domain

You shut down the components of a workload domain that runs virtualized workloads in VCF in a specific order to keep components operational by maintaining the necessary infrastructure, networking, and management services as long as possible before shutdown.

You shut down the management components for the workload domains before you shut down the components for the management domain.

If the NSX Manager cluster and NSX Edge cluster are shared with other workload domains, follow this general order:

1. Shut down the consumer workloads in all workload domains that share the NSX Manager instance. Otherwise, all NSX networking services in the consumer workloads will be interrupted when you shut down NSX Manager.
2. Shut down the workload domain that runs the shared NSX Edge nodes.
3. Shut down the other workload domains.

Shutdown Order for a Workload Domain

Table 944: Shutdown Order for a Workload Domain

Shutdown Order	SDDC Component
1	Virtualized customer workloads
2	VMware Live Recovery for the workload domain
4	NSX Edge nodes for the workload domain
5	NSX Manager nodes for the workload domain
7	ESX hosts in the workload domain
8	vCenter Server for the workload domain

Related Links

[Shut Down the Management Domain on page 2999](#)

You shut down the components of the management domain in VCF in a specific order to keep components operational by maintaining the necessary infrastructure, networking, and management services as long as possible before shutdown.

Shut Down the NSX Edge Nodes in the Workload Domain

You begin shutting down the NSX infrastructure in a workload domain in VCF by shutting down the NSX Edge nodes that provide north-south traffic connectivity between the physical data center networks and the NSX SDN networks

1. Identify the instance that runs nodes.
2. Log in to for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **VMs and Templates** inventory, locate the appliance.
4. Right-click the appliance and select **Power > Shut down Guest OS**.
5. Repeat the steps for the remaining appliances in the domain.

Shut Down the NSX Manager Nodes in a Workload Domain

You continue shutting down the NSX infrastructure for the workload domain by shutting down the NSX Manager VM or cluster by using the vSphere Client.

1. Identify the vCenter instance that runs .
2. Log in to for the management or workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **VMs and Templates** inventory, locate an appliance.
4. Right-click the appliance and select **Power > Shut down Guest OS**.
5. In the confirmation dialog box, click **Yes**.
This operation takes several minutes to complete.
6. Repeat the steps for the remaining NSX Manager appliances.

Shut Down the ESX Hosts in a Workload Domain

You follow specific steps according to the principal storage of the clusters in the workload domain.

Save the Credentials for the ESX Hosts and vCenter for the Workload Domain

Before you shut down the workload domain, get the credentials for the workload domain hosts and vCenter and save them. You might need these credentials to shut down the ESX hosts and then to start them and vCenter back up.

Shut Down vSAN and the ESX Hosts in a Workload Domain

You use the vSAN shutdown cluster wizard in the vSphere Client to shut down gracefully the vSAN clusters in a workload domain. The wizard shuts down the vSAN storage and the ESX hosts added to the cluster.

You perform this operation on all vSAN clusters in all workload domains.

1. Identify the vCenter instance that runs the vCenter VM for this workload domain.

CAUTION:

If the vCenter VM is running on the infrastructure that it manages, note down the cluster in which it is running. This cluster must be stopped last.

2. Log in to for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. For a vSAN cluster, verify the vSAN health and resynchronization status.
 - a) Select the cluster and click the **Monitor** tab.
 - b) In the left pane, navigate to **vSAN > Skyline health** and verify the status of each vSAN health check category.
 - c) In the left pane, under **vSAN > Resyncing objects**, verify that all synchronization tasks are complete.
4. If any member host is in lockdown mode, add the host's **root** account to the Exception Users list.
 - a) Select the host in the inventory and click the **Configure** tab.
 - b) In the left pane, select **System > Security Profile**.
 - c) In the **Lockdown Mode** pane, click the **Edit** button.
 - d) On the **Exception Users** page, enter `root` and click **Add User**.
 - e) Click **OK**.
5. Shut down the vSAN cluster.
 - a) In the inventory, right-click the vSAN cluster and select **vSAN > Shutdown cluster**.
 - b) In the **Shutdown Cluster** wizard, verify that all pre-checks are green and click **Next**.
 - c) Review the **vCenter Server notice** and click **Next**.
 - d) If vCenter is running on the selected cluster, note the orchestration host details.
Connection to vCenter is lost because the vSAN shutdown cluster wizard shuts it down. The shutdown operation is complete after all ESXi hosts are stopped.
 - e) Enter a reason for performing the shutdown, and click **Shutdown**.
6. Repeat Step 3 to Step 5 for other vSAN clusters in the domain.

Shut Down ESX Hosts with NFS or Fibre Channel Storage in a Workload Domain

You use the vSphere Client to shut down gracefully the clusters with non-vSAN in a workload domain.

1. Identify the vCenter instance that runs the vCenter appliance for this workload domain.

CAUTION:

This cluster must be shut down last.

2. Log in to for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **Hosts and Clusters** inventory, for a cluster that is not running the vCenter appliance for the workload domain, put all hosts in the cluster in maintenance mode.
These operations take several minutes to complete.
4. If a cluster is running the vCenter appliance, note down the host on which it is running and put all hosts in the cluster but this one in maintenance mode.
You need the details about this host during the startup process too.
5. Shut down the hosts in maintenance mode
 - a) Select the cluster and click the **Hosts** tab.
 - b) Select all hosts that are in maintenance mode, right-click **Power > Shut Down**.
This operation takes several minutes to complete

6. Repeat Step 3 to Step 4 for all clusters with non-vSAN storage in the domain.
7. If the vCenter appliance is running on one of the ESX hosts in the cluster, shut down the vCenter appliance and the host from the VMware Host Client.
 - a) Log in to the ESX host at https://<esx_fqdn> as a user with the **Administrator** role.
 - b) In the left plane, click **Virtual Machines**.
 - c) Locate the vCenter appliance for the workload domain.
 - d) Right-click the appliance and select **Power > Shut down Guest OS**.
 - e) In the confirmation dialog box, click **Yes**.

This operation takes several minutes to complete.
 - f) In the left plane, right-click on **Host > Shut down**.
 - g) Read the warning and click **Shut down**.

You lose connectivity to the ESX host. The shut down operation takes several minutes to complete.
8. Shut down the storage for the workload domain if there are no other workloads that use it.

Follow vendor 's documentation to check if there are active clients and how to shut it down in a safe manner.

Shut Down the vCenter Appliance for a Workload Domain

If the vCenter appliance is running on a different infrastructure than the one it manages, to shut down the vCenter appliance for a workload domain in VCF, you use the vSphere Client.

Verify that all ESX hosts in all clusters are stopped and are disconnected.

If the vCenter appliance is running on the infrastructure that it manages, you stop it as part of the shutdown operation for ESX hosts.

1. Identify the instance that runs the appliance for this workload domain.
2. Log in to for the management or workload domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
3. Shut down vCenter Server.
 - a) Locate the vCenter Server virtual machine for the workload domain.
 - b) Right-click the virtual machine and select **Power > Shut down Guest OS**.
 - c) In the confirmation dialog box, click **Yes**.

This operation takes several minutes to complete.

Shut Down the Management Domain

You shut down the components of the management domain in VCF in a specific order to keep components operational by maintaining the necessary infrastructure, networking, and management services as long as possible before shutdown.

After you shut down the components in all workload domains, you begin shutting down the management domain.

Shutdown Order for the Management Domain

Note:

If your VCF Instance is deployed without workload domains, shut down any customer workloads or additional virtual machines in the management domain before you proceed with the shutdown order of the management components.

You shut down VMware Live Site Recovery after you shut down the management components that can be failed over between the VCF Instances. You also shut VMware Live Site Recovery down as late as possible to have the management

virtual machines protected as long as possible if a disaster event occurs. The virtual machines in the paired VCF Instance become unprotected after you shut down VMware Live Site Recovery in the current VCF Instance.

Table 945: Shutdown Order for the Management Domain

Shutdown Order	SDDC Component
1	VCF Automation
2	VCF Operations for Networks
3	VCF Operations collector
4	VCF Operations for logs
5	VCF Identity Broker
6	VCF Operations fleet management
7	VCF Operations
8	VMware Live Site Recovery for the management domain
9	NSX Edge nodes
10	NSX Manager
11	SDDC Manager
12	Shut Down vSAN and the ESX Hosts in the Management Domain Shut Down ESX Hosts with NFS or Fibre Channel Storage in the Management Domain

Save the Credentials for the ESX Hosts and vCenter for the Management Domain

Before you shut down the management domain, get the credentials for the management domain hosts and vCenter and save them. You need these credentials to shut down the ESX hosts and then to start them and vCenter back up.

Shutting Down a Management Domain with Infrastructure Services VMs

If the management domain contains virtual machines that are running infrastructure services like Active Directory, NTP, DNS and DHCP servers, stop these virtual machines last, that is, after the vCenter appliance.

- For a vSAN cluster, select **I confirm all VMs below are infrastructure VMs** in the vSAN shutdown wizard.
- For a non-vSAN cluster, migrate all such virtual machines to the first host with vCenter and shut them down after the vCenter appliance is shut down.

Related Links

[Shut Down a Workload Domain on page 2995](#)

You shut down the components of a workload domain that runs virtualized workloads in VCF in a specific order to keep components operational by maintaining the necessary infrastructure, networking, and management services as long as possible before shutdown.

Shut Down VCF Automation

To gracefully power off the VCF Automation, use the VCF Operations user interface.

1. Log in to at the https://<vcf_operations_fqdn> URL as a user with the **Administrator** role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. In the **Components** tab, locate the component that you are starting or stopping and, in the **Action** column, click **Manage**.
4. On the component management page, click the ellipsis icon and select **Power off**.
5. Verify the information and click **Submit**.

This operation takes several minutes to complete.

Shut Down VCF Operations for Networks

To gracefully power off the VCF Operations for Networks, use the VCF Operations user interface.

1. Log in to at the https://<vcf_operations_fqdn> URL as a user with the **Administrator** role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. In the **Components** tab, locate the component that you are starting or stopping and, in the **Action** column, click **Manage**.
4. On the component management page, click the ellipsis icon and select **Power off**.
5. Verify the information and click **Submit**.

This operation takes several minutes to complete.

Shut Down the VCF Operations Collector Appliances

Shut down the VCF Operations collector nodes in the management domain of the VCF Instance by using the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a VCF Operations collector appliance.
3. Right-click the appliance and select **Power > Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.
5. Repeat the steps for the remaining VCF Operations collector appliances.

Shut Down VCF Operations for Logs

Shut down the VCF Operations for logs appliances in the management domain by using the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a VCF Operations for logs appliance.
3. Right-click the VCF Operations for logs appliance and select **Power > Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.
5. Repeat the steps for the remaining VCF Operations for logs appliances.

Shut Down VCF Identity Broker

To gracefully power off the VCF Identity Broker, use the VCF Operations user interface.

CAUTION:

After you stop VCF Identity Broker, you cannot use accounts from external identity providers. All subsequential operations must be done by using local accounts.

1. Log in to at the https://<vcf_operations_fqdn> URL as a user with the **Administrator** role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. In the **Components** tab, locate the component that you are starting or stopping and, in the **Action** column, click **Manage**.
4. On the component management page, click the ellipsis icon and select **Power off**.
5. Verify the information and click **Submit**.

This operation takes several minutes to complete.

Shut Down the VCF Operations Fleet Management Appliance

Shut down the VCF Operations fleet management appliance in the management domain of VCF Instance hosting the VCF from the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate the VCF Operations fleet management appliance.
3. Right-click the VCF Operations fleet management appliance and select **Power > Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.

Shut Down VCF Operations

You shut down VCF Operations by first taking the cluster offline and then shutting down the appliances of the VCF Operations cluster.

1. Log in to the administration UI at the https://<vcf_operations_fqdn>/admin URL as the **admin** local user.
2. Take the cluster offline.
 - a) On the **System status** page, click **Take cluster offline**.
 - b) In the **Take cluster offline** dialog box, provide the reason for the shutdown and click **OK**.
3. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
4. Shut down VCF Operations.

Important:

Shut down the Guest OS of the VCF Operations appliances by following the order described in [Broadcom KB 341964](#).

- a) In the **VMs and Templates** inventory, locate a VCF Operations appliance.
- b) Right-click the appliance and select **Power > Shut down Guest OS**.
- c) In the confirmation dialog box, click **Yes**.

This operations takes several minutes to complete.

- d) Repeat the steps for the remaining VCF Operations appliances by following the order described in [Broadcom KB 341964](#).

Shut Down the NSX Edge Nodes

You begin shutting down the NSX infrastructure in the management domain VMware Cloud Foundation by shutting down the NSX Edge nodes that provide north-south traffic connectivity between the physical data center networks and the NSX SDN networks

1. Log in to for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate the appliance.
3. Right-click the appliance and select **Power > Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.
This operation takes several minutes to complete.
5. Repeat the steps for the remaining NSX Edge nodes for the domain.

Shut Down the NSX Manager Nodes

You continue shutting down the NSX infrastructure in the management domain and a workload domain by shutting down the three-node NSX Manager cluster by using the vSphere Client.

1. Log in to for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. Identify the vCenter instance that runs .
3. In the **VMs and Templates** inventory, locate an appliance.
4. Right-click the appliance and select **Power > Shut down Guest OS**.
5. In the confirmation dialog box, click **Yes**.
This operation takes several minutes to complete.
6. Repeat the steps for the remaining NSX Manager appliances.

Shut Down the SDDC Manager Appliance

Shut down the SDDC Manager appliance in the management domain by using the vSphere Client.

1. Log in to for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Expand the **Management VMs** folder.
4. Right-click the SDDC Manager appliance and click **Power > Shut down Guest OS**.
5. In the confirmation dialog box, click **Yes**.
This operation takes several minutes to complete.

Shut Down the ESX Hosts and vCenter in the Management Domain

In the shutdown order, you must down the ESX hosts and management vCenter last.

Important:

You perform this operation on all clusters in the management domain. The primary cluster which hosts the management vCenter must be stopped last.

Shut Down vSAN and the ESX Hosts in the Management Domain

You use the vSAN shutdown cluster wizard in the vSphere Client to shut down gracefully the vSAN clusters in a management domain. The wizard shuts down the vSAN storage and the ESX hosts added to the cluster.

1. Identify the cluster that hosts the management vCenter for this management domain.
Important:
This cluster must be shut down last.
2. Log in to for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. For a vSAN cluster, verify the vSAN health and resynchronization status.
 - a) In the **Hosts and Clusters** inventory, select the cluster and click the **Monitor** tab.
 - b) In the left pane, navigate to **vSAN > Skyline health** and verify the status of each vSAN health check category.
 - c) In the left pane, under **vSAN > Resyncing objects**, verify that all synchronization tasks are complete.
4. If any member host is in lockdown mode, add the host's **root** account to the Exception Users list.
 - a) Select the host in the inventory and click the **Configure** tab.
 - b) In the left pane, select **System > Security Profile**.
 - c) In the **Lockdown Mode** pane, click the **Edit** button.
 - d) On the **Exception Users** page, enter `root` and click **Add User**.
 - e) Click **OK**.
5. Shut down the vSAN cluster.
 - a) In the inventory, right-click the vSAN cluster and select **vSAN > Shutdown cluster**.
 - b) In the **Shutdown Cluster** wizard, verify that all pre-checks are green and click **Next**.
 - c) Review the **vCenter Server notice** and click **Next**.
 - d) If vCenter is running on the selected cluster, note the orchestration host details.
Connection to vCenter is lost because the vSAN shutdown cluster wizard shuts it down. The shutdown operation is complete after all ESXi hosts are stopped.
 - e) Enter a reason for performing the shutdown, and click **Shutdown**.
6. Repeat Step 3 to Step 5 for the other vSAN clusters in the domain.

Shut Down ESX Hosts with NFS or Fibre Channel Storage in the Management Domain

You use the vSphere Client to shut down gracefully the clusters in a management domain with a principal storage that is different from vSAN.

1. Identify the cluster that hosts the management vCenter for this management domain.
CAUTION:
This cluster must be shut down last.
2. Log in to for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **Hosts and Clusters** inventory, for a cluster that is not running the management vCenter, put all hosts in the cluster in maintenance mode.
These operations take several minutes to complete.
4. If a cluster is running the management vCenter, note down the host on which it is running and put all hosts in the cluster but this one in maintenance mode.
You need the details about this host during the startup process too.
5. Shut down the hosts in maintenance mode.
 - a) Select the cluster and click the **Hosts** tab.
 - b) Select all hosts that are in maintenance mode, right-click **Power > Shut Down**.
This operation takes several minutes to complete.

6. Repeat Step 3 to Step 5 for all clusters with NFS and Fibre Channel storage in the domain.
7. If vCenter is running on one of the ESX hosts in the cluster, shut down the vCenter appliance and the host from the VMware Host Client.
 - a) Log in to the ESX host at https://<esx_fqdn> as a user with the **Administrator** role.
 - b) In the left pane, click **Virtual Machines**.
 - c) Locate the vCenter appliance for the domain.
 - d) Right-click the virtual machine and select **Power > Shut down Guest OS**.
 - e) In the confirmation dialog box, click **Yes**.

This operation takes several minutes to complete.
 - f) In the left pane, right-click on **Host > Shut down**.
 - g) Read the warning and click **Shut down**.

You lose connectivity to the ESX host. The shutdown operation takes several minutes to complete.

8. Shut down the storage for the management domain if there are no other workloads that use it.

Follow vendor 's documentation to check if there are active clients and how to shut it down in a safe manner.

Starting Up VMware Cloud Foundation

To maintain the components integration and avoid operation faults, you follow a specified order to start up the management virtual machines in VMware Cloud Foundation.

- Verify that external services such as Storage, Active Directory, DNS, NTP, SMTP, and FTP or SFTP are available.
- If a vSphere Storage APIs for Data Protection (VADP) based backup solution is deployed on the default management cluster, verify that the solution is properly started and operational according to the vendor guidance.

You start the management components for the management domain first. Then, you start the management components for the workload domains and the customer workloads.

If the NSX Manager cluster and NSX Edge cluster are shared with other workload domains, start the other workload domains first. Start up NSX Manager and NSX Edge nodes as part of the startup of the last workload domain.

Start the Management Domain

You start the management components for the management domain in a specific order to provide the necessary infrastructure, networking, and management services before powering on the components for cloud management.

You start the management components for the management domain first. Then, you start the management components for the workload domains and the customer workloads.

Startup Order for the Management Domain

You start the virtual infrastructure of the management domain that hosts VCF Operations and VCF Automation first.

Table 946: Startup Order for the Management Domain

Startup Order	SDDC Component
1	Start vSAN and the ESX Hosts in the Management Domain Start ESX Hosts with NFS or Fibre Channel Storage in the Management Domain
2	SDDC Manager
3	NSX Manager

Startup Order	SDDC Component
4	NSX Edge
5	VMware Live Site Recovery
6	VCF Operations
	VCF Operations fleet management
	VCF Identity Broker
7	VCF Operations for logs
8	VCF Operations collector
9	VCF Operations for Networks
10	VCF Automation

Related Links

[Start a Workload Domain on page 3010](#)

You start the management components for a workload domain in a specific order to provide the necessary infrastructure, networking, and management services.

Start the ESX Hosts and vCenter in the Management Domain

You start the ESX hosts using an out-of-band management interface, such as, ILO or iDRAC to connect to the hosts and power them on.

Important:

You perform this operation on all clusters in management domain. The primary cluster where the management vCenter must be running must be started first.

Start vSAN and the ESX Hosts in the Management Domain

You start the ESX hosts using an out-of-band management interface, such as, ILO or iDRAC to connect to the hosts and power them on. Then, restarting the vSAN cluster starts automatically vCenter and vSAN.

You perform this operation on all vSAN clusters in the management domain. You must start with the cluster that hosts the management vCenter.

1. Power on the first ESX host in the management domain.
 - a) Log in to the first ESX host in the domain by using the out-of-band management interface.
 - b) Power on the ESX host according to the hardware vendor guide.
2. Repeat the previous step to start all the remaining ESX hosts in the management domain.

This operation takes several minutes to complete.

vCenter is started automatically. Wait until vCenter is running and the vSphere Client is available again.
3. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
4. Restart the vSAN cluster.
 - a) In the **Hosts and Clusters** inventory, right-click the vSAN cluster and select **vSAN > Restart cluster**.
 - b) In the **Restart Cluster** dialog box, click **Restart**.

The **vSAN Services** page on the **Configure** tab changes to display information about the restart process.

5. After the cluster has been restarted, check the vSAN health service and resynchronization status, and resolve any outstanding issues.
 - a) Select the cluster and click the **Monitor** tab.
 - b) In the left pane, under **vSAN > Resyncing objects**, verify that all synchronization tasks are complete.
 - c) In the left pane, navigate to **vSAN > Skyline health** and verify the status of each vSAN health check category.
6. If you have added the **root** user of the ESXi hosts to the Exception Users list for lockdown mode during shutdown, remove the user from the list on each host.
 - a) Select the host in the inventory and click the **Configure** tab.
 - b) In the left pane, select **System > Security Profile**.
 - c) In the **Lockdown Mode** pane, click the **Edit** button.
 - d) On the **Exception Users** page, from the vertical ellipsis menu in front of the **root** user, select **Remove User** and click **OK**.

Start ESX Hosts with NFS or Fibre Channel Storage in the Management Domain

You start the ESX hosts using an out-of-band management interface, such as, ILO or iDRAC to connect to the hosts and power them on. Then, restart the cluster from vCenter.

External storage is running and accessible.

You perform this operation on all non-vSAN clusters in the management domain. You must start with the cluster that hosts the management vCenter.

1. Power on the first ESX host in the management domain.
 - a) Log in to the first ESX host in the domain by using the out-of-band management interface.
 - b) Power on the ESX host according to the hardware vendor guide.

2. Repeat the previous step to start all the remaining ESX hosts in the management domain.

This operation takes several minutes to complete.

3. Log in into the ESX host that hosts vCenter and any infrastructure components.

Important:

You must first start infrastructure components that runs in the management cluster before you start the vCenter.

- a) Log in into ESX host at https://<esx_fqdn> as a user with the **Administrator** role.
 - b) Locate the management vCenter appliance.
 - c) Right-click the appliance and select **Power > Power on**.
4. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
5. Verify that all hosts in the cluster are connected to the management vCenter.
6. Exit maintenance mode for all hosts in the cluster.

This operations takes several minutes to complete.
7. Repeat Step 5 and Step 6 for all clusters with non-vSAN storage in the domain.

Start the SDDC Manager Appliance

Start the SDDC Manager appliance by using the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Expand the **Management VMs** folder.
4. Right-click the SDDC Manager appliance and click **Power > Power On**.
5. In the confirmation dialog box, click **Yes**.

This operation takes several minutes to complete.

Start the NSX Manager Nodes

You begin powering on the NSX infrastructure in the management domain by starting the NSX Manager appliance or cluster by using the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Power on the NSX Manager nodes for the management domain.
 - a) Right-click NSX Manager node and select **Power > Power on**.
 - b) Repeat the steps to power on the remaining NSX Manager nodes.

This operation takes several minutes to complete until the NSX Manager cluster becomes fully operational again and its user interface - accessible.

4. Log in to NSX Manager for the management domain at https://<nsxt_manager_cluster_fqdn> as **admin**.
5. Verify the system status of NSX Manager cluster.
 - a) On the main navigation bar, click **System**.
 - b) In the left pane, navigate to **Configuration > Appliances**.
 - c) On the **Appliances** page, verify that the NSX Manager cluster has a *Stable* status and all NSX Manager nodes are available.

Start the NSX Edge Nodes

You continue powering on the NSX infrastructure in the management domain by starting the NSX Edge nodes by using the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Right-click an NSX Edge node from the edge cluster and select **Power > Power on**.

This operations takes several minutes to complete.
4. Repeat these steps to power on the remaining NSX Edge nodes.

Start VCF Operations

You start VCF Operations by first starting the appliances of the VCF Operations cluster, then taking the cluster online.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. Start VCF Operations.

Important:

Power on the Guest OS of the VCF Operations appliances by following the order described in [Broadcom KB 341964](#).

- a) Locate a VCF Operations appliance.
- b) Right-click the appliance and select **Power > Power On**.
- c) In the confirmation dialog box, click **Yes**.

This operations takes several minutes to complete.

- d) Repeat the steps for the remaining VCF Operations appliances by following the order described in [Broadcom KB 341964](#).

3. Log in to the administration UI at the https://<vcf_operations_fqdn>/admin URL as the **admin** local user.
4. Take the cluster online.
 - a) Log in to the administration interface at the https://<vcf_operations_fqdn>/admin URL as the **admin** local user.
 - b) On the **System status** page, click **Bring Cluster Online**.

This operation might take about an hour to complete.

Start the VCF Operations Fleet Management Appliance

Start the VCF Operations fleet management appliance in the management domain from the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate the VCF Operations fleet management appliance.
3. Right-click the VCF Operations fleet management appliance and select **Power > Power On**.
4. In the confirmation dialog box, click **Yes**.

Start VCF Identity Broker

To start VCF Identity Broker, use the VCF Operations user interface.

Note:

Once you start VCF Identity Broker you will be able to use accounts from external identity providers.

1. Log in to at the https://<vcf_operations_fqdn> URL as a user with the **Administrator** role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. In the **Components** tab, locate the component that you are starting or stopping and, in the **Action** column, click **Manage**.
4. On the component manage page, click the ellipsis icon, and select **Power on**.
5. Verify the information and click **Submit**.

This operation takes several minutes to complete.

Start VCF Operations for Logs

Start the VCF Operations for logs nodes in the management domain by using the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a VCF Operations for logs appliance.
3. Right-click the VCF Operations for logs appliance and select **Power > Power On**.
4. In the confirmation dialog box, click **Yes**.
5. Repeat the steps for the remaining VCF Operations for logs appliances.

Start the VCF Operations Collector Appliances

Start the VCF Operations collector nodes in the management domain by using the vSphere Client.

1. Log in to for the management domain at https://<vcenter_fqdn>/ui as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a VCF Operations collector appliance.
3. Right-click the VCF Operations collector appliance and select **Power > Power On**.
4. In the configuration dialog box, click **Yes**.
5. Repeat the steps on the remaining VCF Operations collector appliances.

Start VCF Operations for Networks

To start VCF Operations for Networks, use the VCF Operations user interface.

1. Log in to at the https://<vcf_operations_fqdn> URL as a user with the **Administrator** role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. In the **Components** tab, locate the component that you are starting or stopping and, in the **Action** column, click **Manage**.
4. On the component manage page, click the ellipsis icon, and select **Power on**.
5. Verify the information and click **Submit**.

This operation takes several minutes to complete.

Start VCF Automation

To start VCF Automation, use the VCF Operations user interface.

1. Log in to at the https://<vcf_operations_fqdn> URL as a user with the **Administrator** role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. In the **Components** tab, locate the component that you are starting or stopping and, in the **Action** column, click **Manage**.
4. On the component manage page, click the ellipsis icon, and select **Power on**.
5. Verify the information and click **Submit**.

This operation takes several minutes to complete.

Start a Workload Domain

You start the management components for a workload domain in a specific order to provide the necessary infrastructure, networking, and management services.

You start the management components for the management domain first. Then, you start the management components for the workload domains and the customer workloads.

If the NSX Manager cluster and NSX Edge cluster are shared across workload domains, follow this general order:

1. Start the other workload domains.
2. Start the workload domain that runs the shared NSX Edge nodes.
3. Start the customer workloads that rely on NSX services.

Startup Order for a Workload Domain

Table 947: Startup Order for a Workload Domain

Startup Order	SDDC Component
1	vCenter for the workload domain
2	ESX hosts for the workload domain
3	Restart Clusters for the workload domain
5	NSX Manager nodes for the workload domain
6	NSX Edge nodes for the workload domain
8	VMware Live Site Recovery for the workload domain
9	Virtualized customer workloads

Related Links

[Start the Management Domain on page 3005](#)

You start the management components for the management domain in a specific order to provide the necessary infrastructure, networking, and management services before powering on the components for cloud management.

Start the vCenter Appliance for a Workload Domain

Use the vSphere Client to power on the vCenter appliance for the workload domain. If the workload domain contains a vSAN cluster, check its health status too.

vCenter startup is done:

- If vCenter is running on a vSAN cluster, automatically from the orchestration host .
- If vCenter is running on a non-vSAN storage, manually following the steps below.

If the vCenter appliance is running on the infrastructure, you must start the ESX hosts first.

1. Log in into the resource that hosts the vCenter appliance.
 - a) For an ESX host, log in to the ESX host at `https://<esx_fqdn>` as a user with the **Administrator** role.
 - b) For the management vCenter, log in to vCenter at the `https://<vcenter_fqdn>/ui` URL as a user with the **Administrator** role.
2. Start vCenter Server.
 - a) Locate the vCenter Server virtual machine for the workload domain.
 - b) Right-click the virtual machine and select **Power > Power on**.

The startup of the virtual machine and the vSphere services takes some time to complete.

Start the ESX Hosts in a Workload Domain

You start the ESX hosts using an out-of-band management interface, such as, ILO or iDRAC to connect to the hosts and power them on.

If you are using non-vSAN storage, the storage must be started and accessible.

1. Power on the first ESX host in the workload domain.
 - a) Log in to the first ESX host in the workload domain by using the out-of-band management interface.
 - b) Power on the ESX host according to the documentation from the hardware vendor.
2. Repeat the step to start all the remaining ESX hosts in the workload domain.

This operation takes several minutes to complete.

Restart Clusters in a Workload Domain

After you start the ESX hosts and the vCenter appliance for a workload domain, restart all clusters to re-initiate the integration with the principal storage.

Restart the vSAN Clusters in a Workload Domain

You start vSAN in a workload domain by restarting the vSAN cluster.

All hosts are started.

Verify that all ESX hosts in the cluster are connected to the workload domain vCenter.

You perform this operation on all vSAN clusters in all workload domains.

1. Log in to vCenter Server for the management or VI workload domain at `https://<vcenter_server_fqdn>/ui` as **administrator@vsphere.local**.
2. In the **Hosts and clusters** inventory, expand the tree of the VI workload domain vCenter Server and expand the data center for the VI workload domain.
3. Restart the vSAN cluster.
 - a) Right-click the vSAN cluster and select **vSAN > Restart cluster**.
 - b) In the **Restart Cluster** dialog box, click **Restart**.
The **vSAN Services** page on the **Configure** tab changes to display information about the restart process.
4. After the cluster has restarted, check the vSAN health service and resynchronization status, and resolve any outstanding issues.
 - a) Select the cluster and click the **Monitor** tab.
 - b) In the left pane, navigate to **vSAN > Skyline health** and verify the status of each vSAN health check category.
 - c) In the left pane, under **vSAN > Resyncing objects**, verify that all synchronization tasks are complete.
5. If you have added the **root** user of the ESXi hosts to the Exception Users list for lockdown mode during shutdown, remove the user from the list on each host.
 - a) Select the host in the inventory and click the **Configure** tab.
 - b) In the left pane, select **System > Security Profile**.
 - c) In the **Lockdown Mode** pane, click the **Edit** button.
 - d) On the **Exception Users** page, from the vertical ellipsis menu in front of the **root** user, select **Remove User** and click **OK**.

Restart the Clusters in a Workload Domains with NFS or Fibre Channel Storage Storage

You use the vSphere Client to restart the clusters with non-vSAN storage in a workload domain.

All hosts are started

Verify that all ESXi hosts in the cluster are connected to the workload domain vCenter Server.

1. Log in to for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. Check that all hosts in the cluster are connected to vCenter.
3. Exit maintenance mode for all hosts in the cluster.
This operations takes several minutes to complete.
4. Repeat Step 2 and Step 3 for all clusters with non-vSAN storage in the domain.

Start the NSX Manager Nodes

You begin powering on the NSX infrastructure in the workload domain by starting the NSX Manager appliance or cluster by using the vSphere Client.

1. Identify the vCenter instance that runs .
2. Log in to for the management or workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
4. Power on the NSX Manager nodes for the workload domain.
 - a) Right-click NSX Manager node and select **Power > Power on**.
 - b) Repeat the steps to power on the remaining NSX Manager nodes.

This operation takes several minutes to complete until the NSX Manager cluster becomes fully operational again and its user interface - accessible.
5. Log in to NSX Manager for the workload domain at `https://<nsx_manager_cluster_fqdn>` as **admin**.
6. Verify the system status of the NSX Manager cluster.
 - a) On the main navigation bar, click **System**.
 - b) In the left pane, navigate to **Configuration > Appliances**.
 - c) On the **Appliances** page, verify that the NSX Manager cluster has a *Stable* status and all the listed NSX Manager nodes are available.

Start the NSX Edge Nodes

You continue powering on the NSX infrastructure in the workload domain by starting the NSX Edge nodes by using the vSphere Client.

1. Identify the instance that runs nodes.
2. Log in to for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **VMs and templates** inventory, expand the tree of workload domain vCenter Server and expand data center for the workload domain.
4. Right-click an NSX Edge node from the edge cluster and select **Power > Power on**.
This operations takes several minutes to complete.
5. Repeat these steps to power on the remaining NSX Edge nodes.

Backup and Restore of VMware Cloud Foundation

Regularly backing up the management components of a VMware Cloud Foundation platform is essential to ensure the resilience and continuity of your VCF private cloud. In the event of data loss or system failure, having up-to-date backups allows for efficient restoration of critical services, minimizing downtime and maintaining operational integrity.

Implementing a consistent backup strategy safeguards your infrastructure against unforeseen disruptions and supports effective disaster.

You implement scheduled backups to prepare for:

- A critical failure of a management component.
- An upgrade of a management component.
- A certificate update of a management component.

In addition, as a best practice, you can take on-demand manual backups in the following use cases:

- After a successful recovery operation.
- After resolving asynchronously reported errors in SDDC components.
- After resolving an incomplete workflow in SDDC Manager.
- After noting the failure of a scheduled backup of an SDDC component.
- Immediately before performing a system upgrade.

It is recommended that you configure an external SFTP server as a backup location for the following reasons:

- An external SFTP server is a prerequisite for restoring SDDC Manager file-based backups.
- Using an external SFTP server provides better protection against failures because it decouples NSX backups from SDDC Manager.

Prerequisites

Before performing backup and restore operations in your VMware Cloud Foundation platform, ensure the following requirements are met:

General Requirements:

- Administrator permissions to VMware Cloud Foundation platform components.
- A stable network connection between the management domain and backup storage.
- Sufficient storage capacity to accommodate backup data.
- A defined backup and retention policy aligned with best practices or organizational requirements.
- Access to an image-level backup solution for required components.
- Sufficient time allowed between backup operations so that a new backup request is not submitted when another backup of the same environment is in progress.

Backup-Specific Requirements:

- Configure and validate an external backup target based on individual management component requirements.
- Ensure all management components, including SDDC Manager, vCenter, NSX, VCF Operations, and VCF Automation, are configured with a regular backup schedule and retention policy.

Restore-Specific Requirements:

- Access to the most recent backup files.
- In place infrastructure for restoring.
- Proper sequencing and verification of component restoration to maintain dependencies and system integrity.

Backup Methods for VMware Cloud Foundation

Category	Component	Supported Backup Method(s)
Core	vCenter	File-based or Image-Base*
	ESX	N/A

Category	Component	Supported Backup Method(s)
	NSX Manager	File-based
	NSX Edge	Edge configuration is included in NSX Manager backup
Management	VCF Operations	Image-Based
	VCF Operations fleet management	Image-Based
	VCF Operations for logs	Image-Based
	VCF Operations for networks	Image-Based
	VCF Identity Broker	File-based
	SDDC Manager	File-based or Image-Based*
Consumption	VCF Automation	File-based

* This document uses file-based backup for this component.

Reconfigure SFTP Backups for SDDC Manager and NSX Manager

By default, SDDC Manager and NSX Manager backups are stored on the SDDC Manager appliance. Change the destination of the backups to an external SFTP server.

- The external SFTP server must support a 256-bit length ECDSA SSH public key.
- The external SFTP server must support a 2048-bit length RSA SSH public key
- You will need the SHA256 fingerprint of RSA key of the SFTP server.
- Host Key algorithms: At least one of rsa-sha2-512 or rsa-sha2-256 and one of ecdsa-sha2-nistp256, ecdsa-sha2-nistp384, or ecdsa-sha2-nistp521.
- Additional pre-requisites when FIPS Security Mode is enabled on SDDC Manager:

Algorithms and Ciphers	Required when FIPS Security Mode is Enabled
Kex Algorithms	At least one of: - diffie-hellman-group-exchange-sha256 - ecdh-sha2-nistp256 - ecdh-sha2-nistp384 - ecdh-sha2-nistp521
Message Authentication Key (MAC) Algorithms	hmac-sha2-256
Ciphers	At least one of: - TLS_DHE_DSS_WITH_AES_128_CBC_SHA256 - TLS_DHE_DSS_WITH_AES_256_CBC_SHA256 - TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 - TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384 - TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 - TLS_RSA_WITH_AES_128_CBC_SHA256 - TLS_RSA_WITH_AES_256_CBC_SHA256 - TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 - TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 - TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 - TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 - TLS_AES_128_GCM_SHA256 - TLS_AES_256_GCM_SHA384

Note: SHA1 algorithms are not supported.

- Log in to the VCF Operations interface at https://<vcf_operations_fqdn> with a user assigned the Administrator role.
- Navigate to **Administration > SDDC Manager**.
- Select the VCF Instance to configure backup.
- In the right hand pane, click the **Backup Settings** tab and then click **Site Settings**.
- On the **Backup Settings** page, enter the settings and click **Save**.

Setting	Value
Host FQDN or IP	The FQDN or IP Address of the SFTP server.
Port	22
Transfer Protocol	SFTP
Username	A service account with privileges to the SFTP server. For example: <code>svc-vcf-bck</code> .
Password	The password for the username provided.
Backup Directory	The directory on the SFTP server where backups are saved. For example: <code>/backups/</code> .
SSH Fingerprint	The SSH Fingerprint is automatically retrieved from the SFTP server, verify the SSH Fingerprint.
Confirm Fingerprint	Selected
Encryption Passphrase	The encryption passphrase used to encrypt the backup data. Note: The encryption passphrase should be stored safely as it is required during the restore process.

- In the **Confirm your changes to backup settings** dialog box, click **Confirm**.

File-Based Backups for SDDC Manager NSX Manager and vCenter

You can use the native file-based backup capabilities of SDDC Manager, vCenter and NSX Manager. SDDC Manager configures the NSX Manager backup during the bring-up process. You configure the file-based backup jobs for SDDC Manager and vCenter.

Verify that you have an SFTP server on the network to serve as a target of the file-based backups.

To ensure that all management components are backed up correctly, you must create a series of backup jobs that capture the state of a set of related components at a common point in time. With some components, simultaneous backups of the component nodes ensure that you can restore the component a state where the nodes are logically consistent with each other and eliminate the necessity for further logical integrity remediation of the component.

Table 948: File-Based Backup Jobs

Component	Recommended Frequency	Recommended Retention	Notes
SDDC Manager	Daily	7 days	You must configure the backup jobs for the SDDC Manager instance and all vCenter instances to start within the same 5-minute window.
vCenter	Daily	7 days	

Component	Recommended Frequency	Recommended Retention	Notes
vSphere Distributed Switch	On-demand	Retain last 3 configurations.	-
NSX Manager	Hourly	7 days	Configured by SDDC Manager during the bring-up process.

Note:

- You must monitor the space utilization on the SFTP server to ensure that you have sufficient storage space to accommodate all backups taken within the retention period.
- Do not make any changes to the `/opt/vmware/vcf` directory on the SDDC Manager appliance. If this directory contains any large files, backups may fail.

Back Up SDDC Manager

You configure file-based daily backups of the SDDC Manager instances using VCF Operations.

Only a user with the **Admin** role can perform this task.

- Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
- Navigate to **Administration > SDDC Manager**.
- Select the VCF Instance to configure a backup schedule.
- In the right hand pane, click the **Backup Settings** tab and then click **SDDC Manager Configurations**.
- Under **Backup Schedule**, click **Edit**.
- On the **Backup Schedule** page, enter the settings and click **Save**.

Setting	Value
Automatic Backup	Enabled
Backup Frequency	Weekly
Days of the Week	All selected
Schedule Time	04:02 AM
Take Backup on State Change	Enabled
Retain Last Backups	7
Retain Hourly Backups for Days	1
Retain Daily Backups for Days	7

- To verify the backup, click **Backup Now**.

The status and the start time of the backup is displayed on the UI. You have set the SDDC Manager backup schedule to run daily at 04:02 AM and after each change of state.

If the backup is unsuccessful, verify if the SFTP server is available and able to provide its SSH fingerprint:

- SSH to the SDDC Manager appliance run the following command as the root user:

```
sftp username@IP of sftp server
```

Enter the SFTP user password when prompted. The following message indicates a successful connection:

```
Connected to username@IP of sftp server.
```

- To check that the SFTP server SSH fingerprint is available, run:

```
ssh-keygen -lf <(ssh-keyscan -t ssh-rsa -p port_numberIP of sftp server 2>/dev/null)
```

Configure a Backup Schedule for vCenter

You configure file-based daily backups of the vCenter instances by using the vCenter Management Interface of each vCenter instance.

- Log in to the vCenter management interface at `https://<vcenter_fqdn>:5480` using the root user.
- In the left navigation pane, click **Backup**.
- In the **Backup schedule** pane, click **Configure**.
- In the **Create backup schedule** dialog box, enter these values and click **Create**.

Setting		Value
Backup location		Enter the backup location from SFTP server. For example: <code>sftp://172.16.11.60/backups/</code>
Backup server credentials	User name	A service account with privileges to the SFTP server. For example: <code>svc-vcf-bck</code> .
	Password	Enter the password for the username provided.
Schedule		Daily 11:00 PM
Encrypt backup	Encryption password	<code>encryption_password</code>
	Confirm password	<code>encryption_password</code>
Number of backups to retain		Retain last 7 backups
Data	Supervisors Control Plane	Select if running vSphere Supervisor on this vCenter.
	Stats, events, and tasks	Selected
	Inventory and configuration	Selected

The backup schedule information appears in the **Backup schedule** pane.

- Repeat the procedure for the other vCenter instances.

Any complete and in-progress backup appears in the **Activity** pane.

On-Demand Back Up of vCenter

Before you upgrade a vCenter instance, you should use the vCenter Management Interface to manually back it up.

- In the vSphere Client, for each vSphere cluster that is managed by the vCenter, note the current vSphere DRS Automation Level setting and then change the setting to **Manual**. After the vCenter backup is complete, you can change the vSphere DRS Automation Level setting back to its original value. See [KB 87631](#) for information about using VMware PowerCLI to change the vSphere DRS Automation Level.
- Ensure that there are no active vMotion tasks.

1. Log in to the vCenter management interface at `https://<vcenter_fqdn>:5480` using the root user.
2. In the left navigation pane, click **Backup**.
3. In the Activity pane, click **Backup Now**.
4. If you already have a backup schedule set up, select **Use backup location and user name from backup schedule** and click **Start**.
5. If you do not already have a backup schedule, enter the following information and click **Start**.

Setting		Value
Backup location		Enter the backup location from SFTP server. For example: <code>sftp://172.16.11.60/backups/</code>
Backup server credentials	User name	A service account with privileges to the SFTP server. For example: <code>svc-vcf-bck</code> .
	Password	Enter the password for the username provided.
Encrypt backup	Encryption password	<i>encryption_password</i>
	Confirm password	<i>encryption_password</i>
Number of backups to retain	Choose to retain all backups or a certain number of backups.	
Data	Supervisors Control Plane	Selected (If using Supervisor Control Plane in this vCenter)
	Stats, events, and tasks	Selected
	Inventory and configuration	Selected

In order to restore vCenter, you will need the vCenter Appliance ISO file that matches the version you backed up.

- Identify the required vCenter version. In the vCenter Management Interface, click **Summary** in the left navigation pane to see the vCenter version and build number.
- Download the vCenter Appliance ISO file for that version from the Broadcom Support Portal.

Export the Configuration of the vSphere Distributed Switches

The vCenter backup includes the configuration of the entire vCenter instance. To have a backup only of the vSphere Distributed Switch and distributed port group configurations, you export a configuration file that includes the validated network configurations. If you want to recover only the vSphere Distributed Switch, you can import this configuration file to the vCenter instance.

You can use the exported file to create multiple copies of the vSphere Distributed Switch configuration on an existing deployment, or overwrite the settings of existing vSphere Distributed Switch instances and port groups.

You must backup the configuration of a vSphere Distributed Switch immediately after each change in configuration of that switch.

1. Log in to the vCenter interface at `https://<vcenter_fqdn>/ui` with a user assigned the Administrator role.
2. Select the Networking inventory.
3. In the inventory expand the vCenter and Datacenter.
4. Right click the vSphere Distributed Switch you want to backup, and select Export Configuration.
5. In the **Export configuration** dialog box, select **Distributed switch and all port groups**.
6. In the **Description** text box enter the date and time of export, and click **OK**.
7. Copy the backup zip file to a secure location from where you can retrieve the file and use it if a failure of the appliance occurs.
8. Repeat the procedure for the other vSphere Distributed Switches.

File-Based Restore for SDDC Manager, vCenter, and NSX

When SDDC Manager, vCenter or NSX Manager in the SDDC fails, you can restore the component to a fully operational state by using its file-based backup. When an NSX Edge node fails, you redeploy the node from the NSX Manager instance.

Use this guidance as appropriate based on the exact nature of the failure encountered within your environment. Sometimes, you can recover localized logical failures by restoring individual components. In more severe cases, such as a complete and irretrievable hardware failure, to restore the operational status of your SDDC, you must perform a complex set of manual deployments and restore sequences. In failure scenarios where there is a risk of data loss, there has already been data loss or where it involves a catastrophic failure, contact Broadcom Support to review your recovery plan before taking any steps to remediate the situation.

Restore SDDC Manager

If SDDC Manager fails, you can restore it from its file-based backup.

- Power off and rename the failed SDDC Manager instance.
- Verify that you have a valid file-based backup of the failed SDDC Manager instance.
To be valid, the backup must be of the same version as the version of the SDDC Manager appliance on which you plan to restore the instance.
- Verify that you have the SFTP server details:
 - SFTP Server IP address
 - SFTP Server Username
 - SFTP Server Password
 - Encryption Password
- The following components must be available before proceeding with the restore of SDDC Manager:
 - vCenter

After a successful recovery, securely delete the decrypted backup files.

Prepare for Restoring SDDC Manager

Before restoring SDDC Manager, you must download and decrypt the encrypted backup file from the SFTP server.

Verify that your host machine with access to the SDDC has OpenSSL installed.

Note:

The procedures have been written based on the host machine being a Linux-based operating system.

The backup file contains sensitive data about your SDDC Manager instance, including passwords in plain text. As a best practice, you must control access to the decrypted files and securely delete them after you complete the restore operation.

1. Identify the backup file for the restore and download it from the SFTP server to your host machine.
2. On your host machine, open a terminal and run the following command to extract the content of the backup file.

```
OPENSSE_FIPS=1 openssl enc -d -aes-256-cbc -md sha256 -in filename-of-restore-file | tar -xz
```
3. When prompted, enter the *encryption_password*.
4. In the extracted folder, locate and open the *metadata.json* file in a text editor.
5. Locate the *sddc_manager_ova_location* value and copy the URL.
6. In a web browser, paste the URL and download the OVA file.
7. In the extracted folder, locate and view the contents of the *security_password_vault.json* file.
8. Locate the *entityType BACKUP* value and record the backup password.

Restore SDDC Manager from a File-Based Backup

First, you deploy a new SDDC Manager appliance by using the OVA file that you downloaded during the preparation for the restore. After that, you restore the file-based backup on the newly deployed SDDC Manager appliance.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu > VMs and templates**.
3. In the inventory expand **vCenter > Datacenter**.
4. Right-click the management folder and select **Deploy OVF template**.
5. On the **Select an OVF template** page, select **Local file**, click **Upload files**, browse to the location of the SDDC Manager OVA file, click **Open**, and click **Next**.
6. On the **Select a name and folder** page, in the **Virtual machine name** text box, enter a virtual machine name, and click **Next**.
7. On the **Select a compute resource** page, click **Next**.
8. On the **Review details** page, review the settings and click **Next**.
9. On the **License agreements** page, accept the license agreement and click **Next**.
10. On the **Select storage** page, select the vSAN datastore and click **Next**.

The datastore must match the *vsan_datastore* value in the *metadata.json* file that you downloaded during the preparation for the restore.

11. On the **Select networks** page, from the **Destination network** drop-down menu, select the management network distributed port group and click **Next**.

The distributed port group must match the *port_group* value in the *metadata.json* file that you downloaded during the preparation for the restore.

12. On the **Customize template** page, enter the following values and click **Next**.

Setting	Description
Enter root user password	You can use the original root user password or a new password.
Enter Local user password	You can use the original Local user password or a new password.
Host Name	The FQDN must match the <i>hostname</i> value in the <i>metadata.json</i> file that you downloaded during the preparation for the restore.
NTP Servers	The NTP server details for the appliance.

Setting	Description
Enable FIPs	Selected
Network 1 IP address	The IP address for the appliance.
Network 1 Subnet Mask	The subnet mask for the appliance.
Network Default Gateway	The default gateway for the appliance.
DNS Domain Name	The domain name for the appliance.
Domain search path	The domain search path(s) for the appliance.
Domain name servers	The DNS servers for the appliance.

13. On the **Ready to complete** page, click **Finish** and wait for the process to complete.
14. Once the SDDC Manager appliance deployment completes, expand the management folder.
15. Right-click the SDDC Manager appliance and select **Snapshots > Take Snapshot**.
16. Right-click the SDDC Manager appliance, select **Power > Power On**.
17. On the host machine, copy the encrypted backup file to the */tmp* folder on the newly deployed SDDC Manager appliance by running the following command. When prompted, enter the *vcf_user_password*.

```
scp filename-of-restore-file vcf@sddc_manager_fqdn:/tmp/
```
18. On the host machine, obtain the authentication token from the SDDC Manager appliance in order to be able to execute the restore process by running the following command:

```
TOKEN='curl https://<sddc_manager_fqdn>/v1/tokens -k -X POST -H "Content-Type: application/json" -d '{"username": "admin@local","password": "<admin@local_password>"}' | awk -F "\"" '{ print $4}''
```
19. On the host machine with access to the SDDC Manager, open a terminal and run the command to start the restore process.

```
curl https://<sddc_manager_fqdn>/v1/restores/tasks -k -X POST -H "Content-Type: application/json" -H "Authorization: Bearer $TOKEN" \
-d '{
  "elements": [ {
    "resourceType" : "SDDC_MANAGER"
  } ],
  "backupFile" : "<backup_file>",
  "encryption" : {
    "passphrase" : "<encryption_password>"
  }
}'
```

The command output contains the ID of the restore task.

20. Record the ID of the restore task.
21. Monitor the restore task by using the following command until the status becomes **Successful**.

```
curl https://<sddc_manager_fqdn>/v1/restores/tasks/<restore_task_id> -k -X GET -H "Content-Type: application/json" -H "Authorization: Bearer $TOKEN"
```

Refresh the SSH keys that are stored in the SDDC Manager inventory. See [VMware Cloud Foundation SDDC Manager Recovery Scripts \(79004\)](#).

Validate the Status of SDDC Manager

After a successful restore of SDDC Manager you must validate its status. You run the health checks by using the *sos* tool.

1. Log in to SDDC Manager by using a Secure Shell (SSH) client as the `vcf` user.

2. Run the health checks by using the `sos` tool.

```
sudo /opt/vmware/sddc-support/sos --health-check
```

3. When prompted, enter the `vcf_password`.

All tests show green when SDDC Manager is in healthy state.

4. Manually delete the snapshot created in [Restore SDDC Manager from a File-Based Backup](#).

Refresh the SSH keys that are stored in the SDDC Manager inventory. See [VMware Cloud Foundation SDDC Manager Recovery Scripts \(79004\)](#).

Restoring a vCenter Instance

If a vCenter instance fails, you can restore it from its file-based backup.

- Power off and rename the failed vCenter instance.
- Verify that you have a valid file-based backup of the failed vCenter instance.
To be valid, the backup must be of the version of the vCenter appliance on which you plan to restore the instance.
- Verify that you have the SFTP server details:
 - SFTP Server IP address
 - SFTP Server username
 - SFTP Server password
 - Encryption password

Prepare for Restoring a vCenter Instance

Before restoring a vCenter instance, you must retrieve the vCenter build number and deployment details, as well as vCenter and ESX credentials from the SDDC Manager inventory.

- The following components must be available before proceeding with the restore of vCenter:
 - SDDC Manager

Retrieve the vCenter Instance Deployment Details

Before restoring a vCenter instance, you must retrieve the vCenter build number and deployment details from the SDDC Manager inventory. The vCenter instances in your system might be running different build numbers if the backups are taken during an upgrade process. You must restore each vCenter instance to its correct version.

Because the management domain vCenter might be unavailable to authenticate the login, you use the SDDC Manager API via the shell to retrieve this information.

1. Log in to SDDC Manager by using a Secure Shell (SSH) client.

2. Run the command to get the list of vCenter instances.

```
curl http://localhost/inventory/vcenters -k | json_pp
```

3. For each vCenter instance, record the values of these settings.

Setting	Value
domainId	Id of the domain
vmName	VM name of the vCenter instance
datastoreForVmDeploymentSourceId	Datastore Id
hostName	FQDN of the vCenter instance

Setting	Value
version	<code>version_number-build_number</code>

4. Verify that the vCenter version retrieved from SDDC Manager is the same as the version associated with the backup file that you plan to restore.

Retrieve the Credentials for Restoring a vCenter Instance

Before restoring a vCenter instance, you must retrieve the vCenter root and vCenter Single Sign-On administrator credentials from the SDDC Manager inventory. Before restoring the management domain vCenter instance, you must also retrieve the credentials of a healthy management domain ESX host.

Note:

If SDDC Manager is not operational, you can retrieve the required vCenter root, vCenter Single Sign-On administrator, and ESX root credentials from the file-based backup of SDDC Manager. See [Prepare for Restoring SDDC Manager](#).

Before you can query the SDDC Manager API, you must obtain an API access token by using **admin@local** account.

1. Log in to your host machine with access to the SDDC and open a terminal.

2. Obtain the API access token.

a) Run the command to obtain an access token by using the **admin@local** credentials.

```
TOKEN=$(curl https://<sddc_manager_fqdn>/v1/tokens -k -X POST -H "Content-Type: application/json" -d '{"username": "admin@local", "password": "admin@local_password"}' | awk -F "\"" '{print $4}')
```

The command returns an access token and a refresh token.

b) Record the access token.

3. Retrieve the vCenter **root** credentials.

a) Run the following command to retrieve the vCenter **root** credentials.

```
curl https://<sddc_manager_fqdn>/v1/credentials?resourceType=VCENTER&accountType=USER -k -X GET \-H "Accept: application/json" -H "Authorization: Bearer $TOKEN" | json_pp
```

The command returns the vCenter **root** credentials.

Setting	Value
domainName	Name of the domain
resourceName	FQDN of the vCenter instance
username	root
password	<code>vcenter_root_password</code>

b) Record the vCenter **root** credentials.

4. Retrieve the vCenter Single Sign-On administrator credentials.

a) Run the following command to retrieve the vCenter Single Sign-On administrator credentials.

```
curl https://<sddc_manager_fqdn>/v1/credentials?resourceType=PSC -k -X GET \-H "Accept: application/json" -H "Authorization: Bearer $TOKEN" | json_pp
```

The command returns the **administrator@vsphere.local** credentials.

Setting	Value
domainName	Name of the domain
resourceName	FQDN of the vCenter instance
username	administrator@vsphere.local

Setting	Value
password	<i>vcenter_sso_admin_password</i>

b) Record the **administrator@vsphere.local** credentials.

5. If you plan to restore the management domain vCenter instance, retrieve the credentials for a healthy management domain ESX host.

a) Run the following command to retrieve the credentials for a management domain ESX host.

```
curl https://<sddc_manager_fqdn>/v1/credentials?resourceType=ESXI -k -X GET -H "Accept: application/json" -H "Authorization: Bearer $TOKEN" | jq -r '.password'
```

The command returns the ESX **root** credentials.

Setting	Value for first ESX host
domainName	management domain name
resourceName	FQDN of the first ESX host
username	root
password	<i>esx_root_password</i>

b) Record the ESX **root** credentials.

Restore a vCenter Instance from a File-Based Backup

If a vCenter instance fails, you can restore it from its file-based backup. If the management domain vCenter instance and the workload domain vCenter instance are both in a failed state, you must restore the management domain vCenter instance before restoring the workload domain vCenter instance.

- Download the vCenter ISO file for the version of the failed instance. See [Retrieve the vCenter Instance Deployment Details](#).
- If you are recovering a workload domain vCenter instance, verify that the management domain vCenter instance is available.

You deploy a new vCenter appliance and perform a file-based restore. If you are restoring the management domain vCenter instance, you deploy the new appliance on a healthy ESX host in the management domain cluster. If you are restoring the workload domain vCenter instance, you deploy the new appliance in the management domain vCenter instance.

1. Mount the vCenter ISO image to your host machine with access to the SDDC and run the UI installer for your operating system.

For example, for a Windows host machine, open the *dvd-drive:\vcsa-ui-installer\win32\installer* application file.

2. Click **Restore**.

3. Complete the **Restore - Stage 1: Deploy vCenter** wizard.

- On the **Introduction** page, click **Next**.
- On the **End user license agreement** page, select the **I accept the terms of the license agreement** check box and click **Next**.
- On the **Enter backup details** page, enter these values and click **Next**.

Setting	Value
Location or IP/hostname	<i>sftp://sftp_server_ip/backups/vCenter/sn_vc_fqdn/backup_folder</i>

Setting	Value
User name	SFTP Server Username
Password	<i>sftp-user-password</i>

d) On the **Review backup information** page, review the backup details, record the **vCenter configuration** information, and click **Next**.

You use the vCenter configuration information at a later step to determine the deployment size for the new vCenter appliance.

e) On the **vCenter deployment target** page, enter the values by using the information that you retrieved during the preparation for the restore, and click **Next**.

Setting	Value for Management Domain vCenter Instance	Value for Workload Domain vCenter Instance
ESX host or vCenter name	FQDN of the first ESX host	FQDN of the management vCenter instance
HTTPS port	443	443
User name	root	administrator@vsphere.local
Password	<i>esx_root_password</i>	<i>vcenter_sso_admin_password</i>

f) In the **Certificate warning** dialog box, click **Yes** to accept the host certificate.

g) On the **Set up target vCenter VM** page, enter the values by using the information that you retrieved during the preparation for the restore, and click **Next**.

Setting	Value
VM name	vCenter instance VM name
Set root password	<i>vcenter_root_password</i>
Confirm root password	<i>vcenter_root_password</i>

h) On the **Select deployment size** page, select the deployment size that corresponds with the vCenter configuration information from Step 3.d and click **Next**.

Refer to vSphere documentation to map CPU count recorded from Step 3.d to a vSphere configuration size.

i) On the **Select datastore** page, select these values, and click **Next**.

Setting	Value
Datastore	Datastore name
Enable thin disk mode	Selected

j) On the **Configure network settings** page, enter the values by using the information that you retrieved during the preparation for the restore, and click **Next**.

Setting	Value
Network	Name of the vSphere distributed switch portgroup
IP version	IPv4
IP assignment	static
FQDN	FQDN of the vCenter
IP address	IP address of the vCenter
Subnet mask or prefix length	24
Default gateway	Default gateway IP address

Setting	Value
---------	-------

- | | |
|-------------|---|
| DNS servers | Comma separated list of DNS server IP addresses |
|-------------|---|
- On the **Ready to complete stage 1** page, review the restore settings and click **Finish**.
 - When stage 1 of the restore process completes, click **Continue**.
- Complete the **Restore - Stage 2: vCenter** wizard.
 - On the **Introduction** page, click **Next**.
 - On the **Backup details** page, in the **Encryption password** text box, enter the encryption password of the SFTP server and click **Next**.
 - On the **Ready to complete** page, review the restore details and click **Finish**.
 - In the **Warning** dialog box, click **OK** to confirm the restore.
 - When stage 2 of the restore process completes, click **Close**.

Refresh the SSH keys that are stored in the SDDC Manager inventory. See [VMware Cloud Foundation SDDC Manager Recovery Scripts \(79004\)](#).

Validate the vCenter Instance State

After restoring a vCenter instance, you must validate the state of the vCenter and vCenter Single Sign-On.

- Log in to the vCenter interface at `https://<vcenter_fqdn>/ui` with a user assigned the Administrator role.
- In the inventory, click the management domain vCenter inventory, click the **Summary** tab, and verify that there are no unexpected vCenter alerts.
- For vCenter instances participating in a shared vCenter Single Sign-On domain, do the following:
 - Click the **Linked vCenter systems** tab and verify that the list contains all other vCenter instances in the vCenter Single Sign-On domain.
 - Log in to the recovered vCenter instance by using a Secure Shell (SSH) client.
 - Run the command to navigate to the bin directory.


```
cd /usr/lib/vmware-vmmdir/bin
```
 - Run the command to list the current replication partners of the vCenter instance with the current replication status between the nodes.


```
./vdcprepadmin -f showpartnerstatus -h localhost -u administrator -w vsphere_admin_password
```
 - Verify that for each partner, the `vdcprepadmin` command output contains `Host available: Yes, Status available: Yes, and Partner is 0 changes behind`.
 - If you observe significant differences, because the resyncing might take some time, wait five minutes and repeat this step.
- Repeat the procedure for the other vCenter instance.

Validate the SDDC Manager State After a vCenter Instance Restore

After a successful vCenter restore, verify that the SDDC Manager inventory is consistent with the recovered VMs and that the vCenter instances are healthy. You use the Supportability and Serviceability tool (SoS) and the SDDC Manager patch/upgrade precheck function.

- Log in to SDDC Manager by using a Secure Shell (SSH) client.
- Run the SoS health check and verify the output.

```
sudo /opt/vmware/sddc-support/sos --health-check
```

All tests show green when SDDC Manager is in a healthy state.

- In a Web browser, log in to SDDC Manager using the user interface.
- In the navigation pane, click **Inventory > Workload Domains**.
- For each workload domain, validate the vCenter status.
 - Click the workload domain name and click the **Updates/Patches** tab.
 - Click **Precheck**.
 - Click **View status** to review the precheck result for the vCenter instance and verify that the status is `Succeeded`.

Restore the Configuration of a vSphere Distributed Switch

To recover the configuration of a vSphere Distributed Switch, you can restore its settings from the configuration file that you previously exported.

This procedure restores only the vSphere Distributed Switch configuration of a vCenter instance.

The restore operation changes the settings on the vSphere Distributed Switch back to the settings saved in the configuration file. The operation overwrites the current settings of the vSphere Distributed Switch and its port groups. The operation does not delete existing port groups that are not a part of the configuration file.

The vSphere Distributed Switch configuration is part of the vCenter backup. If you want to restore the entire vCenter instance, see [Restoring a vCenter Instance](#).

- Log in to the vCenter interface at `https://<vcenter_fqdn>/ui` with a user assigned the Administrator role.
- Select **Menu > Networking**.
- In the inventory expand **vCenter > Datacenter**.
- Right-click the distributed switch and select **Settings > Restore configuration**.
- On the **Restore switch configuration** page, click **Browse**, navigate to the location of the configuration file for the distributed switch, and click **Open**.
- Select the **Restore distributed switch and all port groups** radio-button and click **Next**.
- On the **Ready to complete** page, review the changes and click **Finish**.
- Repeat these steps for the other vSphere Distributed Switch.
- Review the switch configuration to verify that it is as you expect after the restore.

Restore an NSX Manager Cluster Node

If an NSX Manager instance fails, you can restore it from its file-based backup.

- Verify that you have a valid file-based backup of the failed NSX Manager instance.
- Verify that you have the SFTP server details:
 - SFTP Server IP
 - SFTP Server Username
 - SFTP Server Password
 - Encryption Password
- The following components must be available before proceeding with the restore of vCenter:
 - vCenter
 - SDDC Manager

Prepare for Restoring an NSX Manager Cluster Node

Before restoring an NSX Manager node, you must retrieve the NSX Manager build number and deployment details, as well as the credentials from the SDDC Manager inventory.

Retrieve the NSX Manager Version from SDDC Manager

Before restoring a failed NSX Manager instance, you must retrieve its version from the SDDC Manager inventory.

1. In the navigation pane, click **Inventory > Workload Domains**.
2. Click the domain name of the failed NSX Manager instance.
3. Click the **Update/Patches** tab.
4. Under **Current versions**, in the **NSX** panel, locate and record the **NSX upgrade coordinator** value.
5. Verify that the NSX version retrieved from SDDC Manager is the same as the version associated with the backup file that you plan to restore.

Retrieve the Credentials for Restoring NSX Manager from SDDC Manager

Before restoring a failed NSX Manager instance, you must retrieve the NSX Manager **root** and **admin** credentials from the SDDC Manager inventory.

Before you can query the SDDC Manager API, you must obtain an API access token by using an API service account.

1. Log in to your host machine with access to the SDDC and open a terminal.
2. Obtain the API access token.
 - a) Run the command to obtain an access token by using the **admin@local** account credentials.

```
TOKEN=$(curl https://<sddc_manager_fqdn>/v1/tokens -k -X POST -H "Content-Type: application/json" -d '{"username": "admin@local", "password": "admin@local_password"}' | awk -F "\"" '{print $4}')
```

The command returns an access token and a refresh token.

- b) Record the access token.
3. Retrieve the NSX Manager **root** and **admin** credentials.
 - a) Run the command to retrieve the NSX Manager **root** and **admin** credentials.

```
curl 'https://<sddc_manager_fqdn>/v1/credentials?resourceType=NSXT_MANAGER' -k -i -X GET -H 'Accept: application/json' -H 'Authorization: Bearer $TOKEN' | json_pp
```

The command returns the NSX Manager **root** and **admin** credentials.

- b) Record the NSX Manager **root** and **admin** credentials for the instance you are restoring.

Restore the First Node of a Failed NSX Manager Cluster

If all three NSX Manager nodes in an NSX Manager cluster are in a failed state, you begin the restore process by restoring the first cluster node.

Important:

This procedure is not applicable in use cases when there are operational NSX Manager cluster nodes.

- If two of the three NSX Manager nodes in the NSX Manager cluster are in a failed state, you begin the restore process by deactivating the cluster. See [Deactivate the NSX Manager Cluster](#).
- If only one of the three NSX Manager nodes in the NSX Manager cluster is in a failed state, you directly restore the failed node to the cluster. See [Restore an NSX Manager Node to an Existing NSX Manager Cluster](#).

Redeploy the First Node of a Failed NSX Manager Cluster

You deploy a new NSX Manager instance by using the configuration of the first NSX Manager cluster node.

- Download the NSX Manager OVA file for the version of the failed NSX Manager cluster. See [Retrieve the NSX Manager Version from SDDC Manager](#).
- Verify that the backup file that you plan to restore is associated with the version of the failed NSX Manager cluster.

1. In a web browser, log in to the management domain vCenter by using the vSphere Client (https://<vcenter_server_fqdn>/ui).
2. Select **Menu > VMs and Templates**.
3. In the inventory, expand **vCenter > Datacenter**.
4. Right-click the NSX folder and select **Deploy OVF Template**.
5. On the **Select an OVF template** page, select **Local file**, click **Upload files**, navigate to the location of the NSX Manager OVA file, click **Open**, and click **Next**.
6. On the **Select a name and folder** page, enter the VM name and click **Next**.
7. On the **Select a compute resource** page, click **Next**.
8. On the **Review details** page, click **Next**.
9. On the **Configuration** page, select the appropriate size and click **Next**.
For the management domain, select **Medium** and for workload domains, select **Large** unless you changed these defaults during deployment.
10. On the **Select storage** page, select the vSAN datastore, and click **Next**.
11. On the **Select networks** page, from the **Destination network** drop-down menu, select the management network distributed port group, and click **Next**.
12. On the **Customize template** page, enter these values and click **Next**.

Setting	Value for first NSX Manager cluster node
System root user password	<i>nsx_root_password</i>
CLI admin user password	<i>nsx_admin_password</i>
CLI audit user password	<i>nsx_audit_password</i>
Hostname	Enter hostname for the appliance using FQDN format.
Rolename	NSX Manager
Management network IPv4 address	Enter the IP Address for the appliance.
Management network netmask	Enter the subnet mask for the appliance.
Default IPv4 gateway	Enter the default gateway for the appliance.
DNS server list	Enter the DNS servers for the appliance.
NTP server list	Enter the NTP server for the appliance.
Enable SSH	Deselected
Allow root SSH logins	Selected

13. On the **Ready to complete** page, review the deployment details and click **Finish**.

Restore the First Node in a Failed NSX Manager Cluster from a File-Based Backup

You restore the file-based backup of the first NSX Manager cluster node to the newly deployed NSX Manager instance.

1. In a web browser, log in to the NSX Manager node for the domain by using the user interface (https://<nsx_manager_node_fqdn>/login.jsp?local=true)
2. Accept the EULA and click **Continue**.
3. On the **Customer Experience Improvement Program** screen, select **Join the VMware Customer Experience Improvement Program** and click **Save**.
4. On the **Welcome to NSX** quick tour page, click **Skip**.
5. On the main navigation bar, click **System**.
6. In the left navigation pane, under **Lifecycle management**, click **Backup and Restore**.
7. In the **NSX configuration** pane, under **SFTP server**, click **Edit**.
8. In the **Backup configuration** dialog box, enter these values, and click **Save**.

Setting	Value
FQDN or IP address	IP address of SFTP server
Protocol	SFTP
Port	22
Directory path	/backups
Username	Service account user name For example, svc-vcf-bck@rainpole.io
Password	<i>service_account_password</i>
SSH fingerprint	<i>SFTP_ssh_fingerprint</i>

9. Under **Backup History**, select the target backup, and click **Restore**.
10. On the **Restoring NSX Managers** dialog box, click **Continue**.

Note:

The NSX Manager services will restart during the restore. You will need to refresh the NSX Manager UI.

11. During the restore, when prompted, reject adding NSX Manager nodes by clicking **I understand** and **Resume**.

A progress bar displays the status of the restore operation with the current step of the process.

Validate the Status of the First NSX Manager Cluster Node

After you restored the first NSX Manager cluster node, you validate the services state from the VM Web console of the restored node.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu > VMs and Templates**.
3. In the inventory expand **vCenter > Datacenter > NSX Folder**.
4. Click the VM name of the newly deployed first NSX Manager cluster node, click **Launch Web Console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	<i>nsx_admin_password</i>

5. Run the command to view the cluster status.

```
get cluster status
```

The services on the single-node NSX Manager cluster appear as UP.

Deactivate the NSX Manager Cluster

If two of the three NSX Manager cluster nodes are in a failed state or if you restored the first node of a failed NSX Manager cluster, you must deactivate the cluster.

Important:

This procedure is not applicable in use cases when there are two operational NSX Manager cluster nodes.

If only one of the three NSX Manager nodes in the NSX Manager cluster is in a failed state, after you prepared for the restore, you directly restore the failed node to the cluster. See [Restore an NSX Manager Node to an Existing NSX Manager Cluster](#).

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu > VMs and Templates**.
3. In the inventory expand **vCenter > Datacenter > NSX Folder**.
4. Click the VM of the operational NSX Manager node in the cluster, click **Launch Web Console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	<i>nsx_admin_password</i>

5. Run the command to deactivate the cluster

```
deactivate cluster
```

6. On the **Are you sure you want to remove all other nodes from this cluster? (yes/no)** prompt, enter *yes*.

You deactivated the cluster.

Power off and delete the two failed NSX Manager nodes from inventory.

Restore an NSX Manager Node to an Existing NSX Manager Cluster

If only one of the three NSX Manager cluster nodes is in a failed state, you restore the failed node to the existing cluster. If two of the three NSX Manager cluster nodes are in a failed state, you repeat this process for each of the failed nodes.

Detach the Failed NSX Manager Node from the NSX Manager Cluster

Before you recover a failed NSX Manager node, you must detach the failed node from the NSX Manager cluster.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu > VMs and Templates**.
3. In the inventory expand **vCenter > Datacenter > NSX Folder**.
4. Click the VM of an operational NSX Manager node in the cluster, click **Launch Web Console**, and log in by using administrator credentials.

Setting	Value
User name	admin

Setting	Value
Password	<i>nsx_admin_password</i>

5. Retrieve the UUID of the failed NSX Manager node.
 - a) Run the command to view the details of the cluster members.

```
get cluster status
```

The status of the failed node is `Down`.

- b) Record the UUID of the failed NSX Manager node.

6. Run the command to detach the failed node from the cluster

```
detach node failed_node_uuid
```

The detach process might take some time.

7. When the detaching process finishes, run the command to view the cluster status.

```
get cluster status
```

The status of all cluster nodes is `Up`.

Redeploy the Failed NSX Manager Node

You deploy a new NSX Manager instance by using the configuration of the failed node.

Download the NSX Manager OVA file for the version of the failed NSX Manager instance. See [Retrieve the NSX Manager Version from SDDC Manager](#).

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu > VMs and Templates**.
3. In the inventory expand **vCenter > Datacenter**.
4. Right-click the NSX folder and select **Deploy OVF Template**.
5. On the **Select an OVF template** page, select **Local file**, click **Upload files**, navigate to the location of the NSX Manager OVA file, click **Open**, and click **Next**.
6. On the **Select a name and folder** page, in the **Virtual machine name** text box, enter VM name of the failed node, and click **Next**.
7. On the **Select a compute resource** page, click **Next**.
8. On the **Review details** page, click **Next**.
9. On the **Configuration** page, select **Medium**, and click **Next**.
10. On the **Select storage** page, select the vSAN datastore, and click **Next**.
11. On the **Select networks** page, from the **Destination network** drop-down menu, select the management network distributed port group, and click **Next**.
12. On the **Customize template** page, enter these values and click **Next**.

Setting	Value
System root user password	<i>nsx_root_password</i>
CLI admin user password	<i>nsx_admin_password</i>
CLI audit password	<i>nsx_audit_password</i>
Hostname	<i>failed_node_FQDN</i>

Setting	Value
Default IPv4 gateway	Enter the default gateway for the appliance.
Management network IPv4 address	<i>failed_node_IP_address</i>
Management network netmask	Enter the subnet mask for the appliance.
DNS server list	Enter the DNS servers for the appliance.
NTP servers list	Enter the NTP services for the appliance.
Enable SSH	Deselected
Allow root SSH logins	Selected

13. On the **Ready to complete** page, review the deployment details and click **Finish**.

The NSX Manager virtual machine begins to deploy.

Join the New NSX Manager Node to the NSX Manager Cluster

You join the newly deployed NSX Manager node to the cluster by using the virtual machine web console from the vSphere Client.

1. In a web browser, log in to the management domain vCenter by using the vSphere Client (https://<vcenter_server_fqdn>/ui).
2. Select **Menu > VMs and Templates**.
3. In the inventory expand **vCenter > Datacenter > NSX Folder**.
4. Click the VM of an operational NSX Manager node in the cluster, click **Launch web console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	<i>nsx_admin_password</i>

5. Retrieve the ID of the NSX Manager cluster.
 - a) Run the command to view the cluster ID.

```
get cluster config | find Id:
```

 - b) Record the cluster ID.
6. Retrieve the API thumbprint of the NSX Manager API certificate.
 - a) Run the command to view the certificate API thumbprint.

```
get certificate api thumbprint
```

 - b) Record the certificate API thumbprint.
7. Exit the VM Web console.
8. In the vSphere Client, click the VM of the newly deployed NSX Manager node, click **Launch Web console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	<i>nsx_admin_password</i>

9. Run the command to join the new NSX Manager node to the cluster.

```
join existing_node_ip cluster-id cluster_id thumbprint api_thumbprint username admin
```


10. At the prompt, **Data on this node will be lost. Are you sure?**, enter **yes** and press enter.

11. Enter the password for the API user and press enter.

The node will join the cluster and restart all services.

Validate the Status of the NSX Manager Cluster

After you added the new NSX Manager node to the cluster, you must validate the operational state of the NSX Manager cluster.

To view the state of the NSX Manager cluster, you log in to the NSX Manager for the particular domain.

1. In a web browser, log in to the NSX Manager cluster for the domain by using the user interface (https://<nsx_manager_cluster_fqdn>/login.jsp?local=true)
2. On the main navigation bar, click **System**.
3. In the left pane, under **Configuration**, click **Appliances**.
4. Verify that the **Cluster** status is green and **Stable** and that each cluster node is **Available**.

Important:

It can take some time for the cluster to reach a stable state after adding a new node.

Validate the Status of the NSX Manager Cluster

After restoring an NSX Manager node, you must validate the system status of the NSX Manager cluster.

To view the system status of the NSX Manager cluster, you log in to the NSX Manager for the particular domain.

1. In a web browser, log in to the NSX Manager cluster for the domain by using the user interface (https://<nsx_manager_cluster_fqdn>/login.jsp?local=true)
2. On the **Home** page, click **Monitoring Dashboards > System**.
3. Verify that all components are healthy.
4. If the host transport nodes are in a **Pending** state, run **Configure NSX** on these nodes to refresh the UI.

Refresh the SSH keys that are stored in the SDDC Manager inventory. See [VMware Cloud Foundation SDDC Manager Recovery Scripts \(79004\)](#).

Update or Recreate the VM Anti-Affinity Rule for the NSX Manager Cluster Nodes

During the NSX Manager bring-up process, SDDC Manager creates a VM anti-affinity rule to prevent the VMs of the NSX Manager cluster from running on the same ESX host. If you redeployed all NSX Manager cluster nodes, you must recreate this rule. If you redeployed one or two nodes of the cluster, you must add the new VMs to the existing rule.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu > Hosts and Clusters**.
3. In the inventory expand **vCenter > Datacenter**.
4. Click the cluster object.
5. Click the **Configure** tab and click **VM/Host Rules**.
6. Update or recreate the VM anti-affinity rule.
 - If you redeployed one or two nodes of the cluster, add the new VMs to the existing rule.
 - a. Click the VM anti-affinity rule name and click **Edit**.
 - b. Click **Add VM/Host rule member**, select the new NSX Manager cluster nodes, and click **Add**.
 - If you redeployed all NSX Manager cluster nodes, click **Add VM/Host rule**, enter these values to create the rule, and click **OK**.

Setting	Value
Name	Enter the name of the anti-affinity rule
Type	Separate virtual machines
Members	Click Add VM/Host rule member , select the NSX Manager cluster nodes, and click Add .

Validate the SDDC Manager Inventory State

After a successful restore of an NSX Manager cluster, you must verify that the SDDC Manager inventory is consistent with the recovered virtual machines. You run this verification by using the `sos` tool.

1. Log in to SDDC Manager by using a Secure Shell (SSH).
2. Verify the SDDC Manager health.
 - a) Run the command to view the details about the VMware Cloud Foundation system.


```
sudo /opt/vmware/sddc-support/sos --get-vcf-summary
```
 - b) When prompted, enter the `vcf_password`.

All tests show green state.
3. Run the command to collect the log files from the restore of the NSX Manager cluster.


```
sudo /opt/vmware/sddc-support/sos --domain-name domain_name --nsx-logs
```

Refresh the SSH keys that are stored in the SDDC Manager inventory. See [VMware Cloud Foundation SDDC Manager Recovery Scripts \(79004\)](#).

Restoring NSX Edge Cluster Nodes

If one or both NSX Edge cluster nodes fail due to a hardware or software issue, you must redeploy the failed NSX Edge instances. You do not restore the NSX Edge nodes from a backup.

Redeploy an NSX Edge Cluster for VCF Instance Backup and Restore

To restore an NSX edge cluster, use the NSX API to redeploy the edge nodes.

Procedure

1. Retrieve the edge node configuration by running the following API call and copy the output payload of this API.

```
GET /<NSX-Manager-IPaddress>/api/v1/transport-nodes/<edgenode_id>

{"resource_type": "EdgeNode",
  "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
  "display_name": "Edge_TN2",
  "description": "EN",
  "external_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
  "ip_addresses": [
    "10.170.94.240"
  ],
  "_create_user": "admin",
  "_create_time": 1600106319056,
  "_last_modified_user": "admin",
  "_last_modified_time": 1600106907312,
  "_system_owned": false,
  "_protection": "NOT_PROTECTED",
  "_revision": 2
},
{"is_overridden": false,
"failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
"resource_type": "TransportNode",
"id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
"display_name": "Edge_TN2",
"_create_user": "admin",
"_create_time": 1600106319399,
"_last_modified_user": "admin",
"_last_modified_time": 1600106907401,
"_system_owned": false,
"_protection": "NOT_PROTECTED",
"_revision": 1
}
```

2. Redeploy the edge node using the following API call, passing the JSON data retrieved in Step 2 as the body. You do not need to pass any passwords in the JSON file.
- ```
POST /<NSX-Manager-IPaddress>/api/v1/transport-nodes/<edgenode_id>?action=redeploy
```
3. Repeat steps 1-2 for the remaining edge cluster nodes.
4. In NSX Manager, monitor the Configuration Status of the new NSX Edge nodes, until they show **Success**.

Note:

If you encounter an error **Edge redeploy is blocked as an active alarm is found for Edge VM present in NSX Inventory but missing in vCenter**, navigate to the **Alarms** section in the relevant NSX Manager UI, select all relevant alarms whose **Event Type** is **Edge VM Present In NSX Inventory Not Present in vCenter** and from the vertical ellipsis menu, select **Resolve**. Then, retry the redeploy operation.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore

After completing all NSX Edge node redeployments, you must verify the state of the NSX Edge cluster nodes.

1. In a Web browser, log in to NSX Manager for the domain by using the user interface.
2. On the main navigation bar, click **System**.
3. In the left pane, under **Configuration**, click **Fabric > Nodes**.
4. Click the **Edge transport nodes** tab.
5. Verify that all edge transport nodes show these values.

| Setting             | Value   |
|---------------------|---------|
| Configuration state | Success |
| Node status         | Up      |
| Tunnels             | Up      |

Image-Based Backup for VCF Operations

To back up VCF Operations, you create full virtual machine image-based backup jobs by using your VMware vSphere Storage APIs - Data Protection compatible backup solution.

To ensure that all management components are backed up correctly, you must create backup jobs that capture the state of a set of related components at a common point in time. With some components, simultaneous backups of the component nodes ensure that you can restore the component to a state where the nodes are logically consistent with each other and eliminate the necessity for further logical integrity remediation of the component.

| Component                       | Deployment Model  | Nodes                                                                                                                                  | Quiescing | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VCF Operations fleet management | Single appliance  | Fleet management appliance                                                                                                             | Enabled   |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| VCF Operations                  | Simple            | <ul style="list-style-type: none"><li>• Primary Node</li><li>• Collector(s)</li></ul>                                                  | Disabled  | <ul style="list-style-type: none"><li>• You must back up all cluster nodes at the same time.</li><li>• It is recommend to take the cluster offline during backup. If the cluster cannot be taken offline, do not quiesce the file system.</li><li>• Do not perform a backup when dynamic threshold (DT) calculations are running because the backup might lead to performance problems or loss of nodes. By default, DT calculations run at 2 a.m.</li></ul> |
|                                 | High Availability | <ul style="list-style-type: none"><li>• Primary Node</li><li>• Replica Node(s)</li><li>• Data Node(s)</li><li>• Collector(s)</li></ul> | Disabled  |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| VCF Operations for logs         | Simple            | Primary Node                                                                                                                           | Enabled   |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                 | High Availability | <ul style="list-style-type: none"><li>• Primary Node</li><li>• Data Node(s)</li></ul>                                                  | Enabled   |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| VCF Operations for networks     | Simple            | <ul style="list-style-type: none"><li>• Platform Node</li><li>• Collector(s)</li></ul>                                                 | Enabled   |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                 | High Availability | <ul style="list-style-type: none"><li>• Platform Nodes</li><li>• Collector(s)</li></ul>                                                | Enabled   |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

1. Log in to your backup solution by using a set of credentials with permissions to create backup jobs.
2. Create backup jobs by using the configurations from the table above.
3. Schedule the backups to run on a daily basis.
4. Retain each backup for a minimum of three days.
5. Perform an initial on-demand backup to ensure the backups complete successfully.

## Image-Based Restore for VCF Operations

The objective of this procedure is to provide a restore of VCF Operations simple or high availability deployment by using an image-based backup tool.

Prerequisites:

- Backup Data Available: Ensure that you have a recent backup of your VCF Operations cluster.
- VCF Operations Cluster Configuration: Ensure you know the details of your VCF Operations cluster configuration, such as cluster nodes, user credentials.
- Backup Software: Depending on the backup solution you're using, ensure it's compatible with your VCF Operations version.

**Table 949: VCF Operations Failure Scenarios**

| Component                       | Deployment Model  | Failure Scenario  | Recovery Process                             |
|---------------------------------|-------------------|-------------------|----------------------------------------------|
| VCF Operations fleet management | Single appliance  | Single node lost  | Restore image-based backup                   |
| VCF Operations                  | Simple            | Primary node lost | Restore primary node from image-based backup |
|                                 |                   | Collector lost    | Restore collector from image-based backup    |
|                                 | High Availability | Primary node lost | Restore primary node from image-based backup |
|                                 |                   | Replica node lost | Restore replica node from image-based backup |
|                                 |                   | Data node lost    | Restore data node from image-based backup    |
|                                 |                   | Collector lost    | Restore collector from image-based backup    |

1. If there are existing VCF Operations virtual machines, power them off and rename or delete.
2. Access your backup software interface.
3. Locate the backup taken of the VCF Operations nodes.
4. Initiate the restore process.

## Verify VCF Operations State

After a successful restore of VCF Operations, you verify the state of the analytics cluster and each integration adapter

The cluster should be online and all integration adapters should be in a collecting state.

1. Verify the VCF Operations cluster state.
  - a) Log in to the VCF Operations admin console at [https://vcf\\_ops\\_primary-node-fqdn/admin/](https://vcf_ops_primary-node-fqdn/admin/) with the admin user.
  - b) On the system status page, verify that all nodes are in a **Running** state and their status is **Online**.
  - c) Navigate to Cloud Proxies and verify that the **Health Status** for each cloud proxy is **Healthy**.
2. Verify the VCF Operations integration adapters state.
  - a) Log in to the VCF Operations interface at [https://vcf\\_ops\\_primary-node-fqdn/](https://vcf_ops_primary-node-fqdn/) as a user assigned Administrator role.
  - b) Navigate to **Administration > Integrations**.
  - c) Expand each integration and ensure their status is **OK**.

## Configure SFTP Backup Target in VCF Operations

To support backup and restore of fleet-level components, you configure an SFTP target in VCF Operations.

Verify that no VCF Automation or VCF Identity Broker Day-N operations are in progress. Running tasks in parallel can cause the SFTP configuration to fail.

1. Log in to the VCF Operations interface at [https://<vcf\\_operations\\_fqdn>](https://<vcf_operations_fqdn>) as a user assigned Administrator role.
2. Navigate to **Fleet Management > Lifecycle**.
3. In the Lifecycle navigation, select **VCF Management**.
4. On the VCF Management page, select the **Settings** tab.
5. Under Backup & Restore select **SFTP Settings**. Enter the following information, click **Fetch Fingerprint**, and click **Save**.

**Table 950: SFTP Configuration**

| Configuration         | Value                                                                                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Host FQDN or IP       | The FQDN or IP Address of the SFTP server.                                                                                                                                                                          |
| Port                  | 22                                                                                                                                                                                                                  |
| Transfer Protocol     | SFTP                                                                                                                                                                                                                |
| Username              | A service account with privileges to the SFTP server.<br>For example: <code>svc-vcf-bck</code> .                                                                                                                    |
| Password              | The password for the username provided.<br><b>Attention:</b><br>Click the + icon to create the password for the backup user and then click Select Password to select the newly created password.                    |
| Backup Directory      | The directory on the SFTP server where backups are saved.<br>For example: <code>/backups/</code> .<br><b>Important:</b><br>The SFTP user must have read access to all sub-directories of the backup directory path. |
| Encryption Passphrase | The encryption passphrase used to encrypt the backup data.<br><b>Attention:</b><br>Click the + icon to create the encryption passphrase and then click Select Passphrase to select the newly created passphrase.    |

| Configuration | Value                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------|
|               | <b>Note:</b><br>The encryption passphrase should be stored safely as it is required during the restore process. |

**Note:**

Before saving, ensure that the SFTP host, username, and password are correct and validated. Any incorrect information that is entered will propagate to VCF Identity Broker and cause the SFTP configuration to fail.

The SFTP server configuration is automatically pushed to the VCF Automation and VCF Identity Broker instances. In the SFTP Status table below the SFTP Settings settings, the configuration status for each component appears.

**Note:**

If there is a failure, an option to **Retry** appears. Click **Refresh** to fetch the current status of the SFTP configuration and display it in the UI. For additional troubleshooting, see the log file `/var/log/vrlcm/vmware_vrlcm.log` on the fleet management appliance.

## Configure Backup Schedules in VCF Fleet Management

You configure a backup schedule per component to ensure regular backups of VCF Operations fleet management, VCF Automation and VCF Identity Broker.

- Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned Administrator role.
- Navigate to **Fleet Management > Lifecycle**.
- In the Lifecycle navigation, select **VCF Management**.
- On the VCF Management page, select the **Settings** tab.
- Under Backup & Restore select **SFTP Settings**.
- Set a backup schedule and retention for each component.
  - In the action column, click **Edit**.
  - Choose the hour of the day when backups will be taken.
  - Click **Enable retention policy**, if you want to choose how many backups should be kept.

**Note:**

If retention policy is disabled all backups will be kept.

- Click **Save**.

Monitor the task from the task page. If the task fails at synthetic checker for any reason, you can skip that synthetic check execution stage and proceed further on the configuration scheduled info. Parallel scheduled configuration are not supported.

- Repeat for all other deployed components.

## On-Demand Backup of VCF Automation

Before performing maintenance activities, such as upgrades, you should take an on-demand backup of the VCF Automation.

- Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned Administrator role.
- Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
- Under components, click VCF Automation.
- On the component summary page, click the horizontal elipsis and click **Backup & Restore > Backup**.
- On the confirm SFTP page, click **yes** and click **Backup**.

## Restore VCF Automation

The following process can be used to restore a VCF Automation instance in the event of a failure.

- The VCF Automation cluster virtual machines must be in a running state.
- The fleet-level SFTP server, that is configured in fleet management, must be available.
- You must have at least one recent backup.
- The following components must be available before proceeding with the restore of VCF Automation:
  - vCenter
  - VCF Operations
  - VCF Operations fleet management

A list of potential failure scenarios and corresponding recovery processes.

**Table 951: VCF Automation Failure Scenarios**

| Failure Scenario                      | Recovery Process                                                                                                                                                        | Additional Information |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| VCF Automation application corruption | <ul style="list-style-type: none"> <li>Restore from SFTP to existing nodes</li> </ul>                                                                                   |                        |
| Single node lost                      | <ul style="list-style-type: none"> <li>No action required</li> <li>Failed node will be automatically redeployed</li> </ul>                                              |                        |
| Two node lost                         | <ul style="list-style-type: none"> <li>Delete VCF Automation component</li> <li>Deploy a new VCF Automation Instance</li> <li>Restore from SFTP to new nodes</li> </ul> |                        |
| Three nodes lost                      | <ul style="list-style-type: none"> <li>Delete VCF Automation component</li> <li>Deploy a new VCF Automation Instance</li> <li>Restore from SFTP to new nodes</li> </ul> |                        |

- Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned Administrator role.
- Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
- Under components, click VCF Automation.
- On the component summary page, click the horizontal elipsis and click **Backup & Restore > Restore**.
- On the Restore pane, select an option to restore VCF Automation from a specific backup or the latest backup.
  - Restore from the latest backup on the cluster.
  - To restore from a full backup, pass the path to the backup folder using `vcf/backups/cluster-name/version/component-name/timestamp`.
  - To restore from an incremental backup, pass the path to the backup using `vcf/backups/cluster-name/version/component-name/timestamp/<full-backup-file-path.tgz>`.

- Click **Restore**.
- Monitor the restore task from the **Tasks** pane.

## Remediate Passwords for VMware Cloud Foundation Components After a VCF Automation Restore

After a successful restore of VCF Automation, some service accounts may become disconnected.

To manually remediate the VCF Automation service account in vCenter and NSX, use the VCF Automation UI.

- Log in to the VCF Automation interface at [https://<vcf\\_automation\\_fqdn>](https://<vcf_automation_fqdn>) with a user assigned the **System Administrator** role.
- In the left pane, navigate to **VCF Instances**.
- On the **VCF Instance** page, click the vertical ellipsis next to the VCF instance, and, from the drop-down menu, select **Rotate service account**.

VCF Automation deletes and recreates the service account. This task may take a few minutes to complete.

## On-Demand Backup of a VCF Identity Broker Appliance Cluster

Before performing maintenance activities, such as upgrades, you should take an on-demand backup of the VCF Identity Broker cluster.

- Log in to the VCF Operations interface at [https://<vcf\\_operations\\_fqdn>](https://<vcf_operations_fqdn>) as a user assigned Administrator role.
- Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
- Under components, click VCF Identity Broker.
- On the component summary page, click the horizontal ellipsis and click **Backup & Restore > Backup**.
- On the confirm SFTP page, click **yes** and click **Backup**.

## Restore a VCF Identity Broker Appliance Cluster

The following process can be used to restore a VCF Identity Broker cluster in the event of a failure.

- The VCF Identity Broker appliances must be in a running state.
- The fleet-level SFTP server, that is configured in fleet management, must be available.
- You must have at least one recent backup.
- The following components must be available before proceeding with the restore of VCF Identity Broker:
  - vCenter
  - VCF Operations
  - VCF Operations fleet management

A list of potential failure scenarios and corresponding recovery processes.

**Table 952: Failure Scenarios for a VCF Identity Broker Appliance Cluster**

| Failure Scenario                           | Recovery Process                                                                                                           |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| VCF Identity Broker application corruption | <ul style="list-style-type: none"> <li>Restore from SFTP to existing nodes</li> </ul>                                      |
| Single node lost                           | <ul style="list-style-type: none"> <li>No action required</li> <li>Failed node will be automatically redeployed</li> </ul> |
| Two nodes lost                             | <ul style="list-style-type: none"> <li>Deploy a new VCF Identity Broker</li> </ul>                                         |

| Failure Scenario | Recovery Process                                                                                                           |
|------------------|----------------------------------------------------------------------------------------------------------------------------|
|                  | <ul style="list-style-type: none"> <li>Restore from SFTP to new nodes</li> </ul>                                           |
| Three nodes lost | <ul style="list-style-type: none"> <li>Deploy a new VCF Identity Broker</li> <li>Restore from SFTP to new nodes</li> </ul> |

- Log in to the VCF Operations interface at [https://<vcf\\_operations\\_fqdn>](https://<vcf_operations_fqdn>) as a user assigned **Administrator** role.
- Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
- Under **Components**, click VCF Identity Broker.
- On the component summary page, click the horizontal ellipsis and click **Backup & Restore > Restore**.
- On the Restore pane, select an option to restore VCF Identity Broker from a specific backup or the latest backup.
  - Restore from the latest backup on the cluster
  - Pass the backup path using `vcf/backups/cluster-name/version/component-name/timestamp`
  - Pass the backup path using `vcf/backups/cluster-name/version/component-name/timestamp/<full-backup-file-path.tgz>`
- Click **Restore**.
- Monitor the restore task from the **Tasks** pane.

## Verify VCF Identity Broker State

After performing a restore of VCF Identity Broker, you verify its state.

- Log in to the VCF Operations interface at [https://<vcf\\_operations\\_fqdn>](https://<vcf_operations_fqdn>) as a user assigned Administrator role.
- Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
- Under components, click **VCF Identity Broker**.
- On the component summary page, click the horizontal ellipsis and click **Trigger inventory Sync**.
- On the **Trigger inventory Sync** page, click **Submit**.
- Monitor the inventory sync task on the **Tasks** tab.

## VMware Cloud Foundation Instance Backup and Restore

*VMware Cloud Foundation Instance Backup and Restore* provides guidance on restoring a VCF Instance by performing a complete reconstruction from a backup.

This guidance contains detailed instructions for both the management domain and workload domains where you must restore all components.

### Example Failure Scenarios

You may need to restore all components in a VCF Instance if one of the following scenarios occurs:

- Complete VCF Instance failure
- Malware or ransomware attack
- Datastore corruption

### Supported Topologies

Several topologies of VCF exist according to the number of availability zones and VCF Instances. See [Design](#).

The following topologies are supported for restore using this guidance.

| Supported Topology                                                                                                                                                                                                                                                                                                                             | Description                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Single VCF fleet</li> <li>Workload domains with a single or multiple vSphere clusters</li> <li>vSphere clusters with a single or multiple vSphere Distributed Switches</li> <li>vSphere Supervisor with a single or multiple zones</li> <li>vSphere content libraries running on NFS storage</li> </ul> | Workload domains deployed in a single availability zone. |

### VCF Components to Back Up

All important data in a VCF system should be backed up to a remote backup site. Include all VMs in the management domain and VMs in workload domains that require data protection.

After initial deployment, the management domain contains a core set of VMs to manage the VCF system. When you deploy a workload domain, VCF deploys additional VMs to manage the workload domain.

#### Note:

If you have other VMs deployed in the management domain that require backup, for example, Microsoft SQL servers, Microsoft Active Directory servers, backup software VMs, and so on, plan to back them up too.

This guidance does not provide information about backing up VMs in workload domains, but your backup plan should also identify and back up critical VMs in workload domains.

### Documenting the System Configuration of VCF

Keeping detailed as-built documentation on the system configuration eases the restore process if a failure in your VCF system occurs.

While the processes in this guidance retrieve much of the information below, keep a record of the following items. Save this information on a secure secondary storage.

- Topology diagrams of the VCF system
  - Physical networking
  - vSphere Distributed Switch networking
  - NSX networking
  - Workload domain configuration
  - Cluster configuration for each cluster in a workload domain
  - ESX hosts assigned to each cluster
  - vSphere Supervisor configuration
- Networking information
  - ESX host vmnic-to-switch port mappings
  - VM virtual NIC to distributed port group mappings
  - IP address information of the VMkernel interfaces on the ESX hosts
  - IP address information of VMs
  - DNS, NTP, AD, and other well-known services used by the VCF system

### PowerShell Automation for This Guidance

You can perform the procedures in this guide manually or by running cmdlets in an associated PowerShell module.

The PowerShell cmdlets are available in an open-source module as code-based alternatives to completing certain procedures in each SDDC component's user interface. For more information on the PowerShell cmdlets for restore of VMware Cloud Foundation, see the [VMware.CloudFoundation.InstanceRecovery open-source project in GitHub](#).

**Table 953: Prerequisites for CLI Implementation of VMware Cloud Foundation Restore**

| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Microsoft Windows 2022</li> <li>Microsoft PowerShell 7.x</li> <li><a href="#">VMware OVF Tool</a></li> <li>OpenSSL lite version 1.0.2g or higher installed and added to the Windows PATH system variable.</li> <li>PowerCLI <b>DefaultVlServerMode</b> setting must be configured as <b>Multiple</b> for the all <b>AllUsers</b> scope</li> <li>Install the supporting modules from the PowerShell Gallery by running the following commands:           <pre>Install-Module -Name VCF.PowerCLI -MinimumVersion 9.0.0 -Scope AllUsers -Force Install-Module -Name Posh-SSH -RequiredVersion 3.0.8 -Scope AllUsers -Force Install-Module -Name VMware.CloudFoundation.InstanceRecovery -RequiredVersion 9.0.2.1002 -Scope AllUsers -Force</pre> </li> <li>Open a new PowerShell console and run the following command to confirm all the prerequisites are installed.           <pre>Confirm-VCFInstanceRecoveryPreReqs</pre> </li> </ul> |

### Sequence Optimization for Restore of a VCF Instance

When restoring an entire VCF Instance it is possible to run certain tasks in parallel to minimize the time required to restore it.

#### Example Restore Timeline for a VCF Instance

To minimize overall restore time of a VCF instance, you may run multiple restore plans in parallel across VCF domains with the following guidelines:

- If you do not use all of the components shown then you may ignore the associated steps and proceed to the next step in the respective restore plan.
- If you wish to prioritize workload availability over management and operational components you may do so.
- You may not reorder the overall sequence of any given restore plan as that may compromise dependencies.

#### Important:

Where multiple items exist in a single cell, they may be run in parallel. Items across a single row may also be run in parallel.

The following timeline is for a sample setup with the following configuration:

- Two VCF domains.
  - One management domain (which is also running fleet components for the entire fleet)
  - Fleet components are not on an NSX Overlay network
  - One workload domain with an isolated vCenter Single Sign-On (SSO) domain
- VCF 9.0.x only.

#### Attention:

\* If fleet components are on an NSX overlay network then the fleet component restore plan should be not be started until after the appropriate NSX Edges are redeployed.



\*\* Should there be more than one isolated workload domain, you may start the restore plan for all of them in parallel.

| Task | Management Domain Restore Plan                                                                                                                                                              | Fleet Component Restore Plan                                              | Workload Domain Restore Plan                                                                                                                  | Notes                                                                                                          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1    | Prepare for Restore                                                                                                                                                                         |                                                                           | Prepare for Restore                                                                                                                           |                                                                                                                |
| 2    | Deploy ESX Hosts                                                                                                                                                                            |                                                                           | Deploy ESX Hosts                                                                                                                              | Hosts should be deployed with the same FQDNs and IP addresses as before.                                       |
| 3    | Deploy VCF Installer                                                                                                                                                                        |                                                                           |                                                                                                                                               | Can be done in advance, but correct binaries need to be staged and kept up to date as environment is upgraded. |
| 4    | Prepare for Partial VCF Deployment                                                                                                                                                          |                                                                           |                                                                                                                                               |                                                                                                                |
| 5    | Perform Partial VCF Deployment                                                                                                                                                              |                                                                           |                                                                                                                                               |                                                                                                                |
| 6    | <ul style="list-style-type: none"> <li>Deploy vCenter from an OVA File</li> <li>Deploy the first NSX Manager Node from an OVA File</li> <li>Deploy SDDC Manager from an OVA File</li> </ul> |                                                                           | <ul style="list-style-type: none"> <li>Deploy vCenter from an OVA File</li> <li>Deploy the first NSX Manager Node from an OVA File</li> </ul> |                                                                                                                |
| 7    | <ul style="list-style-type: none"> <li>Restore vCenter</li> <li>Restore NSX Manager</li> </ul>                                                                                              |                                                                           | <ul style="list-style-type: none"> <li>Restore vCenter</li> <li>Restore NSX Manager</li> </ul>                                                |                                                                                                                |
| 8    | Restore SDDC Manager                                                                                                                                                                        |                                                                           |                                                                                                                                               |                                                                                                                |
| 9    | Update Extracted SDDC Data                                                                                                                                                                  |                                                                           | Update Extracted SDDC Data                                                                                                                    |                                                                                                                |
| 10   | Back up Cluster Configuration                                                                                                                                                               |                                                                           | Back up Cluster Configuration                                                                                                                 |                                                                                                                |
| 11   | Rebuild the vSphere Cluster                                                                                                                                                                 |                                                                           |                                                                                                                                               |                                                                                                                |
| 12   | Remediate the NSX Installation on the vSphere Cluster                                                                                                                                       | Restore VCF Operations from Image-based backup <sup>[1]</sup>             | Rebuild the vSphere Cluster <sup>[2]</sup>                                                                                                    |                                                                                                                |
| 13   | [Optional] Redeploy remaining nodes of the NSX ManagerCluster <sup>[3]</sup>                                                                                                                | Restore VCF Operations fleet management appliance from Image-based backup | Remediate NSX Installation on the vSphere Cluster                                                                                             | You can start restoring workloads that <i>do not</i> use NSX Edge nodes into your workload domain              |
| 14   | Redeploy NSX Edge Nodes                                                                                                                                                                     | Redeploy VCF Identity Broker                                              | [Optional] Redeploy remaining nodes of the NSX ManagerCluster <sup>[3]</sup>                                                                  |                                                                                                                |
| 15   | [Conditional] Redeploy remaining nodes of the NSX ManagerCluster <sup>[4]</sup>                                                                                                             | Restore VCF Identity Broker from file-based backup                        | Redeploy NSX Edge Nodes                                                                                                                       |                                                                                                                |

| Task | Management Domain Restore Plan            | Fleet Component Restore Plan                  | Workload Domain Restore Plan                                                    | Notes                                                                                         |
|------|-------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 16   |                                           | Redeploy VCF Automation                       | [Conditional] Redeploy remaining nodes of the NSX ManagerCluster <sup>[4]</sup> | You can start restoring workloads that <i>do</i> use NSX Edge nodes into your workload domain |
| 17   |                                           | Restore VCF Automation from file-based backup | Restore Supervisor Control Plane                                                | If present in the domain                                                                      |
| 18   | Perform Post-Restore Tasks <sup>[5]</sup> |                                               | Restore vSphere Pods                                                            | If present in the domain                                                                      |
| 19   |                                           |                                               | Restore Stateless and Stateful Workloads on a VKS cluster                       | If present in the domain                                                                      |
| 20   |                                           |                                               | Restore VM Service VMs                                                          | If present in the domain                                                                      |
|      |                                           |                                               | Perform Post-Restore Tasks <sup>[5]</sup>                                       |                                                                                               |

<sup>[1]</sup> Dependent on the restore of the primary vSphere cluster in the Management domain

<sup>[2]</sup> Dependent on the restore of SDDC Manager being complete

<sup>[3]</sup> If high availability of the NSX Manager nodes is considered less important than time to make workloads available, you can perform this task later.

<sup>[4]</sup> Perform this task if you skipped the deployment of the additional NSX Manager nodes earlier.

<sup>[5]</sup> Restore vSphere DRS, VM Location, VM Overrides and VM Tags only after all workload VMs are restored to the cluster.

#### Post-Restore Cleanup Plan for VMware Cloud Foundation

- Delete the VM snapshots.

### Preparing for Restore of the VCF Instance

Before you can perform a restore, you must perform some preparation tasks.

**Table 954: Preparation Tasks VCF for Restore**

| Task | Task Name                                                                                                                                                                                                                                                                     | Additional Information |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1    | Locate the most recent known good file-based backups for the following components: <ul style="list-style-type: none"> <li>SDDC Manager</li> <li>All vCenter instances</li> <li>All NSX Manager instances</li> <li>VCF Automation</li> <li>VCF Identity Broker</li> </ul>      |                        |
| 2    | Locate the most recent known good image-based backups for the following components: <ul style="list-style-type: none"> <li>VCF Operations</li> <li>VCF Operations fleet management appliance</li> <li>VCF Operations for logs</li> <li>VCF Operations for networks</li> </ul> |                        |
| 3    | <a href="#">Gather Information from SDDC Manager Backup</a>                                                                                                                                                                                                                   |                        |

| Task | Task Name                                                                                                   | Additional Information                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | Locate version and build number for vCenter, NSX Manager, and SDDC Manager.                                 | <ul style="list-style-type: none"> <li>You can see the vCenter build number in the <code>backup-metadata.a.json</code> in the vCenter file-based backup.</li> <li>You can see the NSX Manager version and build number from the file name of the backup file.</li> </ul> |
| 7    | Download the OVA files of vCenter, SDDC Manager, and NSX Manager for the required version and build number. |                                                                                                                                                                                                                                                                          |
| 8    | Download the Refresh SSH Key Script from Broadcom Knowledge Base article <a href="#">79004</a>              |                                                                                                                                                                                                                                                                          |
| 9    | Re-install ESX on the hosts.                                                                                | See <a href="#">Deploying a New VMware Cloud Foundation or VMware vSphere Foundation Platform</a> .                                                                                                                                                                      |

## Management Domain Restore Plan

Perform the following tasks to restore the VCF Management Domain. Refer to [Sequence Optimization for Restore of a VCF Instance](#) for detail on where tasks can be performed in parallel across domains.

### Management Domain Restore Plan Summary

- [Gathering Required Information](#)
- [Restore Management vCenter, NSX Manager and SDDC Manager](#)
- [Rebuild Management vSphere Cluster](#)
- [Restore Management NSX Edges](#)
- [Redeploy Remaining Management NSX Manager Nodes](#)
- [Finalize Management Domain](#)

### Gathering Required Information

| Step | Manual Task                                                     | PowerShell Equivalent                                                                                                                                                                                                                                                                                   |
|------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Gather Information from the SDDC Manager Backup</a> | <pre>## Parameter Block ## \$vcfBackupFilePath = ".\vcf-backup-sfo-vcf01-sfo-rainpole-io-2025-09-22-10-30-55.tar.gz" \$encryptionPassword = "VMw@rel!VMw@rel!"  ## Command Block ## New-ExtractDataFromSDDCBackup -vcfBackupFilePath \$vcfBackupFilePath -encryptionPassword \$encryptionPassword</pre> |

### Restore Management vCenter, NSX Manager and SDDC Manager

| Step | Manual Task                                      | PowerShell Equivalent                                                                                                                                                                                                                                                |
|------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Perform a Partial VCF Deployment</a> | <pre>## Parameter Block ## \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$vcCenterIp = "10.11.10.170" #temporary IP, previously unused \$vcCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #temporary FQDN, previously unused, must resolve to vCenterIp</pre> |

| Step | Manual Task                                                                                                    | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                | <pre>\$transportVlanId = 1114 \$vcCenterServerSize = "small" \$vcfInstaller = "sfo-ins01.sfo.rainpole.io" \$vcfInstallerAdminUser = "admin@local" \$vcfInstallerAdminUserPassword = "VMw@rel!VMw@rel!" \$partialBringupSpecFile = ".\sfo-m01-partial-bringup-spec.json"  ## Command Block ## New-VVFBasedPartialBringupJsonSpec -tempVCenterIp \$vCenterIp -tempVCenterFqdn \$vCenterFqdn -extractedSDDCDataFile \$extractedSDDCDataFile -transportVlanId \$transportVlanId -dedupEnabled \$false -vcCenterServerSize \$vcCenterServerSize  ## Command Block ## New-VVFBasedPartialBringupValidation -vcfInstaller \$vcfInstaller -vcfInstallerAdminUser \$vcfInstallerAdminUser -vcfInstallerAdminUserPassword \$vcfInstallerAdminUserPassword -partialBringupSpecFile \$partialBringupSpecFile  ## Command Block ## New-VVFBasedPartialBringup -vcfInstaller \$vcfInstaller -vcfInstallerAdminUser \$vcfInstallerAdminUser -vcfInstallerAdminUserPassword \$vcfInstallerAdminUserPassword -partialBringupSpecFile \$partialBringupSpecFile</pre> |
| 2    | <a href="#">Deploy a vCenter Appliance from an OVA File for VCF Instance Backup and Restore</a>                | <pre>## Parameter Block ## \$vcCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter created during partial bringup \$vcCenterAdmin = "administrator@vsphere.local" \$vcCenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-m01" \$restoredvCenterDeploymentSize = "small-lstorage" \$vcCenterOvaFile = "F:\OVA\VMware-vCenter-Server-Appliance-9.0.2.0.25148086_OVF10.ova"  ## Command Block ## New-vCenterOvaDeployment -vCenterFqdn \$vcCenterFqdn -vCenterAdmin \$vcCenterAdmin -vCenterAdminPassword \$vcCenterAdminPassword -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -restoredvCenterDeploymentSize \$restoredvCenterDeploymentSize -vCenterOvaFile \$vcCenterOvaFile</pre>                                                                                                                                                                                                                                           |
| 3    | <a href="#">Deploy the First NSX Manager Cluster Node from an OVA File for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$vcCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter created during partial bringup \$vcCenterAdmin = "administrator@vsphere.local" \$vcCenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-m01" \$restoredNsxManagerDeploymentSize = "medium" \$nsxManagerOvaFile = "F:\OVA\nsx-unified-appliance-9.0.2.0.25150388.ova"  ## Command Block ##</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Step | Manual Task                                                                                            | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                        | <pre>New-NSXManagerOvaDeployment -vCenterFqdn \$vcenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -restoredNsxManagerDeploymentSize \$restoredNsxManagerDeploymentSize -nsxManagerOvaFile \$nsxManagerOvaFile</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4    | <a href="#">Deploy the SDDC Manager Appliance from an OVA File for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter created during partial bringup \$vCenterAdmin = "administrator@vsphere.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$sddcManagerOvaFile="F:\OVA\VCF-SDDC-Manager-Appliance-9.0.2.0.25151285.ova" \$rootUserPassword = "VMw@rel!VMw@rel!" \$vcfUserPassword = "VMw@rel!VMw@rel!" \$localUserPassword = "VMw@rel!VMw@rel!" \$basicAuthUserPassword = "VMw@rel!VMw@rel!"  ## Command Block ## New-SDDCManagerOvaDeployment -vCenterFqdn \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -extractedSDDCDataFile \$extractedSDDCDataFile -sddcManagerOvaFile \$sddcManagerOvaFile -rootUserPassword \$rootUserPassword -vcfUserPassword \$vcfUserPassword -localUserPassword \$localUserPassword -basicAuthUserPassword \$basicAuthUserPassword</pre> |
| 5    | <a href="#">Restore vCenter from a File-Based Backup for VCF Instance Backup and Restore</a>           | <pre>## Parameter Block ## \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain ="sfo-m01" \$vCenterBackupPath = "192.168.1.1/F\$/backups/sn_sfo-m01-vc01.sfo.rainpole.io/M_9.0.2.0_20260406-084633_KBWGC5DGN5ZG2VDP-N5WHGICMFRW5LQ" \$locationtype = "SMB" \$locationUser = "Administrator" \$locationPassword = "VMw@rel123!" \$vCenterBackupPassword = "VMw@rel!VMw@rel!"  ## Command Block ## Invoke-vCenterRestore -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -vCenterBackupPath \$vCenterBackupPath -locationtype \$locationtype -locationUser \$locationUser -locationPassword \$locationPassword  <b>Important:</b> If the vCenter backup was taken using a backup password then append the following to the above command  -backupPassword \$vCenterBackupPassword</pre>                                                                                                              |
| 6    | <a href="#">Restore NSX Manager from a File-Based Backup for VCF Instance Backup and Restore</a>       | <pre>## Parameter Block ## \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain ="sfo-m01"</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Step | Manual Task                                                                                                | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                            | <pre>\$sftpServer ="192.168.1.2" \$sftpUser ="svc-vcf-bck" \$sftpPassword ="VMw@rel!VMw@rel!" \$sftpServerBackupPath ="media/backups" \$nsxBackupPassphrase ="VMw@rel!VMw@rel!"  ## Command Block ## Invoke-NSXManagerRestore -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -sftpServer \$sftpServer -sftpUser \$sftpUser -sftpPassword \$sftpPassword -sftpServerBackupPath \$sftpServerBackupPath -backupPassphrase \$nsxBackupPassphrase</pre>                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7    | <a href="#">Verify the State of the First NSX Manager Cluster Node for VCF Instance Backup and Restore</a> | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 8    | <a href="#">Update vCenter SSH Keys in SDDC Manager Backup for VCF Instance Backup and Restore</a>         | <pre>## Parameter Block ## \$rootUserPassword = "VMw@rel!VMw@rel!" \$vcfUserPassword = "VMw@rel!VMw@rel!" \$vcfBackupFilePath = "F:\backup\vvcf-backup-sfo-vcf01-sfo-rainpole-io-2025-09-22-10-30-55.tar.gz" \$encryptionPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = "F:\backup\extracted-sddc-data.json" \$vCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter created during partial bringup \$vCenterAdmin = "Administrator@vsphere.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!"  ## Command Block ## New-UploadAndModifySDDCManagerBackup -rootUserPassword \$rootUserPassword -vcfUserPassword \$vcfUserPassword -backupFilePath \$vcfBackupFilePath -encryptionPassword \$encryptionPassword -extractedSDDCDataFile \$extractedSDDCDataFile -vCenterFqdn \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword</pre> |
| 9    | <a href="#">Restore SDDC Manager from a File-Based Backup for VCF Instance Backup and Restore</a>          | <pre>## Parameter Block ## \$extractedSDDCDataFile = "F:\backup\extracted-sddc-data.json" \$vcfBackupFilePath = "F:\backup\vcf-backup-sfo-vcf01-sfo-rainpole-io-2023-11-21-10-42-38.tar.gz" \$vcfUserPassword = "VMw@rel!VMw@rel!" \$localUserPassword = "VMw@rel!VMw@rel!" \$rootUserPassword = "VMw@rel!VMw@rel!" \$encryptionPassword = "VMw@rel!VMw@rel!"  ## Command Block ## Invoke-SDDCManagerRestore -extractedSDDCDataFile \$extractedSDDCDataFile -backupFilePath \$vcfBackupFilePath -vcfUserPassword \$vcfUserPassword -localUserPassword \$localUserPassword -rootUserPassword \$rootUserPassword -encryptionPassword \$encryptionPassword</pre>                                                                                                                                                                                                                                 |
| 10   | <a href="#">Update Extracted SDDC Data for Instance Backup and Restore</a><br><b>Note:</b>                 | <pre>## Parameter Block ## \$extractedSDDCDataFile = "F:\backup\extracted-sddc-data.json" \$sddcManagerFQDN = "sfo-vcf01.sfo.rainpole.io"</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Step | Manual Task                                                                                                                                                                                       | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | Is only required when using the PowerShell Automation approach. If performing a fully manual restore, this information is gathered by <a href="#">Documenting the System Configuration of VCF</a> | <pre>\$sddcManagerAdmin = "administrator@vSphere.local" \$sddcManagerAdminPassword = "VMw@rel!VMw@rel!" \$restoredVcenterFqdn = "sfo-m01-vc01.sfo.rainpole.io"  ## Command Block ## Update-ExtractedSDCCData -extractedSDCCDataFile \$extractedSDCCDataFile -sddcManagerFQDN \$sddcManagerFQDN -sddcManagerAdmin \$sddcManagerAdmin -sddcManagerAdminPassword \$sddcManagerAdminPassword -vCenterFQDN \$restoredVcenterFqdn</pre> |

### Rebuild Management vSphere Cluster

| Step | Manual Task                                                                                               | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Record the Cluster Settings from the Restored vCenter for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$VcenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VcenterAdmin = "administrator@vSphere.local" \$VcenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-m01-cl01"  ## Command Block ## Connect-VIServer -server \$VcenterFqdn -user \$VcenterAdmin -password \$VcenterAdminPassword Backup-ClusterVMOVERRIDES -clusterName \$clusterName Backup-ClusterVMLOCATIONS -clusterName \$clusterName Backup-ClusterDRSGROUPSANDRULES -clusterName \$clusterName Backup-ClusterVMTAGS -clusterName \$clusterName Disconnect-VIServer * -confirm:\$false</pre>                                                                                                                                                                                    |
| 2    | <a href="#">Remove Non-Responsive ESX Hosts from the Inventory for VCF Instance Backup and Restore</a>    | <pre>## Parameter Block ## \$VcenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VcenterAdmin = "administrator@vSphere.local" \$VcenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-m01-cl01" \$nsxManagerFqdn = "sfo-m01-nsx01.sfo.rainpole.io" #fqdn of restored NSX Manager \$nsxManagerAdmin = "admin" \$nsxManagerAdminPassword = "VMw@rel!VMw@rel!" \$nsxManagerRootPassword = "VMw@rel!VMw@rel!"  ## Command Block ## Remove-NonResponsiveHosts -vCenterFQDN \$VcenterFqdn -vCenterAdmin \$VcenterAdmin -vCenterAdminPassword \$VcenterAdminPassword -clusterName \$clusterName -NsxManagerFQDN \$nsxManagerFqdn -NsxManagerAdmin \$nsxManagerAdmin -NsxManagerAdminPassword \$nsxManagerAdminPassword -NsxManagerRootPassword \$nsxManagerRootPassword</pre> |
| 3    | <a href="#">Configure vSAN to Ignore Cluster Member Updates for VCF Instance Backup and Restore</a>       | <pre>## Parameter Block ## \$VcenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter fqdn of temp vCenter created during partial bringup created during partial bringup</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Step | Manual Task                                                                                                            | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                        | <pre>\$VcenterAdmin = "administrator@vSphere.local" \$VcenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "vcfir-cl01" # Do Not Change \$setting = "enable" \$extractedSDCCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Set-ClusterHostsvsSanIgnoreClusterMemberList -vCenterFQDN \$VcenterFqdn -vCenterAdmin \$VcenterAdmin -vCenterAdminPassword \$VcenterAdminPassword -clusterName \$clusterName -setting \$setting -extractedSDCCDataFile \$extractedSDCCDataFile</pre>                                                                                                        |
| 4    | <a href="#">Migrate Host Networking from vSphere Distributed Switch to vSphere Standard Switch</a>                     | <pre>## Parameter Block ## \$VcenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter fqdn of temp vCenter created during partial bringup created during partial bringup \$VcenterAdmin = "administrator@vSphere.local" \$VcenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDCCDataFile = ".\extracted-sddc-data.json" \$mtu = "9000"  ## Command Block ## Move-ClusterHostNetworkingTovSS -vCenterFqdn \$VcenterFqdn -vCenterAdmin \$VcenterAdmin -vCenterAdminPassword \$VcenterAdminPassword -extractedSDCCDataFile \$extractedSDCCDataFile -mtu \$mtu</pre>                             |
| 5    | <a href="#">Prepare to Migrate the New vSphere Cluster to the Restored vCenter for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$VcenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter fqdn of temp vCenter created during partial bringup created during partial bringup \$VcenterAdmin = "administrator@vSphere.local" \$VcenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "vcfir-cl01" # Do Not Change  ## Command Block ## Connect-VIServer \$VcenterFqdn -user \$VcenterAdmin -password \$VcenterAdminPassword Set-Cluster -cluster \$clusterName -DrsAutomationLevel "Manual" -confirm:\$false Disconnect-VIServer -Server \$global:DefaultVIServers -Force -Confirm:\$false</pre> |
| 6    | <a href="#">Move All Management VMs to a Temporary Port Group on the vSphere Standard Switch</a>                       | <pre>## Parameter Block ## \$VcenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter fqdn of temp vCenter created during partial bringup created during partial bringup \$VcenterAdmin = "administrator@vSphere.local" \$VcenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDCCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Move-MgmtVmsToTempPg -vCenterFQDN \$VcenterFqdn -vCenterAdmin \$VcenterAdmin -vCenterAdminPassword \$VcenterAdminPassword -extractedSDCCDataFile \$extractedSDCCDataFile</pre>                                                                  |

| Step | Manual Task                                                                                                              | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <a href="#">Remove the ESX Hosts from the vSphere Distributed Switch on the Temporary vCenter</a>                        | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter fqdn of temp vCenter created during partial bringup created during partial bringup \$VCenterAdmin = "administrator@vsphere.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!"  ## Command Block ## Remove-ClusterHostsFromVds -vCenterFQDN \$VCenterFqdn -vCenter- Admin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8    | <a href="#">Migrate ESX Hosts from the Temporary vCenter to the Restored vCenter for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$tempvCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" \$tempvCenterAdmin = "administrator@vsphere.local" \$tempvCenterAdminPassword = "VMw@rel!VMw@rel!" \$restoredvCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" \$restoredvCenterAdmin = "administrator@vsphere.local" \$restoredvCenterAdminPassword = "VMw@rel!VMw@rel!" \$restoredClusterName = "sfo-m01-cl01" \$extractedSDDCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Move-ClusterHostsToRestoredVcenter -tempvCenterFqdn \$tempvCenter- Fqdn -tempvCenterAdmin \$tempvCenterAdmin -tempvCenterAdmin- Password \$tempvCenterAdminPassword -restoredvCenterFQDN \$re- storedvCenterFQDN -restoredvCenterAdmin \$restoredvCenterAdmin - restoredvCenterAdminPassword \$restoredvCenterAdminPassword -re- storedClusterName \$restoredClusterName -extractedSDDCDataFile \$extractedSDDCDataFile</pre>                                                                                              |
| 9    | <a href="#">Migrate the ESX Hosts and VMkernel Adapters to the vSphere Distributed Switch on the Restored vCenter</a>    | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@vsphere.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-m01-cl01" \$extractedSDDCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## New-RebuiltVdsConfiguration -vCenterFQDN \$VCenterFqdn -vCenter- Admin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword -clusterName \$clusterName -extractedSDDCDataFile \$extractedSD- DCDataFile  <b>Note:</b> At this point, the NSX installation on each host in the vSphere cluster should self-heal. Monitor the self-healing process until complete in the NSX Manager UI at <b>System &gt; Fabric &gt; Hosts &gt; Clusters</b> before proceeding. It might take several minutes for the process to initiate. If you see an error on the hosts that they are not part of the distributed switch, it just means that the self-healing process is yet to start.</pre> |
| 10   | <a href="#">Configure vSAN Back to Honour Cluster Member Updates for VCF Instance Backup and Restore</a>                 | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@vsphere.local"</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Step | Manual Task                                                              | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                          | <pre>\$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-m01-cl01" \$setting = "disable" \$extractedSDDCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Set-ClusterHostsvsanIgnoreClusterMemberList -vCenterFQDN \$VCenter- Fqdn -vCenterAdmin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword -clusterName \$clusterName -setting \$set- ting -extractedSDDCDataFile \$extractedSDDCDataFile</pre> |
| 11   | <a href="#">Update SSH Host Keys for VCF Instance Backup and Restore</a> | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                            |

### Restore Management NSX Edges

| Step | Manual Task                                                                                        | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Redeploy an NSX Edge Cluster for VCF Instance Backup and Restore</a>                   | <pre>## Parameter Block ## \$nsxManagerFqdn = "sfo-m01-nsx01.sfo.rainpole.io" #fqdn of re- stored NSX Manager \$nsxManagerAdmin = "admin" \$nsxManagerAdminPassword = "VMw@rel!VMw@rel!" \$VCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@vsphere.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-m01-cl01" \$extractedSDDCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Invoke-NSXEdgeClusterRecovery -nsxManagerFqdn \$nsxManagerFqdn - nsxManagerAdmin \$nsxManagerAdmin -nsxManagerAdminPassword \$nsx- ManagerAdminPassword -vCenterFQDN \$VCenterFqdn -vCenterAdmin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword - clusterName \$clusterName -extractedSDDCDataFile \$extractedSDDC- DataFile</pre> |
| 2    | <a href="#">Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore</a> | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Redeploy Remaining Management NSX Manager Nodes

| Step | Manual Task                                                                            | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Redeploy a Failed NSX Manager Node for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@vsphere.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-m01" \$restoredNsxManagerDeploymentSize = "medium"</pre> |

|   |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                       | <pre>\$nsxManagerOvaFile = "F:\OVA\nsx-unified-appliance-9.0.0.0.24733065.ova"  ## Command Block ## New-NSXManagerOvaDeployment -vCenterFqdn \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -restoredNsxManagerDeploymentSize \$restoredNsxManagerDeploymentSize -nsxManagerOvaFile \$nsxManagerOvaFile</pre> |
| 2 | <a href="#">Join NSX Manager Nodes to the NSX Manager Cluster for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$workloadDomain = "sfo-m01" \$extractedSDDCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Add-AdditionalNSXManagers -workloadDomain \$workloadDomain -extractedSDDCDataFile \$extractedSDDCDataFile</pre>                                                                                                                                                                                    |

### Finalize Management Domain

| Step | Manual Task                                                                                                       | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Synchronize the SDDC Manager Service Account on the ESX Hosts for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$vCenterAdmin = "administrator@vSphere.local" \$vCenterAdminPassword = "VMw@re1!VMw@re1!" \$clusterName = "sfo-m01-cl01" \$svcAccountPassword = "VMw@re1231!" \$sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" \$sddcManagerAdmin = "administrator@vSphere.local" \$sddcManagerAdminPassword = "VMw@re1!VMw@re1!"  ## Command Block ## Resolve-PhysicalHostServiceAccounts -vCenterFQDN \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -clusterName \$clusterName -svcAccountPassword \$svcAccountPassword -sddcManagerFqdn \$sddcManagerFqdn -sddcManagerAdmin \$sddcManagerAdmin -sddcManagerAdminPassword \$sddcManagerAdminPassword</pre> |
| 2    | <a href="#">Update the Backup Configuration in SDDC Manager for VCF Instance Backup and Restore</a>               | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3    | <a href="#">Update the Backup Configuration for vCenter for VCF Instance Backup and Restore</a>                   | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4    | <a href="#">Reapply Cluster Settings for the Restored vSphere Cluster for VCF Instance Backup and Restore</a>     | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$vCenterAdmin = "administrator@vSphere.local" \$vCenterAdminPassword = "VMw@re1!VMw@re1!" \$clusterName = "sfo-m01-cl01" \$clusterVMOVERRIDESJsonFile = ".\sfo-m01-cl01-vmOverrides.json"</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Step | Manual Task | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |             | <pre>\$clusterVMLocationsJsonFile = ".\sfo-m01-cl01-vmLocations.json" \$clusterDRSConfigurationJsonFile = ".\sfo-m01-cl01-drsConfiguration.json" \$clusterVMTagsJsonFile = ".\sfo-m01-cl01-vmTags.json"  ## Command Block ## Connect-VIServer -server \$vCenterFqdn -user \$vCenterAdmin -password \$vCenterAdminPassword Restore-ClusterVMOVERRIDES -clusterName \$clusterName -jsonFile \$clusterVMOVERRIDESJsonFile Restore-ClusterVMLocations -clusterName \$clusterName -jsonFile \$clusterVMLocationsJsonFile Restore-ClusterDRSGROUPSANDRULES -clusterName \$clusterName -jsonFile \$clusterDRSConfigurationJsonFile Restore-ClusterVMTags -clusterName \$clusterName -jsonFile \$clusterVMTagsJsonFile Disconnect-VIServer * -confirm:\$false</pre> |

## Fleet Component Restore Plan

Perform the following tasks to restore the fleet components into a VCF Management Domain. Refer to [Sequence Optimization for Restore of a VCF Instance](#) for detail on where tasks can be performed in parallel across domains.

### Fleet Component Restore Plan Summary

- [Restore VCF Operations](#)
- [Restore VCF Identity Broker](#)
- [Restore VCF Automation](#)

### Restore VCF Operations

| Step | Manual Task                                                                                           | PowerShell Equivalent |
|------|-------------------------------------------------------------------------------------------------------|-----------------------|
| 1    | <a href="#">Restore VCF Operations from an Image-Based Backup for VCF Instance Backup and Restore</a> | Not Applicable        |
| 2    | <a href="#">Verify VCF Operations State for VCF Instance Backup and Restore</a>                       | Not Applicable        |

### Restore VCF Identity Broker

| Step | Manual Task                                                                                              | PowerShell Equivalent |
|------|----------------------------------------------------------------------------------------------------------|-----------------------|
| 1    | <a href="#">Prepare to Restore VCF Identity Broker</a>                                                   | Not Applicable        |
| 2    | <a href="#">Deploy a New VCF Identity Broker Instance for VCF Instance Backup and Restore</a>            | Not Applicable        |
| 3    | <a href="#">Restore VCF Identity Broker from a File-Based Backup for VCF Instance Backup and Restore</a> | Not Applicable        |



**Restore VCF Automation**

| Step | Manual Task                                                                                         | PowerShell Equivalent |
|------|-----------------------------------------------------------------------------------------------------|-----------------------|
| 1    | <a href="#">Prepare to Restore VCF Automation</a>                                                   | Not Applicable        |
| 2    | <a href="#">Deploy a New VCF Automation Instance for VCF Instance Backup and Restore</a>            | Not Applicable        |
| 3    | <a href="#">Restore VCF Automation from a File-Based Backup for VCF Instance Backup and Restore</a> | Not Applicable        |
| 4    | <a href="#">Update Connections in VCF Automation for VCF Instance Backup and Restore</a>            | Not Applicable        |

**Workload Domain Restore Plan**

Perform the following tasks to restore a VCF Workload Domain. Refer to [Sequence Optimization for Restore of a VCF Instance](#) for detail on where tasks can be performed in parallel across domains.

**Workload Domain Restore Plan Summary**

- [Restore Workload vCenter and NSX Manager](#)
- [Rebuild Workload vSphere Cluster](#)
- [Restore Workload NSX Edges](#)
- [Restore Remaining Workload NSX Manager Nodes](#)
- [Finalize Workload Domain](#)
- [Restore vSphere Supervisor](#)

**Restore Workload vCenter and NSX Manager**

| Step | Manual Task                                                                                     | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Deploy a vCenter Appliance from an OVA File for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$vcCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter created during partial bringup \$vcCenterAdmin = "administrator@vsphere.local" \$vcCenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-w01" \$restoredvCenterDeploymentSize = "medium" \$vcCenterOvaFile = "F:\OVA\VMware-vCenter-Server-Appliance-9.0.2.0.25148086_OVF10.ova"  <b>Note:</b> These parameters assume you plan to deploy OVAs for the workload domain before the management domain restore is complete. If you choose to fully restore the management domain before this process, then you can use the management vCenter details instead of the temporary vCenter details.  ## Command Block ##</pre> |

| Step | Manual Task                                                                                                    | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                | <pre>New-vCenterOvaDeployment -vCenterFqdn \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -restoredvCenterDeploymentSize \$restoredvCenterDeploymentSize -vCenterOvaFile \$vCenterOvaFile</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2    | <a href="#">Deploy the First NSX Manager Cluster Node from an OVA File for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$vcCenterFqdn = "sfo-m01-vc02.sfo.rainpole.io" #fqdn of temp vCenter created during partial bringup \$vcCenterAdmin = "administrator@vsphere.local" \$vcCenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-w01" \$restoredNsxManagerDeploymentSize = "large" \$nsxManagerOvaFile = "F:\OVA\nsx-unified-appliance-9.0.2.0.25150388.ova"  <b>Note:</b> These parameters assume you plan to deploy OVAs for the workload domain before the management domain restore is complete. If you choose to fully restore the management domain before this process, then you can use the management vCenter details instead of the temporary vCenter details.  ## Command Block ## New-NSXManagerOvaDeployment -vCenterFqdn \$vcCenterFqdn -vCenterAdmin \$vcCenterAdmin -vCenterAdminPassword \$vcCenterAdminPassword -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -restoredNsxManagerDeploymentSize \$restoredNsxManagerDeploymentSize -nsxManagerOvaFile \$nsxManagerOvaFile</pre> |
| 3    | <a href="#">Restore vCenter from a File-Based Backup for VCF Instance Backup and Restore</a>                   | <pre>## Parameter Block ## \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-w01" \$vcCenterBackupPath = "192.168.1.1/F\$/backups/sn_sfo-w01-vc01.sfo.rainpole.io/M_9.0.2.0_20260406-084633_KBWGC5DGN5ZG2VDP-N5WHGICCMFRW5LQ" \$locationType = "SMB" \$locationUser = "Administrator" \$locationPassword = "VMw@rel123!" \$vcCenterBackupPassword = "VMw@rel!VMw@rel!"  ## Command Block ## Invoke-vCenterRestore -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -vcCenterBackupPath \$vcCenterBackupPath -locationType \$locationType -locationUser \$locationUser -locationPassword \$locationPassword  <b>Important:</b> If the vCenter backup was taken using a backup password then append the following to the above command  -backupPassword \$vcCenterBackupPassword</pre>                                                                                                                                                                                                                                                               |

| Step | Manual Task                                                                                                                                                                                                                                                                         | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | <a href="#">Restore NSX Manager from a File-Based Backup for VCF Instance Backup and Restore</a>                                                                                                                                                                                    | <pre>## Parameter Block ## \$extractedSDDCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-w01" \$sftpServer = "192.168.1.2" \$sftpUser = "svc-vcf-bck" \$sftpPassword = "VMw@rel!VMw@rel!" \$sftpServerBackupPath = "/media/backups" \$nsxBkupPassphrase = "VMw@rel!VMw@rel!" ## Command Block ## Invoke-NSXManagerRestore -extractedSDDCDataFile \$extractedSDDCDataFile -workloadDomain \$workloadDomain -sftpServer \$sftpServer -sftpUser \$sftpUser -sftpPassword \$sftpPassword -sftpServerBackupPath \$sftpServerBackupPath -backupPassphrase \$nsxBkupPassphrase</pre> |
| 5    | <a href="#">Verify the State of the First NSX Manager Cluster Node for VCF Instance Backup and Restore</a>                                                                                                                                                                          | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6    | Update Extracted SDDC Data for Instance Backup and Restore<br><br><b>Note:</b><br>Is only required when using the PowerShell Automation approach. If performing a fully manual restore, this information is gathered by <a href="#">Documenting the System Configuration of VCF</a> | <pre>## Parameter Block ## \$extractedSDDCDataFile = "F:\backup\extracted-sddc-data.json" \$sddcManagerFQDN = "sfo-vcf01.sfo.rainpole.io" \$sddcManagerAdmin = "administrator@vsphere.local" \$sddcManagerAdminPassword = "VMw@rel!VMw@rel!" \$restoredVcenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" ## Command Block ## Update-ExtractedSDDCData -extractedSDDCDataFile \$extractedSDDCDataFile -sddcManagerFQDN \$sddcManagerFQDN -sddcManagerAdmin \$sddcManagerAdmin -sddcManagerAdminPassword \$sddcManagerAdminPassword -vCenterFQDN \$restoredVcenterFqdn</pre>                          |

### Rebuild Workload vSphere Cluster

| Step | Manual Task                                                                                               | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Record the Cluster Settings from the Restored vCenter for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$vCenterAdmin = administrator@sfo-w01.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" ## Command Block ## Connect-VIServer -server \$vCenterFqdn -user \$vCenterAdmin -password \$vCenterAdminPassword Backup-ClusterVMOVERRIDES -clusterName \$clusterName Backup-ClusterVMLocations -clusterName \$clusterName Backup-ClusterDRSGroupsAndRules -clusterName \$clusterName Backup-ClusterVMTags -clusterName \$clusterName Disconnect-VIServer * -confirm:\$false</pre> |

| Step | Manual Task                                                                                                           | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | <a href="#">Remove Non-Responsive ESX Hosts from the Inventory for VCF Instance Backup and Restore</a>                | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$vCenterAdmin = "administrator@sfo-w01.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$nsxManagerFqdn = "sfo-w01-nsx01.sfo.rainpole.io" #fqdn of restored NSX Manager \$nsxManagerAdmin = "admin" \$nsxManagerAdminPassword = "VMw@rel!VMw@rel!" \$nsxManagerRootPassword = "VMw@rel!VMw@rel!" ## Command Block ## Remove-NonResponsiveHosts -vCenterFQDN \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -clusterName \$clusterName -NsxManagerFQDN \$nsxManagerFqdn -NsxManagerAdmin \$nsxManagerAdmin -NsxManagerAdminPassword \$nsxManagerAdminPassword -NsxManagerRootPassword \$nsxManagerRootPassword</pre> |
| 3    | <a href="#">Add New Hosts to the Cluster in the Restored Workload Domain vCenter</a>                                  | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" \$vCenterAdmin = "administrator@sfo-w01.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" \$sddcManagerAdmin = "administrator@vsphere.local" \$sddcManagerAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-HostsToCluster -vCenterFQDN \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -clusterName \$clusterName -sddcManagerFqdn \$sddcManagerFqdn -sddcManagerAdmin \$sddcManagerAdmin -sddcManagerAdminPassword \$sddcManagerAdminPassword -extractedSDDCDataFile \$extractedSDDCDataFile</pre>                              |
| 4    | <a href="#">Migrate the ESX Hosts and VMkernel Adapters to the vSphere Distributed Switch on the Restored vCenter</a> | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$vCenterAdmin = "administrator@sfo-w01.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## New-RebuildVdsConfiguration -vCenterFQDN \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -clusterName \$clusterName -extractedSDDCDataFile \$extractedSDDCDataFile</pre> <p><b>Note:</b></p>                                                                                                                                                                                                                                                       |

| Step | Manual Task                                                                                                       | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                   | At this point, the NSX installation on each host in the vSphere cluster should self-heal. Monitor the self-healing process until complete in the NSX Manager UI at <b>System &gt; Fabric &gt; Hosts &gt; Clusters</b> before proceeding. It might take several minutes for the process to initiate. If you see an error on the hosts that they are not part of the distributed switch, it just means that the self-healing process is yet to start.                                                                                                                                                                                                                                                                                                                                                |
| 5    | <a href="#">Add VMkernel Adapters to the ESX Hosts in the Workload Domain for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@sfo-w01.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$sdccManagerFqdn = "sfo-vcf01.sfo.rainpole.io" \$sdccManagerAdmin = "administrator@vsphere.local" \$sdccManagerAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDCCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Add-VMKernelsToHost -vCenterFQDN \$VCenterFQDN -vCenterAdmin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword -clusterName \$clusterName -sdccManagerFqdn \$sdccManagerFqdn -sdccManagerAdmin \$sdccManagerAdmin -sdccManagerAdminPassword \$sdccManagerAdminPassword -extractedSDCCDataFile \$extractedSDCCDataFile</pre> |
| 6    | <a href="#">Recreate a vSAN Datastore</a>                                                                         | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@sfo-w01.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$extractedSDCCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## New-RebuildVsanDatastore -vCenterFqdn \$VCenterFqdn -vCenterAdmin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword -clusterName \$clusterName -extractedSDCCDataFile \$extractedSDCCDataFile</pre>                                                                                                                                                                                                                                                                            |
| 7    | <a href="#">Re-Enable vSphere HA on the Workload Domain Cluster for VCF Instance Backup and Restore</a>           | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@sfo-w01.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-m01-cl01"  ## Command Block ## Connect-VIServer -server \$VCenterFqdn -user \$VCenterAdmin -password \$VCenterAdminPassword   Out-Null Get-Cluster \$clusterName   Set-Cluster -HAEnabled \$false -confirm:\$false   Out-Null Get-Cluster \$clusterName   Set-Cluster -HAEnabled \$true -confirm:\$false   Out-Null Disconnect-VIServer * -confirm:\$false</pre>                                                                                                                                                                                                               |

| Step | Manual Task                                                              | PowerShell Equivalent |
|------|--------------------------------------------------------------------------|-----------------------|
| 8    | <a href="#">Update SSH Host Keys for VCF Instance Backup and Restore</a> | Not Applicable        |

### Restore Workload NSX Edges

| Step | Manual Task                                                                                        | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Redeploy an NSX Edge Cluster for VCF Instance Backup and Restore</a>                   | <pre>## Parameter Block ## \$nsxManagerFqdn = "sfo-w01-nsx01.sfo.rainpole.io" #fqdn of restored NSX Manager \$nsxManagerAdmin = "admin" \$nsxManagerAdminPassword = "VMw@rel!VMw@rel!" \$VCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@sfo-w01.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$extractedSDCCDataFile = ".\extracted-sddc-data.json"  ## Command Block ## Invoke-NSXEdgeClusterRecovery -nsxManagerFqdn \$nsxManagerFqdn -nsxManagerAdmin \$nsxManagerAdmin -nsxManagerAdminPassword \$nsxManagerAdminPassword -vCenterFQDN \$VCenterFQDN -vCenterAdmin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword -clusterName \$clusterName -extractedSDCCDataFile \$extractedSDCCDataFile</pre> |
| 2    | <a href="#">Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore</a> | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### Restore Remaining Workload NSX Manager Nodes

| Step | Manual Task                                                                            | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <a href="#">Redeploy a Failed NSX Manager Node for VCF Instance Backup and Restore</a> | <pre>## Parameter Block ## \$VCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$VCenterAdmin = "administrator@sfo-w01.local" \$VCenterAdminPassword = "VMw@rel!VMw@rel!" \$extractedSDCCDataFile = ".\extracted-sddc-data.json" \$workloadDomain = "sfo-w01" \$restoredNsxManagerDeploymentSize = "large" \$nsxManagerOvaFile = "F:\OVA\nsx-unified-appliance-9.0.0.0.24733065.ova"  ## Command Block ## New-NSXManagerOvaDeployment -vCenterFqdn \$VCenterFqdn -vCenterAdmin \$VCenterAdmin -vCenterAdminPassword \$VCenterAdminPassword -extractedSDCCDataFile \$extractedSDCCDataFile -workloadDomain \$workloadDomain -restoredNsxManagerDeploymentSize \$restoredNsxManagerDeploymentSize -nsxManagerOvaFile \$nsxManagerOvaFile</pre> |

|   |                                                                                       |                                                                                                                                                                                                                                                    |
|---|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | Join NSX Manager Nodes to the NSX Manager Cluster for VCF Instance Backup and Restore | <pre>## Parameter Block ## \$workloadDomain = "sfo-w01" \$extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-AdditionalNSXManagers -workloadDomain \$workloadDomain -extractedSDDCDataFile \$extractedSDDCDataFile</pre> |
|---|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Finalize Workload Domain

| Step | Manual Task                                                                                       | PowerShell Equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Synchronize the SDDC Manager Service Account on the ESX Hosts for VCF Instance Backup and Restore | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$vCenterAdmin = "administrator@sfo-w01.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$svcAccountPassword = "VMw@rel23!" \$sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" \$sddcManagerAdmin = "administrator@vsphere.local" \$sddcManagerAdminPassword = "VMw@rel!VMw@rel!" ## Command Block ## Resolve-PhysicalHostServiceAccounts -vCenterFQDN \$vCenterFqdn -vCenterAdmin \$vCenterAdmin -vCenterAdminPassword \$vCenterAdminPassword -clusterName \$clusterName -svcAccountPassword \$svcAccountPassword -sddcManagerFqdn \$sddcManagerFqdn -sddcManagerAdmin \$sddcManagerAdmin -sddcManagerAdminPassword \$sddcManagerAdminPassword</pre>                                                                                                                            |
| 2    | Update the Backup Configuration for vCenter for VCF Instance Backup and Restore                   | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3    | Reapply Cluster Settings for the Restored vSphere Cluster for VCF Instance Backup and Restore     | <pre>## Parameter Block ## \$vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter \$vCenterAdmin = "administrator@sfo-w01.local" \$vCenterAdminPassword = "VMw@rel!VMw@rel!" \$clusterName = "sfo-w01-cl01" \$clusterVMOOverridesJsonFile = ".\sfo-w01-cl01-vmOverrides.json" \$clusterVMLocationsJsonFile = ".\sfo-w01-cl01-vmLocations.json" \$clusterDRSConfigurationJsonFile = ".\sfo-w01-cl01-drsConfiguration.json" \$clusterVMTagsJsonFile = ".\sfo-w01-cl01-vmTags.json" ## Command Block ## Connect-VIServer -server \$vCenterFqdn -user \$vCenterAdmin -password \$vCenterAdminPassword Restore-ClusterVMOOverrides -clusterName \$clusterName -jsonFile \$clusterVMOOverridesJsonFile Restore-ClusterVMTags -clusterName \$clusterName -jsonFile \$clusterVMTagsJsonFile Restore-ClusterDRSGroupsAndRules -clusterName \$clusterName -jsonFile \$clusterDRSConfigurationJsonFile</pre> |

| Step | Manual Task | PowerShell Equivalent                                                                                                                |
|------|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
|      |             | <pre>Restore-ClusterVMTags -clusterName \$clusterName -jsonFile \$clusterVMTagsJsonFile Disconnect-VIServer * -confirm:\$false</pre> |

### Restore vSphere Supervisor

| Step | Manual Task                                                                           | PowerShell Equivalent |
|------|---------------------------------------------------------------------------------------|-----------------------|
| 1    | Mount the NFS Share for the VKr Content Library for VCF Instance Backup and Restore   | Not Applicable        |
| 2    | Update the VKr Content Library Datastore Mapping for VCF Instance Backup and Restore  | Not Applicable        |
| 3    | Verify vSphere Lifecycle Manager Image Compliance for VCF Instance Backup and Restore | Not Applicable        |
| 4    | Restore the vSphere Supervisor Control Plane for VCF Instance Backup and Restore      | Not Applicable        |
| 5    | Remediate vSphere Supervisor Nodes for VCF Instance Backup and Restore                | Not Applicable        |
| 6    | Recreate the NSX NCP Pods for VCF Instance Backup and Restore                         | Not Applicable        |

## Instance Backup and Restore Task Library

A hierarchy of tasks that are associated with the backup and restore of a VCF Instance, sorted by component or functional area. The order in which tasks are listed in the library is not necessarily the order in which they should be performed. Refer to the relevant restore plan to see the sequence of tasks appropriate to the elements you are restoring.

### Data Gathering Tasks

Tasks associated with collecting data required for restoring a VCF Instance.

### Gather Information from the SDDC Manager Backup

The encrypted SDDC Manager backup contains information required for the restore process.

#### UI Procedure

- Identify a backup file for the restore operation and download it from the SFTP server to your host machine.
- On your host machine, open a terminal and run the following command to extract the content of the backup file.

```
OPENSSL_FIPS=1 openssl enc -d -aes-256-cbc -md sha256 -in filename-of-restore-file | tar -xz
```
- When prompted, enter the `encryption_password`.
- In the extracted folder, locate and view the contents of the `security_password_vault.json` file.

5. Write down the following information for each component:

| Component             | Information                                                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESX hosts             | <ul style="list-style-type: none"> <li>FQDN</li> <li>IP address</li> <li><b>root</b> user password</li> <li>Service account user name</li> </ul>                                      |
| All vCenter instances | <ul style="list-style-type: none"> <li>FQDN</li> <li>IP address</li> <li><b>root</b> user password</li> <li>SSO administrator password</li> </ul>                                     |
| NSX Manager nodes     | <ul style="list-style-type: none"> <li>FQDN</li> <li>IP address</li> <li><b>admin</b> user password</li> <li><b>root</b> user password</li> <li><b>audit</b> user password</li> </ul> |
| NSX Edge nodes        | <ul style="list-style-type: none"> <li>FQDN</li> <li>IP address</li> <li><b>root</b> user password</li> <li><b>admin</b> user password</li> <li><b>audit</b> user password</li> </ul> |
| SDDC Manager          | <ul style="list-style-type: none"> <li>FQDN</li> <li>IP address</li> <li><b>root</b> user password</li> <li>Backup user password</li> <li>FIPS configuration</li> </ul>               |

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

### VCF Installer Restore Tasks

Tasks performed using VCF Installer.

#### Perform a Partial VCF Deployment

You perform a partial deployment to enable the restore of the management domain default cluster using a temporary vCenter instance.

#### Note:

You use the VCF Installer to perform a vSphere Foundation deployment to deploy a temporary vCenter instance and configure the default cluster of the management domain. VCF Operations is not deployed as part of this process as it will be restored from image-based backup at a later step.

#### UI Procedure

Follow [Deploying a New VMware Cloud Foundation or VMware vSphere Foundation Platform](#) for using the modified deployment JSON file.

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

### vCenter Restore Tasks

Tasks related to the restore of vCenter.

#### Deploy a vCenter Appliance from an OVA File for VCF Instance Backup and Restore

To restore a vCenter instance, you first deploy a new appliance from an OVA file on the temporary vCenter, deployed during partial deployment. You then use this appliance to restore the management domain or workload domain.

#### UI Procedure

- Mount the vCenter ISO in Windows Explorer.
- Log in to the temporary vCenter, deployed with the partial deployment operation, and navigate to the default cluster.
- Right-click the cluster and select **Deploy OVF Template**.
- Navigate to the `vcsa` folder on the mounted ISO, select the vCenter OVA file, and click **Next**.
- Enter the virtual machine name and click **Next**.
- On the **Compute resource** page, select the cluster and click **Next**.
- On the **Review details** page, if warnings appear, click **Ignore** and click **Next**.
- On the **License agreements** page, click **I accept all license agreements** and click **Next**.
- On the **Configuration** page, select the correct size for the vCenter, and click **Next**.
- On the **Select storage** page, select the vSAN datastore, and click **Next**.
- On the **Select networks** page, select the management network, and click **Next**.
- On the **Customize template** page, enter the following information, and click **Next**.

| Section                  | Setting                        | Value                             |
|--------------------------|--------------------------------|-----------------------------------|
| Networking Configuration | Host Network IP Address Family | ipv4                              |
|                          | Host Network Mode              | static                            |
|                          | Host Network IP Address        | <vcenter_server_ip_address>       |
|                          | Host Network Prefix            | <management_network_prefix>       |
|                          | Host Network Default Gateway   | <management_network_gateway>      |
|                          | Host Network DNS Servers       | <dns_server_comma_separated_list> |
|                          | Host Network Identity          | <vcenter_server_fqdn>             |
| SSO Configuration        | N/A                            | N/A                               |
| System Configuration     | Root Password                  | <vcenter_server_root_password>    |
| Upgrade Configuration    | N/A                            | N/A                               |
| Miscellaneous            | N/A                            | N/A                               |

| Section               | Setting            | Value                                     |
|-----------------------|--------------------|-------------------------------------------|
| Networking Properties | Domain Name        | <dns_domain_name>                         |
|                       | Domain Search Path | <domain_search_path_comma_separated_list> |

13. On the **Ready to complete** page, review the details and click **Finish**.

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

### Restore vCenter from a File-Based Backup for VCF Instance Backup and Restore

Restore the vCenter from a file-based backup on the appliance that you deployed from an OVA file on the temporary vCenter instance.

#### UI Procedure

1. Log in to the appliance management interface (VAMI) of the vCenter at `https://<vcenter_server_fqdn>:5480` as **root**.
2. Select the **Restore** and click **Next**.
3. Provide the backup location, the SSO administrator user name and password and encryption password (if applicable), and click **Next**.
4. On the **Ready to complete** page, review the details and click **Finish**.

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

### Update the Backup Configuration for vCenter for VCF Instance Backup and Restore

If the backup target for vCenter has changed, you must update the backup configuration for each vCenter instance.

1. Log in to the appliance management interface of vCenter at `https://<vcenter_fqdn>:5480`.
2. Navigate to **Backup** and click **Edit**.
3. Enter the new backup target details and click **Save**.

### vSphere Cluster Restore Tasks

Tasks related to the rebuild of vSphere clusters.

### Record the Cluster Settings from the Restored vCenter for VCF Instance Backup and Restore

Before you can restore vSphere clusters of the restored vCenter, you must first record the settings so they can be reapplied to the restored cluster.

- vCenter has been restored and powered on.
1. Log in to the temporary vCenter at `https://<restored_vcenter_server_fqdn>/ui` by using the vSphere Client.
  2. Record the following settings for every cluster in the restored vCenter.
    - a) Virtual Machine Overrides
    - b) Virtual Machine Locations
    - c) Virtual Machine Tags
    - d) DRS Groups and Rules

### Remove Non-Responsive ESX Hosts from the Inventory for VCF Instance Backup and Restore

Before you can repair a failed vSphere cluster, you must first remove the non-responsive hosts from the cluster.

#### Prerequisites

Ensure you have exported all cluster settings before proceeding.

#### CAUTION:

If your environment uses vSAN storage clusters, then you must rebuild a vSAN storage cluster fully before you attempt to remove non-responsive hosts from a dependent compute-only cluster. Failure to do so will result in the inability to remount the vSAN datastore on the compute-only cluster.

#### UI Procedure

1. Log into the NSX Manager for the workload domain and navigate to **System > Fabric > Hosts > Clusters**.
2. Select the check box next for the relevant vSphere cluster and click **Remove NSX**.
3. Deselect the check box next for the relevant vSphere cluster.
4. Expand the cluster and wait for all hosts in the cluster to go into an **Orphaned** State
5. Select the check box that selects all hosts in the cluster without selecting the cluster object itself and select **Remove NSX**.
6. Select the **Force** option and submit.  
Wait until all hosts show as unconfigured.
7. Log in to the vCenter with the non-responsive hosts and navigate to the cluster.
8. Select the cluster, and in the right pane, navigate to the **Hosts** tab.
9. Select the check box for each non-responsive host, right click the selected hosts and select **Remove from Inventory**.  
If the cluster use vSphere Lifecycle Manager images, wait for about a minute to allow the background tasks in NSX to complete the removal of the NSX solution from the relevant cluster before proceeding to the next step.
10. Log in to the NSX Manager for the workload domain and navigate back to **System > Fabric > Hosts > Clusters**
11. Select the check box next for the relevant vSphere cluster and click **Configure NSX**.
12. Select the relevant Transport Node Profile and Submit.

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)



- [Workload Domain Restore Plan](#)

## Configure vSAN to Ignore Cluster Member Updates for VCF Instance Backup and Restore

To permit moving the vSAN cluster from the temporary vCenter to the restored vCenter, you must configure vSAN to ignore cluster member updates.

### Manual Procedure

1. Connect to each ESX host using SSH and login as **root** user.
2. Run the following command:

```
esxcli system settings advanced set --int-value=1 --option=/VSAN/IgnoreClusterMemberListUpdates
```

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of [here](#).

- [Management Domain Restore Plan](#)

## Migrate Host Networking from vSphere Distributed Switch to vSphere Standard Switch

Before you can move the new vSAN cluster to the restored vCenter during a VCF Instance Backup and Restore, you must first disconnect the hosts from the temporary vSphere Distributed Switch by using a temporary vSphere Standard Switch.

### UI Procedure

1. Disconnect a physical vmnic for each host from the vSphere Distributed Switch on the temporary vCenter.
  - a. Log in to the temporary vCenter at [https://<temporary\\_vcenter\\_server\\_fqdn>/ui](https://<temporary_vcenter_server_fqdn>/ui) by using the vSphere Client.
  - b. In the **Networking** inventory, right-click the distributed switch and select **Add and Manage Hosts**.
  - c. Select the **Manage host networking** task.
  - d. On the **Member hosts** tab, select all hosts and click **Next**.
  - e. On the **Manage physical adapters** page, select one vmnic, for example vmnic1, and then click **Unassign adapter**.
  - f. In the **Confirm Unassign Adapter** dialog box, select **Apply this operation to all other hosts** and then click **Unassign**.
  - g. Click **Next**.
  - h. Click **Next**, **Next** and **Finish**.
2. Create a standard switch on each ESX host.
  - a. In the **Hosts and clusters** inventory, select the first ESX host, and on the Configure tab, and select **Networking > Virtual switches**.
  - b. Click **Add Standard Virtual Switch**.
  - c. Set the vSwitch Name to `vSwitch0`.
  - d. Ensure the MTU is set to 9,000.
  - e. Repeat this task to complete these steps for each ESX host.

3. Connect ESX hosts to vSphere Standard Switch.
  - a. On the **Virtual switches** page for the host, click **Add Networking**.
  - b. In the **Add Networking** wizard, select **Physical Network Adapter** and click **Next**.
  - c. On the **Select target device** page, select **vSwitch0** and click **Next**.
  - d. On the **Add physical network adapter** page, select an unassigned physical adapter, for example vmnic1, from the adapter list, and move it under **Active adapters**, and click **Next**.
  - e. Review the information on the Ready to complete page and click **Finish**.
4. Create a temporary management port group on the temporary vSphere standard switch.
  - a. On the **Virtual switches** page for the host, expand vSwitch0 and click **Add Networking**.
  - b. In the **Add Networking** wizard, select **Virtual Machine Port Group for a Standard Switch** and click **Next**.
  - c. On the **Select target device** page, the standard switch will default to vSwitch0, click **Next**.
  - d. On the **Connection settings** page, change the **Network label** to `temp_mgmt`, update the VLAN ID with the correct VLAN ID for the management VLAN, and click **Next**.
  - e. Review the information on the **Ready to complete** page and then click **Finish**.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of [here](#).

- [Management Domain Restore Plan](#)

## Prepare to Migrate the New vSphere Cluster to the Restored vCenter for VCF Instance Backup and Restore

Before you can migrate the new vSphere cluster to the restored vCenter, you must first prepare the cluster in the temporary vCenter.

### UI Procedure

1. Log in to the temporary vCenter at [https://<temporary\\_vcenter\\_server\\_fqdn>/ui](https://<temporary_vcenter_server_fqdn>/ui) by using the vSphere Client.
2. Set vSphere DRS to manual on the target cluster.
  - a. In the **Hosts and clusters** inventory, select the target cluster.
  - b. On the **Configure** tab, select **vSphere DRS > Edit**.
  - c. Change the DRS level to **Manual** and click **Save**.
3. Migrate all VMs to the first ESX host in the cluster.
  - a. Right-click on each VM and select **Migrate**.
  - b. Select the first host in the cluster as a target and click **Migrate**.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of [here](#).

- [Management Domain Restore Plan](#)

## Move All Management VMs to a Temporary Port Group on the vSphere Standard Switch

Before you can move the new vSAN cluster to the restored vCenter during a VCF Instance Backup and Restore, you must first move all management VMs to a temporary port group on the temporary vSphere Standard Switch.

**UI Procedure**

1. Log in to the temporary vCenter.
2. In the **VMs and templates** inventory, right-click the VM for the temporary vCenter and select **Edit Settings**.
3. Move network adapter 1 to the **temp\_mgmt** port group on the standard switch, and click **OK**.
4. Repeat for all VMs in the temporary vCenter.

**PowerShell Equivalent**

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

**Remove the ESX Hosts from the vSphere Distributed Switch on the Temporary vCenter**

Before you can move the new vSAN cluster to the restored vCenter during a VCF Instance Backup and Restore, you must first remove all ESX hosts from the temporary vSphere Distributed Switch.

**UI Procedure**

1. Disconnect the remaining physical adapters from the vSphere Distributed Switch.
  - a. Log in to the temporary vCenter at [https://<temporary\\_vcenter\\_server\\_fqdn>/ui](https://<temporary_vcenter_server_fqdn>/ui) by using the vSphere Client as a user with the administrator role.
  - b. In the **Networking** inventory, right-click the distributed switch and select **Add and Manage Hosts**.
  - c. Select the **Manage host networking** task.
  - d. On the **Member hosts** tab, select all hosts and click **Next**.
  - e. On the **Manage physical adapters** page, select the remaining physical adapters, for example vmnic0, and then select **Unassign adapter**.
  - f. In the **Confirm Unassign Adapter** dialog box, select **Apply this operation to all other hosts** and then click **Unassign**.  
You are returned to the **Manage physical adapters** screen. Note the vmnic0 is marked as Unassigned.
  - g. On the **Manage physical adapters** page, click **Next**.
  - h. Click **Next**, **Next**, and **Finish**.
2. Remove the ESX hosts from the vSphere Distributed Switch on the temporary vCenter.
  - a. Connect to the temporary vCenter UI and select **Networking**.
  - b. In the **Networking** inventory, right-click the distributed switch and select **Add and Manage Hosts**.
  - c. Select the **Remove hosts** task.
  - d. On the **Member hosts** tab, select all hosts and click **Next**.
  - e. Review the information on the **Ready to complete** page and click **Finish**.

**PowerShell Equivalent**

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

**Migrate ESX Hosts from the Temporary vCenter to the Restored vCenter for VCF Instance Backup and Restore**

You migrate the newly deployed vSAN cluster to the restored vCenter.

**UI Procedure**

1. Log in to the restored vCenter for the management domain by using the vSphere Client.
2. Right-click the restored cluster and click **Add Hosts**.
3. In the **Add hosts** wizard, enter the ESX host names for the management cluster including the user name and password, and click **Next**.  
A security alert dialog box appears because vCenter is not able to verify the certificate thumbprint.
4. In the security alert dialog box, select all hosts and click **OK**.
5. Review the information on the **Host summary** page and click **Next**.
6. Review the information on the **Ready to complete** page and click **Finish**.

**PowerShell Equivalent**

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

**Migrate the ESX Hosts and VMkernel Adapters to the vSphere Distributed Switch on the Restored vCenter**

You connect the new cluster hosts to the vSphere Distributed Switch and migrate the VMkernel adapters during a VCF Instance Backup and Restore.

**UI Procedure**

1. Log in to the restored vCenter by using the vSphere Client.
2. Add the hosts to the vSphere Distributed Switch.
  - a. In **Networking** inventory, right-click the vSphere Distributed Switch and select **Add and Manage Hosts**.
  - b. Select **Add Hosts** and click **Next**.
  - c. On the **Select hosts** page, select all the ESX hosts, and click **Next**.
  - d. On the **Manage physical adapters** page, select a free physical adapter, for example, vmnic0, and click **Assign Uplink**.
  - e. Select uplink1, and click **Next**.
  - f. On the **Manage VMkernel adapters** page, update the following VMkernel adapters to assign them to the appropriate port group on the new distributed switch.

**Table 955: VMkernel Migration by Domain Type**

| Management Domain                    | Workload Domain                      |
|--------------------------------------|--------------------------------------|
| vmk0 – Management Network port group | vmk0 – Management Network port group |
| vmk1 – vMotion Network port group    | N/A - Not yet created                |
| vmk2 – vSAN Network port group       | N/A - Not yet created                |

- g. To assign the VMkernel adapters, select the adapter and under actions for the corresponding port group, click **Assign**.
- h. Click **Next**.
- i. On the **Migrate VM networking** step, click **Next**.
- j. Review the information on the **Ready to complete** page and click **Finish**.

- For a management domain cluster, migrate the management VMs to the original management port group.
  - Right-click the temporary management port group and select **Migrate VMs to Another Network**.
  - For destination network, select the management port group on the vSphere Distributed Switch, for example sfo-m01-vc01-vds01-management, and click **Next**.
  - On the **Select VMs to migrate** page, select all management VMs and click **Next**.
  - On the **Ready to complete** page, click **Finish**.
- Remove the temporary standard switch on each ESX Host.
  - Select the first ESX host and, on the Configure tab, select **Networking > Virtual Switches**.
  - Expand **vSwitch0** and click the horizontal ellipsis.
  - Click **Remove** and click **Yes**.
- Add more host uplinks to the vSphere Distributed Switch.
  - Right-click the distributed switch and select **Add and Manager Hosts**.
  - Select **Manage Host Networking** and click **Next**.
  - On the **Select hosts** step, select all the ESX hosts and click **Next**.
  - On the **Manage physical adapters** step, select one or more required free physical adapters, for example, vmnic1, and from **Assign Uplink** select the desired uplinks to corresponding physical adapters, and click **Next**.
  - Click **Next** and click **Next**.
  - Review the information on the **Ready to complete** page and click **Finish**.

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

### Configure vSAN Back to Honour Cluster Member Updates for VCF Instance Backup and Restore

After the vSAN cluster has been moved to the restored vCenter, you must configure back vSAN to honour cluster member updates.

#### UI Procedure

- Connect to each ESX host using SSH and login as **root**.
- Run the following command.

```
esxcli system settings advanced set --int-value=0 --option=/VSAN/IgnoreClusterMemberListUpdates
```

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

### Reapply Cluster Settings for the Restored vSphere Cluster for VCF Instance Backup and Restore

After all workloads are restored on a vSphere cluster for the management or workload domain, you reapply the cluster settings for virtual machine overrides, locations, and vSphere DRS rules and groups.

- Log in to the restored vCenter at `https://<restored_vcenter_server_fqdn>/ui` by using the vSphere Client.
- Re-apply the following settings for every cluster in the restored vCenter.
  - Virtual Machine Overrides
  - Virtual Machine Locations
  - Virtual Machine Tags
  - DRS Groups and Rules

### Add New Hosts to the Cluster in the Restored Workload Domain vCenter

You add new hosts to the cluster in the restored vCenter instance for the workload domain.

#### UI Procedure

- Log in to the restored vCenter by using the vSphere Client.
- In the **Hosts and clusters** inventory, navigate to the cluster.
- Right-click the cluster and select **Add Hosts**.
- Enter the FQDN, user name, and password for each host to be added, and click **Next**.
- When prompted to accept the SSL certificate for the new hosts, click **Accept**.
- Click **Finish**.

#### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Workload Domain Restore Plan](#)

### Add VMkernel Adapters to the ESX Hosts in the Workload Domain for VCF Instance Backup and Restore

You add vSphere vMotion and vSAN VMkernel adapters to the new ESX hosts in the workload domain.

#### UI Procedure

- Log in to the restored vCenter instance by using the vSphere Client.
- In the **Hosts and clusters** inventory, select the first ESX host and on the Configure tab, select **Networking > VMkernel adapters**.
- Click **Add networking**.
- Select **VMkernel Network Adapter** and click **Next**.
- Select **Select an existing network**, select the port group for the VMkernel type, and click **Next**.
- On the **Port properties** page, configure the following settings, leaving the default values for all other settings, and click **Next**.

| Setting            | vMotion VMkernel | vSAN VMkernel |
|--------------------|------------------|---------------|
| TCP/IP stack       | vMotion          | Default       |
| Available services | N/A              | vSAN          |

- On the **IPv4 settings** page, select **Use static IPV4 settings** and enter the IP details for each VMkernel adapter from your system documentation, and click **Next**.
- On the **Ready to complete** page, click **Finish**.
- Repeat the procedure for each ESX host in the cluster.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Workload Domain Restore Plan](#)

### Recreate a vSAN Datastore

To recreate the vSAN datastore, you claim cache and capacity disks to create disk groups.

#### CAUTION:

Recreate the vSAN datastore only on clusters that host vSAN datastores. Do not perform this operation on compute-only clusters.

If the vSAN datastore is ESA, the datastore will be recreated automatically. Only step 6 is required.

#### UI Procedure

1. Connect to the restored vCenter by using the vSphere Client.
2. In the **Hosts and clusters** inventory, navigate to the cluster and click **Configure**.
3. Under **vSAN**, click **Disk Management**, and click **Claim Unused Disks**.
4. Select the relevant disks for each tier and click **Claim**.

#### CAUTION:

The process to claim the disks and create the vSAN datastore might take some time. Wait for it to complete.

5. After the disk groups are created, right-click the datastore and select **Rename**.
6. Enter the original datastore name and click **OK**.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Workload Domain Restore Plan](#)

### Re-Enable vSphere HA on the Workload Domain Cluster for VCF Instance Backup and Restore

After recreating a vSAN datastore, you turn off and turn on vSphere HA again on the cluster so that vCLS can self-heal.

#### UI Procedure

1. Connect to the restored vCenter by using the vSphere Client, and add the hosts to the vSphere Distributed Switch.
2. In the **Hosts and clusters** inventory, navigate to the cluster and click **Configure**.
3. Under **Services**, click **vSphere Availability**, and under **vSphere HA is Turned ON** click the **Edit** button.
4. Switch off the **vSphere HA** toggle button and Click **OK**.
5. Under **Services**, click **vSphere Availability**, and under **vSphere HA is Turned ON** click the **Edit** button.
6. Switch on the **vSphere HA** toggle button and Click **OK**.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Workload Domain Restore Plan](#)

### NSX Restore Tasks

Tasks related to the restore of NSX.

#### Deploy the First NSX Manager Cluster Node from an OVA File for VCF Instance Backup and Restore

When the three nodes in an NSX Manager cluster are in a failed state, you begin the restore process by restoring the first cluster node from an OVA file.

#### Note:

The first NSX Manager node is defined as the node that took the backup of the NSX Manager cluster. To identify this node, examine the name of the folder that is created during backup. The folder name includes the node's IP address.

#### UI Procedure

1. Log in to the temporary vCenter, deployed with the partial deployment operation, and navigate to the default cluster.
2. Right-click the cluster and select **Deploy OVF Template**.
3. On the **Select an OVF template** page, select **Local file**, click **Upload files**, navigate to the location of the NSX Manager OVA file, click **Open**, and click **Next**.
4. On the **Select a name and folder** page, enter the VM name and click **Next**.
5. On the **Select a compute resource** page, select the cluster and click **Next**.
6. On the **Review details** page, click **Next**.
7. On the **Configuration** page, select the appropriate size and click **Next**.
8. For the management domain, select **Medium**, and for workload domains, select **Large** unless you changed these defaults during deployment.
9. On the **Select storage** page, select the management vSAN datastore, and click **Next**.
10. On the **Select networks** page, from the **Destination network** drop-down menu, select management distributed port group, and click **Next**.
11. On the **Customize template** page, enter these values and click **Next**.

| Setting                         | Value                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------|
| System root user password       | <first_nsx_cluster_node_root_password>                                                                |
| CLI admin user password         | <first_nsx_cluster_node_admin_password><br>You must enter a password. Do not leave the default value. |
| CLI audit user password         | <first_nsx_cluster_node_audit_password><br>You must enter a password. Do not leave the default value. |
| Hostname                        | <first_nsx_cluster_node_fqdn>                                                                         |
| Default IPv4 gateway            | <first_nsx_cluster_node_gw>                                                                           |
| Management network IPv4 address | <first_nsx_cluster_node_ip>                                                                           |
| Management network netmask      | <first_nsx_cluster_node_mask>                                                                         |
| DNS server list                 | <dns_server_list>                                                                                     |
| NTP server list                 | <ntp_server_list>                                                                                     |
| Enable SSH                      | Selected                                                                                              |
| Allow root SSH logins           | Deselected                                                                                            |

12. On the **Ready to complete** page, review the deployment details and click Finish.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

### Restore NSX Manager from a File-Based Backup for VCF Instance Backup and Restore

You restore the file-based backup of the first NSX Manager cluster node to the newly-deployed NSX Manager instance.

#### UI Procedure

1. In a Web browser, log in to the newly-deployed first NSX Manager cluster node by using the user interface as the **admin** user.
2. On the main navigation bar, click **System**.
3. In the left navigation pane, under **Lifecycle management**, click **Backup & Restore**.
4. In the **NSX configuration** pane, under **SFTP server**, click **Edit**.
5. In the **Backup configuration** dialog box, enter the details for your backup server, and click **Save**.
6. Under **Backup history**, select the source backup, and click **Restore**.
7. During the restore, when prompted, reject adding NSX Manager nodes by clicking **I understand** and **Resume**.
8. If a **Confirm CM/VC Connectivity** dialog box appears, click **I understand the message mentioned above and wish to proceed**, and click **Resume**.
9. If a fabric node discovery time-out dialog box appears, click **Resume** to continue the restore.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

### Verify the State of the First NSX Manager Cluster Node for VCF Instance Backup and Restore

After you restore the first NSX Manager cluster node, you verify the services state from the Web console of the restored node VM.

1. In a Web browser, log in to the temporary vCenter at `https://<temp_management_vcenter_server_fqdn>/ui` by using the vSphere Client.
2. Click the VM name of the newly-deployed first NSX Manager cluster node, click **Launch Web Console**, and log in by using administrator credentials.
3. Run the command to view the cluster status.

```
get cluster status
```

The services on the single-node NSX Manager cluster appear as UP.

### Redeploy a Failed NSX Manager Node for VCF Instance Backup and Restore

You deploy a new NSX Manager instance by using the configuration of the failed node.

### UI Procedure

1. In a Web browser, log in to the management domain vCenter by using the vSphere Client.
2. In the **Hosts and clusters** inventory, sight-click the management cluster and select **Deploy OVF Template**.
3. On the **Select an OVF template** page, select **Local file**, click **Upload files**, navigate to the location of the NSX Manager OVA file, click **Open**, and click **Next**.
4. On the **Select a name and folder** page, enter the VM name and click **Next**.
5. On the **Select a compute resource** page, select the cluster and click **Next**.
6. On the **Review details** page, click **Next**.
7. **Note:**  
  
For the management domain, select **Medium**, and for workload domains, select **Large** unless you changed these defaults during deployment.
8. On the **Select storage** page, select the management vSAN datastore, and click **Next**.
9. On the **Select networks** page, from the **Destination network** drop-down menu, select management distributed port group, and click **Next**.
10. On the **Customize template** page, enter these values and click **Next**.

| Setting                         | Value                                    |
|---------------------------------|------------------------------------------|
| System root user password       | <failed_nsx_cluster_node_root_password>  |
| CLI admin user password         | <failed_nsx_cluster_node_admin_password> |
| CLI audit user password         | <failed_nsx_cluster_node_audit_password> |
| Hostname                        | <failed_nsx_cluster_node_fqdn>           |
| Default IPv4 gateway            | <failed_nsx_cluster_node_gw>             |
| Management network IPv4 address | <failed_nsx_cluster_node_ip>             |
| Management network netmask      | <failed_nsx_cluster_node_mask>           |
| DNS server list                 | <dns_server_list>                        |
| NTP server list                 | <ntp_server_list>                        |
| Enable SSH                      | Selected                                 |
| Allow root SSH logins           | Deselected                               |

11. On the **Ready to complete** page, review the deployment details and click **Finish**.
12. Repeat for the remaining failed node.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

### Join NSX Manager Nodes to the NSX Manager Cluster for VCF Instance Backup and Restore

You retrieve the ID and API thumbprint of the NSX Manager cluster, and use it join the newly-deployed NSX Manager instance to the cluster.

#### UI Procedure

1. In a Web browser, log in to the management domain vCenter by using the vSphere Client.

- In the **VMs and templates** inventory, click the VM of an operational NSX Manager node in the cluster, click **Launch web console**, and log in by using **administrator** credentials.
- Retrieve the ID of the NSX Manager cluster.
  - Run the command to view the cluster ID.

```
get cluster config | find Id:
```
  - Write down the cluster ID.
- Retrieve the API thumbprint of the NSX Manager API certificate.
  - Run the command to view the certificate API thumbprint.

```
get certificate api thumbprint
```
  - Write down the certificate API thumbprint.
- Close the VM Web console.
- In the vSphere Client, click the VM of the newly deployed NSX Manager node, click **Launch Web console**, and log in by using **administrator** credentials.
- Run the command to join the new NSX Manager node to the cluster.

```
join new_node_ip cluster-id cluster_id thumbprint api_thumbprint username admin
```
- Repeat for the remaining failed node.

**PowerShell Equivalent**

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

**Redeploy an NSX Edge Cluster for VCF Instance Backup and Restore**

To restore an NSX edge cluster, use the NSX API to redeploy the edge nodes.

**Procedure**

- Retrieve the edge node configuration by running the following API call and copy the output payload of this API.

```
GET /<NSX-Manager-IPaddress>/api/v1/transport-nodes/<edgenode_id>

{
 "resource_type": "EdgeNode",
 "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
 "display_name": "Edge_TN2",
 "description": "EN",
 "external_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
 "ip_addresses": [
 "10.170.94.240"
],
 "_create_user": "admin",
 "_create_time": 1600106319056,
 "_last_modified_user": "admin",
 "_last_modified_time": 1600106907312,
 "_system_owned": false,
 "_protection": "NOT_PROTECTED",
 "_revision": 2
},
{
 "is_overridden": false,
 "failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",

```

```

 "resource_type": "TransportNode",
 "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
 "display_name": "Edge_TN2",
 "_create_user": "admin",
 "_create_time": 1600106319399,
 "_last_modified_user": "admin",
 "_last_modified_time": 1600106907401,
 "_system_owned": false,
 "protection": "NOT_PROTECTED",
 "_revision": 1
 }
}
```

- Redeploy the edge node using the following API call, passing the JSON data retrieved in Step 2 as the body. You do not need to pass any passwords in the JSON file.

```
POST /<NSX-Manager-IPaddress>/api/v1/transport-nodes/<edgenode_id>?action=redeploy
```
- Repeat steps 1-2 for the remaining edge cluster nodes.
- In NSX Manager, monitor the Configuration Status of the new NSX Edge nodes, until they show **Success**.

**Note:**

If you encounter an error **Edge redeploy is blocked as an active alarm is found for Edge VM present in NSX Inventory but missing in vCenter**, navigate to the **Alarms** section in the relevant NSX Manager UI, select all relevant alarms whose **Event Type** is **Edge VM Present In NSX Inventory Not Present In vCenter** and from the vertical ellipsis menu, select **Resolve**. Then, retry the redeploy operation.

**PowerShell Equivalent**

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

**Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore**

After completing all NSX Edge node redeployments, you must verify the state of the NSX Edge cluster nodes.

- In a Web browser, log in to NSX Manager for the domain by using the user interface.
- On the main navigation bar, click **System**.
- In the left pane, under **Configuration**, click **Fabric > Nodes**.
- Click the **Edge transport nodes** tab.
- Verify that all edge transport nodes show these values.

| Setting             | Value   |
|---------------------|---------|
| Configuration state | Success |
| Node status         | Up      |
| Tunnels             | Up      |

**SDDC Manager Restore Tasks**

Tasks related to the restore of SDDC Manager.



## Deploy the SDDC Manager Appliance from an OVA File for VCF Instance Backup and Restore

You deploy a new SDDC Manager appliance by using the OVA file that you downloaded during the preparation for restore.

### UI Procedure

- Log in to the temporary vCenter, deployed with the partial deployment operation, and navigate to the default cluster.
- Right-click the cluster and select **Deploy OVF Template**.
- On the **Select an OVF template** page, select **Local file**, and upload the SDDC Manager OVA file, and click **Next**.
- On the **Select a name and folder** page, in the **Virtual machine name** text box, enter a virtual machine name, and click **Next**.
- On the **Select a compute resource** page, click **Next**.
- On the Review details page, review the settings and click **Next**.
- On the **License agreements** page, accept the license agreement and click **Next**.
- On the **Select storage** page, select the vSAN datastore and click **Next**.  
The datastore must match the `vsan_datastore` value in the `metadata.json` file that you downloaded from the SDDC Manager backup during the preparation for the restore.
- On the **Select networks** page, from the **Destination network** drop-down menu, select the management network distributed port group and click **Next**.  
The distributed port group must match the `port_group` value in the `metadata.json` file that you downloaded during the preparation for the restore.
- On the **Customize template** page, enter the following values and click **Next**.

| Setting                             | Description                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enter root user password            | You can use the original <b>root</b> user password or a new password.                                                                                                                     |
| Enter login (vcf) user password     | You can use the original <b>vcf</b> user password or a new password.                                                                                                                      |
| Enter basic auth user password      | You can use the original <b>admin</b> user password or a new password.                                                                                                                    |
| Enter backup (backup) user password | The backup password that you saved during the preparation for the restore. This password can be changed later.                                                                            |
| Enter Local user password           | You can use the original Local user password or a new password.                                                                                                                           |
| Hostname                            | The FQDN must match the <code>hostname</code> value in the <code>metadata.json</code> file that you downloaded during the preparation for the restore.                                    |
| NTP sources                         | The NTP server details for the appliance.                                                                                                                                                 |
| Enable FIPS                         | The FIPS configuration in the <code>metadata.json</code> file that you saved during the preparation for the restore. You cannot restore a FIPS backup to a non-FIPS configured appliance. |
| PSC Address                         | Leave blank.                                                                                                                                                                              |
| Initial PSC SSH password            | Leave blank.                                                                                                                                                                              |
| SSO Username                        | Leave blank.                                                                                                                                                                              |
| SSO password                        | Leave blank.                                                                                                                                                                              |
| Network 1 IP address                | The IP address for the appliance.                                                                                                                                                         |
| Network 1 Subnet Mask               | The subnet mask for the appliance.                                                                                                                                                        |
| Network Default Gateway             | The default gateway for the appliance.                                                                                                                                                    |
| DNS Domain name                     | The domain name for the appliance.                                                                                                                                                        |
| Domain search path                  | The domain search path(s) for the appliance.                                                                                                                                              |

| Setting             | Description                        |
|---------------------|------------------------------------|
| Domain name servers | The DNS servers for the appliance. |

- On the **Ready to complete** page, click **Finish** and wait for the process to complete.
- When the SDDC Manager appliance deployment completes, expand the management folder.
- Right-click the SDDC Manager appliance, select **Power > Power On**.

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

## Update vCenter SSH Keys in SDDC Manager Backup for VCF Instance Backup and Restore

To enable a restore of SDDC Manager, you must update the SSH keys for the new management vCenter appliance in the backup archive.

### Prerequisites

Verify that the vCenter restore operation is complete.

### Manual Procedure

- Copy the encrypted backup file to the `/tmp` folder on the newly-deployed SDDC Manager appliance.
- Run the following command to extract the backup archive.  

```
OPENSSL_FIPS=1 openssl enc -d -aes-256-cbc -md sha256 -in <backup_archive_name>.tar.gz | tar -xz
```

  
When prompted, enter the `encryption_password`.
- Retrieve the new **ecdsa-sha2-nistp256** and **ssh-rsa** SSH keys for the management vCenter.  

```
ssh-keyscan <management_vc_fqdn>
```
- Open the backup of the known hosts and replace the existing SSH keys with the new SSH keys for the management vCenter from Step 3, and save the file.  

```
vi <backup_archive_folder_name>/appliancemanager_ssh_knownHosts.json
```
- Rename the original backup archive.  

```
mv <backup_archive_name>.tar.gz <backup_archive_name>.tar.gz.original
```
- Re-encrypt the backup folder as an archive using the same archive name.  

```
tar -cz <backup_archive_folder_name> | OPENSSL_FIPS=1 openssl enc -aes-256-cbc -md sha256 -out <backup_archive_name>.tar.gz
```

### PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- [Management Domain Restore Plan](#)

## Restore SDDC Manager from a File-Based Backup for VCF Instance Backup and Restore

You restore the file-based backup on the newly-deployed SDDC Manager appliance.

## Manual Procedure

1. Open an SSH connection to the SDDC Manager appliance as **vcf** user.
2. Change to **root** user and edit the `restore_status.json` file.
3. 

```
su -
cd /opt/vmware/sddc-support/backup
cp restore_status.json restore_status.json.bak
vi restore_status.json
```
4. Delete the “PostRestoreNfsRefresh” task in two places.
5. Save the file.
6. Obtain the authentication token from the SDDC Manager appliance by running the following command so that you can perform the restore process.
 

```
TOKEN=$(curl https://<sddc_manager_fqdn>/v1/tokens -k -X POST -H "Content-Type: application/json" -d '{
 "username": "admin@local",
 "password": "<admin@local_password>"
}' | awk -F "\"" '{ print $4}')
```
7. Run the following command to verify the token.
 

```
echo $TOKEN
```
8. Run the following command to start the restore process.
 The command output contains the ID of the restore task.
 

```
RESTOREID=$(curl https://<sddc_manager_fqdn>/v1/restores/tasks -k -X POST -H "Content-Type: application/json" -H "Authorization: Bearer $TOKEN" \
-d '{
 "elements": [{
 "resourceType": "SDDC_MANAGER"
 }],
 "backupFile": "<backup_file>",
 "encryption": {
 "passphrase": "<encryption_password>"
 }
}' | jq '.id' | cut -d '"' -f 2)
```
9. Monitor the restore task by using the following command until the status becomes Successful.
 

```
curl https://<sddc_manager_fqdn>/v1/restores/tasks/$RESTOREID -k -X GET -H "Content-Type: application/json" -H "Authorization: Bearer $TOKEN" | jq .pp
```

## PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of [here](#).

- [Management Domain Restore Plan](#)

## Update SSH Host Keys for VCF Instance Backup and Restore

For each ESX host that you rebuilt as part of restoring the management cluster or a cluster in a workload domain, you must update the SSH host keys in the `known_hosts` file.

- Download `refreshsshkeys.py` from Broadcom knowledge base article [79004](#).

1. Enable SSH on all hosts in the cluster.
2. Refresh the SSH keys for the hosts saved in SDDC Manager.
  - a) Copy the `refreshsshkeys.py` file to the `/tmp` directory on the SDDC Manager appliance by using a secure file copy utility as the **vcf** user.
  - b) Open an SSH connection to the SDDC Manager appliance as the **vcf** user
  - c) Switch to **root** and run `refreshsshkeys.py`.
 

```
su -
cd /tmp
"yes" | python refreshsshkeys.py
```
  - d) Alternatively, run the following command to update all new SSH keys with the need to accept each individual key.
 

```
python refreshsshkeys.py
```

## Synchronize the SDDC Manager Service Account on the ESX Hosts for VCF Instance Backup and Restore

The service account of SDDC Manager on the hosts must be in sync with the SDDC Manager inventory. You must manually set a new password on each ESX host and perform a password remediate operation in SDDC Manager.

## UI Procedure

1. If the service account does not exist, create a service account on the ESX host.
  - a. Log in to the first ESX host in the cluster using the host client as **root**.
  - b. Navigate to **Manage > Security & users > Users**, and click **Add user**.
  - c. Enter the SDDC Manager service user name and password, and click **Add**.
 

**Note:**

The service account user name must be in the **svc-vcf-ESX\_hostname** format.
2. If the service account exists, set a new password on the ESX host.
  - a. Log in to the first ESX host in the cluster using the host client as **root**.
  - b. In the host client on the first ESX host, navigate to **Manage > Security & users > Users**.
  - c. Select the SDDC Manager service account **svc-vcf-ESX\_hostname** and click **Edit user**.
  - d. Set a new password and click **Save**.
3. Perform password remediation in SDDC Manager.
  - a. Log in to SDDC Manager.
  - b. Navigate to **Security > Password Management > ESX**.
  - c. Locate the service account from Step 1 or Step 2, click the vertical ellipsis, and click **Remediate**.
  - d. Enter the password that is used in Step 1 or Step 2 and click **Remediate**.
 

Wait for the password remediation to complete.
4. Repeat the steps for all hosts in the workload domain.

## PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of [here](#).

- [Management Domain Restore Plan](#)
- [Workload Domain Restore Plan](#)

## Update the Backup Configuration in SDDC Manager for VCF Instance Backup and Restore

If the SFTP backup target for SDDC Manager and NSX has changed, you must update the SSH key for the backup configuration in SDDC Manager.

1. Log in to the SDDC Manager UI.
2. Navigate to **Administration > Backup**
3. On the **Site Settings** tab, click **Edit**.
4. Enter the new backup target details and click **Save**.

## VCF Operations Restore Tasks

Tasks related to the restore of VCF Operations fleet management and VCF Operations nodes.

### Restore VCF Operations from an Image-Based Backup for VCF Instance Backup and Restore

Restore a VCF Operations simple or high availability deployment by using an image-based backup tool.

Prerequisites:

- Available backup data: Ensure that you have a recent backup of your VCF Operations cluster.
- VCF Operations cluster configuration: Ensure that you know the details of your VCF Operations cluster configuration, such as cluster nodes and user credentials.
- Backup software: According to the backup solution you are using, ensure it is compatible with your VCF Operations version.

**Table 956: VCF Operations Failure Scenarios**

| Component                       | Deployment Model  | Failure Scenario  | Restore Process                              |
|---------------------------------|-------------------|-------------------|----------------------------------------------|
| VCF Operations fleet management | Single appliance  | Single node lost  | Restore from image-based backup              |
| VCF Operations                  | Simple            | Primary node lost | Restore primary node from image-based backup |
|                                 |                   | Collector lost    | Restore collector from image-based backup    |
|                                 | High Availability | Primary node lost | Restore primary node from image-based backup |
|                                 |                   | Replica node lost | Restore replica node from image-based backup |
|                                 |                   | Data node lost    | Restore data node from image-based backup    |
|                                 |                   | Collector lost    | Restore collector from image-based backup    |

1. If existing VCF Operations virtual machines are running in the management domain, power them off and rename or delete them.
2. Access your backup software interface.
3. Locate the backup of the VCF Operations nodes.
4. Initiate the restore process.  
Do not power on the VMs until all VMs are restored.

## Verify VCF Operations State for VCF Instance Backup and Restore

After a successful restore of VCF Operations, you verify the state of the analytics cluster and each integration adapter.

The cluster must be online and all integration adapters must be in a collecting state.

1. Verify the VCF Operations cluster state.
  - a) Log in to the VCF Operations admin console at [https://vcf\\_ops\\_primary-node-fqdn/admin/](https://vcf_ops_primary-node-fqdn/admin/) with the admin user.
  - b) On the system status page, verify that all nodes are in a **Running** state and their status is **Online**.
  - c) Navigate to Cloud Proxies and verify that the **Health Status** for each cloud proxy is **Healthy**.
2. Verify the VCF Operations integration adapters state.
  - a) Log in to the VCF Operations interface at [https://vcf\\_ops\\_primary-node-fqdn/](https://vcf_ops_primary-node-fqdn/) as a user assigned Administrator role.
  - b) Navigate to **Administration > Integrations**.
  - c) Expand each integration and ensure that their status is **OK**.

## VCF Identity Broker Restore Tasks

Tasks related to the restore of VCF Identity Broker.

### Prepare to Restore VCF Identity Broker

Before you deploy a VCF Identity Broker to restore to, you must delete the existing VCF Identity Broker environment.

#### Delete VCF Identity Broker

1. Log in to VCF Operations at [https://vcf\\_ops\\_primary-node-fqdn/](https://vcf_ops_primary-node-fqdn/) with the **admin** user.
2. Navigate to **Fleet Management > Lifecycle**, click the vertical ellipsis on the **identity broker** component, and click **Delete**  
**Note:**  
Do not select the checkbox to delete the associated VMs from vCenter.
3. If the delete operation fails, click **Retry** and set **Skip Task** to **true**.

#### Enable VCF Identity Broker Redeployment

Connect to the VCF Operations fleet management appliance using SSH as the **root** user, and run the following command.

```
touch /var/lib/vrlcm/SKIP_VIDB_SPECIFIC_VALIDATION
systemctl restart vrlcm-server
```

### Deploy a New VCF Identity Broker Instance for VCF Instance Backup and Restore

You deploy a new VCF Identity Broker to accommodate the restore from backup.

#### Procedure

Before starting the appliance deployment, download the VCF Identity Broker installation binary to the fleet management appliance. See [Downloading VCF Management Components into Binary Management](#). Then, select **Fleet Management > Lifecycle** and click **Add** on **identity broker**.

The deployment includes the following stages.

1. Deployment
  - Installation type: New Install
  - Version: 9.0.0.0
  - Deployment type: Small.
2. Certificate: Select the certificate for deployment. If a certificate is not available, click **Add Certificate** to create a one.
3. Infrastructure
  - Select vCenter.
    - vCenter servers with any of the following conditions are not listed:
      - VCF Identity Broker deployed in vCenter using the VCF Operations API.
      - vCenter version greater than 9.0
      - vCenter associated with SDDC Manager
      - vCenter used by another VCF Identity Broker instance. To use the vCenter, delete the associated VCF Identity Broker instance.
  - Select Cluster: Select the cluster in the management domain vCenter for deployment.
  - Select Folder
  - Select Resource Pool
  - Select Network
  - Select Datastore
4. Network
  - Domain Name
  - Domain Search Path
  - DNS Servers: If the required DNS server is not listed, add it.
  - Time Sync Mode: Point to an NTP Server or Host Time
  - IPv4 Details:
    - Default IPv4 Gateway
    - IPv4 Netmask
5. Components
  - Install identity broker:
    - Add Password: If you do not have a password, click **Add Password** to create a new one. Password must be a minimum of 15 characters.
    - Add Certificate: Select the certificate for deployment. To create a new certificate, click **Add Certificate**.
  - Component Properties
    - FQDN: Component FQDN
    - Certificate: Accept the automatic selection.
    - Component Password: Needed for the `vmware-system-user` and the `operator` account. Password must be a minimum of 15 characters.
  - Components
    - VM Node Prefix: A unique identifier used to name component nodes. To avoid conflicts and maintain proper node identification within the cluster, ensure that the node prefix is unique.
    - Internal Cluster CIDR: IP Address range used for internal cluster communication within the cluster. Choose a range that does not conflict with existing networks in infrastructure.
    - Primary VIP: Used for accessing component services. Should resolve to the Component Properties FQDN
    - Additional VIPs: (optional) Add up to 2 VIPs.

- Cluster Node IP Pool: The Node IP Pool is a range of IP addresses allocated to cluster nodes. These IP's ensure that each node has a unique address for internal communication within the cluster. Additional IP's from the pool will be used during rolling cluster upgrades.
  - A minimum of 4 IP addresses needed, maximum number of IP's is 7.
  - Three of these IP addresses will be used for deploying cluster nodes hosting VCF Identity Broker component. Spare IP addresses will be used during rolling upgrades and patches.

To run the precheck, click **Next**.

1. Precheck: If the precheck completes successfully, click **Next**.
2. Summary: To trigger deployment after reviewing the summary, click **Submit**.

### Restore VCF Identity Broker from a File-Based Backup for VCF Instance Backup and Restore

The following process can be used to restore a VCF Identity Broker cluster instance if a failure occurs.

- The VCF Identity Broker appliances must be in a running state.
- You must have at least one recent backup.
- The following components must be available before proceeding with the restore of VCF Identity Broker:
  - vCenter
  - VCF Operations
  - VCF Operations fleet management

A list of potential failure scenarios and corresponding restore processes.

**Table 957: VCF Identity Broker Appliance Cluster Failure Scenarios**

| Failure Scenario                           | Restore Process                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| VCF Identity Broker application corruption | <ul style="list-style-type: none"> <li>• Restore from SFTP to existing nodes</li> </ul>                                        |
| Single node lost                           | <ul style="list-style-type: none"> <li>• No action required</li> <li>• Failed node will be automatically redeployed</li> </ul> |
| Two nodes lost                             | <ul style="list-style-type: none"> <li>• Deploy a new VCF Identity Broker</li> <li>• Restore from SFTP to new nodes</li> </ul> |
| Three nodes lost                           | <ul style="list-style-type: none"> <li>• Deploy a new VCF Identity Broker</li> <li>• Restore from SFTP to new nodes</li> </ul> |

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned Administrator role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. Under components, click VCF Identity Broker.
4. On the component summary page, click the horizontal ellipsis and click **Backup & Restore > Restore**.
5. On the Restore pane, select an option to restore VCF Identity Broker from a specific backup or the latest backup.
  - Restore from the latest backup on the cluster
  - Pass the backup path using `vcf/backups/cluster-name/version/component-name/timestamp`
  - Pass the backup path using `vcf/backups/cluster-name/version/component-name/timestamp/<full-backup-file-path.tgz>`

- Click **Restore**.
- Monitor the restore task from the **Tasks** pane.

## VCF Automation Restore Tasks

Tasks related to the restore of VCF Automation.

### Prepare to Restore VCF Automation

Before you can deploy a new VCF Automation to restore to, you must delete the existing VCF Automation deployment.

#### Procedure

- Log in to the VCF Operations console at [https://<vcf\\_ops\\_primary-node-fqdn>/as/admin](https://<vcf_ops_primary-node-fqdn>/as/admin).
- Navigate to **Fleet Management > Lifecycle**, click the vertical ellipsis on the **automation** component, and click **Delete**.

#### Note:

Do not select the checkbox to delete the associated VMs from vCenter.

- If the delete operation fails, click retry and set **Skip Task** to **true**.

### Deploy a New VCF Automation Instance for VCF Instance Backup and Restore

You deploy a new VCF Automation to enable the restore from backup.

#### Procedure

Before starting the appliance deployment, download the VCF Automation install binary to the fleet management appliance. See [Downloading VCF Management Components into Binary Management](#). Then select **Fleet Management > Lifecycle** and click **Add on automation**.

The deployment includes through the following stages.

- Deployment
  - Installation type: New Install
  - Version: 9.0.0.0
  - Deployment type: Small.
- Certificate: Select the certificate for deployment. If a certificate is not available, click **Add Certificate** to create a one.
- Infrastructure
  - Select vCenter.
  - Select Cluster: Select the cluster in the management domain vCenter for deployment.
  - Select Folder
  - Select Resource Pool
  - Select Network
  - Select Datastore

- Network
  - Domain Name
  - Domain Search Path
  - DNS Servers: If the required DNS server is not listed, add it.
  - Time Sync Mode: Point to an NTP Server or Host Time
  - IPv4 Details:
    - Default IPv4 Gateway
    - IPv4 Netmask
- Components
  - Install automation:
    - Add Password: If you do not have a password, click **Add Password** to create a new one. Password must be a minimum of 15 characters.
    - Add Certificate: Select the certificate for deployment. To create a new certificate, click **Add Certificate**.
  - Component Properties
    - FQDN: Component FQDN
    - Certificate: Accept the automatic selection.
    - Component Password: Needed for the `vmware-system-user` and the `operator` account. Password must be a minimum of 15 characters.
  - Components
    - VM Node Prefix: A unique identifier used to name component nodes. To avoid conflicts and maintain proper node identification within the cluster, ensure that the node prefix is unique.
    - Internal Cluster CIDR: IP Address range used for internal cluster communication within the cluster. Choose a range that does not conflict with existing networks in infrastructure.
    - Primary VIP: Used for accessing component services. Should resolve to the Component Properties FQDN
    - Additional VIPs: (optional) Add up to 2 VIPs.
    - Cluster Node IP Pool: The Node IP Pool is a range of IP addresses allocated to cluster nodes. These IP's ensure that each node has a unique address for internal communication within the cluster. Additional IP's from the pool will be used during rolling cluster upgrades.
      - A minimum of 4 IP addresses needed, maximum number of IP's is 7.
      - Three of these IP addresses will be used for deploying cluster nodes hosting VCF Identity Broker component. Spare IP addresses will be used during rolling upgrades and patches.

To run the precheck, click **Next**.

- Precheck: If the precheck completes successfully, click **Next**.
- Summary: To trigger deployment after reviewing the summary, click **Submit**.

### Restore VCF Automation from a File-Based Backup for VCF Instance Backup and Restore

The following process can be used to restore a VCF Automation instance if a failure occurs.

A list of potential failure scenarios and corresponding restore processes.

**Table 958: VCF Automation Failure Scenarios**

| Failure Scenario          | Restore Process                                                                       |
|---------------------------|---------------------------------------------------------------------------------------|
| VCF Automation nodes lost | <ul style="list-style-type: none"> <li>Restore from SFTP to existing nodes</li> </ul> |

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned **Administrator** role.
2. Navigate to **Fleet Management > Lifecycle > VCF Management > Components**.
3. Under components, click VCF Automation.
4. On the component summary page, click the horizontal ellipsis and click **Backup & Restore > Restore**.
5. On the Restore pane, select an option to restore VCF Automation from a specific backup or the latest backup.
  - Restore from the latest backup on the cluster
  - Pass the backup path using `vcf/backups/cluster-name/version/component-name/timestamp`
  - Pass the backup path using `vcf/backups/cluster-name/version/component-name/timestamp/<full-backup-file-path.tgz>`
6. Click **Restore**.
7. Monitor the restore task from the **Tasks** pane.

### Update Connections in VCF Automation for VCF Instance Backup and Restore

After you store VCF Automation as a part of the VCF Instance Backup and Restore operation, to ensure communication between VCF Automation and the VCF Instances it is connected to is active, you refresh the connections.

#### Procedure

1. Log in to the VCF Automation provider portal as a user with the **System Administrator** role.
2. Under Administration, click **VCF Instances**.
3. Click the vertical ellipsis next to the first VCF instance, and click **Rotate Service Account**.
4. Repeat for each VCF Instance.
5. From the left menu, click **Connections**.
6. Verify each vCenter connection is enabled and connected.

### vSphere Supervisor Restore Tasks

Tasks related to the restore of vSphere Supervisor.

### Mount the NFS Share for the VKr Content Library for VCF Instance Backup and Restore

If you are using an NFS datastore for the content library with the VMware vSphere Kubernetes releases (VKrs), mount the relevant NFS share.

#### Procedure

1. Log into vCenter for the workload domain as a user with the **administrator** role.
2. In the **Hosts and Clusters** inventory, right- click the vSphere cluster and select **Storage > New Datastore**.
3. On the **Type** page, select NFS and click **Next**.
4. On the **NFS version** page, select your NFS version and click **Next**.
5. On the **Name and configuration** page, enter the datastore details and click **Next**.
6. On the **Hosts accessibility** page, select all hosts in the cluster and click **Next**.
7. On the **Ready to complete** page, review the settings and click **Finish**.

### Update the VKr Content Library Datastore Mapping for VCF Instance Backup and Restore

The NFS datastore backing the vSphere content library has a new moref-id. You update the vCenter Content Library Datastore Mapping in the vCenter database to reflect the change.

#### Procedure

1. Log in to the vCenter appliance as **root** over SSH or directly to appliance console.
2. Type **shell** to switch to the BASH Shell.
3. Connect to the VCDB by running this command.
 

```
/opt/vmware/vpostgres/current/bin/psql -d VCDB -U postgres
```
4. Find the id of your content library.
 

```
select id from cl_library where name='content-library-name';
```
5. Find the id of the old datastore that corresponds to the content library id from the previous step.
 

```
select * from cl_library_storage;
```
6. Find the storageuri to be updated with the new datastore details.
 

```
select * from cl_storage;
```
7. Get the storageuri for the new NFS datastore.
  - a. In a browser, log into the vCenter server where the new NFS datastore is located.
  - b. Navigate to Inventory > Storage and click on the datastore.
  - c. In the URL select and copy the following string.
 

```
Datastore:datastore-4682698:c7481195-b0a1-4c74-a67f-c5bce8b2145d
```
8. Update the datastore mapping in the `cl_storage` table.
 

```
update from cl_storage SET storageuri='storageuri-from-step-7' where id='storage-id-from-step-5';
```

### Verify vSphere Lifecycle Manager Image Compliance for VCF Instance Backup and Restore

Before you restore the vSphere Supervisor, you must verify each vSphere cluster is in compliance with the assigned vSphere Lifecycle Manager image.

1. In the vSphere Client, go to the **Hosts and Clusters** inventory and select the first cluster.
2. On the **Updates** tab, select **Hosts > Image** verify that the cluster is in compliance with the attached image.
3. If it is not compliant, click **Remediate**.

### Restore the vSphere Supervisor Control Plane for VCF Instance Backup and Restore

You can restore the Supervisors control pane in vCenter from the backup files of the vCenter system itself.

Configure recording the Supervisor state from the vCenter management interface file-based backups.

Restoring the Supervisor control plane in vCenter and restoring the state of vCenter are two different workflows. Restoring vCenter does not lead to restoring the Supervisor control plane.

1. In the vSphere Client, go to **Supervisor Management**.
2. Select **Supervisors**, select the Supervisor to be restored, and click **Restore**.
3. Enter the backup details.

| Option         | Description                                            |
|----------------|--------------------------------------------------------|
| <b>vCenter</b> | Select the vCenter system that manages the Supervisor. |



| Option                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Backup selection</b> | <ul style="list-style-type: none"> <li>Select <b>Browse backup server folder</b> to upload the files that are stored in the root folder of the backup file server configured with this vCenter system.</li> <li>Select <b>Single backup location</b> to upload a particular backup file, and then enter the URL to that backup file.</li> <li>Select <b>Use backup location and user name from vCenter backup schedule</b> to populate the root folder location and user name for the backup location that are configured with vCenter.</li> </ul> |
| <b>Location</b>         | Enter the location of the backup root folder.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Username</b>         | Enter the user name to access the backups                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Password</b>         | Enter the password for that user name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

- Click **Next**.
- Select a backup file to restore from and click **Next** to start the download of the backup file.
- Select the Supervisor to restore and click **Next**.
- Review the settings and click **Finish**.

The Supervisor goes back to configuring state and all the control plane VMs are deleted and re-deployed with the data from the backup file. You can monitor the process by clicking **view** in the **Config Status** column.

What to do next:

- Restore vSphere Pods. See [Backup and Restore vSphere Pods Using the Velero Plugin for vSphere](#).
- Restore Stateless and Stateful Workloads on a vSphere Supervisor Cluster. See [Backing Up and Restoring VKS Cluster Workloads](#).
- Restore VM Service VMs on a vSphere Supervisor cluster. See [Backing Up and Restoring VM Service VMs in vSphere Supervisor](#).

### Remediate vSphere Supervisor Nodes for VCF Instance Backup and Restore

To ensure the vSphere Supervisor spherelet configuration and certificate are pushed to each ESX host, you must remove and add again each host as a Supervisor node.

- Retrieve the Supervisor **root** password from vCenter.
  - Open an SSH connection to vCenter as the **root** user.
  - Run the following command.
 

```
/usr/lib/vmware-wcp/decryptK8Pwd.py
```
- Using the **root** password from the previous step, open an SSH connection to the Supervisor.
- List the Supervisor nodes by running the following command.
 

```
kubectl get nodes
```
- In the vSphere Client, place the first host in the list into maintenance mode.
- In the SSH session to the Supervisor, run the following command.
 

```
kubectl delete node <esx-fqdn>
```

- Return to the vSphere Client and exit the host from maintenance mode.
- Repeat these steps for all Supervisor nodes.

### Recreate the NSX NCP Pods for VCF Instance Backup and Restore

The NSX NCP Pods might be in crashed state after the Supervisor restore and you must restore them back to running state.

#### Procedure

- Retrieve the Supervisor **root** password from vCenter.
  - Open an SSH connection to vCenter as the **root** user.
  - Run the following command.
 

```
/usr/lib/vmware-wcp/decryptK8Pwd.py
```
- Using the **root** password from the previous step, open an SSH connection to the Supervisor.
- List the NSX-NCP Pod by running the following command.
 

```
kubectl get pods -n vmware-system-nsx | grep nsx-ncp
```
- Restart an NSX-NCP Pod by deleting it.
 

When you delete an NSX-NCP Pod, it is automatically recreated.

```
kubectl delete pod <nsx-ncp pod name> -n vmware-system-nsx
```
- After the NSX-NCP Pod is recreated, confirm it is now in a **Running** state:
 

```
kubectl get pods -n vmware-system-nsx | grep nsx-ncp
```
- Repeat these steps for all NSX-NCP Pods.

### Verification and Remediation Tasks

Tasks related to the verification and remediation after the restore of a VCF Instance.

#### Check SDDC Manager Health After VCF Instance Backup and Restore

After rebuilding a cluster, you should perform a health check using the SoS utility on SDDC Manager.

- Open an SSH connection to the SDDC Manager appliance as **vcf**.
- Run the following commands:
 

```
su -
cd /opt/vmware/sddc-support
./sos --health-check --domain-name ALL --skip-cert-check
```
- Remediate any issues reported by the health check.

#### Delete the Temporary vCenter Instance After VCF Instance Backup and Restore

You can now delete the temporary vCenter deployed at partial deployment.

Power off and delete the temporary vCenter.

#### Perform SDDC Manager Prechecks After VCF Instance Backup and Restore

Perform an upgrade pre-check on each workload domain to ensure that the system is in a healthy state.

1. Log in to the SDDC Manager UI as **administrator@vsphere.local**.
2. In the navigation pane, click **Inventory > Workload Domains**.
3. On the **Workload Domains** page, click the workload domain where you want to run the pre-check operation.
4. On the **Summary** tab, click the **Updates** tab.
5. Click **Run Precheck** to validate that the environment is ready to be upgraded.
6. From the **Target Version** drop-down menu, select **General Upgrade Readiness**.
7. Configure the precheck to run on the entire workload domain.
8. Click **Run Precheck**.
9. Click **View Status** to see detailed tasks and their status.
10. To see details for a task, click the Expand arrow.

If a precheck task failed, fix the issue, and click **Retry Precheck** to run the task again. You can also click **Precheck Failed Resources** to retry all failed tasks.

## FIPS Configuration for VCF Components

VMware Cloud Foundation supports the Federal Information Processing Standards (FIPS) 140-2, which specify security requirements for cryptographic modules.

Many VCF components support the FIPS standard inherently, and cannot be configured in ways that do not meet the standard. For example, ESX and VMware NSX. Other components may require enablement for FIPS compliance or have varying behavior depending on how they were deployed.

The following components have FIPS enabled by default:

- ESX
- vCenter
- NSX
- VCF Operations HCX
- VMware Live Recovery
- Data Services Manager

The table below summarizes FIPS enablement for components where FIPS is either not enabled by default or can be disabled in certain circumstances.

| Component               | FIPS Configuration                                                                                                                                                                                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SDDC Manager            | In new VCF 9.0 deployments, FIPS compliance in SDDC Manager is on by default and cannot be turned off.<br>For deployments upgraded from VCF 5.x, the FIPS setting remains the same as it was in the original deployment. This setting cannot be altered after upgrade. |
| VCF Operations          | FIPS compliance is off by default.<br>You can activate FIPS compliance during deployment by selecting the <b>Activate FIPS Mode</b> option during deployment.                                                                                                          |
| VCF Operations for logs | FIPS compliance is off enabled by default.<br>You can enable FIPS compliance during deployment by selecting the <b>On</b> option for <b>FIPS Compliance</b> . See <a href="#">Deploying the VCF Operations for Logs Appliance</a> .                                    |

| Component                                | FIPS Configuration                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VCF Operations for networks              | VCF Operations for networks uses FIPS compliant cryptographic algorithms for internal connections. FIPS compliance for external connections is off by default.<br>You can enable FIPS compliance for external connections by navigating to <b>Settings &gt; System Configuration &gt; FIPS Mode for External Connections</b> and toggling the setting to <b>On</b> .                                                                |
| Cloud Proxy for VCF Operations           | FIPS compliance is not enabled by default. You can enable FIPS compliance for cloud proxies after first enabling FIPS compliance for VCF Operations. For instructions on enabling FIPS compliance, see <a href="#">Using the Cloud Proxy Command-Line Interface</a> .                                                                                                                                                               |
| VCF Automation                           | In new VCF 9.0 deployments, FIPS compliance in VCF Automation is enabled by default and cannot be turned off.<br>For deployments upgraded from VMware Aria Operations 18.x, the FIPS setting remains the same as it was in the original deployment. This setting cannot be altered after upgrade.                                                                                                                                   |
| VMware vSphere Kubernetes Service        | VKS 3.3 and later provide the ability to enable OS FIPS mode by configuring the osConfiguration cluster class variable. See <a href="#">osConfiguration</a> for more details.                                                                                                                                                                                                                                                       |
| VMware Virtual Disk Developer Kit (VDDK) | By default, VDDK does not enable FIPS. You can configure it by setting the option <code>vixDiskLib.ssl.enableSslFIPS=1</code> in the VDDK configuration file. See the instructions provided with the development kit at <a href="https://developer.broadcom.com/sdks/vmware-virtual-disk-development-kit-vddk/latest">https://developer.broadcom.com/sdks/vmware-virtual-disk-development-kit-vddk/latest</a> for more information. |

## FIPS Configuration for VMware vSphere Foundation Components

You can activate Federal Information Processing Standards (FIPS) for the VCF Operations component of VMware vSphere Foundation to make your environment FIPS compliant. You can also activate firewall hardening.

### Activate FIPS

You can activate FIPS in the VCF Operations cluster at the time of installation or after VCF Operations is up and running. Adding FIPS at installation is less intrusive because the cluster has not yet started.

If the cluster is running, to activate FIPS, you must take the cluster offline.

FIPS mode is supported in Cloud Proxy. You can continue using your cloud proxy after enabling FIPS for the VCF Operations cluster.

1. In a Web browser, navigate to the master node administration interface. `https://master-node-name-or-ip-address/admin`.
2. Enter the VCF Operations administrator username of admin.
3. Enter the VCF Operations administrator password and click **Log In**.
4. Click **Administrator Settings**.  
**Note:** The **Activate FIPS** button is deactivated when the cluster is running.
5. Click **Activate FIPS** after you take your cluster offline.  
**Note:** After you activate FIPS, you cannot deactivate the FIPS mode in the current setup. To revert to a FIPS deactivated setup, you must re-deploy VCF Operations.
6. In the **Are you sure you want to activate FIPS** dialog box, read the note and provide your consent for enabling FIPS and then click **Yes**.  
**Note:** After you activate FIPS, the cluster restarts and is not be available during this time. The cluster nodes are rebooted and after the cluster is online, all the nodes are FIPS activated.

### Activate Firewall Hardening

Activating firewall hardening restricts network access to internal services in VCF Operations.

1. In a Web browser, navigate to the master node administration interface. <https://master-node-name-or-ip-address/admin>.
2. Enter the VCF Operations administrator username of admin.
3. Enter the VCF Operations administrator password and click **Log In**.
4. Click **Administrator Settings**, and then click **Security Settings** from the **Administrator Settings** page.
5. Click **Activate Firewall Hardening**.

## Scaling Up and Scaling Out Management Nodes in VCF

Use fleet management to perform Day-N operations such as scaling up and scaling out management nodes in VCF management components.

### Vertical scale up for VCF Operations and VCF Operations for logs

To scale up resource allocations such as vCPUs, RAM, and disk capacity in cluster nodes, you can use VCF Operations fleet management. The **Vertical Scale Up** option is supported for VCF Operations and VCF Operations for logs.

1. Under **Fleet Management > Lifecycle > VCF Management > Components**, select the integrated component that you want to update.
2. On the component details screen that appears, click the ellipses (...) and select **Vertical Scale Up**.
3. The **Proceed to Vertical Scale Up** pop-up window appears. Acknowledge the downtime warning and click **Trigger Inventory Sync** to synchronize the VCF Operations fleet management appliance with the latest data from the cluster.
  - a) When the inventory sync completes successfully, click **Proceed**.
4. Select the nodes to be scaled up.
  - For VCF Operations , you scale up the **operations primary nodes**.
  - For VCF Operations for logs, you scale up the **operations-log primary nodes**.
5. For vertical scale up details, the current cluster size appears. To scale up, provide the following information.
  - Scale up Size: Select a larger size such as Medium, Large, or Extra large.
  - Additional disk size: Provide the additional size in GB.
  - Advanced settings: Select the datastore for nodes and provide an additional disk size in GB.
6. Click **Next**.
7. Click **RUN PRECHECK**.

#### Note:

If the data validation is not successful, perform the recommended actions and run the precheck again.

8. After a successful precheck run, review the summary of your vertical scale up request and click **Submit**.

Vertical scale up begins. During the process, the VCF Operations nodes are restarted. During the downtime, VCF Operations is not available.

### Scale VCF Automation

You use the fleet management appliance to scale a VCF Automation deployment. In addition to increasing the size of the environment from small to medium or medium to large, scale offers options to increase CPU and memory, and to add nodes.

1. Under **Fleet Management > Lifecycle > Components**, select **automation** for the integrated component.
2. Click the ellipses (...) and select **Scale**.
3. On Select Size, choose a larger target deployment type such as Medium or Large.
4. On Configure IP addresses, add the following information and click **Next**.
  - For Additional VIPs: Click Add IPs to your VIP pool. You can add up to two more IPs.
  - For Cluster Node IP Pool: Click to add IPs to the Cluster Node Pool. The deployed nodes host the VCF Automation containers.
    - A small deployment requires a minimum of two IPs.
    - A medium deployment requires a minimum of four IPs.
    - A large deployment requires a minimum of four IPs.
5. Click **RUN PRECHECK**.

#### Note:

If the data validation is not successful, perform the recommended actions and run the precheck again.

6. After a successful precheck run, review the summary of your VCF Automation scale out request and click **Submit**.

### Add Nodes to VCF Operations

You can add nodes to your VCF Operations, VCF Operations for logs, and VCF Operations for networks.

1. On the **Components** tab under **Fleet Management > Lifecycle**, select the integrated component that you want to update.
2. On the component details screen that appears, click **Add Nodes**.
3. The **Proceed to Add Nodes** pop-up window appears. Read the downtime warning and click **Trigger Inventory Sync** to synchronize the VCF Operations fleet management appliance with the latest data from the cluster. Click Submit to start the inventory sync.
  - a) Click **Submit** to start the inventory sync.
  - b) When the inventory sync completes successfully, click **Proceed**.
4. On the Infrastructure pane, review the existing infrastructure and click **Next**.
 

If the Cluster, Network, and Datastore are not selected, you must make selections before clicking Next.
5. On the Network pane, review the existing network information and click **Next**.
6. On the Components pane, click the down arrow for Component Properties to display the existing nodes. Click the plus sign to add a node.
  - For VCF Operations , select from the following node types and complete the FQDN and IP Address fields for each:
    - Operations replica node. Adding a replica node requires a certificate change on the cluster.
    - Operations data node. Adding a data node requires a certificate change on the cluster. See [Managing Certificates in VMware Cloud Foundation](#).
    - Cloud Proxy. Adding a cloud proxy node also requires a Deployment type.
    - Collector groups. Adding a collector group automatically adds a new cloud proxy node to the group. The new cloud proxy node also requires a Deployment type.
  - For VCF Operations for logs, each added node requires a minimum of two operations-logs worker nodes. Complete the FQDN and IP Address fields for the two nodes.

7. Click **RUN PRECHECK**.**Note:**

If the data validation is not successful, perform the recommended actions and run the precheck again.

8. After a successful precheck run, review the summary of your add nodes request and click **Submit**.**Scale Out a Standalone VCF Operations Orchestrator Deployment**

You can increase the availability and scalability of your configured standalone VCF Operations orchestrator deployment by scaling it out.

- Download, deploy, and configure an VCF Operations orchestrator instance. Go to [Deploying VCF Operations Orchestrator](#).
- Download and deploy two additional VCF Operations orchestrator instances.
- Configure a load balancer server, such as Avi Load Balancer. For more information on configuring Avi Load Balancer, go to the [Avi Load Balancer documentation](#).

## 1. Configure the primary node.

- Log in to the appliance of your configured VCF Operations orchestrator deployment as **root**.
- Unregister the current authentication provider by running the `vracli vro authentication unregister` command. For more information, go to [Configuring the VCF Operations Orchestrator Authentication Provider](#).
- To set the load balancer, run the `vracli load-balancer set load_balancer_FQDN` command.
- Configure the authentication provider by running the `vracli vro authentication set` command.

**Note:**

Optionally, you can add the FQDN address of the load balancer server by adding the `--fqdn` parameter to the `vracli authentication set` command. For more information on how to configure the authentication provider, go to [Configuring the VCF Operations Orchestrator Authentication Provider](#).

- Stop all the services of the VCF Operations orchestrator instance by running the `/opt/scripts/deploy.sh --onlyClean` command.
- Optional: If your primary node uses a custom certificate, you must either set the certificate in the appliance or generate a new certificate. For more information, go to [Generate a custom TLS certificate for VCF Operations orchestrator](#).

**Note:** The file containing the certificate chain must be PEM-encoded.

## 2. Join secondary nodes to the configured instance.

- Log in to the VCF Operations orchestrator command line of the secondary node as **root**.
- To join the secondary node to the configured instance, run the `vracli cluster join primary_node_hostname_or_IP` command.

After running the `vracli cluster join`, you are asked for the root password of the secondary node.

- Repeat for the other secondary node.
- Finish the scale-out process.
    - Log in to the VCF Operations orchestrator appliance command line of the configured instance as **root**.
    - To confirm that all nodes are in a ready state, run the `kubectl -n prelude get nodes` command.
    - Run `/opt/scripts/deploy.sh` and wait for the script to finish.

You have scaled out your VCF Operations orchestrator deployment.

**Note:**

For a deployment with three VCF Operations orchestrator instances, the scaled out deployment can withstand one instance failing and still function. Two instances failing renders the VCF Operations orchestrator deployment non-functional.

**Add, Remove, or Move VCF Instances across VCF Fleets**

After you have configured a VCF fleet, you can add existing VCF Instances 5.x or later to that fleet, remove instances from the fleet either permanently or to move them between fleets.

To remove a VCF instance from a VCF fleet, you first disconnect it from the VCF Operations instance in that fleet. If you have VCF Automation in the source fleet consuming resources from the VCF Instance you are removing, you must first delete those objects in VCF Automation.

**Prerequisites**

- Verify that you have administrative access to the VCF Operations instance that manages your VCF fleet.
- Verify that your VCF Instance you want to add to the fleet is running version 5.x or later.

**Procedure**

- (Optional) If you want to move a VCF Instance to another fleet, you first remove it from the fleet it is part of.
  - Log in to the VCF Operations UI for the VCF instance you want to remove, as a user with the admin role.
  - Navigate to **Administration > Integrations**.
  - Expand **VMware Cloud Foundation**.
  - Click the vertical ellipsis for the VCF Instance you want to remove, and click **Delete**.
  - Navigate to **Administration > Cloud Proxies**.
  - Click the vertical ellipsis for the Cloud Proxy you want to remove, and click **Delete**.
  - Navigate to **Fleet Management > Lifecycle > Settings > Deployment Targets**.
  - Click the **Delete vCenter** icon to delete the management vCenter for the VCF Instance you want to remove.

**Important:**

If you want to add additional VCF Instances to this fleet, follow the steps in this KB <https://knowledge.broadcom.com/external/article/424495> to remove the deleted VCF Instance from the VCF fleet management appliance database.
- To add an existing VCF Instance to a VCF fleet, you integrate it with VCF Operations, deploy a cloud proxy and configure it.
  - Log in to the VCF Operations UI for that manages the VCF fleet.
  - Create an integration in VCF Operations for the new VCF instance being added and update the integration to use a system-managed service account.  
See [Configuring a Standalone VCF Account in VCF Operations](#).
  - Add the management domain vCenter of the VCF Instance you add as a deployment target in VCF Operations fleet management.  
See [Add a Deployment Target and Import VCF Operations to VCF Operations fleet management](#).
  - Deploy a VCF Operations collector/cloud proxy to the new instance and update the integration to use the new collector/collector group.  
See [Deploy the VCF Operations collector](#).